

CLASSIFYING $*$ -HOMOMORPHISMS I: UNITAL SIMPLE NUCLEAR C^* -ALGEBRAS

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ABSTRACT. We classify the unital embeddings of a unital separable nuclear C^* -algebra satisfying the universal coefficient theorem into a unital simple separable nuclear C^* -algebra that tensorially absorbs the Jiang–Su algebra.

This gives a new and essentially self-contained proof of the stably finite case of the unital classification theorem: unital simple separable nuclear C^* -algebras that absorb the Jiang–Su algebra tensorially and satisfy the universal coefficient theorem are classified by Elliott’s invariant of K -theory and traces.

OVERVIEW OF RESULTS

This is the first of a series of papers in which we give an abstract approach to classification results for simple separable nuclear C^* -algebras. This line of research, known as the Elliott classification program, has been a large-scale endeavor since the 1990s and seeks to obtain results for C^* -algebras analogous to the celebrated structure and classification theorems for amenable von Neumann algebras.

This paper handles the case of unital C^* -algebras. Our overarching objective is to give a new and comparatively self-contained proof of the unital classification theorem (Theorem A). This says that a suitable class of unital simple separable nuclear C^* -algebras is classified by operator algebraic K -theory (the non-commutative version of Atiyah and Hirzebruch’s topological K -theory) and its pairing with traces (thought of as non-commutative measures on C^* -algebras). We put the hypotheses of the theorem, especially $\mathcal{Z}$ -stability and the universal coefficient theorem (UCT) of Rosenberg and Schochet, into context in Section 1.1. At this point, we emphasize that these two conditions are both necessary: the classifiable class is maximal for such a result. Moreover, powerful tools exist to verify the hypotheses in a wide range of concrete situations (see Section 1.1.4). This classification theorem may be regarded as the C^* -algebraic analog of the Connes–Haagerup classification of injective factors ([48, 136]).

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Theorem A (The unital classification theorem; see Theorem 9.9). *Unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebras satisfying Rosenberg and Schochet’s universal coefficient theorem are classified by Elliott’s invariant consisting of K -theory and traces.*

This theorem was first obtained in 2015 (with slightly different, but equivalent hypotheses; see Section 1.1.4 and Section 1.2.7) by combining the papers [131, 132] (originally made available as [129]), [100, 287, 311, 310, 309] (and the large body of work on which these papers rely) with the Kirchberg–Phillips theorem [172, 236] from the 1990s. It is a capstone result for the Elliott classification program, involving decades of work by large numbers of researchers, and so is often attributed to “many hands”. Our approach follows an entirely different strategy — made possible by the recent developments of [267, 268, 37] — which we believe is both shorter and more conceptual.

Beginning with Elliott’s classification of approximately finite dimensional C^* -algebras, classification results for C^* -algebras have been built on classification results for morphisms. Our strategy in this paper fits the version of this framework pioneered by Rørdam in [253]: the main goal of the paper is the following classification of unital embeddings, from which Theorem A is quickly deduced. We discuss the need for the total invariant $\underline{KT}_u$ further below (in Sections 1.1, 1.2, and 1.3), and describe its components in detail in Sections 2.2, 2.3, 3.1, and 3.2.

Theorem B (Classification of unital embeddings; see Theorem 9.3). *Let A be a unital separable nuclear C^* -algebra satisfying the UCT and B be a unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebra. Up to approximate unitary equivalence, the unital embeddings $A \hookrightarrow B$ correspond bijectively with those morphisms of the total invariant $\underline{KT}_u$ — consisting of enriched K -theoretic and tracial data — mapping traces on B to faithful traces on A .*

Both Theorem A and Theorem B admit a dichotomy thanks to a special case of a beautiful theorem of Kirchberg: a C^* -algebra B as in Theorem B is either *purely infinite* or *stably finite*. Moreover, the tracial component of the invariants used in Theorems A and B detects these conditions: purely infinite simple C^* -algebras are traceless, while a deep theorem of Haagerup ([137]) combines with work of Blackadar and Handelmann ([137, 12]) to give traces on unital simple nuclear stably finite C^* -algebras. Accordingly, both theorems and their proofs split into these two cases. In the purely infinite setting, Theorem A was established independently by Kirchberg and by Phillips in the 1990s ([172, 175, 236]; see also Rørdam’s expository account in [255]). Subsequent work to reach Theorem A was then devoted to the stably finite case. The same is true for our proof. In this paper we only handle the stably finite aspect of Theorem A, and one obtains the entire unital classification theorem by combining this with the Kirchberg–Phillips theorem. Likewise, when B is purely infinite, Kirchberg obtained a classification of embeddings; our contribution is the stably finite counterpart, allowing for the unified statement of Theorem B.

The generality of the stably finite case of Theorem B is the headline new result of this paper. It simultaneously unifies and generalizes earlier classifications of embeddings of nuclear C^* -algebras into simple nuclear stably finite C^* -algebras by K -theoretic data. The key point is the abstract nature of the hypotheses: the theorem allows all separable nuclear C^* -algebras in the UCT class as domains of embeddings, whereas prior results such as [195, 199, 210] rely on internal structure for the

domains (either commutativity, or explicit inductive limit or tracial approximation structure), and often also for the codomains. While this paper was in preparation, Gong, Lin, and Niu independently obtained Theorem B when A is additionally simple ([130], building on [203]). Their work uses the power of the unital classification theorem to obtain very precise internal structure on UHF-stabilizations of A .

In fact, Kirchberg's embedding theorem is more general, with some hypotheses transferred from the domain and codomain to the morphisms. In addition to covering non-unital classification, the second paper in this series will extend the stably finite aspect of Theorem B to Kirchberg's framework. Following the passing of Eberhard Kirchberg, his long-term project [170] describing these, and many more results, will sadly remain unfinished. JG gives a new approach to Kirchberg's classification theorems in [118, 115].

The passage back from Theorem B to Theorem A is via an Elliott intertwining argument, for which it is vital that Theorem B has both existence and uniqueness components. That is, Theorem B specifies exactly which maps between total invariants are realized by injective *-homomorphisms, and then shows uniqueness of these. Moreover, although the total invariant $\underline{KT}_u$ contains more information than K -theory and traces, any map between K -theory and traces can be extended to $\underline{KT}_u$. In this way, the existence aspect of Theorem B yields the following:

Corollary C. *Let A be a unital separable nuclear C^* -algebra satisfying the UCT and B be a unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebra. Any morphism from the Elliott invariant of A to that of B that maps traces on B to faithful traces on A is realized by a unital injective *-homomorphism $A \rightarrow B$.*

We will put the main ingredients in these theorems into context in Section 1.1 of the introduction, connecting these with von Neumann algebras and describing the range of examples covered by Theorem A. We follow this with a brief history in Section 1.2 and then outline our methods in Section 1.3.

We first announced these results at NCGOA in Münster in 2018, and then in Oberwolfach ([286]). A summary is also given in the survey [304].

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1. INTRODUCTION

1.1. Context and examples. From the outset, the classification of simple nuclear C^* -algebras has been intimately tied to von Neumann algebras. Indeed, the first such result — Glimm’s classification of uniformly hyperfinite C^* -algebras ([125]) — is a direct analog of Murray and von Neumann’s uniqueness of the hyperfinite II_1 factor ([219]) allowing for the more refined invariants needed with C^* -algebras.

Later developments in C^* -algebra classification were similarly influenced by the spectacular successes on the von Neumann algebra side. Connes’ groundbreaking

work on injective von Neumann algebras ([48]) gives a clean, testable, and abstract characterization of hyperfiniteness and is a landmark of 20th century mathematics. It provides a game-changing enlargement of the scope of Murray and von Neumann’s uniqueness theorem and led to a complete classification of injective infinite factors (combining [48] with [44, 47, 46, 183] and Haagerup’s subsequent uniqueness of the hyperfinite III_1 factor in [136]). Beyond the impact in von Neumann algebras — which continues to the present day, for example in subfactor theory ([156, 242, 157]) and Popa’s deformation/rigidity theory ([244, 298]) — Connes’ theorem inspired classification results for amenable measurable dynamical systems ([50]) and is vital in developing approximation properties for C^* -algebras.

The unital classification theorem is the C^* -counterpart of the von Neumann algebra classification theorems. The hypotheses of unitality, simplicity, separability, and nuclearity in Theorem A correspond directly to the von Neumann algebra setting. All von Neumann algebras are unital, and factors are the simple von Neumann algebras. In the von Neumann algebra classification, one demands a separable predual; in both cases separability powers intertwining arguments (and is necessary, as shown by [112]). Finally, nuclearity corresponds to injectivity in Connes’ theorem: a C^* -algebra A is nuclear if and only if its bidual A^{**} is injective ([39, 40]).

The other two hypotheses — tensorial absorption of the Jiang–Su algebra $\mathcal{Z}$ and belonging to the UCT class — are of a more subtle nature. We describe these next.

1.1.1. $\mathcal{Z}$ -stability. A key step in Connes’ work is that every separably acting injective II_1 factor $\mathcal{M}$ absorbs the hyperfinite II_1 factor $\mathcal{R}$ tensorially, i.e., $\mathcal{M} \cong \mathcal{M} \bar{\otimes} \mathcal{R}$. This property was first extensively studied in [215], and such II_1 factors are said to be *McDuff*. In contrast, not all infinite dimensional unital simple separable nuclear C^* -algebras factorize as a non-trivial tensor product (examples without such factorizations include Rørdam’s famous construction of a unital simple separable nuclear C^* -algebra with both an infinite and finite projection from [256]). Yet tensorial absorption has been crucial in C^* -classification ever since Kirchberg’s Geneva theorems, one of which shows that for simple separable nuclear C^* -algebras, pure infiniteness is characterized by tensorial absorption of the Cuntz algebra $\mathcal{O}_\infty$. Moreover, “ $\mathcal{O}_\infty$ -stability” is an important ingredient in the Kirchberg–Phillips theorem.

The Jiang–Su algebra $\mathcal{Z}$, introduced in [154], is the stably finite counterpart of $\mathcal{O}_\infty$. A C^* -algebra A is said to be $\mathcal{Z}$ -stable if $A \cong A \otimes \mathcal{Z}$. Both $\mathcal{O}_\infty$ and $\mathcal{Z}$ are infinite dimensional, unital, simple, separable, nuclear, with the same K -theory as the complex numbers. However, $\mathcal{Z}$ has a unique trace, so is stably finite, whereas $\mathcal{O}_\infty$ is purely infinite. Both are tensorially (strongly) self-absorbing. Moreover, $\mathcal{Z}$ -stability is the mildest C^* -tensorial absorption condition akin to the McDuff property for II_1 factors (see [308]). From this point of view, $\mathcal{Z}$ is a natural analog of $\mathcal{R}$.

Just as $\mathcal{O}_\infty$ -stability underpins the Kirchberg–Phillips classification, we use $\mathcal{Z}$ -stability to separate classifiable C^* -algebras from the higher dimensional examples of [299, 300, 256, 289, 294, 124] pioneered by Villadsen. The use of $\mathcal{Z}$ -stability in classification was explicitly predicted by Toms in 2005 ([289]): “Optimistically $\mathcal{Z}$ -stability is an abstraction of slow dimension growth, and the Elliott conjecture will be confirmed for all simple separable nuclear C^* -algebras having this property.” See also [107]. Not only does $\mathcal{Z}$ have the same K -theory and traces as $\mathbb{C}$, but for any C^* -algebra A , the K -theory and traces of A and of $A \otimes \mathcal{Z}$ are isomorphic. For this

reason, $\mathcal{Z}$ -stability is for all intents and purposes necessary for classification by K -theory and traces. We describe the Jiang–Su algebra and collect the consequences of $\mathcal{Z}$ -stability used throughout the paper in Section 4.

1.1.2. The UCT. Kasparov’s KK -theory is a bivariant theory unifying K -theory and K -homology ([163, 164]). It has proved indispensable in both C^* -algebra classification and index theory (see for example [165, 143]). The Kirchberg–Phillips classification of stable simple separable nuclear purely infinite C^* -algebras is expressed directly in terms of KK -theory: two such algebras A and B are isomorphic if and only if they are KK -equivalent. Thinking of KK -equivalence as a very loose notion of homotopy between C^* -algebras, this has the spirit of Mostow’s rigidity theorem in geometric topology.

Inspired by earlier work of Brown ([27]), Rosenberg and Schochet established their universal coefficient theorem (UCT), which computes KK -theory in terms of K -theory for a large class of C^* -algebras ([260]) with generous permanence properties. This class is precisely the C^* -algebras that are KK -equivalent to some abelian C^* -algebra. Such algebras are said to *satisfy the UCT*.

In contrast to $\mathcal{Z}$ -stability, where examples lacking the property are known, it remains a crucial open question whether all (simple) separable nuclear C^* -algebras satisfy the UCT. For naturally occurring, concrete examples, this is less problematic as there are very general tools for verifying the UCT. Notably, all amenable groupoid C^* -algebras satisfy the UCT ([296]). In the setting of group actions, the UCT is preserved by crossed products of nuclear C^* -algebras by countable torsion-free amenable groups ([216]). Whether the same holds for finite groups is open (in fact, equivalent to the UCT problem). See Section 8.1 and Remark 8.2 for more details.

Among the hypotheses, the UCT is the only one that does not appear to have a von Neumann algebraic counterpart, due to its inherently topological nature.

1.1.3. The invariant. The data comprising the *Elliott invariant* of a unital stably finite C^* -algebra evolved over the early stages of the classification program, from the dimension groups appearing in Elliott’s classification of approximately finite dimensional C^* -algebras ([87]), to its present form ([90, 283]; see [255]). This consists of ordered K_0 , K_1 , the class of the unit, traces, and the pairing map. In fact, the order on K_0 is redundant under the hypothesis of $\mathcal{Z}$ -stability (see [257, Corollary 4.10]). While we have stated Theorem A and Corollary C in terms of Elliott’s invariant, in the main body of the paper we use the (formally weaker) invariant KT_u consisting of K -theory, the class of the unit, traces, and the pairing map, but not the order on K_0 . See the discussion in Section 2.1.

The unital classification theorem is complemented by an important range of invariant result: all possible pairs of countable groups and metrizable Choquet simplices are realized by the C^* -algebras of Theorem A (see Remark 2.5). These range of invariant theorems were established via crossed product constructions ([253]) in the purely infinite setting and inductive limit constructions ([92]) in the stably finite one. Understanding the range of the invariant leads to a wealth of other automatic structural results such as the existence of Cartan subalgebras inside the C^* -algebras covered by Theorem A ([187, 42]); equivalently via [246], classifiable C^* -algebras all arise from twisted groupoids.

Turning to Theorem B, examples in the 1990s show that K -theory and traces alone are not enough to classify $*$ -homomorphisms (this was made explicit in [223]).

This forces enlargement of the invariant, which we formalize as the *total invariant*: $\underline{KT}_u$ (see Section 3.2). This is obtained by adjoining a Hausdorffized unitary algebraic K_1 -group developed by Thomsen in [284], which we denote $\overline{K}_1^{\text{alg}}(A)$, and *total K -theory* (Schochet's K -theory with $\mathbb{Z}/n\mathbb{Z}$ -coefficients; [271]) to KT_u .

While our approach does not need a range of invariant result (i.e., existence of objects) to classify C^* -algebras in the unital classification theorem, it is vital in the deduction of Theorem A from Theorem B that the classification of unital embeddings does include a range of invariant result. Any morphism at the level of the invariant $\underline{KT}_u$ in Theorem B that is faithful on traces is realized by a unital $*$ -monomorphism. This is more subtle than the previous sentence might initially suggest. Just as any morphism between the Elliott invariants of C^* -algebras is required to intertwine the pairing maps between K -theory and traces, morphisms between the enlarged invariant $\underline{KT}_u$ must intertwine the natural maps between its components. Most of these are familiar to experts: the Bockstein operations connecting the various K -groups with coefficients, the de la Harpe–Skandalis determinant from [70] relating $\overline{K}_1^{\text{alg}}$ and traces, and the canonical map from $\overline{K}_1^{\text{alg}}$ to K_1 . We review these in Section 2.2 and collect properties of the Hausdorffized algebraic K_1 -group and total K -theory there as well.

There is one more natural family of maps

$$(1.1) \quad \zeta_A^{(n)}: K_0(A; \mathbb{Z}/n\mathbb{Z}) \rightarrow \overline{K}_1^{\text{alg}}(A)$$

through which the Bockstein maps $K_0(A; \mathbb{Z}/n\mathbb{Z}) \rightarrow K_1(A)$ factorize. When $K_1(A)$ is torsion-free, these maps are redundant, but in general, compatibility with $\zeta^{(n)}$ is necessary to construct a morphism between C^* -algebras realizing specified behavior on K -theory, traces, total K -theory, and $\overline{K}_1^{\text{alg}}$. In the setting of the classification of unital embeddings (Theorem B) this is the only additional compatibility needed.

We give an explicit construction of the maps (1.1) in Section 3.1. During the (lengthy) gestation of this paper, Gong, Lin, and Niu also discovered the need for additional compatibility between total and algebraic K -theory to produce morphisms ([130]). They took a different, more abstract approach. See Remark 3.4.

1.1.4. Examples. To apply the classification theorem, it is important to be able to recognize $\mathcal{Z}$ -stability in examples. Firstly, $\mathcal{Z}$ -stability can be described in terms of approximately central sequences, without reference to the Jiang–Su algebra itself ([307, Proposition 2.3], using [259], together with [174] or [292]). This may be regarded as a suitable softening to positive elements of McDuff's criterion for absorption of $\mathcal{R}$ in terms of approximately central matrix embeddings from [215].

Over the last 15 years, a major program of work on the Toms–Winter conjecture (see [107, 293] and [312, Section 5]) has provided both a topological characterization of $\mathcal{Z}$ -stability and Matui and Sato's powerful von Neumann techniques for establishing $\mathcal{Z}$ -stability ([213]). *Nuclear dimension* is a noncommutative generalization of topological covering dimension introduced by Winter and Zacharias ([313]). For unital simple separable nuclear non-elementary C^* -algebras, $\mathcal{Z}$ -stability is equivalent to finiteness of the nuclear dimension ([309, 37, 33], building on [307, 214, 266, 21]). In fact, the 2015 version of Theorem A was stated with a finite nuclear dimension hypothesis, though this was only used directly in one part of the argument, and elsewhere finite nuclear dimension was used to access $\mathcal{Z}$ -stability through [309]. Today, the equivalence gives two paths to classifiability. For examples built from

objects of bounded topological dimension, it has often been possible to estimate the nuclear dimension. However, for many examples outside this setting, direct estimates on the nuclear dimension seem inaccessible, and von Neumann techniques provide a route to $\mathcal{Z}$ -stability.

For example, a unital simple separable approximately subhomogeneous C^* -algebra A is always in the UCT class. The nuclear dimension of subhomogeneous algebras is controlled by the dimension of the spectrum ([305]), so that when A has no dimension growth it also has finite nuclear dimension. When the dimension grows, a direct estimate on the nuclear dimension fails, yet, using [309], Toms proved that A is $\mathcal{Z}$ -stable precisely when it has slow dimension growth ([290]).

Given a finitely generated amenable group G , every irreducible unitary representation π generates a simple C^* -algebra precisely when G is virtually nilpotent (this goes back to [217, 240]; see also [76]). In this case, $C_\pi^*(G)$ is $\mathcal{Z}$ -stable whenever π is infinite dimensional ([78], building on [79], an argument using C^* -classification and work on the Toms–Winter conjecture). Very shortly after the first version of this paper, Eckhardt established the UCT for these examples (extending the earlier [77] for nilpotent groups).

A vast class of examples arise from free minimal actions $G \curvearrowright X$ of a countable discrete amenable group on a compact metrizable space X . Here, the associated crossed product C^* -algebra is unital, simple (by [81, Corollary 5.16]), separable, nuclear, and satisfies the UCT (by [296]). It is increasingly clear that $\mathcal{Z}$ -stability is linked to the dimension of X , or more generally, of the action of G on X .

When X has finite covering dimension, direct estimates on the nuclear dimension are possible when G is virtually nilpotent ([280], going back to [147, 278]). For more general groups, Kerr generalized Matui’s notion of almost finiteness from [211] and used this together with Matui and Sato’s von Neumann algebra transfer technique ([213], in the form given in [145]) to obtain his $\mathcal{Z}$ -stability theorem in [167] for free minimal almost finite actions of amenable groups on finite dimensional spaces. This has been used to show $\mathcal{Z}$ -stability (hence classifiability) of such crossed products when G has locally subexponential growth ([169, 74]), when G is elementary amenable ([168, 220]), or for generic actions of any amenable group G on the Cantor set ([43]). It is conceivable that free minimal actions of countable discrete amenable groups on finite dimensional spaces give classifiable crossed products.

Outside the setting of finite dimensional base spaces, examples of non- $\mathcal{Z}$ -stable free minimal crossed products by $\mathbb{Z}$ were shown to exist by Giol and Kerr ([124]); these all have strictly positive mean dimension in the sense of Gromov, Lindenstrauss, and Weiss ([135, 206]). In the positive direction, Elliott and Niu showed that minimal $\mathbb{Z}$ -actions of mean dimension zero give rise to a $\mathcal{Z}$ -stable, and hence classifiable, crossed product ([103]). This has been extended by Niu to $\mathbb{Z}^d$ -actions in [224]. Combining the Phillips–Toms conjecture (see the introduction to [146]) and the last open implication in the Toms–Winter conjecture, crossed products associated to free minimal actions of amenable groups on compact metrizable spaces are predicted to be classifiable precisely when the mean dimension vanishes.

Another very natural class of examples arises from non-commutative dynamics: crossed products from outer actions $G \curvearrowright A$ of amenable groups on classifiable C^* -algebras. When A has a unique trace, Sato showed that the crossed product $A \rtimes G$ is $\mathcal{Z}$ -stable ([265]), so is classifiable when it satisfies the UCT (see Remark 8.2). This result extends outside the monotracial setting ([265, 121, 120]), although at

present there are restrictions both on the size of the tracial state space, and on the structure of the action of G on traces.

1.2. A history of the unital classification theorem. In the previous section, we took advantage of hindsight to set out the modern context for the unital classification theorem. Here we explain in broad strokes how the subject got to this point. We proceed thematically (rather than strictly chronologically) and focus on classification rather than regularity. Therefore, we do not describe the history of the Toms–Winter conjecture (for which, see [107] and [312, Section 5]).

1.2.1. Early days. The first classification results for simple C^* -algebras were for inductive limits of finite dimensional C^* -algebras (AF algebras) from the late 1950s to the mid-1970s ([125, 73, 23, 86]), inspired by Murray and von Neumann’s uniqueness of the hyperfinite II_1 factor. The range of the invariant (ordered K_0) for AF algebras was described abstractly at the end of the 1970s in [88, 82].

Also at the end of the 1970s came Cuntz’s work on his now eponymous C^* -algebras $\mathcal{O}_n$ ([53, 55], building on an example given by Dixmier [72]), and the introduction of simple purely infinite C^* -algebras. These are C^* -analogues of the type III factors, and important examples include $\mathcal{O}_n$ and simple Cuntz–Krieger algebras ([60]).

The 1980 Kingston meeting on operator algebras is widely regarded as a landmark event in the field. Cuntz’s survey article from this conference gives a nice account of early progress and examples of simple C^* -algebras ([56]). A talk by Effros laid out a number of open problems ([80]) related to the structure and classification of (simple) nuclear C^* -algebras (particularly Problems 5–10). As we set out below, these proved influential over the next phase of the subject.

1.2.2. Purely infinite C^* -algebras. In [58], Cuntz asked whether the Cuntz–Krieger algebras were classified by their K_0 -groups (perhaps along with other invariants), following related results in symbolic dynamics ([114]). Interest in such questions led to classification theorems for inductive limits of building blocks constructed from Cuntz algebras ([251, 204]), followed by breakthrough tensorial absorption results: $\mathcal{O}_2 \otimes M_{2^\infty} \cong \mathcal{O}_2$ (a problem posed by Cuntz in [56]) was obtained using dynamical methods in [25], and Elliott proved that $\mathcal{O}_2 \otimes \mathcal{O}_2 \cong \mathcal{O}_2$ (though his proof remains unpublished following Rørdam’s [252]). In [253], Rørdam introduced an important strategy for classifying C^* -algebras by classifying embeddings, which has been crucial to subsequent results. He considered the class of unital simple separable nuclear purely infinite C^* -algebras satisfying the UCT, for which KL classifies embeddings into simple purely infinite C^* -algebras, and showed that such algebras are classified by K -theory (keeping track of the unit). This class encompasses all simple Cuntz–Krieger algebras ([254]) and exhausts all possible combinations of K -groups ([106]). Accordingly, it is the largest class of simple separable nuclear purely infinite C^* -algebras satisfying the UCT that can be classified by K -theory.

Rørdam’s work reduced the classification problem (in the purely infinite setting) to classifying embeddings by KL . Checking this condition in concrete settings would be a herculean task — an abstract approach was needed. The stage for this had been set by decades of advances, including Arveson’s extension theorem ([4]), the Choi–Effros lifting theorem ([38]), Voiculescu’s theorem ([301]), and Kasparov’s development of KK -theory ([163, 164]).

Building on his deep work on tensor products and exactness ([171, 173]), Kirchberg announced his breakthrough “Geneva theorems” at the ICM satellite meeting in 1994: every separable exact C^* -algebra embeds into $\mathcal{O}_2$; $A \otimes \mathcal{O}_2 \cong \mathcal{O}_2$ for any unital simple separable nuclear C^* -algebra A ; and such an A is purely infinite if and only if $A \otimes \mathcal{O}_\infty \cong A$ (the first published proof was written in collaboration with Phillips in [175]). Using these results, Kirchberg ([170]) and Phillips ([236]) independently showed that simple separable nuclear purely infinite C^* -algebras — now known as *Kirchberg algebras* — satisfying the UCT are classified by K -theory. Consequently, Rørdam’s class consists exactly of the UCT Kirchberg algebras, and [106] is a range of invariant result for the Kirchberg–Phillips classification.

1.2.3. *Approximately type I C^* -algebras.* Stepping back a bit, the late 1980s saw two of Effros’ problems from [80] solved: Blackadar showed that the fixed-point algebra of a $\mathbb{Z}/2\mathbb{Z}$ action on an AF algebra need not be AF ([9]), and Evans and Kishimoto showed that removing an AF tensor factor from an AF algebra need not give an AF algebra ([108]). These and related developments brought renewed interest in classifying inductive limits of more general type I building blocks — again one of Effros’ problems — particularly for building blocks of the form $C([0, 1], F)$ and $C(\mathbb{T}, F)$, where F is finite dimensional, which give rise to the classes of AI and AT algebras respectively. Presenting the CAR algebra as an AT algebra (rather than using an AF model) is instrumental in Blackadar’s construction [9], and AT structure was found in important examples coming from dynamics ([245, 95], the latter of which was also roughly conjectured in Effros’ [80]). In his pioneering classification of AT algebras of real rank zero ([90]), Elliott enlarged his previous invariant, used to classify AF algebras, to include the K_1 -group (in talks at the time, Elliott noted inspiration from the use of K_1 in the classification results of [231] and in the proof that extensions of AF algebras are AF from [26]). This led to his slogan “ K -theory suffices”, and arguably, the Elliott classification program began in earnest at this point.

It soon turned out that, outside the real rank zero setting, topological K -theory alone is not sufficient. In the first of many important contributions, Thomsen showed that any metrizable Choquet simplex can arise as the trace space of an AI algebra whose K_0 -group is $\mathbb{Q}$ with the usual order ([283]). This led to the addition of traces and their pairing with K_0 to the invariant (the pairing is also necessary by examples which can be obtained from [134] — see [255, Proposition 3.1.8] — but the big revelation was the need for traces at all). With this larger invariant, Elliott managed to classify simple AI algebras in [89], building on [15, 283], and then extended this to the AT case in [93]. Thus the Elliott invariant reached its modern form, and Elliott stated his classification conjecture in [91].

The need for additional data beyond the Elliott invariant — total K -theory and the Hausdorffized unitary algebraic K_1 -group — to establish uniqueness theorems for $*$ -homomorphisms was first seen in the setting of inductive limits of building blocks with topological dimension one. The role of total K -theory emerged in [68, 84] and was used by Dadarlat and Loring ([69]) to classify morphisms between certain (not necessarily simple) real rank zero inductive limits previously considered by Elliott. The new ingredient introduced in [69] was the universal multicoefficient theorem (UMCT), which was used to relate maps at the level of total K -theory to KK -classes. This has become a crucial tool in classification.

Subsequently, Nielsen and Thomsen gave a new approach to the classification of simple AT algebras ([223, 222]). They give examples to show that the Elliott invariant alone does not classify the morphisms and prove a uniqueness theorem for *-homomorphisms by adjoining the unitary group modulo the closure of the commutators. As AT algebras have stable rank one, this additional data is precisely the Hausdorffized unitary algebraic K_1 -group previously studied by Thomsen ([284]) using the de la Harpe–Skandalis determinant of [70].

Moving to higher dimensional building blocks, large-scale work culminated in far-reaching classification results for approximately homogeneous (AH) C^* -algebras by Elliott and Gong ([96]) and then by Elliott, Gong, and Li ([126, 98]; preprint versions were circulated in 1998). At this point, the key hypothesis entailing classifiability was given in terms of relative upper bounds on the topological dimension of the spectrum of the building blocks of the inductive limit — *very slow dimension growth* (see [255, Chapter 3] for a discussion and the definition) — which was used in a careful analysis to reduce to inductive limits of very specific building blocks.

At the Fields Institute in 1995, Villadsen announced his work on exotic AH algebras with perforated K -theory ([299, 300]). None of these examples can absorb $\mathcal{Z}$ tensorially, as shown by Gong, Jiang, and Su ([128]), who initiated the abstract study of $\mathcal{Z}$ -stable C^* -algebras, continued in [257]. Through clever adaptations of Villadsen’s constructions, Rørdam ([256]) and Toms ([289]) produced strong counterexamples to Elliott’s classification conjecture in 2002 and 2003. It is around this time that $\mathcal{Z}$ -stability was first suggested as a potential abstract hypothesis for classification; see [289, Section 5] and [257, Page 64].

Inductive limits of subhomogeneous building blocks, known as ASH algebras, provided further challenges (as a warning, some early papers use a more restrictive definition of ASH algebras). These were eventually overcome using the techniques described in Sections 1.2.4 and 1.2.5, and it was not until the unital classification theorem in 2015 when unital simple ASH algebras with slow dimension growth (or, equivalently by [290, 309], simple $\mathcal{Z}$ -stable ASH algebras) were classified. Particularly important examples came earlier: Elliott exhausted the range of the invariant of stably finite classifiable C^* -algebras in [92], using certain ASH algebras (whose building blocks have dimension at most two), and Jiang and Su’s classification of simple inductive limits of sums of dimension drop C^* -algebras underpins their work on $\mathcal{Z}$ and strong self-absorption ([154], building on [97]). It remains an open problem whether every simple separable nuclear stably finite C^* -algebra is ASH.

1.2.4. Tracially AF C^* -algebras. The major task around the turn of the millennium was articulated by Lin in his foundational paper [189]: “... to establish a classification result for [simple nuclear stably finite] C^* -algebras that are not assumed to be direct limits of some special form.”

The impetus for the next phase was given by breakthrough results for quasidiagonal C^* -algebras ([13, 243]). Popa’s approximation property, motivated in part by Effros’ problem of abstractly characterizing AF algebras ([80]), obtains finite dimensional subalgebras from quasidiagonality in the presence of sufficiently many projections. For a unital simple quasidiagonal C^* -algebra A of real rank zero there exist non-zero approximately central projections p with the property that compressions to the resulting corners pAp can be approximated by finite dimensional subalgebras. This was based on Popa’s earlier 2-norm version of this idea for von Neumann algebras, where he could use maximality to take $p = 1$, and recover

Connes' Theorem that injective II_1 factors are hyperfinite ([241]). Likewise, by definition, a C^* -algebra is AF if one can always take $p = 1$ in Popa's condition. Lin's innovation was to ask for $1 - p$ to be small in the Cuntz semigroup (described as p being "tracially large"), introducing the class of *tracially AF* (TAF) C^* -algebras ([189]).

Lin obtained a classification theorem for unital simple separable nuclear tracially AF algebras satisfying the UCT in [188], and using a range of the invariant result, showed that this class coincides with unital simple AH algebras of real rank zero and slow dimension growth ([192]).

The classification of TAF algebras instigated the systematic use of absorption in the stably finite setting, prompting the "stable uniqueness" and "stable existence" theorems of Dadarlat–Eilers ([66]) and Lin ([190]) together with a novel use of the UCT as a way of controlling the complexity of stabilizations in KK -theory. Morally, Lin's TAF uniqueness theorem is obtained from stable uniqueness, hiding the stabilization in tracially small corners via the UCT. The corresponding existence theorem is even more difficult: approximately multiplicative maps $A \rightarrow B$ are built in two stages by factoring through explicitly constructed model algebras C with precise internal structure. Lin uses Blackadar and Kirchberg's work on simple nuclear quasidiagonal C^* -algebras ([13, 14]) to pass from A to C ([193, Theorem 4.2]) and the classification of simple real rank zero AH algebras to get from C to B ([188, Corollary 4.4 and Definition 3.1]). See also Dadarlat's streamlined approach (which obtained stronger results for embeddings, also factoring through models) in [63].

To use Lin's theorem, one needs abstract tools for obtaining tracial approximations. A number of dynamical examples were brought within the scope of this classification by ad hoc means (for example [191, Corollary 3.13, Theorem 3.6], [205]). Winter provided the first systematic approach in 2005 ([306]), via *decomposition rank* ([179]). This is another important non-commutative covering dimension which preceded nuclear dimension. The two notions are subtly related, but having finite decomposition rank is a strong condition, entailing quasidiagonality and being challenging to verify directly, although it does cover ASH algebras with no dimension growth ([305]).

For a unital simple separable C^* -algebra with real rank zero and finite decomposition rank, Winter obtains a "Popa approximation" where the projection p is bounded below in all traces in terms of the decomposition rank ([306], noting that while that reference also requires $\mathcal{Z}$ -stability, this is now redundant due to Winter's first $\mathcal{Z}$ -stability theorem from [307]). Repeating the argument inductively in the complementary corner leads to well-approximated corners of arbitrarily large trace, proving such algebras are tracially AF. In particular, Winter's work gave the first classification of simple ASH algebras of real rank zero and no dimension growth.

1.2.5. Localization and rational tracial approximation. Not all simple nuclear stably finite C^* -algebras have good tracial approximations: indeed, they might have no projections. Winter's "localization technique" ([310], first circulated as a preprint in 2007) overcomes this difficulty via an ingenious use of the Jiang–Su algebra $\mathcal{Z}$. By viewing $\mathcal{Z}$ as an inductive limit of generalized dimension drop algebras with UHF fibers as in [259], Winter's strategy passes from a classification of UHF-stable algebras (where there are many projections, so tracial approximations are conceivable) to a classification of $\mathcal{Z}$ -stable algebras. This localization technique comes at a

price: at the UHF-stable level, one must classify *-isomorphisms up to “strong asymptotic unitary equivalence” instead of approximate unitary equivalence — that is, sequences of unitaries are replaced with continuous one-parameter families of unitaries starting at the unit. There are considerable additional obstructions to this form of uniqueness — for example, one must consider KK instead of its Hausdorffization KL — and correspondingly strong existence results are also required.

Strong asymptotic classification results were nevertheless proved for C^* -algebras with tracial approximations: Lin and Niu achieved this for simple separable nuclear TAF C^* -algebras with the UCT in [196, 202]. There is a quantum leap in difficulty over approximate classification. The fact that KK -theory (in contrast to KL) fails to preserve inductive limits calls for highly intricate analysis involving careful use of “basic homotopy lemmas” ([197]), allowing approximately central unitaries to be connected by approximately central homotopies (a concept going back to [24]).

In 2013, Matui and Sato used the Lin–Niu–Winter results together with breakthrough work on the Toms–Winter conjecture to give the first truly abstract definitive stably finite classification theorem: unital simple separable nuclear $\mathcal{Z}$ -stable quasidiagonal monotracial C^* -algebras satisfying the UCT are classified by Elliott’s invariant ([214]). (Via their earlier breakthrough [213], one could replace $\mathcal{Z}$ -stability by strict comparison here.) Matui and Sato show that the UHF-stabilizations of such a C^* -algebra A (possibly without the UCT) are TAF (A is said to be *rationally TAF*). See [214, Section 6]; in hindsight, this argument parallels the last part of Connes’ proof of injectivity implies hyperfiniteness for II_1 factors (a related discussion is given in [35, Section 6]). The Matui–Sato classification theorem includes C^* -algebras arising from a minimal uniquely ergodic homeomorphism of a finite dimensional compact metrizable space with infinitely many points, using [294] and [237] to prove the regularity and quasidiagonality hypotheses, respectively.

1.2.6. The Gong–Lin–Niu class and classification by embeddings. Stepping back a bit, following Lin’s classification of tracially AF C^* -algebras, there was rapid progress in the classification of C^* -algebras with tracial approximations by more complicated type I building blocks such as interval algebras ([195]). The new target of strong asymptotic classification promoted by localization was first obtained for TAI algebras in [198]. This sparked the line of research culminating in the Gong–Lin–Niu strong asymptotic classification ([131, 132]) of unital simple separable nuclear UHF-stable C^* -algebras satisfying the UCT that are tracially approximated by non-commutative 1-dimensional CW-complexes. This was announced in Oberwolfach in 2012 ([127, 200]) and circulated as a preprint in 2015 ([129]). The relevance of the Gong–Lin–Niu class is that it exhausts the values of the Elliott invariant on unital simple separable nuclear UHF-stable stably finite C^* -algebras. No larger class of building blocks are needed.

Gong, Lin, and Niu’s argument is exceptionally technically demanding and involves detailed analysis and layers of quantification and approximation. Just as in the setting of real rank zero inductive limits of no dimension growth, where the ASH case proved an order of magnitude more challenging than the AH case (and came a decade later through tracial approximations), the subhomogeneous nature of the building blocks present huge technical hurdles. Highly bespoke homotopy lemmas and existence and uniqueness results between the building blocks are needed throughout the paper. The arguments also depend on elaborately constructed explicit models, through which to factor their existence and uniqueness results. It is

an incredible achievement to put all this together and overcome all these difficulties, yet the daunting length and extreme technicality of [131, 132] have made it very difficult for researchers to fully absorb the entire argument.

Given the scope of Gong, Lin, and Niu’s work, research on the stably finite case of the unital classification theorem pivoted to abstractly determining which UHF-stable C^* -algebras fall into the Gong–Lin–Niu class. Shortly after Matui and Sato’s abstract approach to rational TAF structure, Winter introduced a general technique of *classification by embeddings* ([311]), and applied it (together with [248, 276]) to bring C^* -algebras associated to minimal diffeomorphisms on odd spheres within the scope of the Elliott program by means of rational TAI classification. Winter’s result shows that a significantly weaker form of tracial approximation implies rational (Lin-style) tracial approximations in the presence of finite nuclear dimension. Combined with a classification result by Robert ([248]), Winter’s technique only requires appropriate external approximations in order to get (internal) rational tracial approximations. As a proof of concept, these external approximations arise immediately for quasidiagonal monotracial C^* -algebras. (See [311, Corollary 2.4], which at the time was a special case of [214, Theorem 6.1]).

Classification by embeddings was to become the route to the Gong–Lin–Niu class. For example, as precursors to the full classification theorem, it was used by Lin to classify simple $\mathbb{Z}$ -crossed products of finite dimensional spaces ([201]) and by Elliott, Gong, Lin, and Niu to classify simple $\mathcal{Z}$ -stable ASH algebras ([99]).

1.2.7. Stable uniqueness across the interval and quasidiagonality. Beyond these precursor results, major breakthroughs came quickly in 2015. Demonstrating a new technique for obtaining Winter-style external approximations, Niu announced joint work with Elliott ([102]) in April 2015, that C^* -algebras with finite decomposition rank and $K_0(A) \otimes \mathbb{Q} \cong \mathbb{Q}$ are rationally TAI. This was followed by Elliott, Gong, Lin, and Niu’s general result ([100]; circulated as a preprint in July 2015) which removed the K_0 hypothesis, and therefore showed all unital simple separable nuclear C^* -algebras with finite decomposition rank satisfying the UCT, are rationally in the Gong–Lin–Niu class, and so classified by [131, 132]. The $\mathcal{Z}$ -stability needed for classification is obtained from Winter’s first $\mathcal{Z}$ -stability theorem ([307]).

The fundamental new idea in [102, 100] is a *stable uniqueness across the interval* argument to connect two maps ϕ and ψ into the universal UHF algebra (agreeing suitably on invariants) by a homotopy of approximately multiplicative maps with approximately constant tracial behavior. The UCT enters to control the matrix size required in the stable uniqueness theorem. This is then applied to maps arising from quasidiagonality and a range of invariant result to produce Winter-style approximations into (UHF-stabilizations of) Elliott–Thomsen building blocks.

As noted in Section 1.2.4, finite decomposition rank is much harder to verify in examples than finite nuclear dimension. In part, this is because it entails quasidiagonality. Right back in his foundational TAF paper ([189]), Lin noted that “recent developments suggest one may further assume simple (nuclear stably finite) C^* -algebras are quasidiagonal”, and indeed Elliott’s examples exhausting the range of the stably finite invariant ([92]) are all quasidiagonal, and moreover, all their traces are quasidiagonal (a condition first raised by Brown in his memoir [29]).

While there are many deep theorems giving quasidiagonality in examples, in the abstract setting this is a major challenge: Blackadar and Kirchberg’s famous question ([13, Question 7.3.1]) of whether all nuclear stably finite C^* -algebras are

quasidiagonal remains open. In fact, the Elliott–Gong–Lin–Niu theorem isolates quasidiagonality of traces as a key hypothesis: their theorem holds for unital simple separable finite C^* -algebras of finite nuclear dimension (via Winter’s second $\mathcal{Z}$ -stability theorem [309] in place of [307]) satisfying the UCT, such that all traces are quasidiagonal. Later, through [21] and [37, 33], it was shown, without reference to the UCT, that a simple separable C^* -algebra A of finite nuclear dimension has finite decomposition rank precisely when both A and all its traces are quasidiagonal.

A little earlier in 2014, Ozawa, Rørdam, and Sato found an unexpected route to quasidiagonality powered by classification, using Matui and Sato’s [214] to show that elementary amenable groups have quasidiagonal C^* -algebras ([229]). Inspired by this, the Kirchberg–Phillips theorem, and talks on [102, 100], AT, SW, and Winter used a stable uniqueness across the interval argument to prove their quasidiagonality theorem ([287]) over the summer of 2015: faithful traces on a separable nuclear C^* -algebra satisfying the UCT are quasidiagonal. This established Rosenberg’s conjecture — C^* -algebras associated to discrete amenable groups are quasidiagonal — and removed the quasidiagonality hypothesis from [100]. In this way, the combination of [131, 132, 100, 287, 311, 309, 310] and the large body of work these papers rely on comes together to prove the unital stably finite classification theorem (with the hypothesis of finite nuclear dimension replacing that of $\mathcal{Z}$ -stability).

1.2.8. Post-2015: $\mathcal{Z}$ -stability and abstract classification. Developments continued apace post 2015 in numerous directions, including classification results in the non-unital setting, a vast body of work bringing examples within the scope of the unital classification theorem as discussed in Section 1.1.4, and the continued large-scale study of fine structure in the Cuntz semigroup. We do not describe these here.

The present paper has its origins in two results announced at BIRS in 2017 ([104]). Firstly Castillejos, Evington, AT, SW, and Winter showed that unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebras have nuclear dimension at most one ([37]). This allows the unital classification theorem to be accessed from $\mathcal{Z}$ -stability. (As explained in [35, Section 6], the methods of [37] can be combined with those of [214] to give a direct proof of Winter’s classification by embeddings theorem from $\mathcal{Z}$ -stability, bypassing the only use of nuclear dimension in the 2015 proof of the unital classification theorem.) Secondly, building on his abstract approach to the quasidiagonality theorem ([267]), CS announced a precursor of what would become his AF-embedding theorem ([268]): a classification of unital full nuclear embeddings of unital separable nuclear UCT C^* -algebras into the ultrapower of the universal UHF algebra. The techniques provided an abstract classification of unital simple separable nuclear UHF-stable monotracial C^* -algebras satisfying the UCT which does not rely on obtaining any form of tracial approximation. (The version published in [268] assumes $\mathcal{Q}$ -stability and trivial K_1 -group as this allows one to bypass some technicalities and is sufficient to obtain the headline AF-embedding result.) This was the first abstract stably finite classification theorem which does not pass through some kind of internal C^* -algebra structure in the proof.

1.3. Discussion of the methods behind Theorem B. The overarching strategy of our paper is to combine the ideas in [268] and [37] with a new KK -uniqueness theorem (Theorem 1.4). This avoids Winter’s localization technique and directly classifies $\mathcal{Z}$ -stable C^* -algebras without first classifying UHF-stable algebras.

1.3.1. Classifying full approximate embeddings. The existence portion of Theorem B requires producing $*$ -homomorphisms between abstract C^* -algebras. Constructing these directly is hard. Instead, the usual strategy is to classify approximate $*$ -homomorphisms — these are sequences of maps that become $*$ -homomorphisms in the limit. These are often easier to produce. Indeed, for example, quasidiagonality provides a rich supply of approximate $*$ -homomorphisms into matrix algebras.

Let A and B be as in Theorem B. To avoid the need to carefully keep track of quantifiers, we work with the sequence algebra $B_\infty := \ell^\infty(B)/c_0(B)$, and $*$ -homomorphisms $A \rightarrow B_\infty$; these encode approximate $*$ -homomorphisms. Unitary equivalence of $*$ -homomorphisms into B_∞ corresponds to a quantified approximate unitary equivalence of approximate $*$ -homomorphisms into B . Although B is simple, B_∞ is not, so we impose a simplicity condition on the allowed morphisms $\phi: A \rightarrow B_\infty$: we require that $\phi(a)$ generates B_∞ as an ideal for each non-zero $a \in A$. Such maps are called *full*, and (under our hypotheses on B) fullness is readily tested using traces via Cuntz comparison.

With the setup above, the following theorem is the main objective of the paper.

Theorem 1.1 (Classification of unital full approximate embeddings; see Theorem 9.1). *Let A be a unital separable nuclear C^* -algebra satisfying the UCT and B be a unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebra. Up to unitary equivalence, the unital full embeddings $A \rightarrow B_\infty$ are classified by faithful morphisms on $\underline{KT}_u$.*

The deduction of Theorem B from Theorem 1.1 is a standard intertwining argument. For this, it is vital that the classification in Theorem 1.1 contains both uniqueness and existence statements. This is set out in Section 9.

As with Theorem B, the purely infinite case of Theorem 1.1 is a known consequence of the methods behind the Kirchberg–Phillips classification theorem, while our proof handles the finite (i.e., tracial) situation. We proceed by dividing B_∞ into two pieces: a “tracially small” part,

$$(1.2) \quad J_B := \{b \in B_\infty : \tau(b^*b) = 0 \text{ for all } \tau \in T(B_\infty)\} \triangleleft B_\infty,$$

and a “tracially large” part, $B^\infty := B_\infty/J_B$, known as the *trace-kernel ideal* and the *trace-kernel quotient*, respectively. As such, B_∞ fits into the *trace-kernel extension*

$$(1.3) \quad 0 \longrightarrow J_B \xrightarrow{j_B} B_\infty \xrightarrow{q_B} B^\infty \longrightarrow 0,$$

an ultrapower version of which first came to the fore in Matui and Sato’s work on the Toms–Winter conjecture ([213, 214]; see also [178]).

Just as in CS’s $\mathcal{Q}$ -stable monotracial classification theorem ([268]), the trace-kernel extension organizes the proof of Theorem 1.1 into two conceptual steps: classify morphisms into B^∞ , and then classify lifts of these morphisms to B_∞ .

$$(1.4) \quad \begin{array}{c} \text{lifts are classified by} \\ \text{\textit{K}-theoretic data} \end{array} \quad \begin{array}{c} A \\ \swarrow \theta, \text{ classified by traces} \\ B^\infty \end{array} \\ 0 \longrightarrow J_B \xrightarrow{j_B} B_\infty \xrightarrow{q_B} B^\infty \longrightarrow 0$$

The first step only uses tracial data and follows from [37]. The major work we undertake here is the classification of lifts. The precise data needed for the existence of a lift lives in $KK(A, B_\infty)$. On the other hand, the data needed for uniqueness resides in the group $KL(A, J_B)$ (a Hausdorffization of $KK(A, J_B)$).

Theorem 1.2 (Classification of unital lifts along the trace-kernel extension; see Theorems 7.10(i) and 7.12). *Let A be a unital separable nuclear C^* -algebra and B be a unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebra with $T(B) \neq \emptyset$. Suppose that $\theta: A \rightarrow B^\infty$ is a unital full embedding.*

- (i) *There is a unital full embedding $A \rightarrow B_\infty$ lifting θ if and only if there is a lift of $[\theta]$ in $KK(A, B_\infty)$ which preserves the unit of K_0 .*
- (ii) *Any choice of a unital full embedding $A \rightarrow B_\infty$ lifting θ determines a bijection between the unitary equivalence classes of unital full lifts of θ and the elements of $KL(A, J_B)$ that annihilate the unit of $K_0(A)$.*

The invariants in Theorem 1.2 are somewhat unwieldy. Once Theorem 1.2 is proved, our last major task will be to compute $KL(A, J_B)$ using the UCT, allowing the classification of lifts to be reinterpreted in terms of the total invariant.

Next, we provide more detail regarding these components and how they come together to prove Theorem 1.1. We follow this by discussing the role of the UCT and $\mathcal{Z}$ -stability, and a brief comparison of our methods with the original approach.

1.3.2. Classification into the trace-kernel quotient. Maps from separable nuclear C^* -algebras A into finite von Neumann algebras are classified by traces as a consequence of Connes' theorem. When B has a unique trace, this provides the classification of maps into the trace-kernel quotient. This is more often stated with the tracial ultrapower B^ω — which is a II_1 factor — in place of B^∞ , but it is routine to convert between these situations. In general, neither B^∞ nor B^ω is a von Neumann algebra, but the techniques introduced by Castillejos, Evington, AT, SW, and Winter in [37] implicitly show that when B is nuclear and $\mathcal{Z}$ -stable, B^∞ behaves enough like an ultraproduct of von Neumann algebras to give a classification of maps into B^∞ by traces. The precise result we need was recorded in the short sequel [34] to [37] for use in this paper. The nuclearity of the codomain B required to use [37, 34] is the main reason for this hypothesis in our classification theorems (the other use being to ensure that quasitraces are traces on B by [137]).

Theorem 1.3 (Classification of unital morphisms into the trace-kernel quotient; see Theorem 6.5). *Let A be a unital separable nuclear C^* -algebra and B be a unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebra with $T(B) \neq \emptyset$. Up to unitary equivalence, the unital $*$ -homomorphisms $A \rightarrow B^\infty$ are classified by traces.*

We describe the trace-kernel extension in Section 6, giving a little more detail about how one thinks of B^∞ as being “von Neumann-like”.

1.3.3. Classification of lifts along the trace-kernel extension (Theorem 1.2). The strategy for the proof of Theorem 1.2 builds on CS's work on the non-stable KK -theory of the trace-kernel ideal J_B from [267, 268] which essentially proves Theorem 1.2 under the additional hypotheses that B is $\mathcal{Q}$ -stable and monotracial with $K_1(B) = 0$ (see Remark 7.4). Absorption — in the spirit of Voiculescu's Weyl-von Neumann theorem — is the fundamental tool both in [267, 268] and our arguments.

For a unital full embedding $\theta: A \rightarrow B^\infty$, we may take the pullback of the trace-kernel extension along θ to form a commutative diagram

$$(1.5) \quad \begin{array}{ccccccc} e: 0 & \longrightarrow & J_B & \longrightarrow & E & \longrightarrow & A \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \theta \\ 0 & \longrightarrow & J_B & \xrightarrow{j_B} & B_\infty & \xrightarrow{q_B} & B^\infty \longrightarrow 0. \end{array}$$

The extension $\mathbf{e}$ in the top row of (1.5) defines a class in $\text{Ext}(A, J_B)$. As J_B is neither separable, nor even σ -unital, and it is also not stable, one must take some care with KK -theory. Fortunately it is *separably stable* (every separable subalgebra of J_B is contained in a stable separable subalgebra). We will pretend in this introduction that J_B is separable and stable, but the analysis that follows will really be performed on a suitable separable and stable subextension of the top row of (1.5).

Given a KK -lift of θ in $KK(A, B_\infty)$ as in the statement of Theorem 1.2(i), the class of $\mathbf{e}$ in Ext vanishes, so $\mathbf{e}$ stably splits, i.e., there exists a trivial extension $\mathbf{t}$ such that $\mathbf{e} \oplus \mathbf{t}$ splits. This is where absorption first enters in Theorem 1.2(i): if we can show $\mathbf{e}$ is absorbing, then $\mathbf{e} \cong \mathbf{e} \oplus \mathbf{t}$ so that $\mathbf{e}$ itself splits, providing a lift of θ . This strategy is an abstract version of CS's proof in [267] of the quasidiagonality theorem of [287]. Absorption is used again in correcting the KK -class of the lift.

Absorption is equally crucial in Theorem 1.2(ii), but appears in a slightly different guise. A pair of lifts $\phi, \psi: A \rightarrow B_\infty$ of θ gives rise to a class $[\phi, \psi]_{KK(A, J_B)}$ in $KK(A, J_B)$. Our objective is to show that ϕ and ψ are unitarily equivalent if and only if $[\phi, \psi]_{KL(A, J_B)}$ vanishes. This is achieved through a “ $\mathcal{Z}$ -stable KL -uniqueness theorem”. To set this up, note that the trace-kernel extension induces a canonical map $\mu: B_\infty \rightarrow \mathcal{M}(J_B)$, where $\mathcal{M}(J_B)$ is the multiplier algebra of J_B .

$$(1.6) \quad \begin{array}{ccccccc} 0 & \longrightarrow & J_B & \xrightarrow{j_B} & B_\infty & \xrightarrow{q_B} & B^\infty \longrightarrow 0 \\ & & \parallel & & \downarrow \mu & & \downarrow \\ 0 & \longrightarrow & J_B & \longrightarrow & \mathcal{M}(J_B) & \longrightarrow & \mathcal{Q}(J_B) \longrightarrow 0 \end{array}$$

$\begin{array}{c} A \\ \swarrow \phi \quad \searrow \psi \\ B_\infty \end{array}$

Uniqueness theorems for KK and KL intimately involve absorption. For example, via the *stable uniqueness theorems* of Dadarlat–Eilers and Lin from [66] and [190] (and continuing to pretend that J_B is separable and stable), if $[\phi, \psi]_{KL(A, J_B)}$ vanishes, then $\mu \circ \phi$ and $\mu \circ \psi$ become properly approximately unitarily equivalent after adjoining any absorbing representation $\eta: A \rightarrow \mathcal{M}(J_B)$; i.e., $(\mu \circ \phi) \oplus \eta$ and $(\mu \circ \psi) \oplus \eta$ are approximately unitarily equivalent via unitaries in the minimal unitization of J_B . If $\mu \circ \phi$ and $\mu \circ \psi$ are already known to be absorbing, then we can use these for η in two applications of stable uniqueness to obtain proper approximate unitary equivalence of $(\mu \circ \phi) \oplus (\mu \circ \phi)$ and $(\mu \circ \psi) \oplus (\mu \circ \psi)$. Assuming further that B_∞ and J_B absorb the UHF algebra M_{2^∞} tensorially (really one must work with “separable M_{2^∞} -stability”) the two-fold amplification can be removed, giving rise to approximate unitary equivalence of ϕ and ψ with unitaries from the minimal unitization of J_B , and hence in B_∞ , not just in $\mathcal{M}(J_B)$. This “UHF-stable KL -uniqueness theorem” is an essential trick from [268], and with enough absorption achieves our objective whenever B absorbs a UHF algebra of infinite type.

In this paper, some major new ingredients are KK - and KL -uniqueness theorems using $\mathcal{Z}$ -stability in place of UHF-stability. We state the KK -version below; the KL -version is similar (characterizing proper *approximate* unitary equivalence). The strategy behind our KK -uniqueness theorem goes back to ideas in Dadarlat and Eilers’ stable uniqueness theorem ([66]), as discussed further in Section 5.4.

Theorem 1.4 ($\mathcal{Z}$ -stable KK -uniqueness theorem; see Theorem 5.17(i)). *Let A be a separable C^* -algebra and I be a σ -unital and stable C^* -algebra. Suppose that*

$\phi, \psi: A \rightarrow \mathcal{M}(I)$ are absorbing and $\phi(a) - \psi(a) \in I$ for all $a \in A$. Then the following are equivalent:

- (i) $[\phi, \psi]_{KK(A, I)} = 0$
- (ii) $\phi \otimes 1_{\mathcal{Z}}, \psi \otimes 1_{\mathcal{Z}}: A \rightarrow \mathcal{M}(I \otimes \mathcal{Z})$ are properly asymptotically unitarily equivalent.

The Dadarlat–Eilers stable uniqueness theorem gives the equivalence of (i) and

- (iii) $\phi \oplus \eta, \psi \oplus \eta$ are properly asymptotically unitarily equivalent for all absorbing $\eta: A \rightarrow \mathcal{M}(I)$.

Using $\mathcal{Z}$ -stability of B to remove the tensor factor of $1_{\mathcal{Z}}$ (taking due care with issues around separability and tensorial absorption), and lifting the outcome of the $\mathcal{Z}$ -stable KL -uniqueness theorem back to B_{∞} , is exactly what we need to pass from $[\phi, \psi]_{KK(A, J_B)} = 0$ to the desired unitary equivalence — provided we can show that the maps $\mu \circ \phi$ and $\mu \circ \psi$ from (1.6) are suitably absorbing.

1.3.4. Obtaining absorption. Elliott and Kucerovsky’s vast generalization ([101]) of Voiculescu’s theorem provides a criterion — “pure largeness” — that abstractly characterizes absorption in the presence of nuclearity. The precise definition of pure largeness is not so important to our paper as it can be identified in terms of fullness under mild regularity conditions (much weaker than $\mathcal{Z}$ -stability). The relevant result for us is obtained by combining [101, 184, 227, 257]: an extension of a separable nuclear C^* -algebra by a σ -unital stable $\mathcal{Z}$ -stable ideal is absorbing if and only if the (forced) unitization of the extension is full.

Using the version of the Elliott–Kucerovsky theorem described above requires separable stability and $\mathcal{Z}$ -stability of J_B . The latter condition follows directly from the $\mathcal{Z}$ -stability of B . Hjelmberg and Rørdam characterized stability for σ -unital C^* -algebras in [148]; this also characterizes separable stability. In this paper we use Cuntz semigroup methods to verify the Hjelmberg–Rørdam conditions for J_B from strict comparison (and hence from $\mathcal{Z}$ -stability). This replaces the ad hoc notion of an admissible kernel CS introduced to obtain separable stability of J_B (also via Hjelmberg–Rørdam) when $B = \mathcal{Q}$ ([267]) and more generally, when B is a unital simple $\mathcal{Q}$ -stable AF algebra with unique trace ([268]).

The final detail is a “de-unitization” trick. Unital extensions can never be absorbing, as they do not absorb non-unital extensions. As in [267, 268], we resolve this by taking the direct sum with the zero extension — see Section 7.4.

It is tempting to try and replace the trace-kernel quotient B^{∞} with a von Neumann algebra $\mathcal{M}$, such as an ultrapower of the finite part of the bidual of B . In this case, $\mathcal{M}$ would encode all the traces on B , and one has the classification of maps $A \rightarrow \mathcal{M}$ by traces directly from Connes’ theorem, and Theorem 1.3 (and the work of [37]) would not be needed. However, the ideal in the ultrapower B_{ω} corresponding to the quotient onto $\mathcal{M}$ is not separably stable, obstructing the route to absorption (see Remark 6.12). This prevents us from lifting the classification back from $\mathcal{M}$ to B_{ω} . Loosely speaking, we must work with 2-norm estimates that are uniform over the traces in order to take advantage of strict comparison in B .

1.3.5. Computing KL -theory of the trace-kernel ideal. The major remaining task is to compute the group $KL(A, J_B)$ by combining the universal multicoefficient theorem (UMCT) with total K -theory computations for the trace-kernel extension.

For any C^* -algebra D , the Kasparov product induces a natural map

$$(1.7) \quad KL(A, D) \longrightarrow \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(D)),$$

where $\text{Hom}_\Lambda(\underline{K}(A), \underline{K}(D))$ is the group of morphisms in total K -theory. Dadarlat and Loring's UMCT shows that (1.7) is an isomorphism when A satisfies the UCT. Applying this to the trace-kernel extension allows $KL(A, J_B)$ to be computed using the resulting exact sequence:

$$(1.8) \quad \begin{array}{c} 0 \longrightarrow \ker \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(j_B)) \longrightarrow KL(A, J_B) \longrightarrow \\ \searrow \hspace{10em} \longrightarrow \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(B_\infty)) \longrightarrow \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(B^\infty)). \end{array}$$

The first and fourth terms above can be simplified using the von Neumann algebra-like behavior of B^∞ : $K_1(B^\infty) = 0$, and the pairing map $\rho_{B^\infty}: K_0(B^\infty) \rightarrow \text{Aff } T(B^\infty)$ is an isomorphism just as the K_0 -group of a II_1 factor is identified with $\mathbb{R}$ using the trace. Consequently, $\underline{K}(B^\infty)$ and $\ker \underline{K}(j_B)$ are concentrated in their K_0 and K_1 components respectively, so the last entry in (1.8) is isomorphic to $\text{Hom}(K_0(A), \text{Aff } T(B^\infty))$ (see Proposition 8.6 and (8.19), noting that in the main body we identify $\text{Aff } T(B_\infty)$ and $\text{Aff } T(B^\infty)$ as the regularity hypothesis forces $\text{Aff } T(q_B)$ to be an isomorphism). This allows us to determine $KL(A, J_B)$ by computing the maps in (1.8) explicitly.

These computations can be cleanly interpreted in terms of a Hausdorffized unitary algebraic K_1 -group developed by Thomsen. For a C^* -algebra D , $\overline{K}_1^{\text{alg}}(D)$ is defined as the quotient of the infinite unitary group of D by the closure (in the inductive limit topology) of its commutator subgroup, and there is a natural surjection $\sharp_D: \overline{K}_1^{\text{alg}}(D) \rightarrow K_1(D)$. Thomsen used the de la Harpe–Skandalis determinant to provide a natural sequence

$$(1.9) \quad K_0(D) \xrightarrow{\rho_D} \text{Aff } T(D) \xrightarrow{\text{Th}_D} \overline{K}_1^{\text{alg}}(D) \xrightarrow{\sharp_D} K_1(D) \longrightarrow 0$$

that is exact at $\overline{K}_1^{\text{alg}}(D)$ and satisfies $\ker \text{Th}_D = \overline{\text{im } \rho_D}$. Here, ρ_D is the pairing between K_0 and traces, and the *Thomsen map* Th_D can be constructed from an inverse of the de la Harpe–Skandalis determinant (see Section 2.2, particularly (2.17)). When we take $D = B_\infty$, a fragment of the six-term exact sequence of the trace-kernel extension corresponds to Thomsen's sequence as follows.

Theorem 1.5 (Calculating $K_1(J_B)$; see Theorem 6.8). *Let B be a unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebra with $T(B) \neq \emptyset$. Then there is a natural isomorphism $\omega_B: K_1(J_B) \rightarrow \overline{K}_1^{\text{alg}}(B_\infty)$ that (together with the computations $K_0(B^\infty) \cong \text{Aff } T(B_\infty)$ and $K_1(B^\infty) = 0$) gives the commutative diagram*

$$(1.10) \quad \begin{array}{ccccccc} K_0(B_\infty) & \xrightarrow{K_0(q_B)} & K_0(B^\infty) & \xrightarrow{\partial} & K_1(J_B) & \xrightarrow{K_1(j_B)} & K_1(B_\infty) \longrightarrow 0 \\ \parallel & & \downarrow \cong & & \downarrow \omega_B & & \parallel \\ K_0(B_\infty) & \xrightarrow{\rho_{B_\infty}} & \text{Aff } T(B_\infty) & \xrightarrow{\text{Th}_{B_\infty}} & \overline{K}_1^{\text{alg}}(B_\infty) & \xrightarrow{\sharp_{B_\infty}} & K_1(B_\infty) \longrightarrow 0. \end{array}$$

Putting this all together, and keeping track of the maps explicitly, allows us to compute $KL(A, J_B)$ in terms of the enriched invariant $\underline{K}T_u(\cdot)$.

Theorem 1.6 (Calculating $KL(A, J_B)$; see Theorem 8.9). *Let A be a unital separable C^* -algebra satisfying the UCT and let B be a unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebra with $T(B) \neq \emptyset$. Then the sequence in (1.8) is exact. Moreover:*

- (i) *The third arrow in (1.8) is a group homomorphism from $KL(A, J_B)$ onto the set of $\underline{KT}_u$ -morphisms $\underline{KT}_u(A) \rightarrow \underline{KT}_u(B_\infty)$ that vanish on the trace.*
- (ii) *Given a full *-homomorphism $\psi: A \rightarrow B_\infty$, the kernel of the above homomorphism identifies with the set of $\underline{KT}_u$ -morphisms $\underline{KT}_u(A) \rightarrow \underline{KT}_u(B_\infty)$ that agree with ψ on total K -theory and traces.*

Compatibility with $\zeta^{(n)}$ is essential in establishing surjectivity in Theorem 1.6(ii). In slightly more detail, given a $\underline{KT}_u$ -morphism $(\underline{\alpha}, \beta, \gamma): \underline{KT}_u(A) \rightarrow \underline{KT}_u(B_\infty)$ that agrees with ψ on total K -theory and traces, $\beta - \overline{K}_1^{\text{alg}}(\psi)$ gives rise to a group morphism $r: K_1(A) \rightarrow \ker K_1(j_B)$ via (1.10). In order to extend this map to total K -theory, as in the first term of (1.8), r must vanish on torsion. This is equivalent to compatibility with $\zeta^{(n)}$. See Sections 3.1 and 8.3.

Theorem 1.1 follows by combining the existence and uniqueness parts of Theorems 1.2, 1.3, and 1.6.

1.3.6. The role of the UCT and $\mathcal{Z}$ -stability. The UCT appears twice in the proof. Firstly, it is used to compute $KL(A, J_B)$ in terms of $KK(A, B_\infty)$ and traces, via the UMCT, as described in Section 1.3.5 above. Secondly, the UCT is used to show that the $\text{Ext}(A, J_B)$ obstruction in Theorem 1.2(i) vanishes, by producing a required KK -lifting. This second application of the UCT is analogous to the role it plays in CS's approach ([267]) to the quasidiagonality theorem (see Remark 7.3). Assuming the existence of a “trace-compatible” element of $KK(A, B_\infty)$ does not appear to be sufficient for this step, as the UCT would still be needed to show this element lifts a given morphism $A \rightarrow B^\infty$ in $KK(A, B_\infty)$.

In the classification of UCT Kirchberg algebras by K -theory, the UCT is only needed to lift an isomorphism at the level of K -theory to a KK -equivalence. The more subtle uses of the UCT in the stably finite setting described above mean that classification currently requires the UCT and cannot (yet) be formulated in terms of a combination of KK -theory and traces.

In contrast, regularity — i.e., $\mathcal{Z}$ -stability of the codomain B — is ever-present in the argument. Jiang–Su stability is a rich hypothesis, with many ramifications. In the main body of the paper, we will typically state results with $\mathcal{Z}$ -stability as the hypothesis, even if only a consequence of $\mathcal{Z}$ -stability is needed. Here, we describe which aspects of $\mathcal{Z}$ -stability are used at different points in the argument.

Two consequences of $\mathcal{Z}$ -stability are pertinent: unital simple finite $\mathcal{Z}$ -stable C^* -algebras have *strict comparison* and *uniform property Γ* . Both are analogs of properties of von Neumann algebras. For a unital simple separable nuclear C^* -algebra B with $T(B) \neq \emptyset$, $\mathcal{Z}$ -stability is now known to be equivalent to the combination of strict comparison and uniform property Γ (using [257] and [35], which builds on [213, 264, 291, 178, 228]). Strict comparison is a Cuntz semigroup condition, which can be used to obtain properties of the trace-kernel ideal — in particular, the route to absorption really runs through comparison — and to control the traces on B_∞ . In the main body of the paper, we use separable $\mathcal{Z}$ -stability of J_B (inherited from $\mathcal{Z}$ -stability of B) as one of the two ingredients to access absorption via Theorem 5.15, but this is a matter of convenience. We could have worked with the weaker condition of unperforation of the Cuntz semigroup of J_B and obtained

this from strict comparison of B . Likewise, we use $\mathcal{Z}$ -stability to control traces on B_∞ via Ozawa's theorem ([228]), but we could have used the harder [221, Theorem 1.2] to obtain this from strict comparison. Similarly, although it doesn't appear explicitly in this paper, it is really uniform property Γ rather than the full force of $\mathcal{Z}$ -stability that gives B^∞ its von Neumann-like behavior. We also use that unital simple finite $\mathcal{Z}$ -stable C^* -algebras have stable rank one as a technical convenience (to access cancellation) in the proof.

In contrast, our proof of the $\mathcal{Z}$ -stable KK - and KL -uniqueness theorems use the full force of $\mathcal{Z}$ -stability, not its consequences. We use both an explicit construction of $\mathcal{Z}$ (in the proof of Jiang's theorem; see Section 4.2) and strong self-absorption. This is the only place where we directly use strong self-absorption of $\mathcal{Z}$.

1.3.7. Comparison with classification via tracial approximations. The $\mathcal{Z}$ -stable KK - and KL -uniqueness theorem provides a major shortcut. It allows us to bypass Winter's localization techniques, and obtain a classification of morphisms into $\mathcal{Z}$ -stable codomains up to approximate unitary equivalence directly. A significant amount of the work in Gong, Lin, and Niu's argument ([131, 132]) is obtaining the strong asymptotic classification required for localization. As we do not need asymptotic classification results, we do not prove any in this paper (with some extra work and invariants, it is possible to do this abstractly, and we will return to this point in future) allowing us to work with simpler invariants throughout.

Another fundamental difference is that our approach does not require explicit models of classifiable C^* -algebras nor internal approximation structure such as inductive limits or tracial approximations. Indeed, the only place where internal structure arises in the argument is through Connes' theorem that injective von Neumann algebras (in this case, the bidual of the nuclear domain C^* -algebras) are hyperfinite, and so are inductive limits of finite dimensional algebras. This is crucial in the folklore result that maps from separable nuclear C^* -algebras into finite von Neumann algebras are classified (up to strong*-approximate unitary equivalence) by traces. The classification into the trace-kernel quotient (Theorem 1.3) is obtained by gluing this observation uniformly across the traces.

The absence of tracial approximations allows us to obtain our embedding results in vast generality: in this paper, a separable nuclear C^* -algebra satisfying the UCT is allowed as a domain, as opposed to the simplicity requirement found in [130] (see the discussion in Remark 9.6). In future work we will obtain embedding results for exact domains and quite general codomains, with the simplicity and nuclearity requirements at the level of the map. With our approach, internal C^* -structure — such as arising as an ASH algebra or enjoying tracial approximations — can be viewed as a consequence of C^* -classification, rather than an ingredient.

Our proof essentially contains a copy of CS's proof ([267]) of the quasidiagonality theorem, and so we no longer need this result nor the UCT-powered stable uniqueness across the interval arguments used to obtain tracial approximations [100] (and in the original proof of the quasidiagonality theorem, [287]). These are replaced by the abstract lifting result Theorem 1.2(i) in combination with the UCT to verify the hypotheses. The use of the UCT in [267] is more direct than that in [287] and is similar to its use here.

Our work also has points of contact with the work of Gong, Lin, and Niu, and many others in the long history of C^* -classification. Classifying into the trace-kernel quotient is spiritually connected to classifying tracially large maps into C^* -algebras

with good tracial approximations, and our use of the Elliott–Kucerovsky absorption theorem plays a role analogous to the use of stable uniqueness theorems in [131, 132]. As with [131, 132], our existence result is proven in two stages. We first prove the existence of an approximate embedding with prescribed tracial information (through Theorem 1.2(i)), and then modify the K -theoretic behavior (in Theorem 1.2(ii)). This two-step process dates back to Lin’s classification of TAF algebras ([188]), which is made explicit in Dadarlat’s account ([63]).

Finally, while we do not use any of the results from the major papers establishing the 2015 classification theorem during our work, we carefully studied the invariants and rotation maps from [129]. This led us to expect that a suitable combination of total K -theory, traces, and Hausdorffized unitary algebraic K_1 would provide the classification invariant found in Theorem B.

1.4. Structure of the paper, notation, and references. We hope to clarify the new ingredients of the invariants used in our classification in the next two sections and review their essential properties (with certain facts about total K -theory being collated in Appendix A for completeness). Section 3 is devoted to the maps $\zeta^{(n)}: K_0(\cdot; \mathbb{Z}/n) \rightarrow \overline{K}_1^{\text{alg}}(\cdot)$, and formalizes the total invariant $\underline{KT}_u$. Section 4 is concerned with the Jiang–Su algebra $\mathcal{Z}$. We give a short self-contained proof of Jiang’s unpublished theorem that $\mathcal{Z}$ -stable C^* -algebras are K_1 -injective. This result is a vital component of the $\mathcal{Z}$ -stable KK - and KL -uniqueness theorems. We also collect results on strict comparison and the Cuntz semigroup in Section 4, which are used later in absorption results.

Absorption and KK -theory strike back in Section 5. Since we wish to use $KK(A, J_B)$ and $KL(A, J_B)$ and the trace-kernel ideal is not σ -unital, we will separabilize and treat these KK -groups as limits of the KK - and KL -groups corresponding to separable subalgebras J of J_B . This is explained in Section 5.1, with the proofs of facts that are standard in the σ -unital case deferred to Appendix B. The rest of Section 5 is devoted to absorption and the proof of the $\mathcal{Z}$ -stable KK - and KL -uniqueness theorems.

In Section 6, we describe the trace-kernel extension, and compute the K -theory of the trace-kernel quotient. This enables us to prove Theorem 1.5. Section 6.4 shows that the trace-kernel extension J_B is separably stable for use in absorption. Section 7 analyzes lifts along the trace-kernel extension, proving Theorem 1.2 (which we deduce from non-unital versions).

The final act of the paper sees the return of the UCT and its cousin, the UMCT. We review these results in Section 8, and compute $KL(A, J_B)$ to prove Theorem 1.6. In particular, we do not use that our domain algebras A satisfy the UCT in any results before Section 8. Finally we put everything together in Section 9, which proves the classification of full approximate *-homomorphisms (Theorem 1.1) and deduces Theorems A and B.

1.4.1. Notation. Throughout the paper we endeavor to consistently use A for C^* -algebras appearing as the domain of the classification of unital embeddings (Theorem B). When A is used in other results (for example, appearing as the first variable in KK -theory) it is because we envisage applying these results to the domains of Theorem B. Likewise, we use B for the codomains of the classification of embeddings, with the corresponding notation for the trace-kernel extension in (1.3). For that reason we typically write D or E for a general C^* -algebra, and I or J for a

general ideal (often the second variable in KK -theory). We will write $\mathcal{M}(I)$ for the multiplier algebra of I , and $\mathcal{Q}(I)$ for its corona. The symbols α, β, γ will be reserved for the components of a morphism between invariants — see Section 3.2. We also use κ for a KK -element, λ for a KL -element, and typically use ϕ, ψ, θ for $*$ -homomorphisms between C^* -algebras ($\kappa_*^{(m,n)}$ also appears briefly as one of the Bockstein maps; no ambiguity is possible). For $D \subseteq E$ an inclusion of C^* -algebras, we write $\iota_{D \subseteq E}: D \rightarrow E$ for the inclusion map, and $D^\dagger$ denotes the forced unitization of D (when D is already unital, a new unit is added).

1.4.2. References underpinning the argument. We have attempted to provide both context throughout, and also to track down the intellectual origins of the ideas we use. However, from a formal point of view, the vast majority of the (many) papers we cite do not form part of our argument for the stably finite parts of Theorems A and B, which are relatively self-contained. Concretely, in addition to material in textbooks and short stand-alone lemmas which will be found straightforward by experts (this includes every instance where we appeal to one of [21, 268, 287] for a result rather than for context), we only rely on the following works:

- [70, 284] for properties of algebraic K_1 and the de la Harpe–Skandalis determinant;
- [69, 271] for total K -theory;
- [154, 228, 257, 259, 292] for the Jiang–Su algebra and consequences of $\mathcal{Z}$ -stability;
- [137] for the fact that quasitraces on exact C^* -algebras are traces (we only need this for the simple nuclear $\mathcal{Z}$ -stable codomain algebras B ; there is a short proof of this fact for C^* -algebras with finite nuclear dimension [31], although we don’t know how to do this from $\mathcal{Z}$ -stability directly);
- [148, 227, 250] for the Cuntz semigroup, strict comparison, and its use in obtaining stability, and the corona factorization property;
- [65, 66, 101, 116, 184] for the absorption theorem; [62, 65] for ingredients in the KK - and KL -uniqueness theorems (including a convenient form of Paschke duality);
- [34, 37] for the von Neumann-like behavior of B^∞ (i.e., the complemented partition of unity technique from [37, Sections 2,3]) and Theorem 1.2 (the full force of the finite nuclear dimension theorem from [37] is not required);
- [64, 69] for the universal multicoefficient theorem, and aspects of KL ;
- [118] for intertwining via reparameterizations.

1.5. Addendum. Since the first version of this paper, KK -uniqueness has seen major progress. Firstly, Farah and Szabó (building on Farah’s [111]) established $\mathcal{Z}$ -saturation for the multiplier algebra of a $\mathcal{Z}$ -stable algebra. Combining this with our approach to Theorems 1.4 and 5.17(i), their result improves the $\mathcal{Z}$ -stable KK -uniqueness theorem to give a clean statement assuming the ideal I is $\mathcal{Z}$ -stable ([113, Theorem 5.5]). Using this in place of Theorem 5.17 makes things a little bit tidier, but does not result in major simplifications.

Secondly, in [279], Szabó obtained a definitive KK -uniqueness theorem, answering Question 5.21 below positively. Following this breakthrough, we believe that the best abstract approach to the unital classification theorem is now to use [279, Theorem 2.10] in place of Theorem 1.4. Doing so enables one to use the combination of strict comparison and the existence of complemented partitions of unity as

codomain hypotheses in place of $\mathcal{Z}$ -stability. As such, it will be possible to avoid any use of strong self-absorption of $\mathcal{Z}$ (or indeed an explicit construction of $\mathcal{Z}$) in the unital classification theorem (very much in the spirit of the recent work of Bouwen and JG [22] in the purely infinite case).

Additionally, after we first arXived this paper, together with Castillejos and Evington, we have extended the gluing technique from [37] to apply without requiring nuclearity in the preprint [32]. This will be particularly relevant in combination with Szabó's KK -uniqueness theorem. We will set out the effect of all the advances above on the classification of embeddings theorem in the next paper in this series.

We have not modified the main body of the paper to take account of these advances; the argument given here still uses our $\mathcal{Z}$ -stable KK -uniqueness theorem, not least because the material in Section 5.4 is the primary motivation for [278]. However, we have removed our original Section 5.5, as the now obsolete $\mathcal{Z}$ -stable KK_{nuc} - and KL_{nuc} -uniqueness theorems there were designed for further work.

In another direction, CS has built on the ideas in this paper, to take the first steps towards a classification in terms of KK -theory paired with traces ([269]). One result is that finite strongly self-absorbing C^* -algebras are isomorphic if they are unitally KK -equivalent ([269, Corollary C]).

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2. INVARIANTS FOR UNITAL C^* -ALGEBRAS

In this section, we review the invariants used for the classification of unital C^* -algebras and their $*$ -homomorphisms.

2.1. K -theory and traces. Elliott's invariant for the classification of simple nuclear C^* -algebras consists of ordered K -theory and traces. Write $K_* = (K_0, K_1)$ for the operator algebraic K -theory functor from C^* -algebras to pairs of abelian groups (see [258] for an overview). When A is a unital C^* -algebra, the element $[1_A]_0$ represents a distinguished element in $K_0(A)$. The positive elements $K_0(A)_+$ of $K_0(A)$ are precisely those classes that can be realized by a projection in $M_n(A)$ for some $n \geq 1$, so that $(K_0(A), K_0(A)_+)$ is an ordered group whenever A is stably finite and unital (see [10, Proposition 6.3.3], for example).

Since our focus is unital C^* -algebras, by a *trace* we always mean a tracial state. The set of all traces on a unital C^* -algebra A is denoted by $T(A)$. This is convex, and unitality ensures that $T(A)$ is compact in the relative weak*-topology. In fact, though this plays no role in our approach to classification, $T(A)$ is a Choquet simplex (see [262, Theorem 3.1.18], for example) that is metrizable when A is separable. Since $A \mapsto T(A)$ is contravariant, we prefer to work with the function system $\text{Aff } T(A)$ of continuous real-valued affine functions on $T(A)$ rather than with $T(A)$ itself. The duality between compact convex subsets of locally convex spaces and Archimedean order unit spaces (which are concretely realized as function systems) is known as *Kadison duality*, and was set out in [158]. This works as follows: every continuous affine map $\gamma^*: T(B) \rightarrow T(A)$ induces a unital positive linear map $\gamma: \text{Aff } T(A) \rightarrow \text{Aff } T(B)$ given by $\gamma(f)(\tau) := f(\gamma^*(\tau))$ for $f \in \text{Aff } T(A)$ and $\tau \in T(B)$. Conversely, $T(A)$ is naturally recovered as the states on $\text{Aff } T(A)$ (see [133, Theorem 7.1]). That is, $T(A)$ is identified with order preserving group homomorphisms $\text{Aff } T(A) \rightarrow \mathbb{R}$ sending 1 to 1. Therefore, unital positive linear maps $\gamma: \text{Aff } T(A) \rightarrow \text{Aff } T(B)$ give rise to continuous affine maps $\gamma^*: T(B) \rightarrow T(A)$.

The relationship between the function system $\text{Aff } T(A)$ and the space A_{sa} of self-adjoint elements of A goes back to the work of Cuntz and Pedersen ([61]). Every $a \in A_{\text{sa}}$ induces a continuous affine function $\hat{a}: T(A) \rightarrow \mathbb{R}$ by $\hat{a}(\tau) = \tau(a)$ for $\tau \in T(A)$. Write $[a, b] := ab - ba$ and $[A, A]$ for the span of the set $\{ab - ba : a, b \in A\}$, so a trace is precisely a state that vanishes on $[A, A]$. We need the following result, which is often attributed to [61] but does not appear there explicitly (a strategy for deducing it from [61] is indicated at the beginning of the proof of [284, Theorem 3.1]). We give a short proof for completeness (alternatively, one can deduce the result from [178, Lemma 6.2(iii)] or [228, Theorem 5], both of which factor through the enveloping von Neumann algebra).

Proposition 2.1. *Suppose A is a unital C^* -algebra.*

- (i) *If $f \in \text{Aff } T(A)$, then there exists $a \in A_{\text{sa}}$ such that $f = \hat{a}$ (i.e., $f(\tau) = \tau(a)$ for all $\tau \in T(A)$), and a is unique modulo $\overline{[A, A]}$. Moreover, given $\epsilon > 0$, we may choose a with $\|a\| < \|f\| + \epsilon$.*
- (ii) *$[A, A] \cap A_{\text{sa}}$ is spanned by $\{[x, x^*] : x \in A\}$.*

Proof. Let $T_{\mathbb{C}}(A) := \{\tau \in A^* : \tau|_{[A, A]} = 0\}$ be the space of bounded tracial functionals on A , so that there is a canonical isometric isomorphism $T_{\mathbb{C}}(A) \rightarrow (A/\overline{[A, A]})^*$ preserving the weak* topology.

Given a continuous affine function $f: T(A) \rightarrow \mathbb{R}$, we may use the Jordan decomposition to extend f to a bounded self-adjoint linear functional $\tilde{f} \in (T_{\mathbb{C}}(A))^* \cong (A/[\overline{A, A}])^{**}$. Note that $\tilde{f}$ is also weak*-continuous: indeed, the intersection of $\ker \tilde{f}$ with any multiple of the closed unit ball in $T_{\mathbb{C}}(A)$ is weak*-closed (using weak*-compactness, one may replace a bounded weak* convergent net in $T_{\mathbb{C}}(A)$ with one whose Jordan decompositions also converge), so that $\ker \tilde{f}$ is weak*-closed by the Krein–Smulian Theorem ([51, Corollary V.12.6]).

As $\tilde{f}$ is weak*-continuous, there is an $a \in A$ with $\tilde{f}(\tau) = \tau(a)$ for all $\tau \in T(A)$. In fact, because $\tilde{f}$ is self-adjoint, we may choose a to be self-adjoint. Uniqueness in (i) follows from $(T_{\mathbb{C}}(A))^* \cong (A/[\overline{A, A}])^{**}$.

The claim in (ii) follows from the polarization identity applied to the sesquilinear form $(a, b) \mapsto [a, b^*]$. $\square$

The relationship between K_0 and traces is encoded in the natural *pairing map* $\rho_A: K_0(A) \rightarrow \text{Aff } T(A)$. For $n \in \mathbb{N}$ and $\tau \in T(A)$, let τ_n be the canonical extension of τ to a tracial functional on $M_n(A)$ with $\tau_n(1_{M_n(A)}) = n$, and define

$$(2.1) \quad \rho_A([p]_0 - [q]_0)(\tau) := \tau_n(p - q)$$

for all $\tau \in T(A)$ and all projections $p, q \in M_n(A)$.

When A is unital and exact, every state on $(K_0(A), K_0(A)_+, [1_A]_0)$ arises from a trace (i.e., every order-preserving homomorphism $\phi: K_0(A) \rightarrow \mathbb{R}$ with $\phi([1_A]_0) = 1$ is of the form $\phi(x) = \rho_A(x)(\tau)$ for some $\tau \in T(A)$). This combines Haagerup's celebrated [137] with [16]. In particular, unital nuclear stably finite C^* -algebras always have traces.

Combining all of these ingredients gives the Elliott invariant of a unital stably finite C^* -algebra. By using $\text{Aff } T(A)$ to record the tracial component of the invariant, $\text{Ell}(\cdot)$ is a covariant functor.

Definition 2.2. If A is a unital stably finite C^* -algebra, then its *Elliott invariant*, $\text{Ell}(A)$, is

$$(2.2) \quad \text{Ell}(A) := (K_0(A), K_0(A)_+, [1_A]_0, K_1(A), \text{Aff } T(A), \rho_A).$$

Historically the Elliott invariant emerged from Elliott's classification of AF-algebras, and so includes the order on K_0 . For reasons explained below, we prefer to work with a weaker invariant, KT_u (defined below), that does not keep track of the order on K_0 . While we are primarily interested in the case of C^* -algebras with traces, the definition also makes sense when $T(A) = \emptyset$. In this case, $\text{Aff } T(A) = 0$, and $KT_u(A)$ reduces to $K_*(A)$ together with the position of the unit — which is precisely the information used in the classification of unital Kirchberg algebras satisfying the UCT by Kirchberg ([172]) and by Phillips ([236]).

Definition 2.3. Let KT_u be the functor on the category of unital C^* -algebras with unital *-homomorphisms that assigns to a unital C^* -algebra A the quadruple

$$(2.3) \quad KT_u(A) = (K_*(A), [1_A]_0, \text{Aff } T(A), \rho_A).$$

A morphism $(\alpha_*, \gamma): KT_u(A) \rightarrow KT_u(B)$ consists of a pair $\alpha_* = (\alpha_0, \alpha_1)$ of homomorphisms $\alpha_i: K_i(A) \rightarrow K_i(B)$ (for $i = 0, 1$) with $\alpha_0([1_A]_0) = [1_B]_0$, and a positive

unital linear map $\gamma: \text{Aff } T(A) \rightarrow \text{Aff } T(B)$ such that

$$(2.4) \quad \begin{array}{ccc} K_0(A) & \xrightarrow{\rho_A} & \text{Aff } T(A) \\ \downarrow \alpha_0 & & \downarrow \gamma \\ K_0(B) & \xrightarrow{\rho_B} & \text{Aff } T(B) \end{array}$$

commutes. Formally, the objects of the target category of KT_u are quadruples $((G_0, G_1), e, X, \rho)$, where G_0 and G_1 are abelian groups with $e \in G_0$, X is an Archimedean order unit space (the order unit is part of the data), and $\rho: G_0 \rightarrow X$ a group homomorphism mapping e to the order unit. Morphisms between general objects are pairs (α_*, γ) satisfying the analogous commutative diagram to (2.4).

In the case where A is exact and has real rank zero, $T(A)$ can be recovered as states on the ordered group $(K_0(A), K_0(A)_+, [1_A]_0)$ (see [255, Theorem 1.1.11 and Proposition 1.1.12]), so $K_0(A)_+$ can be used in place of $\text{Aff } T(A)$ and ρ_A . Outside this setting, traces are needed for classification, and the pairing map, which first arose explicitly in the invariant in [89], carries data that cannot always be recovered from $T(A)$ and $(K_0(A), K_0(A)_+)$ (see the example on page 29 of [255]). On the other hand, for unital simple nuclear finite $\mathcal{Z}$ -stable C^* -algebras, the order on $K_0(A)$ is induced by the pairing map, (this is a direct consequence of Proposition 4.11(ii) given below, and due to Rørdam from [257]), and so KT_u and the Elliott invariant carry exactly the same information for these algebras.

In general, the order on $K_0(A)$ may fail to be weakly unperforated, in which case, it cannot arise from a pairing map. This phenomenon was first exhibited for a simple nuclear stably finite C^* -algebra by Villadsen in [299]. We prefer to use the invariant KT_u for two reasons: to emphasize that our proof makes no explicit reference to the order structure on $K_0(A)$ and because $KT_u(A) \cong KT_u(A \otimes \mathcal{Z})$ canonically for any C^* -algebra A (see Proposition 4.5 below) whereas, for unital simple A , $\text{Ell}(A \otimes \mathcal{Z}) \cong \text{Ell}(A)$ if and only if $(K_0(A), K_0(A)_+)$ is weakly unperforated ([128, Theorem 1]).

Recall that a trace τ is *faithful* if $\tau(a^*a) = 0$ implies $a = 0$. When B is simple, an embedding $\theta: A \hookrightarrow B$ and a trace $\tau \in T(B)$, gives rise to a faithful trace $\tau \circ \theta$ on A . This gives a necessary condition for a map between invariants to arise from an embedding into a simple C^* -algebra.

Definition 2.4. Let A and B be unital C^* -algebras. We say that $(\alpha_*, \gamma): KT_u(A) \rightarrow KT_u(B)$ is *faithful* if the induced map $\gamma^*: T(B) \rightarrow T(A)$ (given by Kadison duality) satisfies that $\gamma^*(\tau)$ is faithful for all $\tau \in T(B)$. This can be phrased directly in terms of γ as follows: if $f \in \text{Aff } T(A)$ is strictly positive on faithful traces on A , then $\gamma(f)$ is strictly positive (as if there is $a \in A_+ \setminus \{0\}$ with $\gamma^*(\tau)(a) = 0$, then $\hat{a} \in \text{Aff } T(A)$ is strictly positive on faithful traces but $\gamma(\hat{a}) \in \text{Aff } T(B)$ is not).

Remark 2.5. As our approach to classification does not require comparing C^* -algebras with concrete models, the range of the invariant KT_u does not play a role in our proof. However, (in contrast to the Elliott invariant), the range of KT_u on unital C^* -algebras is completely understood. When $T(A) = \emptyset$, $K_0(A)$ and $K_1(A)$ can be arbitrary countable abelian groups, $[1_A]_0$ can be any element of $K_0(A)$, and the pairing map can be disregarded since there are no traces. When $T(A) \neq \emptyset$, $K_1(A)$ can be an arbitrary countable abelian group, $K_0(A)$ is any countable abelian group with a specified element $[1_A]_0$ of infinite order, $T(A)$ can be an arbitrary metrizable

Choquet simplex, and the pairing map $\rho_A: K_0(A) \rightarrow \text{Aff } T(A)$ can be any unit-preserving group homomorphism. Moreover, each possible invariant is realized by some unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebra in the UCT class.

The results described above reinterpret the range of invariant theorems from [253, 92], which predate $\mathcal{Z}$ -stability. While all the examples of [253, 92] are $\mathcal{Z}$ -stable (through $\mathcal{O}_\infty$ -stability for [253] and applying Winter's $\mathcal{Z}$ -stability theorem, [309], for [92]), one can avoid checking this by replacing them with their $\mathcal{Z}$ -stabilizations. It is in realizing the range of the invariant that it is important that $T(A)$ is a Choquet simplex, and hence an inverse limit of finite dimensional simplices ([185]). This allows the models of [92] to be built as inductive limits.

2.2. Unitary algebraic K_1 . The role of the quotient of the infinite unitary group by the closure of the commutator subgroup in the classification program goes back to work of Nielsen and Thomsen ([223]), who showed that it is needed to determine the approximate unitary equivalence classes of maps between simple AT algebras. Although this group and its structure is known to experts, we provide a guide for less familiar readers, both to clarify some topological points and to collect a number of details of the construction — notably, the de la Harpe–Skandalis determinant — for subsequent use.

For a unital C^* -algebra A , write $U_n(A)$ for the group of unitaries in $M_n(A)$ with the norm topology and $DU_n(A)$ for the derived subgroup of $U_n(A)$, i.e., the subgroup generated by commutators uvu^*v^* for $u, v \in U_n(A)$ (which is automatically normal). Each $U_n(A)$ is included into $U_{n+1}(A)$ via the standard connecting maps, and we write $U_\infty(A)$ for the direct limit, which is equipped with the topological inductive limit topology: $V \subseteq U_\infty(A)$ is open if and only if $V \cap U_n(A)$ is open for all $n \in \mathbb{N}$. Let $DU_\infty(A) := \bigcup_n DU_n(A)$ and note that $DU_\infty(A)$ is contained in $U_\infty^{(0)}(A)$. Moreover, any commutator in $U_\infty(A)$ is already a commutator of elements in $U_\infty^{(0)}(A)$: for $u, v \in U_\infty(A)$, one has

$$(2.5) \quad uvu^*v^* = \text{diag}(u, u^*, 1) \text{diag}(v, 1, v^*) \text{diag}(u^*, u, 1) \text{diag}(v^*, 1, v).$$

As multiplication is separately continuous on $U_\infty(A)$ and inversion is continuous, the closure $\overline{DU_\infty(A)}$ of $DU_\infty(A)$ in the inductive limit topology is a normal subgroup of $U_\infty(A)$. Taking the quotient gives the version of algebraic K_1 we need for the uniqueness component of the approximate classification of embeddings.

Definition 2.6. Write

$$(2.6) \quad \overline{K}_1^{\text{alg}}(A) := U_\infty(A) / \overline{DU_\infty(A)}$$

and call this the *Hausdorffized unitary algebraic K_1 -group* of A . Write $[u]_{\text{alg}}$ for the class of $u \in U_\infty(A)$ in $\overline{K}_1^{\text{alg}}(A)$. For a unital $*$ -homomorphism $\phi: A \rightarrow B$, we get an induced map $\overline{K}_1^{\text{alg}}(\phi): \overline{K}_1^{\text{alg}}(A) \rightarrow \overline{K}_1^{\text{alg}}(B)$ given by $\overline{K}_1^{\text{alg}}([u]_{\text{alg}}) := [\phi^{(n)}(u)]_{\text{alg}}$ for $u \in U_n(A)$. In this way $\overline{K}_1^{\text{alg}}$ is a functor from unital C^* -algebras to abelian groups that is invariant under approximate unitary equivalence of morphisms.

Remark 2.7. The algebraic K_1 group of a unital ring A is $GL_\infty(A)/DGL_\infty(A)$. In the C^* -context, it is natural to use unitaries in place of invertibles. The other difference is the quotient by the closure of $DU_\infty(A)$, to ensure invariance under approximate unitary equivalence of morphisms. As an example comparison, the algebraic K_1 -group of $\mathbb{C}$ is $\mathbb{C}^\times$, whereas $\overline{K}_1^{\text{alg}}(\mathbb{C}) = \mathbb{T}$. The Hausdorffized algebraic

K_1 -group (using invertibles) is naturally recovered from $\underline{KT}_u$; see [94, 263] for this and other relations between these algebraic K_1 -groups.

Remark 2.8. When the natural maps $\pi_0(U(A)) \rightarrow K_1(A)$ and $\pi_1(U(A)) \rightarrow K_0(A)$ are bijections, then so is $U(A)/\overline{DU(A)} \rightarrow \overline{K}_1^{\text{alg}}(A)$ ([284, Corollary 3.4]). This is the case, in particular, for C^* -algebras with stable rank one ([247, Theorem 3.3]) and for $\mathcal{Z}$ -stable C^* -algebras ([153, Theorem 3]). This is typically why $U(\cdot)/\overline{DU(\cdot)}$ is used in classification results (for example, in [131]). However, while our codomain algebras from Theorem B are $\mathcal{Z}$ -stable, there is no such restriction on the domains, and so, in general, we cannot work with $U(A)/\overline{DU(A)}$ (or even $U_k(A)/\overline{DU_k(A)}$) in place of $\overline{K}_1^{\text{alg}}(A)$.

The structure of $\overline{K}_1^{\text{alg}}(A)$ was investigated by Thomsen in [284], building on de la Harpe and Skandalis' determinant map and analysis of quotients by commutator subgroups in $GL_\infty(A)$ ([70]). Importantly, it is related to $\text{Aff } T(A)$ and $K_1(A)$ through natural group homomorphisms defined below:

$$(2.7) \quad \text{Aff } T(A) \xrightarrow{\text{Th}_A} \overline{K}_1^{\text{alg}}(A) \xrightarrow{\not\!A} K_1(A).$$

Definition 2.9. Let A be a unital C^* -algebra. The map $\not\!A: \overline{K}_1^{\text{alg}}(A) \rightarrow K_1(A)$ is the canonical surjection given by $\not\!A([u]_{\text{alg}}) := [u]_1$ for $u \in U_\infty(A)$ (noting that $\overline{DU_\infty(A)} \subseteq U_\infty^{(0)}(A)$). The *Thomsen map* Th_A is given by writing an element $f \in \text{Aff } T(A)$ as $f = \hat{a}$ for some $a \in A_{\text{sa}}$, and then setting

$$(2.8) \quad \text{Th}_A(f) := [e^{2\pi i a}]_{\text{alg}}, \quad f = \hat{a}, \quad a \in A_{\text{sa}}.$$

To see that the Thomsen map is well-defined, first note that

$$(2.9) \quad e^{ia}e^{ib} = e^{i(a+b)} \text{ modulo } \overline{DU(A)}$$

for $a, b \in A_{\text{sa}}$ (using the Lie–Trotter formula as in [284, Equation (1.1)]), so that the map $a \mapsto [e^{2\pi i a}]_{\text{alg}}$ is a continuous group homomorphism $A_{\text{sa}} \rightarrow \overline{K}_1^{\text{alg}}(A)$. It then suffices to show $[e^{2\pi i(v^*v - vv^*)}]_{\text{alg}} = 0$ for every $v \in A$ (using Proposition 2.1). This follows by combining (2.9) with the polar decomposition trick found in the last few lines of the proof of [284, Lemma 3.1], which shows that for any $v \in M_n(A)$, $e^{2\pi i(v^*v - vv^*)} = 1_A$ modulo $\overline{DU_n(A)}$. Note that (2.9) also shows that the Thomsen map is a homomorphism. Since $\ker \not\!A$ is generated by exponentials, the definition (2.8) ensures that (2.7) is exact, and we have established the first two out of the following three fundamental properties of the Thomsen map. (One of the inclusions in the last property will be established immediately below, and the second is given after Proposition 2.11).

Properties 2.10. For a unital C^* -algebra A ,

- (i) $\text{Th}_A(\hat{a}) = [e^{2\pi i a}]_{\text{alg}}$ for $a \in A_{\text{sa}}$, where $\hat{a}$ is the continuous affine function $\hat{a}(\tau) = \tau(a)$ for $\tau \in T(A)$;
- (ii) $\text{im } \text{Th}_A = \ker \not\!A$;
- (iii) $\ker \text{Th}_A = \overline{\text{im } \rho_A}$.

For the inclusion of $\overline{\text{im } \rho_A} \subseteq \ker \text{Th}_A$ in (iii), note we can work equally well with elements $a \in M_n(A)_{\text{sa}}$ in (2.8) (following our scaling conventions so that $\hat{1}_{M_n(A)}$ is

the constant function n on $T(A)$). Then, given projections $p, q \in M_n(A)$, we have $\widehat{p - q} = \rho_A([p]_0 - [q]_0)$, and so

$$(2.10) \quad \text{Th}_A(\rho_A([p]_0 - [q]_0)) = [e^{2\pi i(p-q)}]_{\text{alg}} = [e^{2\pi i p}]_{\text{alg}} + [e^{-2\pi i q}]_{\text{alg}} = 0.$$

Given $f \in \text{Aff } T(A)$, one can control the norm of a choice of a satisfying $\hat{a} = f$ (see Proposition 2.1), and therefore the Thomsen map is continuous as a map into $U_1(A)/\overline{DU_1(A)}$, so $\overline{\text{im } \rho_A} \subseteq \ker \text{Th}_A$.

The reverse inclusion $\ker \text{Th}_A \subseteq \overline{\text{im } \rho_A}$ uses the de la Harpe–Skandalis determinant from [70]. For a piecewise smooth path $u: [0, 1] \rightarrow U_n(A)$, de la Harpe and Skandalis defined $\tilde{\Delta}_A(u) \in \text{Aff } T(A)$ by

$$(2.11) \quad \tilde{\Delta}_A(u)(\tau) := \frac{1}{2\pi i} \int_0^1 \tau_n\left(\left(\frac{d}{dt}u(t)\right)u(t)^*\right) dt, \quad \tau \in T(A).$$

We record the basic properties of this determinant in a slightly unusual fashion below, designed for use in Section 3.1 (note that in Proposition 2.11(iv) the proof shows that $\det_A$ actually factors through the more precise quotient $\text{Aff } T(A)/\overline{\text{im } \rho_A}$; we do not need this).

Proposition 2.11. *Let A be a unital C^* -algebra. The de la Harpe–Skandalis determinant map gives a continuous group homomorphism*

$$(2.12) \quad \tilde{\Delta}_A: U_\infty(C([0, 1], A)) \longrightarrow \text{Aff } T(A)$$

such that:

- (i) $\tilde{\Delta}_A(u)$ depends only on the class of u through homotopies of maps $v: C([0, 1]) \rightarrow U_\infty(A)$ with $v(0) = u(0)$ and $v(1) = u(1)$. We call these endpoint-fixed homotopies.
- (ii) for a self-adjoint $h \in M_n(C([0, 1], A))$,

$$(2.13) \quad \tilde{\Delta}_A(e^{2\pi i h})(\tau) = \tau_n(h(1) - h(0)), \quad \tau \in T(A)$$

where τ_n is the canonical non-normalized extension of τ to $M_n(A)$;

- (iii) identifying homotopy classes of $U_\infty(C_0((0, 1), A)^\dagger)$ with $K_0(A)$ via the Bott map, the homomorphism $U_\infty(C_0((0, 1), A)^\dagger) \rightarrow \text{Aff } T(A)$ induced by (i) is precisely the pairing map ρ_A ;

- (iv) there is a continuous group homomorphism

$$(2.14) \quad \det_A: U_\infty^{(0)}(A) \longrightarrow \text{Aff } T(A)/\overline{\text{im } \rho_A}$$

given by $\det_A(u) := \tilde{\Delta}_A(v) + \overline{\text{im } \rho_A}$, where $v \in U_\infty(C([0, 1], A))$ has $v(0) = 1_A$ and $v(1) = u$.

Proof. Observe that $U_\infty(C([0, 1], A))$ and $C([0, 1], U_\infty(A))$ agree, as every compact subset of $U_\infty(A)$ lies in $U_n(A)$ for some n (see [138, Proposition A.1]). Therefore, (2.11) defines $\tilde{\Delta}_A(u)$ for piecewise smooth paths in $U_\infty(C([0, 1], A))$.

For such a path, [70, Lemme 1(c)] shows that $\tilde{\Delta}_A(u)(\tau)$ depends only on the homotopy class of u (through endpoint-fixed homotopies, but not necessarily made up of piecewise smooth paths). Moreover, by dividing $[0, 1]$ into small subintervals where $\|u(s_1) - u(s_2)\| < 2$, every $u \in C([0, 1], U_\infty(A))$ has an endpoint-fixed homotopy to a piecewise smooth path.

Therefore, the definition of $\tilde{\Delta}_A$ extends to all continuous paths $u: [0, 1] \rightarrow U_n(A)$, by applying $\tilde{\Delta}_A$ to a piecewise smooth path homotopic to u (again with endpoints

fixed). Evidently, (i) holds. Moreover, this definition of $\tilde{\Delta}_A$ is a homomorphism by [70, Lemme 1(a)].

Condition (ii) is immediate when h is piecewise smooth, and so follows in general by replacing h with a piecewise smooth path homotopic to h with the same endpoints. Note that the Bott map associates a projection $p \in M_n(A)$ with the unitary $u \in U_n((C_0(0, 1), A)^\dagger)$ given by $u(t) := e^{2\pi i t p}$ for $t \in [0, 1]$, so (iii) follows from (ii). Continuity of $\tilde{\Delta}_A$ on the collection of piecewise smooth paths in $U_n(A)$ is immediate, and then extends to $U_n(C([0, 1], A))$. As this holds for all n , $\tilde{\Delta}_A$ is continuous on $U_\infty(C([0, 1], A))$.

Condition (iv) is essentially [70, Lemme 1(d)] and the subsequent definitions. Once one knows $\det_A$ is well-defined, it is a continuous homomorphism because $\tilde{\Delta}_A$ is. Since $U_\infty^{(0)}(A) = \bigcup_n U_n^{(0)}(A)$ (as every compact subset of $U_\infty(A)$ is contained in some $U_n(A)$) given $u \in U_\infty^{(0)}(A)$, we can find $n \in \mathbb{N}$ and $v \in U_n(C[0, 1], A)$ with $v(0) = 1_{M_n(A)}$ and $v(1) = u$. If $v_1, v_2 \in U_n(C([0, 1], A))$ are two such paths, then $v_1 v_2^*$ can be viewed as an element of $U_n(C_0((0, 1), A)^\dagger)$ and so, by (iii), $\tilde{\Delta}_A(v_1 v_2^*) \in \text{im } \rho_A$. Therefore $\tilde{\Delta}_A(v_1) + \overline{\text{im } \rho_A} = \tilde{\Delta}_A(v_2) + \overline{\text{im } \rho_A}$. $\square$

With this in place, we complete the proof of Properties 2.10.

Proof that $\ker \text{Th}_A \subseteq \overline{\text{im } \rho_A}$ in Property 2.10(iii). Since $\det_A$ is a continuous group homomorphism, $DU_\infty^{(0)}(A) \subseteq \ker \det_A$, and hence $\det_A$ induces a continuous group homomorphism

$$(2.15) \quad \overline{\det}_A: \ker \not\Delta_A = U_\infty^{(0)}(A) / \overline{DU_\infty^{(0)}(A)} \longrightarrow \text{Aff } T(A) / \overline{\text{im } \rho_A}.$$

A computation shows $\overline{\det}_A(\text{Th}_A(f)) = f + \overline{\text{im } \rho_A}$ for all $f \in \text{Aff } T(A)$ (using that $f = \hat{a}$ for some $a \in A_{\text{sa}}$). This implies $\ker \text{Th}_A \subseteq \overline{\text{im } \rho_A}$. $\square$

Combining this with Properties 2.10(ii), we obtain an isomorphism

$$(2.16) \quad \overline{\text{Th}}_A: \text{Aff } T(A) / \overline{\text{im } \rho_A} \xrightarrow{\cong} U_\infty^{(0)}(A) / \overline{DU_\infty(A)} \subset \overline{K_1^{\text{alg}}(A)},$$

and the computation in the proof above shows that

$$(2.17) \quad \overline{\det}_A = \overline{\text{Th}}_A^{-1}.$$

In this way there is a natural assignment of a short exact sequence to a unital C^* -algebra.

Definition 2.12. The *Thomsen extension* of a unital C^* -algebra A is the short exact sequence

$$(2.18) \quad 0 \longrightarrow \text{Aff } T(A) / \overline{\text{im } \rho_A} \xrightarrow{\overline{\text{Th}}_A} \overline{K_1^{\text{alg}}(A)} \xrightarrow{\not\Delta_A} K_1(A) \longrightarrow 0.$$

Note that $\text{Aff } T(A) / \overline{\text{im } \rho_A}$ is a divisible group (for every $f + \overline{\text{im } \rho_A}$, and $n \in \mathbb{N}$, we have $n \cdot (f/n + \overline{\text{im } \rho_A}) = (f + \overline{\text{im } \rho_A})$). For an abelian group, this is equivalent to injectivity (see [303, Corollary 2.3.2], for example), and so there is a splitting $\overline{K_1^{\text{alg}}(A)} \rightarrow \text{Aff } T(A) / \overline{\text{im } \rho_A}$. In this way (2.18) splits unnaturally, and $\overline{K_1^{\text{alg}}(A)}$ can be identified as

$$(2.19) \quad \overline{K_1^{\text{alg}}(A)} \cong \text{Aff } T(A) / \overline{\text{im } \rho_A} \oplus K_1(A)$$

(see [284, Corollary 3.3]). Consequently, any morphism between K -theory and traces can be extended to a map between the $\overline{K}_1^{\text{alg}}$ -groups (as in the following proposition), albeit in a non-canonical way. Therefore, the additional data encoded by $\overline{K}_1^{\text{alg}}$, beyond that held in KT_u , is found not in $\overline{K}_1^{\text{alg}}(A)$ but in the behavior of $\overline{K}_1^{\text{alg}}$ on morphisms. To our knowledge, the first explicit result extending KT_u -morphisms to $\overline{K}_1^{\text{alg}}$ was set out by Nielsen in [222, Section 3] for $U(A)/DU(A)$ in the context of simple AT algebras (see also [223, Lemma 3.2]), although the argument works generally. As it is essential in deducing the unital classification theorem (Theorem A) from the classification of embeddings (Theorem B), we give a short proof of this extension result for completeness.

Proposition 2.13. *Let A and B be unital C^* -algebras and $(\alpha_*, \gamma): KT_u(A) \rightarrow KT_u(B)$ be a morphism. Then there exists a group homomorphism $\beta: \overline{K}_1^{\text{alg}}(A) \rightarrow \overline{K}_1^{\text{alg}}(B)$ such that*

$$(2.20) \quad \begin{array}{ccccccc} K_0(A) & \xrightarrow{\rho_A} & \text{Aff } T(A) & \xrightarrow{\text{Th}_A} & \overline{K}_1^{\text{alg}}(A) & \xrightarrow{\not\alpha_A} & K_1(A) \\ \downarrow \alpha_0 & & \downarrow \gamma & & \downarrow \beta & & \downarrow \alpha_1 \\ K_0(B) & \xrightarrow{\rho_B} & \text{Aff } T(B) & \xrightarrow{\text{Th}_B} & \overline{K}_1^{\text{alg}}(B) & \xrightarrow{\not\alpha_B} & K_1(B) \end{array}$$

commutes. Moreover, if (α_, γ) is an isomorphism, then any such β is an isomorphism.*

Proof. As γ is continuous, since it is unital, positive, and linear, the maps (α_*, γ) induce the diagram

$$(2.21) \quad \begin{array}{ccccccc} 0 & \longrightarrow & \text{Aff } T(A)/\overline{\text{im } \rho_A} & \xrightarrow{\overline{\text{Th}}_A} & \overline{K}_1^{\text{alg}}(A) & \xrightarrow{\not\alpha_A} & K_1(A) \longrightarrow 0 \\ & & \downarrow \bar{\gamma} & & \downarrow \beta & & \downarrow \alpha_1 \\ 0 & \longrightarrow & \text{Aff } T(B)/\overline{\text{im } \rho_B} & \xrightarrow{\overline{\text{Th}}_B} & \overline{K}_1^{\text{alg}}(B) & \xrightarrow{\not\alpha_B} & K_1(B) \longrightarrow 0 \end{array}$$

with exact rows. Both rows of this diagram split because $\text{Aff } T(A)/\overline{\text{im } \rho_A}$ and $\text{Aff } T(B)/\overline{\text{im } \rho_B}$ are divisible abelian groups (as discussed above). Let

$$(2.22) \quad p_A: \overline{K}_1^{\text{alg}}(A) \rightarrow \text{Aff } T(A)/\overline{\text{im } \rho_A} \quad \text{and} \quad i_B: K_1(B) \rightarrow \overline{K}_1^{\text{alg}}(B)$$

be splittings of $\overline{\text{Th}}_A$ and $\not\alpha_B$, respectively. Defining the dashed arrow in (2.21) by

$$(2.23) \quad \beta := i_B \circ \alpha_1 \circ \not\alpha_A + \overline{\text{Th}}_B \circ \bar{\gamma} \circ p_A$$

ensures that (2.21), and hence (2.20), commutes. If (α_*, γ) is an isomorphism, then γ is an isomorphism and thus $\bar{\gamma}$ is an isomorphism. By the five lemma, any map β making (2.21) commute is necessarily an isomorphism. $\square$

We end this section by noting a few topological facts. Firstly, as pointed out to us by George Elliott, $DU_\infty(A)$ is norm-dense in $U_\infty^{(0)}(A)$ (this can be seen using a trick in the proof of [141, Theorem 2.4.7] that goes back to [233]). Thus, it is important to take the closure of $DU_\infty(A)$ in the inductive limit topology when defining $\overline{K}_1^{\text{alg}}(A)$. Secondly, while the inverse map on $U_\infty(A)$ is continuous, multiplication is in general only separately continuous, so $U_\infty(A)$ is not a topological group with the inductive limit topology (in fact, using [281, Theorem 2.7] and [314, Theorem 4] it follows

Proposition 2.14. *Let A be a unital C^* -algebra. Then $\overline{K}_1^{\text{alg}}(A)$ is a Hausdorff topological group in the topology induced by the inductive limit topology on $U_\infty(A)$. Moreover, given a unital C^* -algebra B and a KT_u -morphism $(\alpha_*, \gamma): KT_u(A) \rightarrow KT_u(B)$, any group homomorphism $\beta: \overline{K}_1^{\text{alg}}(A) \rightarrow \overline{K}_1^{\text{alg}}(B)$ making (2.20) commute is automatically continuous.*

$$(2.24) \quad \overline{\text{Th}}_A: \text{Aff } T(A)/\overline{\text{im } \rho_A} \longrightarrow \ker \not\partial_A.$$

For the second statement, note that any map β making (2.20) commute also makes (2.21) commute, so that by applying the continuous map $\det_A$, it follows that $\beta|_{\ker \not{A}_A}$ is continuous. Since $\ker \not{A}_A$ is open and $\overline{K}_1^{\text{alg}}(A)$ is a topological group, it follows that β is continuous. $\square$

The topology $\mathcal{T}_g$ is defined as the strongest topological group topology on $U_\infty(A)$ for which all the inclusions $U_n(A) \rightarrow (U_\infty(A), \mathcal{T}_g)$ are continuous, and consequently has the universal property that if H is a topological group and $\phi_n: U_n(A) \rightarrow H$ are compatible continuous group homomorphisms, then there is a unique $\mathcal{T}_g$ -continuous group homomorphism $\phi: U_\infty(A) \rightarrow H$ extending the ϕ_n 's. Taking $H = \overline{K}_1^{\text{alg}}(A)$, the resulting ϕ annihilates $\overline{DU_\infty(A)}^{\mathcal{T}_g}$ (as $\overline{K}_1^{\text{alg}}(A)$ is abelian and Hausdorff), so descends to a continuous group homomorphism $\bar{\phi}: \overline{K}_1^{\text{alg},g}(A) \rightarrow \overline{K}_1^{\text{alg}}(A)$. The inverse arises as the maps $U_n(A) \rightarrow \overline{K}_1^{\text{alg},g}(A)$ are continuous group homomorphisms and

therefore they extend to a continuous map $(U_\infty(A), \mathcal{T}) \rightarrow \overline{K}_1^{\text{alg},g}(A)$ (by the universal property of the inductive limit topology), which is a group homomorphism, and hence induces a continuous group homomorphism $\overline{K}_1^{\text{alg}}(A) \rightarrow \overline{K}_1^{\text{alg},g}(A)$ inverse to $\bar{\phi}$.

2.3. Total K -theory. Total K -theory comprises K -theory, K -theory with coefficients in $\mathbb{Z}/n\mathbb{Z}$ (written $\mathbb{Z}/n$ from now on) for all $n \geq 2$, and the natural connecting maps between these groups — the *Bockstein operations* — satisfying a number of 6-term exact sequences. To our knowledge, the first explicit use of K -theory with coefficients in the classification program was by Dadarlat and Loring as an obstruction to the classification of certain non-simple C^* -algebras using the Elliott invariant ([67]). Eilers subsequently used K -theory with coefficients (equipped with an order structure) to classify certain AD algebras [83].

For $n \geq 2$ and $i \in \{0, 1\}$, the groups $K_i(A; \mathbb{Z}/n)$ and the Bockstein operations were introduced by Schochet in [271] and have subsequently been expressed in a number of equivalent forms. We will use two pictures of total K -theory in the paper. For the purpose of constructing our new pairing map connecting $K_0(A; \mathbb{Z}/n)$ to $\overline{K}_1^{\text{alg}}(A)$ in Section 3.1, we use the definition

$$(2.25) \quad K_i(A; \mathbb{Z}/n) := K_{1-i}(A \otimes \mathbb{I}_n),$$

where $\mathbb{I}_n$ is the algebra

$$(2.26) \quad \mathbb{I}_n := \{f \in C([0, 1], M_n) : f(0) \in \mathbb{C}1_{M_n}, f(1) = 0\}.$$

This form of the definition is obtained in the joint work of Cuntz and Schochet found in [271, Section 6]; as noted there, one can define $K_i(\cdot; \mathbb{Z}/n) = K_i(\cdot \otimes C_n)$ using any separable nuclear C^* -algebra C_n in the UCT class with $K_*(C_n) \cong (\mathbb{Z}/n, 0)$. In our case, we use $C_n := C_0((0, 1)) \otimes \mathbb{I}_n$ and use Bott periodicity to replace suspensions by index-shifts in (2.25). Other possible choices for C_n include the Cuntz algebras $\mathcal{O}_{n+1}$ (as we shall briefly use in Lemma 3.16) or — as in Schochet's original definition — the mapping cone of a degree n map on $C_0(0, 1)$.

Later in the paper, when we need Dadarlat and Loring's universal multicoefficient theorem, we will use the formulation in terms of $KK^i(\mathbb{I}_n, A)$ from [69, Section 1]. While it is well-known to experts that all pictures of total K -theory are naturally isomorphic, this is not easily extracted from the literature. We handle this point in Appendix A.

Using the definition in (2.25) one obtains natural Bockstein maps

$$(2.27) \quad \mu_{i,A}^{(n)} : K_i(A) \longrightarrow K_i(A; \mathbb{Z}/n) \text{ and } \nu_{i,A}^{(n)} : K_i(A; \mathbb{Z}/n) \longrightarrow K_{1-i}(A)$$

from the short exact sequence $0 \rightarrow C_0((0, 1), M_n) \rightarrow \mathbb{I}_n \rightarrow \mathbb{C} \rightarrow 0$, which after tensoring with A induces a 6-term exact sequence

$$(2.28) \quad \begin{array}{ccccc} K_0(A) & \xrightarrow{\mu_{0,A}^{(n)}} & K_0(A; \mathbb{Z}/n) & \xrightarrow{\nu_{0,A}^{(n)}} & K_1(A) \\ \times n \uparrow & & & & \downarrow \times n \\ K_0(A) & \xleftarrow{\nu_{1,A}^{(n)}} & K_1(A; \mathbb{Z}/n) & \xleftarrow{\mu_{1,A}^{(n)}} & K_1(A) \end{array}$$

that is natural in A . For this, apply the natural identification $K_i(C_0(0, 1) \otimes M_n \otimes A) \cong K_{1-i}(A)$ from Bott periodicity and stability of K -theory. It is easily checked that the boundary maps are multiplication by n . These Bockstein maps are often

denoted by ρ and β in the literature, but as we consistently use ρ and β for other maps, we denote them by μ and ν .

For $n, m \geq 2$, the remaining Bockstein operations

$$(2.29) \quad \begin{aligned} \kappa_{i,A}^{(nm,n)} &: K_i(A; \mathbb{Z}/n) \rightarrow K_i(A; \mathbb{Z}/nm) \quad \text{and} \\ \kappa_{i,A}^{(n,nm)} &: K_i(A; \mathbb{Z}/nm) \rightarrow K_i(A; \mathbb{Z}/n) \end{aligned}$$

are induced by the canonical inclusions $\mathbb{I}_n \xrightarrow{\cong} \mathbb{I}_n \otimes 1_{M_m} \hookrightarrow \mathbb{I}_{nm}$ and $\mathbb{I}_{nm} \hookrightarrow \mathbb{I}_n \otimes M_m$ respectively. These maps interact with the rows of (2.28) through the following commutative diagram (which is easily checked directly from the definitions):

$$(2.30) \quad \begin{array}{ccccc} K_i(A) & \xrightarrow{\times m} & K_i(A) & \xlongequal{\quad} & K_i(A) \\ \downarrow \mu_{i,A}^{(n)} & & \downarrow \mu_{i,A}^{(nm)} & & \downarrow \mu_{i,A}^{(n)} \\ K_i(A; \mathbb{Z}/n) & \xrightarrow{\kappa_{i,A}^{(nm,n)}} & K_i(A; \mathbb{Z}/nm) & \xrightarrow{\kappa_{i,A}^{(n,nm)}} & K_i(A; \mathbb{Z}/n) \\ \downarrow \nu_{i,A}^{(n)} & & \downarrow \nu_{i,A}^{(nm)} & & \downarrow \nu_{i,A}^{(n)} \\ K_{1-i}(A) & \xlongequal{\quad} & K_{1-i}(A) & \xrightarrow{\times m} & K_{1-i}(A) \end{array}$$

Definition 2.16. The *total K-theory*, $\underline{K}(A)$, of a C^* -algebra A is the collection of abelian groups $K_i(A)$ and $K_i(A; \mathbb{Z}/n)$ for $i = 0, 1$ and $n = 2, 3, \dots$, together with all the Bockstein maps

$$(2.31) \quad \{\mu_{i,A}^{(n)}, \nu_{i,A}^{(n)}, \kappa_{i,A}^{(n,nm)}, \kappa_{i,A}^{(nm,n)} : n, m \geq 2, i \in \{0, 1\}\}.$$

A Λ -*morphism* $\underline{\alpha}: \underline{K}(A) \rightarrow \underline{K}(B)$ consists of homomorphisms

$$(2.32) \quad \alpha_i: K_i(A) \rightarrow K_i(B) \quad \text{and} \quad \alpha_i^{(n)}: K_i(A; \mathbb{Z}/n) \rightarrow K_i(B; \mathbb{Z}/n)$$

for all $i \in \{0, 1\}$ and $n \geq 2$ that intertwine all the Bockstein operations. The collection of these Λ -morphisms is written $\text{Hom}_\Lambda(\underline{K}(A), \underline{K}(B))$.

As noted in [271, Proposition 1.8], the sequence (2.28) collapses to a short exact sequence

$$(2.33) \quad 0 \longrightarrow K_i(A) \otimes \mathbb{Z}/n \xrightarrow{\overline{\mu}_{i,A}^{(n)}} K_i(A; \mathbb{Z}/n) \xrightarrow{\overline{\nu}_{i,A}^{(n)}} \text{Tor}(K_{1-i}(A), \mathbb{Z}/n) \longrightarrow 0.$$

Alternatively, as $\mathbb{I}_n$ satisfies the UCT, (2.33) can be obtained from the Künneth exact sequence for $K_{1-i}(A \otimes \mathbb{I}_n)$ from [260].

Explicitly, $\text{Tor}(K_{1-i}(A), \mathbb{Z}/n)$ consists of those elements $x \in K_{1-i}(A)$ with $nx = 0$. The sequence (2.33) splits unnaturally ([271, Proposition 2.4]). Moreover, as is known to experts, these splittings may be chosen so that they preserve the Bockstein operations. This goes back to Bödigheimer ([19, 20]) in a purely abstract framework and is noted in the C^* -algebraic setting in [83, Lemma 2.2.8] and [85, Remark 4.6], resulting in Proposition 2.17 below. We explain how to extract this precise statement from the literature in Appendix A.3 (the proof follows that of Lemma A.5) since it is not entirely straightforward.

In order to be clear about exactly where the UCT assumption is used in our approach to the classification theorem, we highlight that Proposition 2.17 is a purely algebraic result, and does not require the UCT. However, in the proof of the classification theorem, we will only apply the proposition when A satisfies the UCT. In this case, as will be familiar to many readers, the result quickly follows from the UCT assumption: briefly, any $\alpha_*: K_*(A) \rightarrow K_*(B)$ lifts to an element of $KK(A, B)$

by the UCT, and this induces a Λ -morphism $\underline{\alpha}: \underline{K}(A) \rightarrow \underline{K}(B)$, which restricts to α_* . The map $\underline{\alpha}$ is an isomorphism if α_* is (by applying the five lemma to (2.28)). For this reason we do not include [19, 20] in Section 1.4.2 in the list of references we rely on for our proofs of Theorems A and B.

Proposition 2.17. *Let A and B be C^* -algebras. Any morphism $\alpha_*: K_*(A) \rightarrow K_*(B)$ can be extended to a Λ -morphism $\underline{\alpha}: \underline{K}(A) \rightarrow \underline{K}(B)$, which is an isomorphism when α_* is.*

As with $\overline{K}_1^{\text{alg}}(A)$, the additional information captured by $\underline{K}$ — compared with K_* — lies in its behavior on morphisms, and not in the constituent groups of $\underline{K}(A)$ (which are completely determined by $K_*(A)$).

We end this subsection with the following calculation using the Bockstein maps, which will be applied to the trace-kernel extension as part of the proof of Theorem 1.6. The relevant hypotheses will be verified in Section 6.3. If G is an abelian group, we write $\text{Tor}(G)$ for the torsion subgroup of G consisting of those elements in G of finite order; i.e., $\text{Tor}(G) = \bigcup_{n \geq 2} \text{Tor}(G, \mathbb{Z}/n)$.

Lemma 2.18. *Let A be a C^* -algebra and let*

$$(2.34) \quad 0 \longrightarrow I \xrightarrow{j} E \xrightarrow{q} D \longrightarrow 0$$

be an extension of C^ -algebras such that $K_1(D) = 0$ and $K_0(D)$ is uniquely divisible, that is, for each $x \in K_0(D)$ and $n \in \mathbb{Z}$ with $n \neq 0$, there exists a unique $y \in K_0(D)$ with $ny = x$. Then a Λ -morphism $\underline{\alpha}: \underline{K}(A) \rightarrow \underline{K}(I)$ belongs to $\ker \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(j))$ if and only if the only non-zero component of $\underline{\alpha}$ is α_1 , $K_1(j) \circ \alpha_1 = 0$, and α_1 vanishes on $\text{Tor}(K_1(A))$. In fact, there is an isomorphism*

$$(2.35) \quad \ker \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(j)) \xrightarrow{\cong} \text{Hom}(K_1(A)/\text{Tor}(K_1(A)), \ker K_1(j))$$

which is natural in both A and in extensions satisfying the specified hypotheses.

Proof. Since $K_0(D)$ is uniquely divisible, multiplication by n is an automorphism of $K_0(D)$ for all $n \geq 2$. This, together with $K_1(D) = 0$ and the exactness of (2.28), gives $K_i(D; \mathbb{Z}/n) = 0$ for all $i = 0, 1$ and $n \geq 2$. Since $K_*(\cdot; \mathbb{Z}/n)$ takes short exact sequences to 6-term exact sequences (this follows from its definition as $K_{1-*}(\cdot \otimes \mathbb{I}_n)$), it follows that $\ker K_i(j; \mathbb{Z}/n) = 0$ for all $n \geq 2$ and $i \in 0, 1$. Similarly, $\ker K_0(j) = 0$ since $K_1(D) = 0$.

Let $\underline{\alpha}: \underline{K}(A) \rightarrow \underline{K}(I)$ be in $\ker \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(j))$ with components α_i and $\alpha_i^{(n)}$ for $i = 0, 1$ and $n \geq 2$. As these homomorphisms factor through $\ker K_i(j)$ and $\ker K_i(j; \mathbb{Z}/n)$ respectively, it follows that $\alpha_0 = 0$ and $\alpha_i^{(n)} = 0$, for $i = 0, 1$. Because $\underline{\alpha}$ is a Λ -morphism, we have $\alpha_1 \circ \nu_{0,A}^{(n)} = \nu_{0,I}^{(n)} \circ \alpha_0^{(n)} = 0$. It follows that α_1 vanishes on the image of $\nu_{0,A}^{(n)}$ for every $n \geq 2$. By exactness of (2.28), the image of $\nu_{0,A}^{(n)}$ consists precisely of the n -torsion elements of $K_1(A)$, and thus α_1 vanishes on $\text{Tor}(K_1(A))$. Thus the map sending $\underline{\alpha} \in \ker \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(j))$ to the map $K_1(A)/\text{Tor}(K_1(A)) \rightarrow K_1(I)$ induced by α_1 is a well-defined injection.

For surjectivity in (2.35), suppose $\alpha'_1: K_1(A)/\text{Tor}(K_1(A)) \rightarrow \ker K_1(j)$ is a homomorphism, and let α_1 be the composition

$$(2.36) \quad K_1(A) \longrightarrow K_1(A)/\text{Tor}(K_1(A)) \xrightarrow{\alpha'_1} \ker K_1(j) \subseteq K_1(I).$$

We define $\underline{\alpha}$ by setting the other components, that is, $\alpha_0: K_0(A) \rightarrow K_0(I)$ and $\alpha_i^{(n)}: K_i(A; \mathbb{Z}/n) \rightarrow K_i(I; \mathbb{Z}/n)$, to be the zero maps. This will certainly lie in the kernel of $\text{Hom}_\Lambda(\underline{K}(A), \underline{K}(I))$ once we have shown that these choices of components do intertwine the Bockstein operations, i.e., that $\underline{\alpha}$ is a Λ -morphism.

Since $\alpha_0 = 0$ and $\alpha_i^{(n)} = 0$, we only need to verify intertwining involving α_1 . That is, we must verify the commutativity of

$$(2.37) \quad \begin{array}{ccc} K_1(A) & \xrightarrow{\mu_{1,A}^{(n)}} & K_1(A; \mathbb{Z}/n) \\ \downarrow \alpha_1 & & \downarrow \alpha_1^{(n)}=0 \\ K_1(I) & \xrightarrow{\mu_{1,I}^{(n)}} & K_1(I; \mathbb{Z}/n) \end{array} \quad \text{and} \quad \begin{array}{ccc} K_0(A; \mathbb{Z}/n) & \xrightarrow{\nu_{0,A}^{(n)}} & K_1(A) \\ \downarrow \alpha_0^{(n)}=0 & & \downarrow \alpha_1 \\ K_0(I; \mathbb{Z}/n) & \xrightarrow{\nu_{0,I}^{(n)}} & K_1(I) \end{array}.$$

As noted above, the image of $\nu_{0,A}^{(n)}$ consists of n -torsion elements so is contained in $\text{Tor}(K_1(A))$. Thus $\alpha_1 \circ \nu_{0,A}^{(n)} = 0$ and the right square in (2.37) commutes. For commutativity of the left square, we note that since α_1 takes values in $\ker K_1(j)$, naturality of the Bockstein maps $\mu_1^{(n)}$ gives

$$(2.38) \quad K_1(j; \mathbb{Z}/n) \circ \mu_{1,I}^{(n)} \circ \alpha_1 = \mu_{1,E}^{(n)} \circ K_1(j) \circ \alpha_1 = 0.$$

By injectivity of $K_1(j; \mathbb{Z}/n)$ (as observed at the beginning the proof), it follows that $\mu_{1,I}^{(n)} \circ \alpha_1 = 0$, so the left hand square of (2.37) also commutes. $\square$

3. THE TOTAL INVARIANT

This section assembles the total invariant $\underline{KT}_u(\cdot)$. We start out by constructing natural maps

$$(3.1) \quad \zeta_A^{(n)}: K_0(A; \mathbb{Z}/n) \rightarrow \overline{K}_1^{\text{alg}}(A)$$

relating total and algebraic K -theory in Section 3.1. Such a system of maps was independently constructed by Gong, Lin, and Niu in [130] by other means; see Remark 3.4. Section 3.2 formalizes $\underline{KT}_u(\cdot)$ and proves an “extension of invariants” theorem (Theorem 3.9): (iso)morphisms at the level of $\underline{KT}_u(\cdot)$ lift to $\underline{KT}_u(\cdot)$. This is crucial in deducing Theorem A and Corollary C from Theorem B. We end by observing that the inclusion of a C^* -algebra into its sequence algebra is injective on invariants (Lemma 3.16), for use in establishing Theorem B from Theorem 1.1.

3.1. Factorizing the Bockstein maps. Let A be a unital C^* -algebra. There is a well-known bijection between projections $p \in A$ and self-adjoint unitaries $u \in A$ given by $p \mapsto 1 - 2p = e^{2\pi i p/2}$. Since self-adjoint unitaries are trivial in K_1 , this identification is not interesting at the level of K -theory. But the map $[p]_0 \mapsto [1 - 2p]_{\text{alg}} = [e^{2\pi i p/2}]_{\text{alg}} = \text{Th}_A(\rho_A([p]_0)/2)$ does give a well-defined natural map $K_0(A) \rightarrow \overline{K}_1^{\text{alg}}(A)$ that factorizes through $K_0(A) \otimes \mathbb{Z}/2$. More generally, for any $n \geq 2$, we have a map $K_0(A) \rightarrow \overline{K}_1^{\text{alg}}(A)$ given by $[p]_0 \mapsto [e^{2\pi i p/n}]_{\text{alg}}$ that factors through $K_0(A) \otimes \mathbb{Z}/n$. Recall that the Bockstein maps induce the short exact sequence

$$(3.2) \quad 0 \longrightarrow K_0(A) \otimes \mathbb{Z}/n \xrightarrow{\tilde{\mu}_{0,A}^{(n)}} K_0(A; \mathbb{Z}/n) \xrightarrow{\tilde{\nu}_{0,A}^{(n)}} \text{Tor}(K_1(A), \mathbb{Z}/n) \longrightarrow 0$$

that splits unnaturally (as a consequence of the unnatural splitting of the Künneth formula, [10, Theorem 23.1.3]). This allows the maps above to be extended (a priori unnaturally) from $K_0(A) \otimes \mathbb{Z}/n$ to $K_0(A; \mathbb{Z}/n)$.

In the next two paragraphs and Proposition 3.1, we show that this can, in fact, be done naturally. Then, in Proposition 3.2, we show that the resulting map is compatible with the other structure maps.

Fix $n \geq 2$. We identify $(A \otimes \mathbb{I}_n)^\dagger$ with

$$(3.3) \quad \{f \in C([0, 1], A \otimes M_n) : f(0) \in A \otimes 1_{M_n}, f(1) \in \mathbb{C}1_{A \otimes M_n}\}.$$

Write $\text{ev}_A^{(0,n)} : (A \otimes \mathbb{I}_n)^\dagger \rightarrow A$ for the map induced by evaluation at 0. (In particular, note that $\text{ev}_A^{(0,n)}(1_{(A \otimes \mathbb{I}_n)^\dagger}) = 1_A$ and not $1_{A \otimes M_n}$.) We will abuse notation and also write $\text{ev}_A^{(0,n)}$ for the induced map when taking matrix amplifications. Then, using the definition $K_0(A; \mathbb{Z}/n) = K_1(A \otimes \mathbb{I}_n)$, the Bockstein map $\nu_{0,A}^{(n)} : K_0(A; \mathbb{Z}/n) \rightarrow K_1(A)$ is the map $K_1(\text{ev}_A^{(0,n)})$.

Let $u \in U_\infty((A \otimes \mathbb{I}_n)^\dagger)$; in spirit, we would like to define $\zeta_A^{(n)}([u]_1)$ to be $[\text{ev}_A^{(0,n)}(u)]_{\text{alg}}$. However, this would not be well-defined since the class $[\text{ev}_A^{(0,n)}(u)]_{\text{alg}}$ in $\overline{K}_1^{\text{alg}}(A)$ is not invariant under homotopy, and thus we need a correcting term built using the de la Harpe–Skandalis determinant. Write $\text{ev}_A^{(1,n)} : (A \otimes \mathbb{I}_n)^\dagger \rightarrow \mathbb{C}1_A \subseteq A$ for the map induced by evaluation at 1 (again with the normalization convention $\text{ev}_A^{(1,n)}(1_{(A \otimes \mathbb{I}_n)^\dagger}) = 1_A$). Regarding u as element of $U_\infty(C([0, 1], A))$, one obtains $\tilde{\Delta}_A(u) \in \text{Aff } T(A)$ as in Proposition 2.11. Now define $\tilde{\zeta}_A^{(n)} : U_\infty((A \otimes \mathbb{I}_n)^\dagger) \rightarrow \overline{K}_1^{\text{alg}}(A)$ by

$$(3.4) \quad \tilde{\zeta}_A^{(n)}(u) := [\text{ev}_A^{(0,n)}(u)]_{\text{alg}} - [\text{ev}_A^{(1,n)}(u)]_{\text{alg}} + \text{Th}_A\left(\frac{1}{n}\tilde{\Delta}_A(u)\right).$$

The factor of $\frac{1}{n}$ arises in (3.4) to account for the fact that when $u \in (A \otimes \mathbb{I}_n)^\dagger$, the computation in the last term is done viewing u as a path in $M_n(A)$, while the normalization conventions use elements of A for the first two terms.

Note that the last two terms in (3.4) lie in the kernel of $\not\! \! \! \mathbb{A}_A$ (as $\text{ev}_A^{(1,n)}(u) \in U_\infty(\mathbb{C})$, and Th_A takes values in $\ker \not\! \! \! \mathbb{A}_A$). Accordingly,

$$(3.5) \quad \not\! \! \! \mathbb{A}_A(\tilde{\zeta}_A^{(n)}(u)) = [\text{ev}_A^{(0,n)}(u)]_1 = \nu_{0,A}^{(n)}([u]_1).$$

Proposition 3.1. *Let A be a unital C^* -algebra and let $n \geq 2$. Then the map $\tilde{\zeta}_A^{(n)}$ in (3.4) induces a group homomorphism $\zeta_A^{(n)} : K_0(A; \mathbb{Z}/n) \rightarrow \overline{K}_1^{\text{alg}}(A)$ that is natural in A and factorizes the Bockstein map $\nu_{0,A}^{(n)}$ through $\overline{K}_1^{\text{alg}}(A)$ as $\nu_{0,A}^{(n)} = \not\! \! \! \mathbb{A}_A \circ \zeta_A^{(n)}$.*

Proof. As $\tilde{\Delta}_A$ gives a homomorphism from $U_\infty(C([0, 1], A))$ to $\text{Aff } T(A)$ by Proposition 2.11, we have $\tilde{\zeta}_A^{(n)}(uv) = \tilde{\zeta}_A^{(n)}(u) + \tilde{\zeta}_A^{(n)}(v)$ for all $u, v \in U_\infty((A \otimes \mathbb{I}_n)^\dagger)$. Therefore, to show $\tilde{\zeta}_A^{(n)}$ induces a well-defined homomorphism $\zeta_A^{(n)}$ on $K_0(A; \mathbb{Z}/n)$ it suffices to check that $\tilde{\zeta}_A^{(n)}(e^{2\pi i h}) = 0$ in $\overline{K}_1^{\text{alg}}(A)$ for a self-adjoint $h \in M_k(A \otimes \mathbb{I}_n)^\dagger$ and $k \in \mathbb{N}$. For $\tau \in T(A)$, Proposition 2.11(ii) gives

$$(3.6) \quad \frac{1}{n}\tilde{\Delta}_A(e^{2\pi i h})(\tau) = \frac{1}{n}\tau_{nk}(h(1) - h(0)) = \tau_k(\text{ev}_A^{(1,n)}(h) - \text{ev}_A^{(0,n)}(h))$$

where τ_{nk} and τ_k are the non-normalized extensions of τ to $M_{kn}(A)$ and $M_k(A)$, and the second equality is a consequence of our normalization conventions. Thus, $\text{ev}_A^{(1,n)}(h) - \text{ev}_A^{(0,n)}(h)$ induces the affine function $\frac{1}{n}\tilde{\Delta}_A(e^{2\pi i h})$, and so the definition of the Thomsen map in (2.8) (working at the matrix amplification level as discussed

in the text above (2.10)) gives

$$(3.7) \quad \begin{aligned} \mathrm{Th}_A(\tfrac{1}{n}\tilde{\Delta}_A(e^{2\pi i h})) &= [e^{2\pi i(\mathrm{ev}_A^{(1,n)}(h) - \mathrm{ev}_A^{(0,n)}(h))}]_{\mathrm{alg}} \\ &\stackrel{(2.9)}{=} [\mathrm{ev}_A^{(1,n)}(e^{2\pi i h})]_{\mathrm{alg}} - [\mathrm{ev}_A^{(0,n)}(e^{2\pi i h})]_{\mathrm{alg}}. \end{aligned}$$

Hence $\tilde{\zeta}_A^{(n)}(e^{2\pi i h}) = 0$ and so $\zeta_A^{(n)}$ is well-defined. Now $\nu_{0,A}^{(n)} = \sharp_A \circ \zeta_A^{(n)}$ follows from (3.5).

Naturality of $\zeta_A^{(n)}$ follows as all components of the construction of $\tilde{\zeta}^{(n)}$ in (3.4) are natural. $\square$

The next proposition shows that the maps $\zeta_A^{(n)}$ satisfy three further compatibility relations with the other structure maps.

Proposition 3.2. *If A is a unital C^* -algebra and $m, n \geq 2$, then the diagrams*

$$(3.8) \quad \begin{array}{ccccc} K_0(A) & \xrightarrow{\mu_{0,A}^{(n)}} & K_0(A; \mathbb{Z}/n) & \xrightarrow{\nu_{0,A}^{(n)}} & K_1(A) \\ \downarrow \frac{1}{n}\rho_A & & \downarrow \zeta_A^{(n)} & & \parallel \\ \mathrm{Aff} T(A) & \xrightarrow{\mathrm{Th}_A} & \overline{K}_1^{\mathrm{alg}}(A) & \xrightarrow{\sharp_A} & K_1(A) \end{array}$$

and

$$(3.9) \quad \begin{array}{ccccc} K_0(A; \mathbb{Z}/nm) & \xrightarrow{\kappa_{0,A}^{(n,nm)}} & K_0(A; \mathbb{Z}/n) & \xrightarrow{\kappa_{0,A}^{(nm,n)}} & K_0(A; \mathbb{Z}/m) \\ & \searrow m\zeta_A^{(nm)} & \downarrow \zeta_A^{(n)} & \swarrow \zeta_A^{(nm)} & \\ & & \overline{K}_1^{\mathrm{alg}}(A) & & \end{array}$$

commute.

Proof. Proposition 3.1 already shows that the right square in (3.8) commutes.

We first show the commutativity of the left square of (3.8). Recall that after identifying $K_0(A) \cong K_1(C_0((0,1), A))$ by Bott periodicity, $\mu_{0,A}^{(n)}$ is given by the inclusion of $C((0,1), A \otimes M_n)$ into $A \otimes \mathbb{I}_n$. Therefore, given a unitary $u \in U_\infty(C((0,1), A)^\dagger)$, and regarding it as a path $[0,1] \rightarrow U_\infty(A)$ with $u(0) = u(1) \in U_\infty(\mathbb{C})$, $\mu_{0,A}^{(n)}([u]_1)$ is the class of u in $K_1(A \otimes \mathbb{I}_n) = K_0(A; \mathbb{Z}/n)$. By Proposition 2.11(iii), $\tilde{\Delta}_A(u) = \rho_A([u]_1)$ (where we again identify $K_0(A) \cong K_1(C_0((0,1), A))$). Since $\mathrm{ev}_A^{(0,n)}(u) = \mathrm{ev}_A^{(1,n)}(u)$, we have

$$(3.10) \quad \zeta_A^{(n)}(\mu_{0,A}^{(n)}([u]_1)) = \tilde{\zeta}_A^{(n)}(u) = \mathrm{Th}_A(\tfrac{1}{n}\rho_A([u]_1)).$$

Next we deal with the right triangle of (3.9). Recall that $\kappa_A^{(nm,n)}$ is induced by the canonical inclusion $\iota: \mathbb{I}_n \xrightarrow{\cong} \mathbb{I}_n \otimes 1_{M_m} \hookrightarrow \mathbb{I}_{nm}$. For $u \in U_\infty((A \otimes \mathbb{I}_n)^\dagger)$ and $i = 0, 1$, we have $\mathrm{ev}_A^{(i,nm)}(\iota(u)) = \mathrm{ev}_A^{i,n}(u)$. Since $\tilde{\Delta}_A(u \otimes 1_{M_m}) = m\tilde{\Delta}_A(u)$, it follows that $(\zeta_A^{(nm)} \circ \kappa_A^{(nm,n)})([u]_1) = \zeta_A^{(n)}([u]_1)$.

Finally, we handle the left-hand triangle of (3.9). Since $\kappa_A^{(n,nm)}$ is induced by the inclusion $\mathbb{I}_{nm} \subseteq \mathbb{I}_n \otimes M_m$, for $u \in U_k((A \otimes \mathbb{I}_{nm})^\dagger)$, our normalization conventions

$$\begin{aligned}
(3.11) \quad & \zeta_A^{(n)} \left(\kappa_A^{(n,nm)}([u]_1) \right) \\
&= [\mathrm{ev}_{M_m(A)}^{(0,n)}(u)]_{\mathrm{alg}} - [\mathrm{ev}_{M_m(A)}^{(1,n)}(u)]_{\mathrm{alg}} + \mathrm{Th}_A \left(\frac{1}{n} \widetilde{\Delta}_A(u) \right) \\
&= m \left([\mathrm{ev}_A^{(0,nm)}(u)]_{\mathrm{alg}} - [\mathrm{ev}_A^{(1,nm)}(u)]_{\mathrm{alg}} + \mathrm{Th}_A \left(\frac{1}{nm} \widetilde{\Delta}_A(u) \right) \right) \\
&= m \zeta_A^{(nm)}([u]_1),
\end{aligned}$$
$$(3.12) \quad \zeta_A^{(n)}(\mu_{0,A}^{(n)}([p]_0)) = \mathrm{Th}_A(\tfrac{1}{n}\hat{p}) = [e^{2\pi i p/n}]_{\mathrm{alg}}.$$

Remark 3.4. As noted in the introduction, while this paper was being written, Gong, Lin, and Niu independently discovered the need for compatibility with natural maps $K_0(A; \mathbb{Z}/n) \rightarrow \overline{K}_1^{\text{alg}}(A)$ in [130] — see Remark 6.6 therein. The notation in [130] is in terms of the unitary group modulo the closure of the commutators, and there is often a finite stable rank condition which allows them to work with $U_n(A)$ instead of $U_\infty(A)$ (see Remark 2.8). (It is a happy coincidence that the same symbol ζ was adopted in [130] and by us.)

Gong, Lin, and Niu construct their maps abstractly through a splitting of the Thomsen extension for $A \otimes \mathbb{I}_n$. It turns out that the two maps are identical (taking care of our different conventions regarding the orientation of the interval in $\mathbb{I}_n$). In fact, since this paper was submitted, Munkholm has shown a general uniqueness statement for maps making the right square of (3.8) commute ([218]), which immediately demonstrates that our maps agree with those of Gong, Lin, and Niu.

3.2. The total invariant. With all the ingredients in place, we now formalize the invariant needed to classify embeddings. Recall our convention that the Bockstein maps $\mu_{i,A}^{(n)}, \nu_{i,A}^{(n)}, \kappa_{i,A}^{(m,n)}$ form part of $\underline{K}(A)$.

Definition 3.5. Let A be a unital C^* -algebra. The *total invariant* of A , denoted $\underline{KT}_u(A)$, is

$$(3.13) \quad \underline{KT}_u(A) := (\underline{K}(A), \text{Aff } T(A), \overline{K}_1^{\text{alg}}(A), [1_A]_0, \rho_A, \text{Th}_A, \mathfrak{A}_A, (\zeta_A^{(n)})_{n \geq 2})$$

with the convention that $\text{Aff } T(A) = 0$ if $T(A) = \emptyset$. A morphism $(\underline{\alpha}, \beta, \gamma) : \underline{KT}_u(A) \rightarrow \underline{KT}_u(B)$ consists of

- $\underline{\alpha} \in \text{Hom}_{\Lambda}(\underline{K}(A), \underline{K}(B))$ with $\alpha_0([1_A]_0) = [1_B]_0$,
- a positive linear map $\gamma: \text{Aff } T(A) \rightarrow \text{Aff } T(B)$, and
- a group homomorphism $\beta: \overline{K}_1^{\text{alg}}(A) \rightarrow \overline{K}_1^{\text{alg}}(B)$,

such that

$$(3.14) \quad \begin{array}{ccccccc} K_0(A) & \xrightarrow{\rho_A} & \text{Aff } T(A) & \xrightarrow{\text{Th}_A} & \overline{K}_1^{\text{alg}}(A) & \xrightarrow{\not\!A_A} & K_1(A) \\ \downarrow \alpha_0 & & \downarrow \gamma & & \downarrow \beta & & \downarrow \alpha_1 \\ K_0(B) & \xrightarrow{\rho_B} & \text{Aff } T(B) & \xrightarrow{\text{Th}_B} & \overline{K}_1^{\text{alg}}(B) & \xrightarrow{\not\!A_B} & K_1(B) \end{array}$$

and

$$(3.15) \quad \begin{array}{ccc} K_0(A; \mathbb{Z}/n) & \xrightarrow{\zeta_A^{(n)}} & \overline{K}_1^{\text{alg}}(A) \\ \downarrow \alpha_0^{(n)} & & \downarrow \beta \\ K_0(B; \mathbb{Z}/n) & \xrightarrow{\zeta_B^{(n)}} & \overline{K}_1^{\text{alg}}(B) \end{array}$$

commute.

As with KT_u -morphisms, we say that $(\underline{\alpha}, \beta, \gamma): \underline{KT}_u(A) \rightarrow \underline{KT}_u(B)$ is *faithful* if the induced map $\gamma^*: T(B) \rightarrow T(A)$ (given by Kadison duality) satisfies that $\gamma^*(\tau)$ is faithful for all $\tau \in T(B)$.

Note that the tracial component of a $\underline{KT}_u$ -morphism $(\underline{\alpha}, \gamma)$ is unital as

$$(3.16) \quad \gamma(1) = \gamma(\rho_A([1_A]_0)) \stackrel{(3.14)}{=} \rho_B(\alpha_0([1_A]_0)) = \rho_B([1_B]_0) = 1.$$

It is important for us that $\underline{KT}_u$ is a functor; the point is not the exact specification of the target category (this can be done in similar fashion to the end of Definition 2.3), but rather that unital $*$ -homomorphisms do induce $\underline{KT}_u$ -morphisms by naturality of the $\zeta^{(n)}$ (together with functoriality of KT_u). This is vital in the proof of the existence portion of the classification of full approximate embeddings in the presence of torsion in K_1 . Moreover, $\underline{KT}_u(\cdot)$ is invariant under approximate unitary equivalence as each component is.

Proposition 3.6. *Given a unital $*$ -homomorphism $\phi: A \rightarrow B$,*

$$(3.17) \quad \underline{KT}_u(\phi) := (\underline{K}(\phi), \overline{K}_1^{\text{alg}}(\phi), \text{Aff } T(\phi)): \underline{KT}_u(A) \rightarrow \underline{KT}_u(B)$$

is a $\underline{KT}_u$ -morphism. In this way, the assignment of $\underline{KT}_u(A)$ to a unital C^ -algebra A provides a functor from the category of unital C^* -algebras and unital $*$ -homomorphisms. Furthermore, if $\phi, \psi: A \rightarrow B$ are approximately unitarily equivalent, then $\underline{KT}_u(\phi) = \underline{KT}_u(\psi)$.*

Remark 3.7. Let B be a unital simple purely infinite C^* -algebra. Then $T(B) = \emptyset$ and exactness of (2.18) shows that $\not\!A_B$ is an isomorphism. Thus, for these codomains, the functors $\underline{KT}_u(\cdot)$ and $(\underline{K}(\cdot), [1]_0)$ carry the same information. More precisely, given any unital C^* -algebra A , there is a one-to-one correspondence between $\underline{\alpha}: \underline{K}(A) \rightarrow \underline{K}(B)$ with $\alpha_0([1_A]_0) = [1_B]_0$ and morphisms $(\underline{\alpha}, \beta, \gamma): \underline{KT}_u(A) \rightarrow \underline{KT}_u(B)$, given by defining $\beta := \not\!A_B^{-1} \circ \alpha_1 \circ \not\!A_A$ and $\gamma := 0$. This allows the classification of nuclear embeddings of unital separable exact C^* -algebras with the UCT into unital Kirchberg algebras by total K -theory (see [253, 236, 170, 115] — in this level of generality, the result is recorded as [115, Theorem 8.12]) to be viewed as a classification using $\underline{KT}_u$.

Remark 3.8. For each integer $n \geq 2$, there are examples of triples $(\underline{\alpha}, \beta, \gamma)$ that satisfy (3.14) but not (3.15). Set $A := \mathbb{I}_n^\dagger$ and let $B := \mathcal{Z}$. Let $\psi: \mathbb{I}_n^\dagger \rightarrow \mathcal{Z}$ be

an embedding inducing the Lebesgue trace on $\mathbb{I}_n^\dagger$ (such an embedding exists, for example, by [270, Theorem 4.11]). Define $\underline{\alpha} := \underline{K}(\psi)$ and $\gamma := \text{Aff } T(\psi)$. As

$$(3.18) \quad K_1(\mathbb{I}_n^\dagger) \cong \mathbb{Z}/n \quad \text{and} \quad \text{Aff } T(\mathcal{Z})/\overline{\text{im } \rho_{\mathcal{Z}}} \cong \mathbb{R}/\mathbb{Z},$$

there is an embedding $r: K_1(\mathbb{I}_n^\dagger) \rightarrow \text{Aff } T(\mathcal{Z})/\overline{\text{im } \rho_{\mathcal{Z}}}$. Define

$$(3.19) \quad \beta := \overline{K}_1^{\text{alg}}(\psi) + \overline{\text{Th}}_{\mathcal{Z}} \circ r \circ \not\!A_{\mathbb{I}_n^\dagger}: \overline{K}_1^{\text{alg}}(\mathbb{I}_n^\dagger) \rightarrow \overline{K}_1^{\text{alg}}(\mathcal{Z}).$$

Then β satisfies (3.14). Also, by (3.8) and the fact that $\underline{K}T_u(\psi)$ is an $\underline{K}T_u$ -morphism (Proposition 3.6), we have

$$(3.20) \quad \beta \circ \zeta_{\mathbb{I}_n^\dagger}^{(n)} = \zeta_{\mathcal{Z}}^{(n)} \circ \alpha_0^{(n)} + \overline{\text{Th}}_{\mathcal{Z}} \circ r \circ \nu_{0, \mathbb{I}_n^\dagger}^{(n)}.$$

Since $K_1(\mathbb{I}_n^\dagger) \cong \mathbb{Z}/n$, exactness of (2.28) implies that $\nu_{0, \mathbb{I}_n^\dagger}^{(n)}$ is surjective. As $\overline{\text{Th}}_{\mathcal{Z}}$ is injective and $r \neq 0$, it follows that $\overline{\text{Th}}_{\mathcal{Z}} \circ r \circ \nu_{0, \mathbb{I}_n^\dagger}^{(n)} \neq 0$. Hence $\beta \circ \zeta_{\mathbb{I}_n^\dagger}^{(n)} \neq \zeta_{\mathcal{Z}}^{(n)} \circ \alpha_0^{(n)}$; i.e., (3.15) does not commute.

For unital C^* -algebras A and B , a morphism $(\underline{\alpha}, \beta, \gamma): \underline{K}T_u(A) \rightarrow \underline{K}T_u(B)$ certainly restricts to a morphism $(\alpha_*, \gamma): KT_u(A) \rightarrow KT_u(B)$. To obtain classification results for C^* -algebras using KT_u from those for $\underline{K}T_u$, we need to be able to go in the other direction and extend (in a non-canonical fashion) KT_u -(iso)morphisms to $\underline{K}T_u$ -(iso)morphisms. Results of this form (often using the UCT to extend morphisms from K -theory to total K -theory rather than via B  digheimer's result, Proposition 2.17), have been used in the classification program previously in cases where $\underline{K}$ or $\overline{K}_1^{\text{alg}}$ is needed to obtain uniqueness of morphisms — see the proof of [131, Theorem 21.9], for example. The new ingredient we need is to construct an extension that intertwines the maps $\zeta^{(n)}$ of the previous subsection.

Theorem 3.9 (Extension of (iso)morphisms from KT_u to $\underline{K}T_u$). *Let A and B be unital C^* -algebras. Then any morphism $(\alpha_*, \gamma): KT_u(A) \rightarrow KT_u(B)$ can be extended to a morphism $(\underline{\alpha}, \beta, \gamma): \underline{K}T_u(A) \rightarrow \underline{K}T_u(B)$. In addition, if (α_*, γ) is an isomorphism, then any such extension $(\underline{\alpha}, \beta, \gamma)$ is an isomorphism.*

Proof. By Proposition 2.17, we may extend α_* to $\underline{\alpha}: \underline{K}(A) \rightarrow \underline{K}(B)$, and by Proposition 2.13, there exists $\beta': \overline{K}_1^{\text{alg}}(A) \rightarrow \overline{K}_1^{\text{alg}}(B)$ such that

$$(3.21) \quad \beta' \circ \text{Th}_A = \text{Th}_B \circ \gamma \quad \text{and} \quad \not\!A_B \circ \beta' = \alpha_1 \circ \not\!A_A.$$

Therefore, β' makes the first diagram (3.14) commute from the definition of a $\underline{K}T_u$ -morphism, but it need not make (3.15) commute. The rest of the proof consists of identifying an appropriate correcting term to address this.

For $n \geq 2$, consider the maps

$$(3.22) \quad \beta' \circ \zeta_A^{(n)} - \zeta_B^{(n)} \circ \alpha_0^{(n)}: K_0(A; \mathbb{Z}/n) \rightarrow \overline{K}_1^{\text{alg}}(B).$$

The plan is to prove that these maps factor through maps $s_n: \text{Tor}(K_1(A), \mathbb{Z}/n) \rightarrow \ker \not\!A_B$ as compositions $s_n \circ \overline{\nu}_{0, A}^{(n)}$, and moreover that the resulting s_n are compatible, and so define a homomorphism $s: \text{Tor}(K_1(A)) \rightarrow \ker \not\!A_B$. This will provide the needed correcting term.

Fix $n \geq 2$ for the next two paragraphs. Since $\underline{\alpha}$ is a Λ -homomorphism and (α_*, γ) is a morphism on KT_u , we get

$$(3.23) \quad \begin{aligned} \zeta_B^{(n)} \circ \alpha_0^{(n)} \circ \mu_{0,A}^{(n)} &\stackrel{(\Lambda)}{=} \zeta_B^{(n)} \circ \mu_{0,B}^{(n)} \circ \alpha_0 \\ &\stackrel{(3.8)}{=} \text{Th}_B \circ \frac{1}{n} \rho_B \circ \alpha_0 \\ &\stackrel{(KT_u)}{=} \text{Th}_B \circ \frac{1}{n} \gamma \circ \rho_A. \end{aligned}$$

Since γ is $\mathbb{R}$ -linear, it follows that $\frac{1}{n} \gamma = \gamma \circ \frac{1}{n}$. Hence, continuing the computations, we get

$$(3.24) \quad \begin{aligned} \text{Th}_B \circ \frac{1}{n} \gamma \circ \rho_A &= \text{Th}_B \circ \gamma \circ \frac{1}{n} \rho_A \\ &\stackrel{(3.21)}{=} \beta' \circ \text{Th}_A \circ \frac{1}{n} \rho_A \\ &\stackrel{(3.8)}{=} \beta' \circ \zeta_A^{(n)} \circ \mu_{0,A}^{(n)}. \end{aligned}$$

Hence the map $\beta' \circ \zeta_A^{(n)} - \zeta_B^{(n)} \circ \alpha_0^{(n)}$ vanishes on the image of $\mu_{0,A}^{(n)}$. By exactness of (2.28), this image is $\ker \nu_{0,A}^{(n)}$, and so the map factors through $\text{Tor}(K_1(A), \mathbb{Z}/n)$ via the map $\bar{\nu}_{0,A}^{(n)}: K_0(A; \mathbb{Z}/n) \rightarrow \text{Tor}(K_1(A), \mathbb{Z}/n)$ from (2.33).

Similarly, we find that

$$(3.25) \quad \begin{aligned} \not\alpha_B \circ \zeta_B^{(n)} \circ \alpha_0^{(n)} &\stackrel{(3.8)}{=} \nu_{0,B}^{(n)} \circ \alpha_0^{(n)} \\ &\stackrel{(\Lambda)}{=} \alpha_1 \circ \nu_{0,A}^{(n)} \\ &\stackrel{(3.8)}{=} \alpha_1 \circ \not\alpha_A \circ \zeta_A^{(n)} \\ &\stackrel{(3.21)}{=} \not\alpha_B \circ \beta' \circ \zeta_A^{(n)} \end{aligned}$$

and so the range of $\beta' \circ \zeta_A^{(n)} - \zeta_B^{(n)} \circ \alpha_0^{(n)}$ is contained in $\ker \not\alpha_B \subseteq \overline{K}_1^{\text{alg}}(B)$. It follows that there is a unique homomorphism

$$(3.26) \quad s_n: \text{Tor}(K_1(A), \mathbb{Z}/n) \rightarrow \ker \not\alpha_B \subseteq \overline{K}_1^{\text{alg}}(B)$$

such that

$$(3.27) \quad s_n \circ \bar{\nu}_{0,A}^{(n)} = \beta' \circ \zeta_A^{(n)} - \zeta_B^{(n)} \circ \alpha_0^{(n)}.$$

Allowing n to vary, we identify the torsion subgroup $\text{Tor}(K_1(A))$ of $K_1(A)$ with the union $\bigcup_{n \geq 2} \text{Tor}(K_1(A), \mathbb{Z}/n)$. Let $s_{nm,n}$ denote the restriction of s_{nm} to $\text{Tor}(K_1(A), \mathbb{Z}/n)$. We will show that $s_{nm,n} = s_n$ for all $n, m \geq 2$, so that these maps induce a homomorphism $s: \text{Tor}(K_1(A)) \rightarrow \ker \not\alpha_B$. By uniqueness of s_n in (3.27), it suffices to show that

$$(3.28) \quad s_{nm,n} \circ \bar{\nu}_{0,A}^{(n)} = \beta' \circ \zeta_A^{(n)} - \zeta_B^{(n)} \circ \alpha_0^{(n)}.$$

As noted in (2.30), $\nu_{0,A}^{(nm)} \circ \kappa_{0,A}^{(nm,n)} = \nu_{0,A}^{(n)}: K_0(A; \mathbb{Z}/n) \rightarrow K_1(A)$, so it follows that $\bar{\nu}_{0,A}^{(n)}$ is the corestriction of $\bar{\nu}_{0,A}^{(nm)} \circ \kappa_A^{(nm,n)}$ to $\text{Tor}(K_1(A), \mathbb{Z}/n)$. Hence

$$(3.29) \quad \begin{aligned} s_{nm,n} \circ \bar{\nu}_{0,A}^{(n)} &= s_{nm} \circ \bar{\nu}_{0,A}^{(nm)} \circ \kappa_A^{(nm,n)} \\ &\stackrel{(3.27)}{=} (\beta' \circ \zeta_A^{(nm)} - \zeta_B^{(nm)} \circ \alpha_0^{(nm)}) \circ \kappa_A^{(nm,n)}. \end{aligned}$$

By commutativity of (3.9), we have $\beta' \circ \zeta_A^{(nm)} \circ \kappa_A^{(nm,n)} = \beta' \circ \zeta_A^{(n)}$. Also

$$(3.30) \quad \zeta_B^{(nm)} \circ \alpha_0^{(nm)} \circ \kappa_{0,A}^{(nm,n)} \stackrel{(\Lambda)}{=} \zeta_B^{(nm)} \circ \kappa_{0,B}^{(nm,n)} \circ \alpha_0^{(n)} \stackrel{(3.9)}{=} \zeta_B^{(n)} \circ \alpha_0^{(n)}.$$

Combining these last two equations with (3.29), we obtain (3.28), as desired. In conclusion, the maps s_n for $n \geq 2$ induce a homomorphism

$$(3.31) \quad s: \operatorname{Tor}(K_1(A)) \longrightarrow \ker \not\!A_B.$$

Since $\operatorname{Aff} T(B)/\overline{\operatorname{im} \rho_B}$ is divisible, and hence injective (see [303, Corollary 2.3.2]), so is $\ker \not\!A_B$, because the two groups are isomorphic via $\overline{\operatorname{Th}_B}$. Accordingly, s can be extended to a homomorphism

$$(3.32) \quad \hat{s}: K_1(A) \longrightarrow \ker \not\!A_B \subseteq \overline{K_1}^{\operatorname{alg}}(B).$$

We now use this to correct the definition of β' by setting

$$(3.33) \quad \beta := \beta' - \hat{s} \circ \not\!A_A: \overline{K_1}^{\operatorname{alg}}(A) \rightarrow \overline{K_1}^{\operatorname{alg}}(B).$$

It remains to prove that $(\underline{\alpha}, \beta, \gamma)$ is a morphism of $\underline{KT}_u$. We already know that (α_*, γ) is a morphism of KT_u , and that $\underline{\alpha}$ is a Λ -morphism, so we only need to show that the last two squares of (3.14) commute and that (3.15) commutes. Since $\not\!A_A \circ \overline{\operatorname{Th}_A} = 0$, we have

$$(3.34) \quad \beta \circ \operatorname{Th}_A \stackrel{(3.33)}{=} \beta' \circ \operatorname{Th}_A \stackrel{(3.21)}{=} \operatorname{Th}_A \circ \gamma.$$

Moreover,

$$(3.35) \quad \not\!A_B \circ \beta \stackrel{(3.32)}{=} \not\!A_B \circ \beta' \stackrel{(3.21)}{=} \alpha_1 \circ \not\!A_A.$$

Therefore, the last two squares of (3.14) commute.

For commutativity of (3.15), fix $n \geq 2$. Recall that $\hat{s}$ is an extension of s_n and that $\overline{\nu}_{0,A}^{(n)}$ is the corestriction of $\nu_{0,A}^{(n)}$ to $\operatorname{Tor}(K_1(A), \mathbb{Z}/n)$, which is the domain of s_n . It follows that $\hat{s} \circ \nu_{0,A}^{(n)} = s_n \circ \overline{\nu}_{0,A}^{(n)}$. Hence

$$(3.36) \quad \hat{s} \circ \not\!A_A \circ \zeta_A^{(n)} \stackrel{(3.8)}{=} s_n \circ \overline{\nu}_{0,A}^{(n)} \stackrel{(3.27)}{=} \beta' \circ \zeta_A^{(n)} - \zeta_B^{(n)} \circ \alpha_0^{(n)},$$

and so

$$(3.37) \quad \begin{aligned} \beta \circ \zeta_A^{(n)} &\stackrel{(3.33)}{=} (\beta' - \hat{s} \circ \not\!A_A) \circ \zeta_A^{(n)} \\ &\stackrel{(3.36)}{=} \beta' \circ \zeta_A^{(n)} - (\beta' \circ \zeta_A^{(n)} - \zeta_B^{(n)} \circ \alpha_0^{(n)}) = \zeta_B^{(n)} \circ \alpha_0^{(n)}, \end{aligned}$$

establishing commutativity of (3.15). Therefore $(\underline{\alpha}, \beta, \gamma)$ defines a well-defined morphism $\underline{KT}_u(A) \rightarrow \underline{KT}_u(B)$.

Finally, Propositions 2.13 and 2.17 ensure that $\underline{\alpha}$ and β are isomorphisms if α_* and γ are, which implies that $(\underline{\alpha}, \beta, \gamma)$ is an isomorphism whenever (α_*, γ) is. $\square$

Remark 3.10. Note that the proof of Theorem 3.9 shows that given a KT_u -morphism $(\alpha_*, \gamma): KT_u(A) \rightarrow KT_u(B)$ and any $\underline{\alpha}: \underline{K}(A) \rightarrow \underline{K}(B)$ extending α_* , there is an $\underline{KT}_u$ -morphism of the form $(\underline{\alpha}, \beta, \gamma)$.

The following lemma is used in the computation of $KL(A, J_B)$ in Theorem 1.6 to show that every $\underline{KT}_u$ -morphism in part (ii) comes from an appropriate KL -class. Its proof illustrates the role played by the maps $\zeta^{(n)}$.

Lemma 3.11. *Let A and B be unital C^* -algebras. If*

$$(3.38) \quad (\underline{\alpha}, \beta, \gamma), (\underline{\alpha}', \beta', \gamma'): \underline{KT}_u(A) \longrightarrow \underline{KT}_u(B)$$

are morphisms such that $\underline{\alpha} = \underline{\alpha}'$ and $\gamma = \gamma'$, then there is a unique homomorphism

$$(3.39) \quad r: K_1(A)/\operatorname{Tor}(K_1(A)) \longrightarrow \ker \not\!A_B$$

such that $\beta - \beta': \overline{K}_1^{\text{alg}}(A) \rightarrow \overline{K}_1^{\text{alg}}(B)$ is the composition

$$(3.40) \quad \overline{K}_1^{\text{alg}}(A) \xrightarrow{\not\!A_A} K_1(A) \twoheadrightarrow K_1(A)/\text{Tor}(K_1(A)) \xrightarrow{r} \ker \not\!A_B \hookrightarrow \overline{K}_1^{\text{alg}}(B).$$

Proof. We have $(\beta - \beta') \circ \text{Th}_A = \text{Th}_B \circ (\gamma - \gamma') = 0$. It follows that $\beta - \beta'$ vanishes on $\text{im } \text{Th}_A = \ker \not\!A_A$ and thus factors uniquely as $\beta - \beta' = \hat{r} \circ \not\!A_A$, for some map $\hat{r}: K_1(A) \rightarrow \overline{K}_1^{\text{alg}}(B)$. It suffices to show that $\hat{r}$ vanishes on $\text{Tor}(K_1(A))$ and takes values in $\ker \not\!A_B$. Since for any $n \geq 2$ we have

$$(3.41) \quad \begin{aligned} \hat{r} \circ \nu_{0,A}^{(n)} &\stackrel{(3.8)}{=} \hat{r} \circ \not\!A_A \circ \zeta_A^{(n)} = (\beta - \beta') \circ \zeta_A^{(n)} \\ &\stackrel{(3.15)}{=} \zeta_B^{(n)} \circ (\alpha_0^{(n)} - \alpha_0'^{(n)}) = 0, \end{aligned}$$

it follows that $\hat{r}$ vanishes on the image of $\nu_{0,A}^{(n)}$. By exactness of (2.28), this image is $\text{Tor}(K_1(A), \mathbb{Z}/n)$, and as such, $\hat{r}$ vanishes on $\text{Tor}(K_1(A))$. Notice that

$$(3.42) \quad \not\!A_B \circ \hat{r} \circ \not\!A_A = \not\!A_B \circ (\beta - \beta') \stackrel{(3.21)}{=} (\alpha_1 - \alpha_1') \circ \not\!A_A = 0.$$

Since $\not\!A_A$ is surjective, it follows that $\hat{r}$ takes values in $\ker \not\!A_B$. $\square$

Although we will not use it in this paper, we note that compatibility with the maps $\zeta^{(n)}$ is automatic when $K_1(A)$ is torsion-free. Gong, Lin, and Niu made a similar observation, by means of a different approach in their framework (see [130, Corollary 5.13 with condition (4)]).

Proposition 3.12. *Suppose that A and B are unital C^* -algebras with $K_1(A)$ torsion-free. Then the condition of commutativity of (3.15) is automatic in the definition of a $\underline{KT}_u$ -morphism.*

Proof. Let $(\underline{\alpha}, \beta, \gamma)$ be as in the definition of a $\underline{KT}_u$ -morphism, except we only assume that (3.14) commutes. We will show that commutativity of (3.15) follows. Fix $n \geq 2$. We have

$$(3.43) \quad \beta \circ \zeta_A^{(n)} \circ \mu_{0,A}^{(n)} \stackrel{(3.8)}{=} \beta \circ \text{Th}_A \circ \frac{1}{n} \rho_A \stackrel{(3.14)}{=} \text{Th}_B \circ \gamma \circ \frac{1}{n} \rho_A.$$

Since γ is $\mathbb{R}$ -linear we have $\gamma \circ \frac{1}{n} = \frac{1}{n} \gamma$. Therefore, continuing the computation above, we get

$$(3.44) \quad \begin{aligned} \text{Th}_B \circ \frac{1}{n} \gamma \circ \rho_A &\stackrel{(3.14)}{=} \text{Th}_B \circ \frac{1}{n} \rho_B \circ \alpha_0 \\ &\stackrel{(3.8)}{=} \zeta_B^{(n)} \circ \mu_{0,B}^{(n)} \circ \alpha_0 \stackrel{(\Lambda)}{=} \zeta_B^{(n)} \circ \alpha_0^{(n)} \circ \mu_{0,A}^{(n)}. \end{aligned}$$

Multiplication by n is an injective endomorphism on $K_1(A)$ because $K_1(A)$ is torsion-free, and so $\mu_{0,A}^{(n)}: K_0(A) \rightarrow K_0(A; \mathbb{Z}/n)$ is surjective by exactness of (2.28). Therefore, $\beta \circ \zeta_A^{(n)} = \zeta_B^{(n)} \circ \alpha_0^{(n)}$ as desired. $\square$

3.3. The sequence algebra. In this short section we define the sequence algebra B_∞ of a C^* -algebra, record Kirchberg's ϵ -test for later use, and observe that $B \hookrightarrow B_\infty$ is injective on $\underline{KT}_u$.

Definition 3.13. Let B be a C^* -algebra. The *sequence algebra* of B , denoted B_∞ , is the quotient $\ell^\infty(B)/c_0(B)$ of bounded sequences modulo those tending to zero. There is a canonical embedding

$$(3.45) \quad \iota_B: B \rightarrow B_\infty$$

induced by embedding B into $\ell^\infty(B)$ as constant sequences.

As is standard in the literature, we will typically use representative sequences in $\ell^\infty(B)$ to denote elements of B_∞ . One important feature of the sequence algebra is that one can often use reindexing arguments to convert approximate properties into exact ones. Our preference is to formulate this using a version of Kirchberg's ϵ -test from [174] suited to sequence algebras. The form we quote below can be proven by an easy modification of the proof of [174, Lemma A.1] or [178, Lemma 3.1].

Lemma 3.14 (Kirchberg's ϵ -test). *Let $(X_n)_{n=1}^\infty$ be a sequence of non-empty sets and write $X := \prod_{n=1}^\infty X_n$. Given functions $f_n^{(k)}: X_n \rightarrow [0, \infty]$ for $k, n \in \mathbb{N}$, define functions $f^{(k)}: X \rightarrow [0, \infty]$ by $f^{(k)}((x_n)_{n=1}^\infty) := \limsup_n f_n^{(k)}(x_n)$. Suppose that for all $\epsilon > 0$ and $k_0 \in \mathbb{N}$, there exists $x \in X$ with $f^{(k)}(x) < \epsilon$ for $k = 1, \dots, k_0$. Then there exists $x \in X$ with $f^{(k)}(x) = 0$ for all $k \in \mathbb{N}$.*

The following application of the ϵ -test is completely standard. (See the proof of [21, Lemma 1.17], for example.)

Lemma 3.15. *Let A and B be C^* -algebras with A separable, and let $\phi, \psi: A \rightarrow B_\infty$ be $*$ -homomorphisms. Then ϕ and ψ are approximately unitarily equivalent if and only if they are unitarily equivalent.*

We end by checking that the inclusion $\iota_B: B \hookrightarrow B_\infty$ is injective on the total invariant.

Lemma 3.16. *If B is a unital C^* -algebra, then*

$$(3.46) \quad \underline{KT}_u(\iota_B): \underline{KT}_u(B) \longrightarrow \underline{KT}_u(B_\infty)$$

is a monomorphism.

Proof. It suffices to prove that each of $\underline{K}(\iota_B)$, $\overline{K}_1^{\text{alg}}(\iota_B)$, and $\text{Aff } T(\iota_B)$ is injective.

For $\text{Aff } T(\iota_B)$, when $T(B) = \emptyset$, the result is vacuous, so assume $T(B) \neq \emptyset$. Let $\tau \in T(B)$, let ω be a free ultrafilter on $\mathbb{N}$ and define $\tau_\omega: B_\infty \rightarrow \mathbb{C}$ by $\tau_\omega((b_n)_{n=1}^\infty) := \lim_{n \rightarrow \omega} \tau(b_n)$. Then $\tau_\omega \in T(B_\infty)$ and $\tau_\omega \circ \iota_B = \tau$, proving that $T(\iota_B)$ is surjective, and hence $\text{Aff } T(\iota_B)$ is injective by duality.

For $\overline{K}_1^{\text{alg}}(\iota_B)$, take a unitary $u \in M_r(B)$ with $[\iota_B(u)]_{\text{alg}} = 0$ in $\overline{K}_1^{\text{alg}}(B_\infty)$. Given $\epsilon > 0$, there are positive integers k and s and unitaries $v_1, \dots, v_k, w_1, \dots, w_k$ in $M_{r+s}(B_\infty)$ such that

$$(3.47) \quad \|\iota_B(u) \oplus 1_{M_s(B_\infty)} - v_1 w_1 v_1^* w_1^* \cdots v_k w_k v_k^* w_k^*\| < \epsilon.$$

For $i \in \{1, \dots, k\}$, let $(v_{i,n})_{n=1}^\infty$ and $(w_{i,n})_{n=1}^\infty$ be sequences of unitaries in $M_{r+s}(B)$ lifting v_i and w_i , respectively. Then, for sufficiently large n , we have

$$(3.48) \quad \|u \oplus 1_{M_s(B)} - v_{1,n} w_{1,n} v_{1,n}^* w_{1,n}^* \cdots v_{k,n} w_{k,n} v_{k,n}^* w_{k,n}^*\| < \epsilon.$$

Thus, $[u]_{\text{alg}} = 0$ in $\overline{K}_1^{\text{alg}}(B)$, proving that $\overline{K}_1^{\text{alg}}(\iota_B)$ is injective.

For injectivity of $\underline{K}(\iota_B)$, we start by establishing injectivity of $K_1(\iota_B)$. Suppose that r and s are positive integers, that $u \in U_r(B)$, and that $\iota_B(u) \oplus 1_{M_s(B_\infty)}$ is homotopic to the identity in $U_{r+s}(B_\infty)$. Then there is a finite sequence $v_1, \dots, v_k$ in $U_{r+s}(B_\infty)$ with $v_1 = u \oplus 1_{M_s(B_\infty)}$ and $v_k = 1_{M_{r+s}(B_\infty)}$ such that $\|v_{j+1} - v_j\| < 2$ for each $j = 1, \dots, k-1$. Lifting $v_2, \dots, v_{k-1}$ to sequences $(v_{2,n})_{n=1}^\infty, \dots, (v_{k-1,n})_{n=1}^\infty$ in $U_{r+s}(B)$, and setting $v_{k,n} := 1_{M_{r+s}(B)}$, $v_{1,n} := u \oplus 1_{M_s(B)}$, we can find some $n \in \mathbb{N}$, such that $\|v_{j+1,n} - v_{j,n}\| < 2$ for all $j = 1, \dots, k-1$, so that $u \oplus 1_{M_s(B)}$ is homotopic to the identity in $U_{r+s}(B)$, and hence $[u]_1 = 0$ in $K_1(B)$. Hence $K_1(\iota_B)$ is injective. A very similar argument shows that $K_0(\iota_B)$ is injective.

As we noted immediately after (2.26), we could equally well have defined total K -theory in terms of $K_i(A \otimes \mathcal{O}_{n+1})$: the functors $K_i(\cdot; \mathbb{Z}/n)$ and $K_i(\cdot \otimes \mathcal{O}_{n+1})$ are naturally isomorphic by [271, Theorem 6.4]. Since $\mathcal{O}_{n+1}$ is nuclear, the canonical maps from B_∞ and $\mathcal{O}_{n+1}$ to $(B \otimes \mathcal{O}_{n+1})_\infty$ with commuting ranges combine to yield a $*$ -homomorphism $\theta: B_\infty \otimes \mathcal{O}_{n+1} \rightarrow (B \otimes \mathcal{O}_{n+1})_\infty$. Then we have the factorization $\iota_{B \otimes \mathcal{O}_{n+1}} = \theta \circ (\iota_B \otimes \text{id}_{\mathcal{O}_{n+1}})$. Since $\mathcal{O}_{n+1}$ is unital (in contrast to $\mathbb{I}_n$), the above argument implies that $K_i(\iota_{B \otimes \mathcal{O}_{n+1}})$ is injective, and thus so is $K_i(\iota_B; \mathbb{Z}/n)$. $\square$

4. $\mathcal{Z}$ -STABILITY AND CUNTZ SEMIGROUP TECHNIQUES

The key regularity hypothesis in the classification of embeddings theorem (Theorem B) is $\mathcal{Z}$ -stability of the codomain B , i.e., $B \cong B \otimes \mathcal{Z}$, where $\mathcal{Z}$ is the Jiang–Su algebra from [154]. We give a brief review of the Jiang–Su algebra in Section 4.1 and then turn to $\mathcal{Z}$ -stability, setting out in Sections 4.1–4.4 the various consequences that we use elsewhere in the paper. Section 4.2 is devoted to a modern proof of the unpublished result of Jiang ([153]) that $\mathcal{Z}$ -stable C^* -algebras are K_1 -injective, for use in Section 5.

The road from $\mathcal{Z}$ -stability to absorption passes through the Cuntz semigroup and the corona factorization property. We provide the relevant background in Section 4.3. Finally, Section 4.4 collates the various structural properties of sequence algebras of $\mathcal{Z}$ -stable C^* -algebras that we need from the literature.

4.1. The Jiang–Su algebra. The Jiang–Su algebra $\mathcal{Z}$ was originally constructed in [154] as an inductive limit of prime dimension drop algebras (certain subhomogeneous C^* -algebras over the interval whose K -theory is $(\mathbb{Z}, 0)$) with unital connecting maps chosen carefully to ensure simplicity and uniqueness of trace. In this way $KT_u(\mathcal{Z}) \cong KT_u(\mathbb{C})$. See [277, Chapter 15] for an expository account of the original construction of $\mathcal{Z}$.

The only place where we need a somewhat explicit description of $\mathcal{Z}$ rather than its properties is in Section 4.2. For this we use the approach of Rørdam and Winter from [259] (which also powers Winter’s localization technique from [310]).

Let $p, q \geq 2$ be coprime, and consider the UHF algebras M_{p^∞} and M_{q^∞} . Define the generalized dimension drop algebra

$$(4.1) \quad \mathcal{Z}_{p^\infty, q^\infty} \subseteq C([0, 1], M_{p^\infty} \otimes M_{q^\infty})$$

to be the subalgebra of functions $f \in C([0, 1], M_{p^\infty} \otimes M_{q^\infty})$ satisfying

$$(4.2) \quad f(0) \in M_{p^\infty} \otimes \mathbb{C}1_{M_{q^\infty}} \quad \text{and} \quad f(1) \in \mathbb{C}1_{M_{p^\infty}} \otimes M_{q^\infty}.$$

For a unital endomorphism $\theta: \mathcal{Z}_{p^\infty, q^\infty} \rightarrow \mathcal{Z}_{p^\infty, q^\infty}$ which is trace-collapsing (i.e., $\tau_1 \circ \theta = \tau_2 \circ \theta$ for all $\tau_1, \tau_2 \in \mathcal{Z}_{p^\infty, q^\infty}$), the Jiang–Su algebra $\mathcal{Z}$ arises as the stationary inductive limit:

$$(4.3) \quad \mathcal{Z} \cong \varinjlim (\mathcal{Z}_{p^\infty, q^\infty} \xrightarrow{\theta} \mathcal{Z}_{p^\infty, q^\infty} \xrightarrow{\theta} \mathcal{Z}_{p^\infty, q^\infty} \xrightarrow{\theta} \cdots).$$

Such a trace-collapsing embedding θ was constructed in a self-contained fashion in [270, Section 4] (in their work, Rørdam and Winter used Jiang and Su’s construction of $\mathcal{Z}$ and its subsequent properties to build a trace-collapsing θ).

One of the most important facts about the Jiang–Su algebra — and a major black box in this paper — is that it is *strongly self-absorbing* (a notion formalized by Toms and Winter in [292]).

Theorem 4.1 (Jiang–Su, [154]). *The Jiang–Su algebra $\mathcal{Z}$ is strongly self-absorbing; i.e., there is an isomorphism $\mathcal{Z} \rightarrow \mathcal{Z} \otimes \mathcal{Z}$ that is approximately unitarily equivalent to $\text{id}_{\mathcal{Z}} \otimes 1_{\mathcal{Z}}: \mathcal{Z} \rightarrow \mathcal{Z} \otimes \mathcal{Z}$.*

Jiang and Su use their original explicit inductive limit model together with their classification results for limits of dimension drop algebras to show that $\mathcal{Z}$ has an approximately inner half flip ([154, Proposition 8.3]), there is a unital embedding $\mathcal{Z} \otimes \mathcal{Z} \rightarrow \mathcal{Z}$ ([154, Proposition 8.5]), and $\mathcal{Z}$ embeds approximately centrally into itself ([154, Corollary 8.6]). From these ingredients they deduced what is now called strong self-absorption of $\mathcal{Z}$. Subsequently, Toms and Winter showed these three conditions give strong self-absorption in an abstract setting ([292, Proposition 1.10]).

Following a direction outlined in Winter’s 2011 CBMS lectures, Schemaitat ([270]) directly established strong self-absorption from the Rørdam–Winter picture of $\mathcal{Z}$ using more elementary classification results (asymptotic classification of maps between UHF algebras). With hindsight, it is possible to take the Rørdam–Winter picture as the definition of $\mathcal{Z}$ and use Winter’s work ([308]) to show that such an algebra embeds unitaly in all strongly self-absorbing algebras (from which it follows that it must be isomorphic to the original construction). One can also obtain all the other properties of $\mathcal{Z}$ needed in the classification theorem from this viewpoint. In a different direction, Ghasemi ([123], building on [75, 209]) shows how to obtain strong self-absorption of $\mathcal{Z}$ from approximately inner flip (obtained by Jiang and Su) via the model-theoretic notion of Fraïssé limits.

Now we turn to $\mathcal{Z}$ -stability. First, the definition.

Definition 4.2. A C^* -algebra B is $\mathcal{Z}$ -stable (or $\mathcal{Z}$ -absorbing) if $B \cong B \otimes \mathcal{Z}$.

One immediate consequence of (strong) self-absorption is that $\mathcal{Z}$ is itself $\mathcal{Z}$ -stable. Moreover, the strong form of the isomorphism $\mathcal{Z} \cong \mathcal{Z} \otimes \mathcal{Z}$ immediately strengthens the isomorphism in Definition 4.2 as follows (see [292, Theorem 2.2]).

Proposition 4.3. *Let B be a unital C^* -algebra. The following are equivalent:*

- (i) B is $\mathcal{Z}$ -stable.
- (ii) *There exists an isomorphism $B \rightarrow B \otimes \mathcal{Z}$ that is approximately unitarily equivalent to the first-factor embedding $\text{id}_B \otimes 1_{\mathcal{Z}}: B \rightarrow B \otimes \mathcal{Z}$.*

It is using this (or similar formulations, such as [257, Lemma 4.4]) that strong self-absorption of $\mathcal{Z}$ is generally used to obtain structural properties of $\mathcal{Z}$ -stable C^* -algebras, such as Proposition 4.11 below (due to Rørdam as [257, Theorem 4.5]). This form of the isomorphism is also a key ingredient in the proof of the $\mathcal{Z}$ -stable KK -uniqueness theorem in Section 5.4.

The first use of $\mathcal{Z}$ -stability in the unital classification theorem is through the dichotomy it provides between purely infinite and stably finite simple C^* -algebras, as shown in [128, Theorem 3]. This is a special case of a more general dichotomy result of Kirchberg for simple C^* -algebras that can be written non-trivially as a tensor product ([17, Corollary 3.9(ii)]; see also [255, Theorem 4.1.10(ii)]).

Theorem 4.4. *Let B be a simple exact $\mathcal{Z}$ -stable C^* -algebra. Then B is either purely infinite or stably finite.*

We end the subsection by recording that tensoring with $\mathcal{Z}$ leaves the invariant $KT_u(\cdot)$ unchanged. The KK -equivalence fact below was observed by Jiang and Su in [154, Lemma 2.11]. In the separable case, the isomorphism of invariants follows from this and uniqueness of trace on $\mathcal{Z}$; the non-separable case follows by continuity.

Proposition 4.5. *For any unital C^* -algebra D , the first-factor embedding induces both an isomorphism $KT_u(D) \cong KT_u(D \otimes \mathcal{Z})$, and (when D is separable) a KK -equivalence.*

4.2. K_1 -injectivity. In this section we give a short proof of an unpublished result of Jiang that we will use in the $\mathcal{Z}$ -stable KK -uniqueness theorem in Section 5.4. Recall that a unital C^* -algebra D is K_1 -injective if whenever $u \in U_1(D)$ vanishes in K_1 , then $u \in U_1^{(0)}(D)$. That is, no matrix amplification is needed to construct a homotopy from u to 1_D .

Theorem 4.6 (Jiang, [153]). *If D is a unital C^* -algebra, then $D \otimes \mathcal{Z}$ is K_1 -injective.*

Before proving the theorem, it is helpful to isolate the following fact regarding the non-stable K -theory of C^* -algebras that absorb a UHF algebra of infinite type.

Lemma 4.7. *Let D be a unital C^* -algebra and $n \in \mathbb{N}$ with $n \geq 2$. Then $D \otimes M_{n^\infty}$ is K_1 -injective and $K_0(D \otimes M_{n^\infty})$ is generated by*

$$(4.4) \quad \{[p]_0 : p \text{ is a projection in } D \otimes M_{n^\infty}\}.$$

Proof. Write $B := D \otimes M_{n^\infty}$ and suppose $u \in U_1(B)$ satisfies $[u]_1 = 0$. Then there is an $m \geq 1$ such that $u \oplus 1_B^{\oplus(m-1)}$ is homotopic to $1_B^{\oplus m}$ in $U_m(B)$. Enlarging m if necessary, we may assume $m = n^k$ for some $k \geq 1$. We have

$$(4.5) \quad u^{\oplus m} = (u \oplus 1_B^{\oplus(m-1)})(1_B \oplus u^{\oplus(m-1)}) \sim_h 1_B \oplus u^{\oplus(m-1)},$$

and by repeating this, it follows that $u^{\oplus m}$ is homotopic to the identity in $U_m(B)$. Put another way, this says that $u \otimes 1_{M_{n^k}}$ is homotopic to the identity in $U_1(B \otimes M_{n^k})$. Therefore, $u \otimes 1_{M_{n^\infty}}$ is homotopic to the identity in the unitary group of $B \otimes M_{n^\infty}$.

Use strong self-absorption of M_{n^∞} (which is straightforward for M_{n^∞}) to fix an isomorphism $\theta: B \rightarrow B \otimes M_{n^\infty}$ that is approximately unitarily equivalent to the embedding $x \mapsto x \otimes 1_{M_{n^\infty}}$. Then take a unitary $v \in B \otimes M_{n^\infty}$ with

$$(4.6) \quad \|\theta(u) - v(u \otimes 1_{M_{n^\infty}})v^*\| < 1$$

so that $\theta(u)$ is homotopic to $v(u \otimes 1_{M_{n^\infty}})v^*$ in $U_1(B \otimes M_{n^\infty})$, which in turn, is homotopic to $1_{B \otimes M_{n^\infty}}$. Applying θ^{-1} shows that u is homotopic to 1_B in $U_1(B)$.

The statement regarding K_0 follows in a similar fashion. If $p \in B \otimes M_m$ is a projection, we may again assume $m = n^k$ for some k . This time, take an isomorphism $\theta: B \rightarrow B \otimes M_{n^k}$ that is approximately unitarily equivalent to the embedding $x \mapsto x \otimes 1_{M_{n^k}}$ (such a θ exists as $B \cong B \otimes M_{n^\infty} \cong B \otimes M_{n^k} \otimes M_{n^\infty} \cong B \otimes M_{n^k}$, where the first and last isomorphisms are approximately unitarily equivalent to the first-factor embeddings). Set $q := \theta^{-1}(p) \in B$. As $p = \theta(q)$ is approximately unitarily equivalent to $q \otimes 1_{M_{n^k}}$ in $B \otimes M_{n^k}$, $[p]_0 = n^k[q]_0$, and so $[p]_0$ is in the group generated by K_0 -classes of projections in B . $\square$

Proof of Theorem 4.6. $D \otimes \mathcal{Z}$ is a stationary inductive limit of the algebra $D \otimes \mathcal{Z}_{2^\infty, 3^\infty}$; see (4.1). It is therefore enough to show $D \otimes \mathcal{Z}_{2^\infty, 3^\infty}$ is K_1 -injective. To this end, choose $u \in U_1(D \otimes \mathcal{Z}_{2^\infty, 3^\infty})$ with $[u]_1 = 0$ in $K_1(D \otimes \mathcal{Z}_{2^\infty, 3^\infty})$.

Let SM_{6^∞} be the suspension $C_0((0, 1), M_{6^\infty}) \subseteq \mathcal{Z}_{2^\infty, 3^\infty}$. As each of SM_{6^∞} , $\mathcal{Z}_{2^\infty, 3^\infty}$, and $M_{2^\infty} \oplus M_{3^\infty}$ is nuclear, we get an exact sequence

$$(4.7) \quad 0 \rightarrow D \otimes SM_{6^\infty} \rightarrow D \otimes \mathcal{Z}_{2^\infty, 3^\infty} \xrightarrow{\sigma} D \otimes (M_{2^\infty} \oplus M_{3^\infty}) \rightarrow 0.$$

Note that $[\sigma(u)]_1 = 0$ in $K_1(D \otimes (M_{2^\infty} \oplus M_{3^\infty}))$. As $D \otimes (M_{2^\infty} \oplus M_{3^\infty}) \cong D \otimes M_{2^\infty} \oplus D \otimes M_{3^\infty}$, and Lemma 4.7 applies to both $D \otimes M_{2^\infty}$ and $D \otimes M_{3^\infty}$, it

follows that $\sigma(u)$ is in the path component of the identity in the unitary group of this algebra. Hence, $\sigma(u)$ lifts to a unitary v in the path component of the identity in $U_1(D \otimes \mathcal{Z}_{2^\infty, 3^\infty})$ (i.e., $\sigma(v) = \sigma(u)$). After replacing u with uv^* , we may assume $\sigma(u) = 1_{D \otimes (M_{2^\infty} \oplus M_{3^\infty})}$ so that u is a unitary in $(D \otimes SM_{6^\infty})^\dagger$.

Since $[u]_1 = 0$ in $K_1(D \otimes \mathcal{Z}_{2^\infty, 3^\infty})$, the 6-term exact sequence gives an element $x \in K_0(D \otimes (M_{2^\infty} \oplus M_{3^\infty}))$ such that $\exp(x) = [u]_1$ in $K_1(D \otimes SM_{6^\infty})$. As above, another application of Lemma 4.7 shows there are projections $p_1, \dots, p_n, q_1, \dots, q_n$ in $D \otimes (M_{2^\infty} \oplus M_{3^\infty})$ with $x = \sum_j ([p_j]_0 - [q_j]_0)$. Then, in $K_1(D \otimes SM_{6^\infty})$, we have

$$(4.8) \quad [u]_1 = [e^{2\pi i h_1} \dots e^{2\pi i h_n} e^{-2\pi i k_1} \dots e^{-2\pi i k_n}]_1$$

where the h_j, k_j are positive contractions in $D \otimes \mathcal{Z}_{2^\infty, 3^\infty}$ lifting p_j, q_j , respectively. Replacing u with

$$(4.9) \quad u e^{-2\pi i h_1} \dots e^{-2\pi i h_n} e^{2\pi i k_1} \dots e^{2\pi i k_n},$$

if necessary, we may assume that $u \in (D \otimes SM_{6^\infty})^\dagger$ with $[u]_1 = 0$ in $K_1(D \otimes SM_{6^\infty})$ and with scalar part equal to 1.

With these reductions in place, we will now show u is in the path component of the identity in the unitary group of $(D \otimes SM_{6^\infty})^\dagger$. We identify $D \otimes SM_{6^\infty}$ with the subalgebra of $C(\mathbb{T}, D \otimes M_{6^\infty})$ consisting of functions vanishing at 1. Then u defines an element in $C(\mathbb{T}, D \otimes M_{6^\infty})$ with trivial K_1 -class. By a third, and final, application of Lemma 4.7, u is in the path component of the identity of $U_1(C(\mathbb{T}, D \otimes M_{6^\infty}))$. Let $(v_t)_{t \in [0, 1]}$ be a path of unitaries in $C(\mathbb{T}, D \otimes M_{6^\infty})$ with $v_0 = 1_{D \otimes M_{6^\infty}}$ and $v_1 = u$. Define $u_t = v_t(1)^* v_t \in C(\mathbb{T}, D \otimes M_{6^\infty})$. Then $u_t(1) = 1_{D \otimes M_{6^\infty}}$ for all $t \in [0, 1]$, so $(u_t)_{t \in [0, 1]}$ is a path of unitaries in $(D \otimes SM_{6^\infty})^\dagger$. Moreover, $u_0 = 1_{(D \otimes SM_{6^\infty})^\dagger}$ and $u_1 = u$, as required. $\square$

4.3. The Cuntz semigroup and the corona factorization property. Considerable information about a C^* -algebra is carried in an algebraic object — the Cuntz semigroup — constructed from its positive elements. The ideas for this semigroup originated in [54] and were subsequently developed in [12, 250, 235, 177, 257, 30, 52] and other works; see the survey [2].

Given a C^* -algebra D and positive elements $a, b \in D \otimes \mathcal{K}$, write $a \precsim b$ if there exists a sequence $(x_n)_{n=1}^\infty$ in $D \otimes \mathcal{K}$ with $x_n^* b x_n \rightarrow a$. Say a and b are *Cuntz equivalent*, written $a \sim b$, if $a \precsim b$ and $b \precsim a$. The *Cuntz semigroup*, $\text{Cu}(D)$, is defined by $(D \otimes \mathcal{K})_+ / \sim$. This is an abelian semigroup (with addition given by identifying $\mathcal{K} \cong M_2(\mathcal{K})$ and then adding representatives diagonally) that inherits an order $\leq$ from $\precsim$. When consulting the literature, it is important to distinguish between the complete Cuntz semigroup $\text{Cu}(D)$ introduced in [52] and the earlier incomplete version $W(D)$, which is defined in a similar fashion as above using $\bigcup_{n=1}^\infty M_n(D)_+$ in place of $(D \otimes \mathcal{K})_+$. Note that $\text{Cu}(D) \cong W(D \otimes \mathcal{K})$. In order to accurately cite the results we need, we use both constructions.

Given $\tau \in T(D)$ and $a \in (D \otimes \mathcal{K})_+$ defining

$$(4.10) \quad d_\tau(a) := \lim_{r \rightarrow \infty} \tau(a^{1/r}),$$

where τ is extended to a densely defined lower semicontinuous trace on $D \otimes \mathcal{K}$, gives states $d_\tau: W(D) \rightarrow [0, \infty)$ and $d_\tau: \text{Cu}(D) \rightarrow [0, \infty]$, i.e., order-preserving semigroup homomorphisms which map $[1_D]$ to 1 (this notion of a state can also be extended to the non-unital case). (See [105, Proposition 4.2], which shows, using work going back to [54, 12], that all states on $\text{Cu}(D)$ are of this form for suitable

lower semicontinuous extended quasitraces on D .) A particularly useful concept, strict comparison, allows Cuntz comparison to be recovered from tracial data. This idea goes back to [8, Section 6], which gives an account of comparison conditions for simple C^* -algebras in the spirit of Murray and von Neumann’s comparison theory for von Neumann algebra factors. There are several variations of “strict comparison” in the literature. We use the following form, which appears in [21, Definition 1.5].

Definition 4.8. Let D be a C^* -algebra. Say that D has *strict comparison with respect to bounded traces* when for all $n \in \mathbb{N}$ and $a, b \in M_n(D)_+$, we have

$$(4.11) \quad d_\tau(a) < d_\tau(b) \text{ for all } \tau \in T(D) \implies a \precsim b.$$

An important well-known application of strict comparison is to catalyze checking the fullness of $*$ -homomorphisms. We recall both the definition of fullness, and the relevant result below. The first statement of Lemma 4.10 is trivial, and the second is relatively standard; see [287, Lemma 2.2], for example.

Definition 4.9. Let B, D be C^* -algebras. A $*$ -homomorphism $\phi: B \rightarrow D$ is *full* if $\phi(b)$ generates D as an ideal, for every non-zero $b \in B$.

Lemma 4.10. *Let $\theta: B \rightarrow D$ be a $*$ -homomorphism between C^* -algebras. If θ is full, then $\tau \circ \theta$ is faithful for every $\tau \in T(D)$. The converse holds if D is unital and has strict comparison with respect to bounded traces.*

While the order structure of the Cuntz semigroup of a general C^* -algebra — even a simple one — can be quite badly behaved (see [299, 288]), Rørdam showed the situation is much better for a $\mathcal{Z}$ -stable C^* -algebra. In the second part of the following proposition, Rørdam uses Haagerup’s theorem ([137]) that quasitraces are traces for exact C^* -algebras. This is the place where this fundamental result appears in our proof of the unital classification theorem.

Proposition 4.11 (Rørdam, [257]). *Let D be a $\mathcal{Z}$ -stable C^* -algebra. Then*

- (i) *$\text{Cu}(D)$ is almost unperforated: whenever $x, y \in \text{Cu}(D)$ satisfy $nx \leq my$ for some $n > m$ in $\mathbb{N}$, we have $x \leq y$.*
- (ii) *If, in addition, D is unital, simple, and exact with $T(D) \neq \emptyset$, then D has strict comparison of positive elements with respect to bounded traces.*

Proof. Theorem 4.5 of [257] shows that the incomplete Cuntz semigroup $W(D)$ is almost unperforated when D is $\mathcal{Z}$ -stable. Part (i) follows by applying this to $D \otimes \mathcal{K}$, since $\text{Cu}(D) \cong W(D \otimes \mathcal{K})$. Part (ii) follows by applying [257, Corollary 4.6] to each $M_n(D)$. $\square$

Now we turn to the corona factorization property of Kucerovsky and Ng from [184], which we use to verify absorption in Section 5 and beyond. A σ -unital C^* -algebra D has the *corona factorization property* if every (norm-)full projection in the multiplier algebra of $D \otimes \mathcal{K}$ is properly infinite. This is a relatively mild regularity hypothesis, and is implied by $\mathcal{Z}$ -stability, although a tight reference for this seems difficult to come by. The route through the literature that we give below uses Ortega, Perera, and Rørdam’s characterization of the corona factorization property in terms of the complete Cuntz semigroup ([227, Theorem 5.11]).

Proposition 4.12. *Let D be a σ -unital $\mathcal{Z}$ -stable C^* -algebra. Then D has the corona factorization property.*

Proof. By Proposition 4.11(i), $\text{Cu}(D)$ is almost unperforated. By [227, Remark 2.4], this is equivalent to 0-comparison for $\text{Cu}(D)$ in the sense of [227, Definition 2.8]. In particular, $\text{Cu}(D)$ has the ω -comparison property of [227, Definition 2.11] (see the sentence immediately following that definition). Then we can combine [227, Proposition 2.17] and [227, Theorem 5.11] to conclude that D has the corona factorization property. $\square$

4.4. Sequence algebras of $\mathcal{Z}$ -stable C^* -algebras. We end this section by recording some structural properties that a sequence algebra B_∞ enjoys when the underlying C^* -algebra B is $\mathcal{Z}$ -stable. Such a sequence algebra cannot be $\mathcal{Z}$ -stable (by [122], where the SAW* hypothesis needed to use this result is obtained for B_∞ by a standard application of Kirchberg ϵ -test). Accordingly, we work with the following version of $\mathcal{Z}$ -stability in terms of separable subalgebras, in the spirit of Blackadar's notion of separable inheritability ([11, Section II.8.5]).

Definition 4.13 (cf. [268, Definition 1.4]). A C^* -algebra D is *separably $\mathcal{Z}$ -stable* if, for every separable C^* -subalgebra D_0 of D , there exists a separable $\mathcal{Z}$ -stable C^* -subalgebra $D_1 \subseteq D$ containing D_0 .

When B is separable and unital, an intertwining argument which seems to be originally due to Elliott (see [177, Theorem 8.2], [252, Page 34], and [292, Section 2]) shows that $\mathcal{Z}$ -stability of B can be characterized in terms of an embedding of $\mathcal{Z}$ into the central sequence algebra $B_\infty \cap B'$. This is in the spirit of McDuff's characterization of separably acting II_1 factors absorbing the hyperfinite II_1 factor tensorially. It gives rise to a local characterization of $\mathcal{Z}$ -stability for unital separable C^* -algebras, that extends to characterize separable $\mathcal{Z}$ -stability; this is why separable $\mathcal{Z}$ -stability is well-suited to working with non-separable C^* -algebras (see [268, Lemma 1.11]). As a consequence, one has the following.

Proposition 4.14 ([268, Proposition 1.12]). *If B is a unital separable $\mathcal{Z}$ -stable C^* -algebra, then B_∞ is separably $\mathcal{Z}$ -stable.*

Now we turn to traces on sequence algebras, with the aim of using Kirchberg's ϵ -test to observe that the image of $K_0(B_\infty)$ is closed in $\text{Aff } T(B_\infty)$ when B satisfies the codomain assumptions in the classification of embeddings theorem. Once this is done, Thomsen's work described in Section 2.2 (in particular, Properties 2.10) will give an exact sequence

$$(4.12) \quad K_0(B_\infty) \xrightarrow{\rho_{B_\infty}} \text{Aff } T(B_\infty) \xrightarrow{\text{Th}_{B_\infty}} \overline{K}_1^{\text{alg}}(B_\infty) \xrightarrow{\not\#_{B_\infty}} K_1(B_\infty) \rightarrow 0$$

which we use in the computation of $KL(A, J_B)$ in Section 8. First, we recall the notion of limit traces; these are the traces on B_∞ that are computationally tractable.

Definition 4.15. The set of *limit traces* on B_∞ , written $T_\infty(B)$, consists of all $\tau \in T(B_\infty)$ of the form $\tau((b_n)_{n=1}^\infty) = \lim_{n \rightarrow \omega} \tau_n(b_n)$, where $(\tau_n)_{n=1}^\infty$ is a sequence in $T(B)$ and ω is a free ultrafilter on $\mathbb{N}$.

Proposition 4.16 (Ozawa, cf. [221, Theorem 1.2]). *If B is an exact $\mathcal{Z}$ -stable C^* -algebra, then the convex hull of $T_\infty(B)$ is weak*-dense in $T(B_\infty)$.*

Proof. Ozawa's [228, Theorem 8] shows this for ultraproducts of $\mathcal{Z}$ -stable separable exact C^* -algebras, and the case of sequence algebras is essentially identical. One follows the proof of [228, Theorem 8] verbatim with $A := B_\infty$ and $\Sigma := \text{co } T_\infty(B)$ and changing "Let $I \in \mathcal{U}$ (or $I = \mathbb{N}$ in case $A = \prod A_n$)" to "Let I be cofinite". $\square$

The following proposition collects additional properties of the sequence algebra that will be useful later.

Proposition 4.17. *Let B be a unital simple separable $\mathcal{Z}$ -stable C^* -algebra with $T(B) \neq \emptyset$. Then*

- (i) B_∞ has stable rank one, and
- (ii) when B is additionally exact, B_∞ has strict comparison of positive elements with respect to bounded traces.

Proof. (i) follows directly from [208, Lemma 19.2.2], since B has stable rank one by [257, Theorem 6.7].

For (ii), note that B has strict comparison of positive elements by bounded traces by Rørdam's result recalled in Proposition 4.11(ii). Using this, the version of (ii) for the ultrapower B_ω has been observed in [21, Lemma 1.23]. The result for sequence algebras is obtained by following this short proof verbatim, with the following changes: replace B_ω by B_∞ , $T_\omega(B_\omega)$ by $T_\infty(B)$, and $I \in \omega$ in [21, (1.38)] by “ I is cofinite.” $\square$

Now we can show exactness of (4.12).

Proposition 4.18. *Let B be a unital simple separable exact $\mathcal{Z}$ -stable C^* -algebra with $T(B) \neq \emptyset$. Then $\rho_{B_\infty}(K_0(B_\infty))$ is closed in $\text{Aff } T(B_\infty)$, and so (4.12) is exact.*

Proof. Let $g \in \overline{\rho_{B_\infty}(K_0(B_\infty))}$. By adding a suitably large $n = \rho_{B_\infty}([1_{M_n(B_\infty)}]_0)$ to g , we can assume that $0 < g < m$ for some $m \in \mathbb{N}$. As a consequence of Proposition 2.1, there exists a self-adjoint $c = (c_n)_{n=1}^\infty \in B_\infty$ such that $\tau(c) = g(\tau)$ for all $\tau \in T(B_\infty)$.

Let X be the unit ball of $M_m(B)$ and define $f_n: X \rightarrow [0, \infty)$ by

$$(4.13) \quad f_n(b) := \|b - b^*\| + \|b^2 - b\| + \sup_{\tau \in T(B)} |\tau(b - c_n)|,$$

and then define $f: \prod_{n=1}^\infty X \rightarrow [0, \infty)$ by

$$(4.14) \quad f((b_n)_{n=1}^\infty) := \limsup_{n \rightarrow \infty} f_n(b_n).$$

For a projection $p = (p_n)_{n=1}^\infty \in M_m(B_\infty)$, one computes

$$(4.15) \quad f((p_n)_{n=1}^\infty) = \sup_{\tau \in T_\infty(B)} |\tau(p) - g(\tau)| \stackrel{\text{Prop. 4.16}}{=} \sup_{\tau \in T(B_\infty)} |\tau(p) - g(\tau)|.$$

By hypothesis, for any $\epsilon > 0$, there exists $x \in K_0(B_\infty)$ such that $\|\rho_{B_\infty}(x) - g\|_\infty < \epsilon$. Write $x = [r]_0 - [s]_0$ for projections r, s in matrices over B_∞ . We may assume ϵ is small enough that $0 < \rho_{B_\infty}(x) < m$. As Cuntz subequivalent projections are Murray–von Neumann subequivalent ([250, Proposition 2.1]), we can use strict comparison by bounded traces (Proposition 4.17(ii)) first to see that s is subequivalent to r and hence x is the class of some projection, and then again to compare this projection with $1_{M_m(B_\infty)}$. In this way, x is the class of a projection $p \in M_m(B_\infty)$ and this p has $f(p) < \epsilon$.

By Kirchberg's ϵ -test (Lemma 3.14), there exists $b \in \prod_{n=1}^\infty X$ such that $f(b) = 0$. This represents a projection in $M_m(B_\infty)$ such that $\tau(b) = g(\tau)$ for all $\tau \in T(B_\infty)$. Therefore, (4.12) is exact at $\text{Aff } T(B)$, and hence exact by Properties 2.10. $\square$

5. ABSORPTION AND KK -EXISTENCE AND UNIQUENESS THEOREMS

The main goal of this section is to prove the $\mathcal{Z}$ -stable KK - and KL -uniqueness theorems (collected together as Theorem 5.17) in Section 5.4. This proves Theorem 1.4 from the introduction. We start by setting out the Cuntz–Thomsen picture of KK -theory in Section 5.1, leaving certain proofs for the non-separable setting to Appendix B. We follow this by recalling salient facts about absorption in Section 5.2 and record the KK -existence theorem in Section 5.3.

5.1. The Cuntz–Thomsen picture of Kasparov’s KK -theory. We start by collecting those aspects of KK - and KL -theory that we need. As was the case in Kasparov’s original definition of KK -theory ([163]) using Fredholm modules, $KK(A, I)$ is typically defined for separable A and σ -unital I . However, we need to allow the non- σ -unital trace-kernel ideal J_B in the second variable. As suggested by Skandalis in [273, Section 3] (see also [274, Definition 5.5]), this can be achieved by taking inductive limits over separable C^* -subalgebras. We set out how to do this here. All definitions and results in this subsection are standard when the second variable is σ -unital; we provide proofs of how to extend these to the general case in Appendix B.

Of the many equivalent ways of defining KK -theory (see [10, Chapter 17], or [152]), the Cuntz–Thomsen picture is particularly well-suited to classification problems (as demonstrated, for example, by Dadarlat and Eilers in [65, 66]). Originally, this approach was developed by means of reductions to Kasparov’s Fredholm module construction. This is implicit in [57, Section 5] and set out in [10, 17.6.2] — see also [65, Section 3.1]. Thomsen later gave a self-contained treatment in [282].

Definition 5.1. Let A and I be C^* -algebras with A separable. An (A, I) -Cuntz pair is a pair (ϕ, ψ) of $*$ -homomorphisms $A \rightarrow E$ for some C^* -algebra E containing I as an ideal, such that $\phi(a) - \psi(a) \in I$ for all $a \in A$. We will often write $(\phi, \psi): A \rightrightarrows E \supset I$ for such a Cuntz pair. We call (ϕ, ψ) *unital* when all of A , E , ϕ , and ψ are unital.

Definition 5.2. Let A and I be C^* -algebras with A separable and I σ -unital. Write $\mathcal{M}(I \otimes \mathcal{K})$ for the multiplier algebra of the stabilization $I \otimes \mathcal{K}$. Let $C_\sigma([0, 1], \mathcal{M}(I \otimes \mathcal{K}))$ denote the algebra of strictly continuous functions $[0, 1] \rightarrow \mathcal{M}(I \otimes \mathcal{K})$ (these are automatically bounded by the principle of uniform boundedness). By [1, Corollary 3.4], one has

$$(5.1) \quad C_\sigma([0, 1], \mathcal{M}(I \otimes \mathcal{K})) \cong \mathcal{M}(C([0, 1], I \otimes \mathcal{K})).$$

Two Cuntz pairs

$$(5.2) \quad (\phi_i, \psi_i): A \rightrightarrows \mathcal{M}(I \otimes \mathcal{K}) \supset I \otimes \mathcal{K}, \quad i = 0, 1,$$

are said to be *homotopic* if there is a Cuntz pair

$$(5.3) \quad (\phi, \psi): A \rightrightarrows C_\sigma([0, 1], \mathcal{M}(I \otimes \mathcal{K})) \supset C([0, 1], I \otimes \mathcal{K})$$

such that evaluation at i induces the Cuntz pairs (ϕ_i, ψ_i) for $i = 0, 1$. The Cuntz–Thomsen picture of KK -theory is then

$$(5.4) \quad KK(A, I) := \{(\phi, \psi): A \rightrightarrows \mathcal{M}(I \otimes \mathcal{K}) \supset I \otimes \mathcal{K}\} / \text{homotopy}.$$

We write $[\phi, \psi]_{KK(A, I)}$ for the class in $KK(A, I)$ of $(\phi, \psi): A \rightrightarrows \mathcal{M}(I \otimes \mathcal{K}) \triangleright I \otimes \mathcal{K}$. More generally, given a Cuntz pair $(\phi, \psi): A \rightrightarrows E \triangleright I$, there is a canonical map $\mu: E \otimes \mathcal{K} \rightarrow \mathcal{M}(I \otimes \mathcal{K})$. Writing e for a rank one projection in $\mathcal{K}$ gives a Cuntz pair

$$(5.5) \quad (\mu \circ (\phi \otimes e), \mu \circ (\psi \otimes e)): A \rightrightarrows \mathcal{M}(I \otimes \mathcal{K}) \triangleright I \otimes \mathcal{K}.$$

In this way, (ϕ, ψ) defines an element

$$(5.6) \quad [\phi, \psi]_{KK(A, I)} := [\mu \circ (\phi \otimes e), \mu \circ (\psi \otimes e)]_{KK(A, I)} \in KK(A, I).$$

An important special case is when $\phi: A \rightarrow I$ is a $*$ -homomorphism, so that $(\phi, 0)$ is an (A, I) -Cuntz pair. We write $[\phi]_{KK(A, I)}$ for the induced class in $KK(A, I)$.

Addition in $KK(A, I)$ is defined by means of an orthogonal direct sum. There is a copy of $\mathcal{B}(\mathcal{H})$ in $\mathcal{M}(I \otimes \mathcal{K})$, so one can find $s_1, s_2 \in \mathcal{M}(I \otimes \mathcal{K})$ satisfying $s_i^* s_i = s_1 s_1^* + s_2 s_2^* = 1_{\mathcal{M}(I \otimes \mathcal{K})}$. Given $\phi_1, \phi_2: A \rightarrow \mathcal{M}(I \otimes \mathcal{K})$, write

$$(5.7) \quad (\phi_1 \oplus \phi_2)(a) := s_1 \phi_1(a) s_1^* + s_2 \phi_2(a) s_2^*, \quad a \in A.$$

As a map, $\phi_1 \oplus \phi_2$ depends on the choice of s_1 and s_2 . However, this sum is well-defined up to unitary equivalence (which implies homotopy equivalence, since the unitary group of $\mathcal{M}(I \otimes \mathcal{K})$ is path-connected in the strict topology by [10, Proposition 12.2.2], for example). Note that $\oplus$ is associative up to unitary equivalence. In this way, the sum in $KK(A, I)$ is well-defined by

$$(5.8) \quad [\phi_1, \psi_1]_{KK(A, I)} + [\phi_2, \psi_2]_{KK(A, I)} := [\phi_1 \oplus \phi_2, \psi_1 \oplus \psi_2]_{KK(A, I)}$$

(using the same choice of Cuntz isometries s_1 and s_2 to define both direct sums). This turns $KK(A, I)$ into an abelian group. The zero element is $[\phi, \phi]_{KK(A, I)}$ where $\phi: A \rightarrow \mathcal{M}(I \otimes \mathcal{K})$ is any $*$ -homomorphism ([152, Lemma 4.1.5]).

Remark 5.3. If I is a C^* -algebra and $s_1, s_2 \in \mathcal{M}(I \otimes \mathcal{K})$ are isometries with $s_1 s_1^* + s_2 s_2^* = 1_{\mathcal{M}(I \otimes \mathcal{K})}$, then there is an isomorphism

$$(5.9) \quad I \otimes \mathcal{K} \xrightarrow{\cong} M_2(I \otimes \mathcal{K}): x \mapsto \begin{pmatrix} s_1^* \\ s_2^* \end{pmatrix} x \begin{pmatrix} s_1 & s_2 \end{pmatrix},$$

which extends to an isomorphism $\mathcal{M}(I \otimes \mathcal{K}) \rightarrow M_2(\mathcal{M}(I \otimes \mathcal{K}))$. Further, for $*$ -homomorphisms $\phi, \psi: A \rightarrow \mathcal{M}(I \otimes \mathcal{K})$, this isomorphism conjugates the Cuntz sum

$$(5.10) \quad s_1 \phi(\cdot) s_1^* + s_2 \psi(\cdot) s_2^*: A \rightarrow \mathcal{M}(I \otimes \mathcal{K})$$

to the usual diagonal direct sum

$$(5.11) \quad \begin{pmatrix} \phi & 0 \\ 0 & \psi \end{pmatrix}: A \rightarrow M_2(\mathcal{M}(I \otimes \mathcal{K})).$$

For this reason, we use the notation $\phi \oplus \psi$ for both maps.

A key feature is that $KK(\cdot, \cdot)$ is a bifunctor. Contravariance in the first variable is straightforward: a $*$ -homomorphism $\theta: A_1 \rightarrow A_2$ induces a homomorphism $KK(\theta, I): KK(A_2, I) \rightarrow KK(A_1, I)$ by precomposing Cuntz pairs (and homotopies thereof) by θ . A direct definition of $KK(A, \theta)$ (for $\theta: I_1 \rightarrow I_2$) is possible, though subtle, in the Cuntz–Thomsen picture (see [282] or [152, Section 4.1]), though is straightforward in the pictures introduced by Cuntz ([57, 59]). All of our computations involving functoriality in the second variable will use the statement in Proposition 5.5(iv) below.

We now (re)define $KK(A, I)$ to cover the case that I is not σ -unital.

Definition 5.4. Let A and I be C^* -algebras with A separable. Then

$$(5.12) \quad KK(A, I) := \varinjlim_{I_0 \text{ sep.}} KK(A, I_0),$$

where I_0 ranges over all separable C^* -subalgebras of I , ordered by inclusion, and with connecting maps $KK(A, \iota_{I_0 \subseteq I_1})$ for $I_0 \subseteq I_1$.

Given a Cuntz pair $(\phi, \psi): A \rightrightarrows E \triangleright I$, we define $[\phi, \psi]_{KK(A, I)}$ to be the image of $[\phi|^{E_0}, \psi|^{E_0}]_{KK(A, I_0)}$ in the right-hand side of (5.12), where $E_0 \subseteq E$ and $I_0 \subseteq I$ are separable C^* -subalgebras such that $I_0 \triangleleft E_0$, $\phi(A) \cup \psi(A) \subseteq E_0$, and $(\phi - \psi)(A) \subseteq I_0$. We will check this is well-defined in Proposition B.4.

Definition 5.4 allows the functoriality in the second variable to be extended to non- σ -unital I , giving a bifunctor. We note that when I is σ -unital but not separable, a priori Definitions 5.2 and 5.4 give two competing definitions of $KK(A, I)$. In Proposition B.6, we show that these two definitions naturally agree.

The following facts are standard when I is separable. The proof that they extend to general I is deferred to Proposition B.9.

Proposition 5.5. *Let A and I be C^* -algebras with A separable, let E be a C^* -algebra containing I as an ideal, and let $(\phi, \psi): A \rightrightarrows E \triangleright I$ be an (A, I) -Cuntz pair.*

- (i) *Let $\iota_I^{(2)}: I \rightarrow M_2(I)$ be the top-left corner inclusion. Then the induced map $KK(A, \iota_I^{(2)}): KK(A, I) \rightarrow KK(A, M_2(I))$ is an isomorphism taking $[\phi, \psi]_{KK(A, I)}$ to $[\iota_E^{(2)} \circ \phi, \iota_E^{(2)} \circ \psi]_{KK(A, M_2(I))}$.*
- (ii) *Given a unitary $u \in I^\dagger$,*

$$(5.13) \quad [\phi, \psi]_{KK(A, I)} = [\text{Ad } u \circ \phi, \psi]_{KK(A, I)}.$$

- (iii) $[\phi, \psi]_{KK(A, I)} + [\psi, \phi]_{KK(A, I)} = 0$.

- (iv) *If $J \triangleleft F$ and $\theta: I \rightarrow J$ is a $*$ -homomorphism that extends to a $*$ -homomorphism $\bar{\theta}: E \rightarrow F$, then $KK(A, \theta)([\phi, \psi]_{KK(A, I)}) = [\bar{\theta} \circ \phi, \bar{\theta} \circ \psi]_{KK(A, J)}$. In particular, $KK(A, \iota_{I \subseteq E})([\phi, \psi]_{KK(A, I)}) = [\phi]_{KK(A, E)} - [\psi]_{KK(A, E)}$ and if $\phi: A \rightarrow I$ and $\theta: I \rightarrow J$ are $*$ -homomorphisms, then*

$$(5.14) \quad KK(A, \theta)([\phi]_{KK(A, I)}) = [\theta \circ \phi]_{KK(A, J)}.$$

- (v) *Let A be nuclear, and suppose*

$$(5.15) \quad \mathbf{e}: 0 \longrightarrow I \xrightarrow{j_e} E \xrightarrow{q_e} D \longrightarrow 0$$

is an extension of C^ -algebras. Then the sequence*

$$(5.16) \quad KK(A, I) \xrightarrow{KK(A, j_e)} KK(A, E) \xrightarrow{KK(A, q_e)} KK(A, D)$$

is exact. The same holds when A is KK -equivalent to a nuclear C^ -algebra.*

There is a well-known action of KK on K -theory,

$$(5.17) \quad \Gamma^{(A, I)}: KK(A, I) \rightarrow \text{Hom}(K_*(A), K_*(I))$$

such that if $\phi: A \rightarrow I$ is a $*$ -homomorphism, then

$$(5.18) \quad \Gamma_i^{(A, I)}([\phi]_{KK(A, I)}) = K_i(\phi).$$

This map is defined using the Kasparov product in the case where I is σ -unital, and it is extended to the non-separable case by taking the limit; this is laid out (and (5.18) is justified) in (B.32).

Next, we recall the KL -groups, first conceived to help classify $*$ -homomorphisms up to approximate unitary equivalence. Rørdam's original definition ([253, Section 5]) required a UCT assumption. This was relaxed by Dadarlat in [64], who defined $KL(A, I)$ to be the quotient of $KK(A, I)$ by the closure of $\{0\}$ in various equivalent topologies on $KK(A, I)$. We use a direct description of this closure, also from [64], which we give without any separability assumptions on I .

Definition 5.6 (cf. [64, Theorem 3.5, Theorem 4.1, and Section 5]). Let A and I be C^* -algebras with A separable. Write $\overline{\mathbb{N}} := \mathbb{N} \cup \{\infty\}$ for the one-point compactification of $\mathbb{N}$ and ev_n for evaluation at a point $n \in \overline{\mathbb{N}}$. Define

$$(5.19) \quad Z_{KK(A, I)} := \left\{ KK(A, \text{ev}_\infty)(\kappa) : \begin{array}{l} \kappa \in KK(A, C(\overline{\mathbb{N}}, I)) \text{ and} \\ KK(A, \text{ev}_n)(\kappa) = 0 \ \forall n \in \mathbb{N} \end{array} \right\}$$

and

$$(5.20) \quad KL(A, I) := KK(A, I) / Z_{KK(A, I)}.$$

With this definition, the bifunctor structure descends from $KK(\cdot, \cdot)$ to $KL(\cdot, \cdot)$, just as it does when I is separable. Likewise $\Gamma^{(A, I)}$ descends to a map

$$(5.21) \quad \widetilde{\Gamma} : KL(A, I) \rightarrow \text{Hom}(K_*(A), K_*(I)),$$

(see the more general statement in the first part of Theorem 8.5, proved for non-separable I in Appendix B.3). We can also express KL as an inductive limit over separable subalgebras, analogous to how we defined KK in Definition 5.4; we prove the following in Appendix B, immediately following Lemma B.8.

Proposition 5.7. *Let A and I be C^* -algebras with A separable. Then*

$$(5.22) \quad KL(A, I) \cong \varinjlim_{I_0 \text{ sep.}} KL(A, I_0)$$

where I_0 ranges over all separable C^* -subalgebras of I , ordered by inclusion, and the isomorphism is induced by the inclusions $I_0 \rightarrow I$.

Bootstrapping standard facts for KK (from Proposition 5.5), we arrive at the following KL version (see Proposition B.9).

Proposition 5.8. *Let A and I be C^* -algebras with A separable, let E be a C^* -algebra containing I as an ideal, and let $(\phi, \psi) : A \rightrightarrows E \triangleright I$ be an (A, I) -Cuntz pair. Then:*

- (i) *Let $\iota_I^{(2)} : I \rightarrow M_2(I)$ be the top-left corner inclusion. Then the induced map $KL(A, \iota_I^{(2)}) : KL(A, I) \rightarrow KL(A, M_2(I))$ is an isomorphism taking $[\phi, \psi]_{KL(A, I)}$ to $[\iota_E^{(2)} \circ \phi, \iota_E^{(2)} \circ \psi]_{KL(A, M_2(I))}$.*
- (ii) *Given a unitary in $I^\dagger$,*

$$(5.23) \quad [\phi, \psi]_{KL(A, I)} = [\text{Ad } u \circ \phi, \psi]_{KL(A, I)}.$$

- (iii) *$[\phi, \psi]_{KL(A, I)} + [\psi, \phi]_{KL(A, I)} = 0$.*

- (iv) *If $J \triangleleft F$ and $\theta : I \rightarrow J$ is a $*$ -homomorphism that extends to a $*$ -homomorphism $\theta : E \rightarrow F$, then $KL(A, \theta)([\phi, \psi]_{KL(A, I)}) = [\theta \circ \phi, \theta \circ \psi]_{KL(A, J)}$. In particular, $KL(A, \iota_{I \subseteq E})([\phi, \psi]_{KL(A, I)}) = [\phi]_{KL(A, E)} - [\psi]_{KL(A, E)}$, and if $\phi : A \rightarrow I$ and $\theta : I \rightarrow J$ are $*$ -homomorphisms, then*

$$(5.24) \quad KL(A, \theta)([\phi]_{KL(A, I)}) = [\theta \circ \phi]_{KL(A, J)}.$$

5.2. Absorption. Now we turn to absorption. Loosely speaking, an absorbing *-homomorphism is one for which the conclusion of Voiculescu’s non-commutative Weyl–von Neumann Theorem holds. The relevance of absorption can be seen in the early developments of the theory: in [238], for example, Pimsner, Popa, and Voiculescu prove an absorption theorem for extensions of $C(X) \otimes \mathcal{K}$ for X finite dimensional, in part to generalize Brown–Douglas–Fillmore theory; Kasparov’s generalization of Voiculescu’s theorem in [162] provides (nuclearly) absorbing *-homomorphisms.

Whereas we set out KK -theory without restriction on the second variable, absorption results are another matter. For these, it is vital that one has a countable approximate unit.

Throughout the rest of the section, I will be σ -unital and stable. We write $\mathcal{Q}(I)$ for the corona algebra $\mathcal{M}(I)/I$, and $q_I: \mathcal{M}(I) \rightarrow \mathcal{Q}(I)$ for the quotient map. We can form the direct sum of two maps $\phi_1, \phi_2: A \rightarrow \mathcal{M}(I)$ as in (5.7). Likewise, given maps $\psi_1, \psi_2: A \rightarrow \mathcal{Q}(I)$, we can form the direct sum $\psi_1 \oplus \psi_2: A \rightarrow \mathcal{Q}(I)$ (using isometries coming from $\mathcal{M}(I)$). Both of these are well defined up to unitary equivalence (in the latter case by unitaries from $\mathcal{M}(I)$). Therefore, the notions of absorption in Definition 5.9 below do not depend on the isometries chosen. The second part of the definition goes back to Kasparov (phrased in the language of extensions) as [163, p. 560, Definition 2]. To our knowledge, absorbing *-homomorphisms into general multiplier algebras (as in part (i) of the definition) started to appear explicitly around the turn of the millennium, and are formalized in [285].

Definition 5.9. Let A and I C^* -algebras with A separable and I σ -unital and stable.

- (i) A *-homomorphism $\phi: A \rightarrow \mathcal{M}(I)$ is *absorbing* if for every *-homomorphism $\psi: A \rightarrow \mathcal{M}(I)$, there is a sequence of unitaries $(u_n)_{n=1}^\infty \subseteq \mathcal{M}(I)$ such that, for each $a \in A$, we have
- $$(5.25) \quad (n \mapsto u_n(\phi(a) \oplus \psi(a))u_n^* - \phi(a)) \in C_0(\mathbb{N}, I).$$
- (ii) A *-homomorphism $\theta: A \rightarrow \mathcal{Q}(I)$ is *absorbing* if for every *-homomorphism $\psi: A \rightarrow \mathcal{M}(I)$ there is a unitary $u \in \mathcal{M}(I)$ such that $\text{Ad}(q_I(u)) \circ (\theta \oplus (q_I \circ \psi)) = \theta$.

The next three results are not new, but they are not entirely straightforward to extract from the literature. The proof below follows the computations in [65] and [66], which in turn uses ideas from [162]. The “asymptotic absorption” condition in (iii) below originates in [65], and Corollary 5.12 below is implicit there.

Proposition 5.10. Let A and I be C^* -algebras with A separable and I σ -unital and stable, and let $\phi: A \rightarrow \mathcal{M}(I)$ be a *-homomorphism. The following are equivalent:

- (i) ϕ is absorbing;
 - (ii) for all *-homomorphisms $\psi: A \rightarrow \mathcal{M}(I)$, there is a unitary $u \in \mathcal{M}(I)$ such that
- $$(5.26) \quad u(\phi(a) \oplus \psi(a))u^* - \phi(a) \in I, \quad a \in A;$$
- (iii) for all *-homomorphisms $\psi: A \rightarrow \mathcal{M}(I)$, there is a norm-continuous path $(u_t)_{t \geq 0} \subseteq \mathcal{M}(I)$ of unitaries such that
- $$(5.27) \quad (t \mapsto u_t(\phi(a) \oplus \psi(a))u_t^* - \phi(a)) \in C_0([0, \infty), I), \quad a \in A.$$

Proof. The implications (iii) $\Rightarrow$ (i) $\Rightarrow$ (ii) are clear. Now assume that (ii) holds and fix a $*$ -homomorphism $\psi: A \rightarrow \mathcal{M}(I)$. Because I is stable, one has $1_{\mathcal{M}(I)} \otimes \mathcal{B}(\mathcal{H}) \subseteq \mathcal{M}(I \otimes \mathcal{K})$, so we may find a sequence $(v_n)_{n=1}^\infty \subseteq \mathcal{M}(I)$ of isometries such that $\sum_{n=1}^\infty v_n v_n^* = 1_{\mathcal{M}(I)}$ with convergence in the strict topology. In particular,

$$(5.28) \quad v_n^* x v_n \rightarrow 0, \quad x \in I.$$

Define $*$ -homomorphisms $\psi_\infty, (\psi_\infty)_\infty: A \rightarrow \mathcal{M}(I)$ by

$$(5.29) \quad \psi_\infty(a) := \sum_{n=1}^\infty v_n \psi(a) v_n^*, \quad a \in A,$$

and

$$(5.30) \quad (\psi_\infty)_\infty(a) := \sum_{n=1}^\infty v_n \psi_\infty(a) v_n^* = \sum_{m,n=1}^\infty v_n v_m \psi(a) v_m^* v_n^*, \quad a \in A.$$

Note that, for each $n \in \mathbb{N}$ and $a \in A$,

$$(5.31) \quad v_n^* (\psi_\infty)_\infty(a) v_n = \psi_\infty(a).$$

Now fix isometries $s_1, s_2 \in \mathcal{M}(I)$ with $s_1 s_1^* + s_2 s_2^* = 1_{\mathcal{M}(I)}$ so that

$$(5.32) \quad (\phi \oplus (\psi_\infty)_\infty)(a) = s_1 \phi(a) s_1^* + s_2 (\psi_\infty)_\infty(a) s_2^*, \quad a \in A.$$

Using (ii), let $u \in \mathcal{M}(I)$ be a unitary with

$$(5.33) \quad u(\phi(a) \oplus (\psi_\infty)_\infty(a)) u^* - \phi(a) \in I, \quad a \in A,$$

and note that this implies

$$(5.34) \quad s_2^* u^* \phi(a) u s_2 - (\psi_\infty)_\infty(a) \in I, \quad a \in A.$$

Then $w_n := u s_2 v_n$ is an isometry for each $n \geq 1$ and, for $n \neq m$, we have $w_n^* w_m = v_n^* v_m = 0$. Furthermore, for each $a \in A$,

$$(5.35) \quad n \mapsto w_n^* \phi(a) w_n - \psi_\infty(a) \stackrel{(5.31)}{=} v_n^* (s_2^* u^* \phi(a) u s_2^* - (\psi_\infty)_\infty(a)) v_n$$

lies in $C_0(\mathbb{N}, I)$ by (5.34) and (5.28). Then, for all $a \in A$, we have

$$(5.36) \quad \begin{aligned} & \left(n \mapsto (w_n \psi_\infty(a) - \phi(a) w_n)^* (w_n \psi_\infty(a) - \phi(a) w_n) \right) \\ &= \left(n \mapsto \psi_\infty(a^*) (\psi_\infty(a) - w_n^* \phi(a) w_n) \right. \\ & \quad + (\psi_\infty(a^*) - w_n^* \phi(a^*) w_n) \psi_\infty(a) \\ & \quad \left. + (w_n^* \phi(a^* a) w_n - \psi_\infty(a^* a)) \right) \in C_0(\mathbb{N}, I). \end{aligned}$$

Therefore,

$$(5.37) \quad (n \mapsto w_n \psi_\infty(a) - \phi(a) w_n) \in C_0(\mathbb{N}, I)$$

by the C^* -identity. Condition (iii) now follows from [65, Lemma 2.3] (which uses [66, Lemma 2.16]). $\square$

The two notions of absorption in Definition 5.9 are closely related.

Corollary 5.11. *Suppose A and I are C^* -algebras with A separable and I σ -unital and stable. Then a $*$ -homomorphism $\phi: A \rightarrow \mathcal{M}(I)$ is absorbing if and only if $q_I \circ \phi: A \rightarrow \mathcal{Q}(I)$ is absorbing.*

Proof. The forward direction is clear. The backward direction follows from (ii) $\Rightarrow$ (i) of Proposition 5.10. $\square$

The asymptotic absorption in Proposition 5.10(iii) leads to the following asymptotic uniqueness result for absorbing *-homomorphisms, which we use in the $\mathcal{Z}$ -stable KK - and KL -uniqueness theorems.

Corollary 5.12. *Suppose A and I are C^* -algebras with A separable and I σ -unital and stable. If $\phi, \psi: A \rightarrow \mathcal{M}(I)$ are absorbing representations, then there is a norm-continuous path $(u_t)_{t \geq 0} \subseteq \mathcal{M}(I)$ of unitaries such that*

$$(5.38) \quad (t \mapsto u_t \phi(a) u_t^* - \psi(a)) \in C_0([0, \infty), I).$$

Proof. By Proposition 5.10(iii) applied to both ϕ and ψ , there are norm-continuous paths $(v_t)_{t \geq 0}$ and $(w_t)_{t \geq 0}$ of unitaries in $\mathcal{M}(I)$ such that for $a \in A$,

$$(5.39) \quad \begin{aligned} (t \mapsto v_t(\phi(a) \oplus \psi(a))v_t^* - \phi(a)) &\in C_0([0, \infty), I), \quad \text{and} \\ (t \mapsto w_t(\phi(a) \oplus \psi(a))w_t^* - \psi(a)) &\in C_0([0, \infty), I). \end{aligned}$$

The result follows with $u_t := w_t v_t^*$. $\square$

We record the standard fact that relative commutants of absorbing maps are properly infinite for later reference.

Proposition 5.13. *Let A and I be C^* -algebras with A separable and I σ -unital and stable. Given an absorbing *-homomorphism $\phi: A \rightarrow \mathcal{M}(I)$ and $\theta: A \rightarrow \mathcal{Q}(I)$, the relative commutants $\mathcal{M}(I) \cap \phi(A)'$ and $\mathcal{Q}(I) \cap \theta(A)'$ are properly infinite.*

Proof. Since ϕ is absorbing, ϕ is unitarily equivalent to $\phi \oplus \phi$ modulo I by Corollary 5.12, and if u is a unitary implementing this equivalence and s_1 and s_2 are the Cuntz isometries defining the direct sum, then us_1 and us_2 are isometries in $\mathcal{M}(I) \cap \phi(A)'$ with orthogonal range projections. Exactly the same argument (with $q_I(us_1)$ and $q_I(us_2)$) handles the case of $\theta: A \rightarrow \mathcal{Q}(I)$. $\square$

Next we turn to tools for verifying absorption of maps $A \rightarrow \mathcal{Q}(I)$ when A is separable and I is σ -unital and stable. These date back to Voiculescu's theorem ([301]) for the case $I = \mathcal{K}$, which was then generalized by Kasparov to produce absorbing maps when at least one of A or I is nuclear ([162, Theorem 6]). The same result when I is separable but without the nuclearity assumption was later obtained by Thomsen ([285]). Kirchberg first characterized absorption for *-homomorphisms under certain pure infiniteness criteria ([172, Theorem 6]; see also [255, Theorem 8.3.1]). In the presence of nuclearity, Elliott and Kucerovsky used Kirchberg's work (see [101, Lemma 11]) to give a beautiful characterization ([101, Theorem 6]) of unitaly absorbing *-homomorphisms (defined by requiring all maps in Definition 5.9 to be unital) by a condition they call *pure largeness*. We will explore pure largeness in more detail in our subsequent work, but, for the maps appearing in this paper, Kucerovsky and Ng's corona factorization property ([184]) allows a clean characterization of absorption in terms of unitizably full maps. Recall that $A^\dagger$ denotes the forced unitization of A , i.e., with a new unit added when A is unital.

Definition 5.14. A *-homomorphism $\phi: A \rightarrow D$ between C^* -algebras with D unital is said to be *unitizably full* if the unitized map $\phi^\dagger: A^\dagger \rightarrow D$ is full.

With this in place, the characterization we use is as follows. The short proof is deceiving as the real work is outsourced to references, especially [101, Theorem 6], which characterizes unitaly nuclearly absorbing maps; the non-unital version [116, Theorem 2.6] used here is an immediate corollary of the unital case. It is fundamental to our abstract approach to C^* -algebra classification in just the same way that a corresponding version for C^* -algebras that absorb the universal UHF algebra was vital in [267, 268].

Theorem 5.15. *If A is a separable nuclear C^* -algebra and I is a σ -unital stable $\mathcal{Z}$ -stable C^* -algebra, then a $*$ -homomorphism $A \rightarrow \mathcal{M}(I)$ or $A \rightarrow \mathcal{Q}(I)$ is absorbing if and only if it is unitizably full.*

Proof. By Proposition 4.12, I has the corona factorization property. The statement for a $*$ -homomorphism $\phi: A \rightarrow \mathcal{Q}(I)$ is [116, Theorem 2.6] (which is phrased in the language of extensions) after noting that, since A is nuclear, absorption and nuclear absorption of ϕ are equivalent. The multiplier version follows from Corollary 5.11 and the standard fact that (by stability of I) an element $x \in \mathcal{M}(I)$ is full if and only if its image in $\mathcal{Q}(I)$ is full. To see this fact, take $x \in \mathcal{M}(I)$ with full image in $\mathcal{Q}(I)$, so there are $y_1, \dots, y_n, z_1, \dots, z_n \in \mathcal{M}(I)$ such that $c := 1_{\mathcal{M}(I)} - \sum_{i=1}^n y_i x z_i \in I$. As I is stable we may pick an isometry $v \in \mathcal{M}(I)$ such that $\|v^* c v\| < 1$. Hence $\|1_{\mathcal{M}(I)} - \sum_{i=1}^n v^* y_i x z_i v\| = \|v^* c v\| < 1$ which implies that x is full in $\mathcal{M}(I)$. $\square$

5.3. KK -existence. The KK -existence theorem we use comes in two forms, both of which are essentially folklore to experts. The first shows that all KK -classes can be realized by Cuntz pairs with a fixed absorbing $*$ -homomorphism in the second variable, while the second uses a result of Dadarlat to rephrase a standard fact from extension theory in terms of KK -theory.

Theorem 5.16 (KK -existence). *Suppose A is a separable C^* -algebra and I is a σ -unital stable C^* -algebra.*

- (i) *If $\psi: A \rightarrow \mathcal{M}(I)$ is an absorbing $*$ -homomorphism and $\kappa \in KK(A, I)$, then there is an absorbing $*$ -homomorphism $\phi: A \rightarrow \mathcal{M}(I)$ such that (ϕ, ψ) is an (A, I) -Cuntz pair and $[\phi, \psi]_{KK(A, I)} = \kappa$.*
- (ii) *Suppose A is nuclear. If $\theta: A \rightarrow \mathcal{Q}(I)$ is an absorbing $*$ -homomorphism, then there is a (necessarily) absorbing $*$ -homomorphism $\phi: A \rightarrow \mathcal{M}(I)$ lifting θ if and only if $[\theta]_{KK(A, \mathcal{Q}(I))} = 0$.*

Part (i) is standard — a version for weakly nuclear, nuclearly absorbing $*$ -homomorphisms is given in [268, Proposition 2.6], for example, and the same proof works here. The result has its origins in Higson’s Paschke duality result ([142]) for K -homology, showing that if $\kappa \in KK(A, \mathcal{K})$ and $\psi: A \rightarrow \mathcal{M}(\mathcal{K})$ is absorbing, then there is a unitary $u \in \mathcal{M}(\mathcal{K})$ commuting with $\psi(A)$ modulo $\mathcal{K}$ such that $\kappa = [\psi, \psi, u]_{KK(A, \mathcal{K})}$ in the Fredholm picture of KK -theory. The result for general I was obtained by Thomsen in [285] in the proof of his Paschke duality result (particularly, the proof of surjectivity in [285, Theorem 3.2]).

Part (ii) is a consequence of Dadarlat’s isomorphism between $\text{Ext}(A, I)$ and $KK(A, \mathcal{Q}(I))$ for A separable and nuclear and I stable and σ -unital from [62]. In the proof below, we identify extensions with their Busby invariants. Recall that, under the hypotheses of Theorem 5.16, an extension of A by I with Busby invariant $\theta: A \rightarrow \mathcal{Q}(I)$ splits if and only if θ lifts to $\phi: A \rightarrow \mathcal{M}(I)$. Moreover, the class of

θ in $\text{Ext}(A, I)$ vanishes precisely when $\theta \oplus \psi$ splits for some split extension ψ . See [10, Chapter 15] for details.

Proof of Theorem 5.16(ii). If θ lifts to a *-homomorphism $\phi: A \rightarrow \mathcal{M}(I)$, then $[\theta]_{KK(A, \mathcal{Q}(I))} = 0$ since $KK(A, \mathcal{M}(I)) = 0$ by [62, Proposition 4.1]. For the converse, if $[\theta]_{KK(A, \mathcal{Q}(I))}$ vanishes, then the extension with Busby invariant θ has the trivial class in $\text{Ext}(A, I)$, by applying the isomorphism between $KK(A, \mathcal{Q}(I))$ and $\text{Ext}(A, I)$ from [62, Proposition 4.2]. Thus there is a split extension, say with Busby invariant $q_I \circ \psi: A \rightarrow \mathcal{Q}(I)$ for some *-homomorphism $\psi: A \rightarrow \mathcal{M}(I)$, such that $\theta \oplus (q_I \circ \psi)$ splits, and so has a lift to $\mathcal{M}(I)$. But since θ is absorbing, there is a unitary $u \in \mathcal{M}(I)$ with $q_I(u)(\theta \oplus (q_I \circ \psi))q_I(u)^* = \theta$. Thus θ has a lift $\phi: A \rightarrow \mathcal{M}(I)$, which is necessarily absorbing by Corollary 5.11, since θ is absorbing. $\square$

5.4. KK -uniqueness. Now we turn to our $\mathcal{Z}$ -stable KK - and KL -uniqueness theorems. The first part of the following result is (i) $\Rightarrow$ (ii) of Theorem 1.4 from the introduction. The other implication will be proved at the end of this subsection.

Theorem 5.17 ($\mathcal{Z}$ -stable KK - and KL -uniqueness). *Let A and I be C^* -algebras with A separable and I σ -unital and stable. Let $(\phi, \psi): A \rightrightarrows \mathcal{M}(I) \triangleright I$ be a Cuntz pair with ϕ and ψ absorbing.*

(i) *If $[\phi, \psi]_{KK(A, I)} = 0$, then there is a norm-continuous path $(u_t)_{t \geq 0}$ of unitaries in $(I \otimes \mathcal{Z})^\dagger$ such that*

$$(5.40) \quad \|u_t(\phi(a) \otimes 1_{\mathcal{Z}})u_t^* - \psi(a) \otimes 1_{\mathcal{Z}}\| \rightarrow 0, \quad a \in A.$$

(ii) *If $[\phi, \psi]_{KL(A, I)} = 0$, then there is a sequence $(u_n)_{n=1}^\infty$ of unitaries in $(I \otimes \mathcal{Z})^\dagger$ such that*

$$(5.41) \quad \|u_n(\phi(a) \otimes 1_{\mathcal{Z}})u_n^* - \psi(a) \otimes 1_{\mathcal{Z}}\| \rightarrow 0, \quad a \in A.$$

Our proof of (i) follows the same strategy as Dadarlat and Eilers' original KK -uniqueness theorem for $I = \mathcal{K}$ and no $\mathcal{Z}$ -stabilization ([65, Theorem 3.12]), making heavy use of Paschke duality ([230], and generalizations [285, Theorem 3.2], [65, Lemma 3.5]). More precisely, given a Cuntz pair $(\phi, \psi): A \rightrightarrows \mathcal{M}(I) \triangleright I$ of absorbing representations, take a unitary path $(u_t)_{t \geq 0}$ as in Corollary 5.12. Then $q_I(u_0)$ commutes with $q_I(\phi(A))$. Paschke duality ensures that $[\phi, \psi]_{KK(A, I)} = 0$ if and only if the class of $q_I(u_0)$ is trivial in $K_1(\mathcal{Q}(I) \cap q_I(\phi(A))')$. If $q_I(u_0)$ is in the path component of the identity in $\mathcal{Q}(I) \cap q_I(\phi(A))'$, then ϕ and ψ are properly asymptotically unitarily equivalent; i.e., there is a continuous path $(u_t)_{t \geq 0}$ of unitaries in $I^\dagger$ with $u_t \phi(a) u_t^* \rightarrow \psi(a)$ for all $a \in A$. This is extracted from the proof of [65, Theorem 3.12] as Lemma 5.18(i) below. Accordingly, this strategy gives KK -uniqueness whenever $\mathcal{Q}(I) \cap q_I(\phi(A))'$ is K_1 -injective.

Dadarlat and Eilers' stable KK -uniqueness theorem (recalled in the remarks after Theorem 1.4) avoids the K_1 -injectivity problem by using an extra summand. As $\mathcal{Q}(I) \cap q_I(\phi(A))'$ is properly infinite (noted in Proposition 5.13), $[q_I(u_0)]_1 = 0$ implies that $q_I(u_0) \oplus 1_{\mathcal{Q}(I)}$ is in the path component of the identity in $U_2(\mathcal{Q}(I) \cap q_I(\phi(A))')$ (see [258, Exercise 8.11(ii)]). The requirement for the direct summand of $1_{\mathcal{Q}(I)}$ in this argument leads to the summand η in condition (iii) following Theorem 1.4, which describes the stable uniqueness theorem (in the case when ϕ and ψ are absorbing).

For our $\mathcal{Z}$ -stable KK -uniqueness theorem, we bypass the problem of whether $\mathcal{Q}(I) \cap q_I(\phi(A))'$ is K_1 -injective, by working with its $\mathcal{Z}$ -stabilization — which is

K_1 -injective by Jiang's result (Theorem 4.6), or alternatively the later unpublished result from Rohde's PhD thesis [249] that properly infinite $\mathcal{Z}$ -stable C^* -algebras are K_1 -injective. This allows us to conclude that $u_0 \otimes 1_{\mathcal{Z}}$ is in the path component of the identity in the $\mathcal{Z}$ -stabilization of $\mathcal{Q}(I) \cap q_I(\phi(A))'$. So in spirit, our $\mathcal{Z}$ -stable KK -uniqueness theorem is obtained by tensoring on a copy of $1_{\mathcal{Z}}$, whereas the Dadarlat–Eilers theorem adjoins a direct summand of the identity.

The KL -uniqueness theorem (Theorem 5.17(ii)) follows from the KK -uniqueness theorem (part (i)). The strategy, which goes back to Lin ([190, 194]) and Dadarlat ([64]), is to relate vanishing in KL with approximate vanishing in KK . This is exemplified in [64, Theorem 5.1], where Dadarlat uses a reduction to KK -classification to reprove that KL detects approximate uniqueness of morphisms between Kirchberg algebras; our argument for (ii) is in the same spirit.

We now give an abstract version of Dadarlat and Eilers' argument. The strategy has its origins in the large body of work on automorphisms and derivations on operator algebras undertaken in the 1960s ([159, 261, 160, 226]; see also [234, Sections 8.6 and 8.7]).

Lemma 5.18. *Let A be a unital separable C^* -algebra, and let*

$$(5.42) \quad 0 \longrightarrow I \longrightarrow E \xrightarrow{q} D \longrightarrow 0$$

be an extension of C^ -algebras with E unital. Suppose $\phi, \psi: A \rightarrow E$ are unital $*$ -homomorphisms such that $q \circ \phi = q \circ \psi$ and that this composition is injective. Suppose further that there is a continuous path $(v_t)_{t \geq 0}$ of unitaries in E such that*

$$(5.43) \quad (t \mapsto v_t \phi(a) v_t^* - \psi(a)) \in C_0([0, \infty), I), \quad a \in A.$$

- (i) *If the path $(v_t)_{t \geq 0}$ satisfying (5.43) can be chosen such that $q(v_0)$ lies in the path component of the identity in the unitary group of $D \cap q(\phi(A))'$, then it can be chosen with $v_t \in I^\dagger \subseteq E$ for all $t \geq 0$.*
- (ii) *If the path $(v_t)_{t \geq 0}$ satisfying (5.43) can be chosen with $[q(v_0)]_1 = 0$ in $K_1(D \cap q(\phi(A))')$, then there is a path $(u_t)_{t \geq 0}$ of unitaries in $(I \otimes \mathcal{Z})^\dagger$ such that for all $a \in A$,*

$$(5.44) \quad (t \mapsto u_t(\phi(a) \otimes 1_{\mathcal{Z}}) u_t^* - \psi(a) \otimes 1_{\mathcal{Z}}) \in C_0([0, \infty), I \otimes \mathcal{Z}).$$

Proof. (i): A path $(v_t)_{t \geq 0}$ of unitaries satisfying (5.43) such that $q(v_0)$ is in the path component of the identity in $D \cap q(\phi(A))'$ can be adjusted so that $v_0 \in I^\dagger$. Indeed, the hypothesis gives self-adjoint elements $\bar{h}_1, \dots, \bar{h}_n \in D \cap q(\phi(A))'$ such that $q(v_0) = e^{i\bar{h}_1} \dots e^{i\bar{h}_n}$. Lifting these to self-adjoints $h_1, \dots, h_n \in E$ with $q(h_i) = \bar{h}_i$ for all $i = 1, \dots, n$, define $v_t := v_0 e^{it h_n} \dots e^{it h_1}$ for $-1 \leq t < 0$. Then $(v_t)_{t \geq -1}$ is a continuous path of unitaries in E with $v_{-1} \in I^\dagger$ and

$$(t \mapsto v_t \phi(a) v_t^* - \psi(a)) \in C_0([-1, \infty), I).$$

Hence by shifting the index, we assume the given path $(v_t)_{t \geq 0}$ satisfies $v_0 \in I^\dagger$.

As

$$(5.45) \quad \begin{aligned} v_0^* v_t \phi(a) v_t^* v_0 - \phi(a) &= v_0^* (v_t \phi(a) v_t^* - \psi(a)) v_0 \\ &+ (v_0^* \psi(a) v_0 - \phi(a)) \in I, \quad a \in A, \end{aligned}$$

$\phi(A) + I$ is a C^* -subalgebra of $\mathcal{M}(I)$ which is invariant under $\text{Ad}(v_0^* v_t)$ for all $t \geq 0$. Then, since A is separable, we can find a separable C^* -subalgebra $C \subseteq \phi(A) + I$ containing $\phi(A)$ such that C is invariant under $\text{Ad}(v_0^* v_t)$ for all $t \geq 0$. To see this, let $(t_n)_{n=1}^\infty$ be an enumeration of $[0, \infty) \cap \mathbb{Q}$. Starting with $C_0 = \phi(A)$, choose separable

C^* -subalgebras $C_0 \subseteq C_1 \subseteq \dots$ of $\phi(A) + I$ such that for each $n \in \mathbb{N}$, C_n is invariant under $\text{Ad}(v_0^* v_{t_m})$ for $m = 1, \dots, n$. The limit $C := \overline{\bigcup_{n=1}^{\infty} C_n}$ provides the required separable invariant subalgebra. Then $(\text{Ad}(v_0^* v_t))_{t \geq 0}$ is a norm-continuous path of automorphisms of C starting at id_C . By [65, Proposition 2.15], there is a continuous path $(w'_t)_{t \geq 0}$ in $C \subseteq \phi(A) + I$ of unitaries such that

$$(5.46) \quad \|\text{Ad}(w'_t)(c) - \text{Ad}(v_0^* v_t)(c)\| \rightarrow 0, \quad c \in C.$$

Set $w_t := v_0 w'_t$. Then w_t is a unitary in $\phi(A) + I$ for all $t \geq 0$ and

$$(5.47) \quad \|w_t \phi(a) w_t^* - \psi(a)\| \rightarrow 0, \quad a \in A.$$

Write $w_t = \phi(a_t) + x_t$, where $a_t \in A$ and $x_t \in I$, so that $q(\phi(a_t)) = q(w_t)$ is a unitary. Since $q \circ \phi$ is injective, a_t (and hence x_t) are uniquely determined by w_t , and $(a_t)_{t \geq 0}$ is a continuous path of unitaries in A . Also, since $q \circ \phi = q \circ \psi$, by (5.47) we have

$$(5.48) \quad \|q(\phi(a_t))q(\phi(a))q(\phi(a_t))^* - q(\phi(a))\| \rightarrow 0, \quad a \in A,$$

so another application of injectivity of $q \circ \phi$ gives

$$(5.49) \quad \|a_t a a_t^* - a\| \rightarrow 0, \quad a \in A.$$

Now, define

$$(5.50) \quad u_t := w_t \phi(a_t)^* = (1_{I^\dagger} + x_t \phi(a_t)^*) \in I^\dagger.$$

Then $(u_t)_{t \geq 0}$ is a continuous path of unitaries in $I^\dagger$, and combining (5.47) and (5.49), we have, for all $a \in A$,

$$(5.51) \quad \begin{aligned} \|u_t \phi(a) u_t^* - \psi(a)\| &\leq \|w_t(\phi(a_t^*) \phi(a) \phi(a_t)) w_t^* - w_t \phi(a) w_t^*\| \\ &\quad + \|w_t \phi(a) w_t^* - \psi(a)\| \rightarrow 0, \end{aligned}$$

as required.

(ii): Let $(v_t)_{t \geq 0}$ be a path of unitaries as in (ii). By nuclearity of $\mathcal{Z}$, we have a short exact sequence

$$(5.52) \quad 0 \longrightarrow I \otimes \mathcal{Z} \longrightarrow E \otimes \mathcal{Z} \xrightarrow{q \otimes \text{id}_{\mathcal{Z}}} D \otimes \mathcal{Z} \longrightarrow 0.$$

We will complete the proof by applying (i) to this extension, the *-homomorphisms $\phi \otimes 1_{\mathcal{Z}}$ and $\psi \otimes 1_{\mathcal{Z}}$, and the path $(v_t \otimes 1_{\mathcal{Z}})_{t \geq 0}$.

One has a containment of relative commutants

$$(5.53) \quad \theta: (D \cap q(\phi(A)))' \otimes \mathcal{Z} \hookrightarrow (D \otimes \mathcal{Z}) \cap (q \otimes \text{id}_{\mathcal{Z}})((\phi \otimes 1_{\mathcal{Z}})(A))'.$$

Since $q(v_0)$ has trivial class in $K_1(D \cap q(\phi(A)))'$, its image $q(v_0) \otimes 1_{\mathcal{Z}}$ is also trivial in $K_1((D \cap q(\phi(A)))' \otimes \mathcal{Z})$. Jiang's K_1 -injectivity of $\mathcal{Z}$ -stable C^* -algebras (Theorem 4.6) implies that $q(v_0) \otimes 1_{\mathcal{Z}}$ is in the path component of the identity in $(D \cap q(\phi(A)))' \otimes \mathcal{Z}$. Applying θ , it follows that $(q \otimes \text{id}_{\mathcal{Z}})(v_0 \otimes 1_{\mathcal{Z}})$ is in the path component of the identity in $(D \otimes \mathcal{Z}) \cap (q \otimes \text{id}_{\mathcal{Z}})(\phi \otimes 1_{\mathcal{Z}})(A)'$. The result now follows from (i). $\square$

Remark 5.19. Using the completely positive approximation property for $\mathcal{Z}$, one can check that the map θ in (5.53) is surjective and hence an isomorphism, though we do not need this.

The following lemma is a restatement of [65, Lemma 3.5].

Lemma 5.20. *Let A and I be C^* -algebras with A separable and I σ -unital and stable, and let $\phi: A \rightarrow \mathcal{M}(I)$ be a unital $*$ -homomorphism. Suppose $u \in \mathcal{M}(I)$ is a unitary such that $q_I(u) \in \mathcal{Q}(I) \cap q_I(\phi(A))'$, and note that $(\phi, \text{Ad}(u) \circ \phi): A \rightrightarrows \mathcal{M}(I) \triangleright I$ is a Cuntz pair. If $[\phi, \text{Ad}(u) \circ \phi]_{KK(A,I)} = 0$, then there is a unital $*$ -homomorphism $\theta: A \rightarrow \mathcal{M}(I)$ such that $1_{\mathcal{Q}(I)} \oplus 1_{\mathcal{Q}(I)}$ and $q_I(u) \oplus 1_{\mathcal{Q}(I)}$ are homotopic in the unitary group of $M_2(\mathcal{Q}(I)) \cap q_{M_2(I)}((\phi \oplus \theta)(A))'$.*

Proof. In order to correctly cite [65, Lemma 3.5], we need the Fredholm picture of KK -theory. We briefly explain how to translate between this and the Cuntz–Thomsen picture used here following the discussion in [65, Section 3.1]. A Cuntz pair (ψ_1, ψ_2) gives the Fredholm triple $(\psi_1, \psi_2, 1_{\mathcal{M}(I)})$, and a Fredholm triple (ψ_1, ψ_2, v) with v a unitary in $\mathcal{M}(I)$ corresponds to the Cuntz pair $(\text{Ad}(v) \circ \psi_1, \psi_2)$. Therefore, the Cuntz pair $(\phi, \text{Ad}(u) \circ \phi)$ corresponds to the cycle (ϕ, ϕ, u^*) and, since $[\phi, \text{Ad}(u) \circ \phi]_{KK(A,I)} = 0 = [\phi, \phi]_{KK(A,I)}$, we can apply [65, Lemma 3.5] to the Fredholm triples (ϕ, ϕ, u) and $(\phi, \phi, 1_{\mathcal{M}(I)})$. This gives the desired θ and homotopy. $\square$

We now prove the $\mathcal{Z}$ -stable KK - and KL -uniqueness theorems.

Proof of Theorem 5.17. (i): In this proof, we use $\oplus$ to denote the usual direct sum (using a matrix amplification), rather than the definition in (5.7); see Remark 5.3. Since ϕ and ψ are absorbing, Corollary 5.12 provides a norm-continuous path $(u_t)_{t \geq 0} \subseteq \mathcal{M}(I)$ of unitaries such that

$$(5.54) \quad (t \mapsto u_t \phi(a) u_t^* - \psi(a)) \in C_0([0, \infty), I).$$

By Lemma 5.18(ii), it suffices to show $[q_I(u_0)]_1 = 0$ in $K_1(\mathcal{Q}(I) \cap q_I(\phi^\dagger(A^\dagger))')$.

Now, $(\phi, \text{Ad}(u_0) \circ \phi)$ is a Cuntz pair, and it is homotopic (via $(\phi, \text{Ad}(u_t) \circ \phi)$) to (ϕ, ψ) (this is [65, Lemma 3.1]), so that $[\phi, \text{Ad}(u_0) \circ \phi]_{KK(A,I)} = [\phi, \psi]_{KK(A,I)} = 0$. By considering the split exact sequence

$$(5.55) \quad 0 \longrightarrow KK(\mathbb{C}, I) \xleftarrow{\quad} KK(A^\dagger, I) \longrightarrow KK(A, I) \longrightarrow 0,$$

it follows that $[\phi^\dagger, \text{Ad}(u_0) \circ \phi^\dagger]_{KK(A^\dagger, I)} = 0$. This follows, as the maps $KK(A^\dagger, I) \rightarrow KK(A, I)$ and $KK(A^\dagger, I) \rightarrow KK(\mathbb{C}, I)$ arise from the inclusions of A and $\mathbb{C}$ into $A^\dagger$, and so are given by restricting Cuntz pairs on $A^\dagger$ to A and $\mathbb{C}$ respectively (by definition). Thus $[\phi^\dagger, \text{Ad}(u_0) \circ \phi^\dagger]_{KK(A^\dagger, I)}$ has image $[\phi, \text{Ad}(u_0) \circ \phi]_{KK(A,I)} = 0$ in $KK(A, I)$ and image $[1_{\mathcal{M}(I)}, 1_{\mathcal{M}(I)}]_{KK(\mathbb{C}, I)} = 0$ in $KK(\mathbb{C}, I)$. Hence $[\phi^\dagger, \text{Ad}(u_0) \circ \phi^\dagger]_{KK(A^\dagger, I)} = 0$.

Lemma 5.20 gives rise to a norm-continuous path $(v_t)_{0 \leq t \leq 1} \subseteq M_2(\mathcal{M}(I))$ with $v_0 = 1_{\mathcal{M}(I)} \oplus 1_{\mathcal{M}(I)}$ and $v_1 = u_0 \oplus 1_{\mathcal{M}(I)}$ such that each $q_{M_2(I)}(v_t)$ is a unitary in $M_2(\mathcal{Q}(I))$ commuting with $q_{M_2(I)}((\phi^\dagger \oplus \theta)(A))'$. Thus $q_I(u_0) \oplus 1_{\mathcal{Q}(I)}$ is homotopic to $1_{\mathcal{Q}(I)} \oplus 1_{\mathcal{Q}(I)}$ in $M_2(\mathcal{Q}(I)) \cap q_{M_2(I)}((\phi^\dagger \oplus \theta)(A))'$. Taking the direct sum of this homotopy with $1_{\mathcal{Q}(I)}$, gives a homotopy between $q_I(u_0) \oplus 1_{\mathcal{Q}(I)} \oplus 1_{\mathcal{Q}(I)}$ and $1_{\mathcal{Q}(I)} \oplus 1_{\mathcal{Q}(I)} \oplus 1_{\mathcal{Q}(I)}$ in $M_3(\mathcal{Q}(I)) \cap q_{M_3(I)}((\phi^\dagger \oplus \theta \oplus \phi^\dagger)(A))'$.

Since ϕ is absorbing and I is stable, $\theta|_A \oplus \phi$ and ϕ are unitarily equivalent modulo I , and hence so too are $\theta \oplus \phi^\dagger$ and $\phi^\dagger$. It follows that $q_I(u_0) \oplus 1_{\mathcal{Q}(I)}$ is homotopic to $1_{\mathcal{Q}(I)} \oplus 1_{\mathcal{Q}(I)}$ in the unitary group of

$$(5.56) \quad M_2(\mathcal{Q}(I)) \cap ((q_I \circ \phi^\dagger) \oplus (q_I \circ \phi^\dagger))(A^\dagger)' = M_2(\mathcal{Q}(I) \cap q_I(\phi^\dagger(A^\dagger))').$$

Accordingly, $[q_I(u_0)]_1 = 0$ in $K_1(\mathcal{Q}(I) \cap q_I(\phi^\dagger(A^\dagger))')$ as required.

(ii): Suppose $[\phi, \psi]_{KK(A, I)} = 0$. Then there exists $\kappa \in KK(A, C(\overline{\mathbb{N}}, I))$ such that $KK(A, \text{ev}_n)(\kappa) = 0$ for $n \in \mathbb{N}$ and $KK(A, \text{ev}_\infty)(\kappa) = [\phi, \psi]_{KK(A, I)}$ (see Definition 5.6, and recall that $\overline{\mathbb{N}} = \mathbb{N} \cup \{\infty\}$). By [1, Corollary 3.4],

$$(5.57) \quad \mathcal{M}(C(\overline{\mathbb{N}}, I)) \cong C_\sigma(\overline{\mathbb{N}}, \mathcal{M}(I)),$$

where the right hand side is the C^* -algebra of (norm-bounded) strictly continuous functions from $\overline{\mathbb{N}}$ to $\mathcal{M}(I)$ (see Definition 5.2). Let $\Psi: A \rightarrow C_\sigma(\overline{\mathbb{N}}, \mathcal{M}(I))$ be given by $\Psi(a)(n) := \psi(a)$ for all $n \in \overline{\mathbb{N}}$. It is absorbing by [64, Proposition 3.2] (noting that the proof only requires that I is σ -unital instead of separable.)

The KK -existence theorem (Theorem 5.16), gives an absorbing $*$ -homomorphism $\Phi: A \rightarrow \mathcal{M}(C(\overline{\mathbb{N}}, I))$ with $[\Phi, \Psi]_{KK(A, C(\overline{\mathbb{N}}, I))} = \kappa$. Viewing Φ as taking values in $C_\sigma(\overline{\mathbb{N}}, \mathcal{M}(I))$, let $\phi_n := \text{ev}_n \circ \Phi: A \rightarrow \mathcal{M}(I)$. Then (ϕ_n, ψ) is a Cuntz pair for all $n \in \overline{\mathbb{N}}$. Since ψ is absorbing, Corollary 5.11 (applied twice) implies that ϕ_n is also absorbing for all $n \in \overline{\mathbb{N}}$.

Fix a finite subset $\mathcal{F}$ of A and $\epsilon > 0$. As A is separable, it suffices to find a unitary $u \in (I \otimes \mathcal{Z})^\dagger$ such that $\|u(\phi(a) \otimes 1_{\mathcal{Z}})u^* - \psi(a) \otimes 1_{\mathcal{Z}}\| < \epsilon$ for all $a \in \mathcal{F}$. A priori, for each $a \in A$, $\phi_n(a) \rightarrow \phi_\infty(a)$ strictly in $\mathcal{M}(I)$. However, since $\Phi(a) - \Psi(a) \in C(\overline{\mathbb{N}}, I)$, we have

$$(5.58) \quad \phi_n(a) - \phi_\infty(a) = (\phi_n(a) - \psi(a)) + (\psi(a) - \phi_\infty(a)) \in I, \quad a \in A,$$

so the convergence $\phi_n(a) \rightarrow \phi_\infty(a)$ actually happens in norm. Then we can find $n \in \mathbb{N}$ such that $\|\phi_n(a) - \phi_\infty(a)\| < \epsilon/3$ for $a \in \mathcal{F}$. Since $[\phi_n, \psi]_{KK(A, I)} = KK(A, \text{ev}_n)(\kappa) = 0$ (by Proposition 5.5(iv)), we can use part (i) to find a unitary $v \in (I \otimes \mathcal{Z})^\dagger$ with

$$(5.59) \quad \|v(\phi_n(a) \otimes 1_{\mathcal{Z}})v^* - \psi(a) \otimes 1_{\mathcal{Z}}\| < \epsilon/3, \quad a \in \mathcal{F}.$$

Also

$$(5.60) \quad [\phi_\infty, \psi]_{KK(A, I)} = KK(A, \text{ev}_\infty)(\kappa) = [\phi, \psi]_{KK(A, I)}.$$

Accordingly, $[\phi_\infty, \phi]_{KK(A, I)} = 0$ (by Proposition 5.5(iii)) and so another application of part (i) gives a unitary $w \in (I \otimes \mathcal{Z})^\dagger$ with

$$(5.61) \quad \|w(\phi(a) \otimes 1_{\mathcal{Z}})w^* - \phi_\infty(a) \otimes 1_{\mathcal{Z}}\| < \epsilon/3, \quad a \in \mathcal{F}.$$

Set $u := vw \in (I \otimes \mathcal{Z})^\dagger$. Then

$$(5.62) \quad \|u(\phi(a) \otimes 1_{\mathcal{Z}})u^* - \psi(a) \otimes 1_{\mathcal{Z}}\| < \epsilon, \quad a \in \mathcal{F}. \quad \square$$

We do not know if either the tensor factor $1_{\mathcal{Z}}$ in Theorem 1.4(ii) or the direct summand η in condition (iii) following Theorem 1.4 are necessary. This leads to the *KK -uniqueness problem*.

Question 5.21 (The KK -uniqueness problem). Let A and I be C^* -algebras with A separable and I σ -unital and stable, and let $(\phi, \psi): A \rightrightarrows \mathcal{M}(I) \triangleright I$ be a Cuntz pair with ϕ and ψ absorbing and $[\phi, \psi]_{KK(A, I)} = 0$. Must there be a norm-continuous path $(u_t)_{t \geq 0}$ of unitaries in $I^\dagger$ such that

$$(5.63) \quad \|u_t \phi(a) u_t^* - \psi(a)\| \rightarrow 0$$

for all $a \in A$?

The first result in this direction is Dadarlat and Eilers' theorem ([65, Theorem 3.12]), giving a positive solution when $I = \mathcal{K}$. We point out the modifications needed in the treatment above to obtain this result, but we emphasize that the proof is essentially the same as that given in [65]. Paschke proved in [230] that the C^* -algebra $\mathcal{Q}(\mathcal{K}) \cap q_{\mathcal{K}}(\phi(A))'$ is K_1 -injective. Apply this in the final line of the proof of Theorem 5.17(i) to conclude that u_0 is in the path component of the identity in $U(\mathcal{Q}(\mathcal{K}) \cap q_{\mathcal{K}}(\phi(A))')$, and then apply part (i) of Lemma 5.18 in place of (ii).

As noted above, Dadarlat and Eilers' strategy gives a positive answer to the KK -uniqueness problem for (A, I) , whenever $\mathcal{Q}(I) \cap q_I(\phi(A))'$ is K_1 -injective for some (equivalently, any) absorbing $*$ -homomorphism $\phi: A \rightarrow \mathcal{M}(I)$ (cf. [186, Theorem 2.11]). Using this strategy, a partial result was obtained in [207]. We would like to reiterate the following question from [18]. As the relative commutants in question are properly infinite (Proposition 5.13), a positive answer would imply a positive answer to the KK -uniqueness problem.

Question 5.22 ([18, Question 2.9]). Is every properly infinite C^* -algebra K_1 -injective?

We end this subsection by recording the easy implication of Theorem 1.4 — the hard direction follows from the KK -uniqueness theorem.

Proof of Theorem 1.4. The implication (i) $\Rightarrow$ (ii) is Theorem 5.17(i). For (ii) $\Rightarrow$ (i), note that (ii) implies that $[\phi \otimes 1_{\mathcal{Z}}, \psi \otimes 1_{\mathcal{Z}}]_{KK(A, I \otimes \mathcal{Z})} = 0$. The inclusion $\iota: I \rightarrow I \otimes \mathcal{Z}$ is a KK -equivalence by Proposition 4.5 and $[\phi \otimes 1_{\mathcal{Z}}, \psi \otimes 1_{\mathcal{Z}}]_{KK(A, I \otimes \mathcal{Z})} = KK(A, \iota)([\phi, \psi]_{KK(A, I)})$, by Proposition 5.5(iv), so it follows that $[\phi, \psi]_{KK(A, I)} = 0$. $\square$

Remark 5.23. Since the submission of this paper, Szabó has given a positive answer to Question 5.21 ([279]) by establishing K_1 -injectivity of suitable relative commutants. Question 5.22 remains open. See Addendum 1.5.

6. THE TRACE-KERNEL EXTENSION

We recall the trace-kernel extension in Section 6.1 and describe the background behind Theorem 1.3 in Section 6.2. In Section 6.3, we use this to compute the K -theory of the trace-kernel quotient (under the codomain hypotheses of Theorem A) and prove Theorem 1.5. We end with Section 6.4, in which we show that the trace-kernel ideal of a simple exact $\mathcal{Z}$ -stable C^* -algebra is separably stable.

6.1. The trace-kernel extension. Matui and Sato's breakthrough work ([213, 214]) on the Toms–Winter conjecture involved a new way of using von Neumann algebras to solve C^* -algebraic problems, which was later pushed further with extension and KK -theoretic arguments in [267, 268]. Matui and Sato introduced what we now call the trace-kernel ideal J_B in the sequence algebra B_{∞} as the intersection of all the ideals associated to limit traces on B_{∞} (recall Definition 4.15). It appears implicitly in [213] and explicitly (in an ultrapower formulation) in [178] and [214].

Definition 6.1. Let B be a unital separable C^* -algebra satisfying $T(B) \neq \emptyset$. The *trace-kernel ideal* of B is

$$(6.1) \quad J_B := \{b \in B_{\infty} : \tau(b^*b) = 0 \text{ for all } \tau \in T_{\infty}(B)\},$$

and the *trace-kernel extension* is the short exact sequence

$$(6.2) \quad 0 \longrightarrow J_B \xrightarrow{j_B} B_\infty \xrightarrow{q_B} B^\infty \longrightarrow 0,$$

where j_B is the canonical inclusion, $B^\infty := B_\infty/J_B$, called the *trace-kernel quotient*, and q_B is the quotient map.

Note that given $x = (x_n)_{n=1}^\infty \in B_\infty$, we have $x \in J_B$ if and only if

$$(6.3) \quad \lim_{n \rightarrow \infty} \sup_{\tau \in T(B)} \tau(x_n^* x_n) = 0.$$

As with B_∞ , we denote elements of B^∞ by representatives from $\ell^\infty(B)$.

Note too that the limit traces descend from B_∞ to give traces on B^∞ ; we will use the notation $T_\infty(B)$ in both cases. For a positive element $x \in B^\infty$, if $\tau(x) = 0$ for all $\tau \in T_\infty(B)$, then $x = 0$.

Remark 6.2. A unital *-homomorphism $\phi: B \rightarrow C$ induces a map $\phi_\infty: B_\infty \rightarrow C_\infty$. As ϕ is unital, any trace $\tau \in T_\infty(C) \subseteq T(C_\infty)$ induces a trace $\phi_\infty^*(\tau) \in T_\infty(B)$. Therefore, $\phi_\infty(J_B) \subseteq J_C$, and hence ϕ also induces a map $\phi^\infty: B^\infty \rightarrow C^\infty$. In this way, the assignment of the trace-kernel extension e_B to B is functorial for unital C^* -algebras with unital *-homomorphisms.

As recorded in Proposition 4.16, convex combinations of limit traces are dense in $T(B_\infty)$ when B is $\mathcal{Z}$ -stable and exact. When this holds, it descends to the trace-kernel quotient B^∞ . While we prove this using Proposition 4.16 here, it is possible for the convex combinations of the limit traces to be dense in $T(B^\infty)$ even when they are not dense in $T(B_\infty)$; see [34, Proposition 2.5].

Proposition 6.3. *Let B be a unital simple separable exact $\mathcal{Z}$ -stable C^* -algebra with $T(B) \neq \emptyset$.*

- (i) *The map $T(q_B): T(B^\infty) \rightarrow T(B_\infty)$ induced by the quotient map $q_B: B_\infty \rightarrow B^\infty$ is an affine homeomorphism. In particular, the convex hull of the limit traces $T_\infty(B)$ is weak*-dense in $T(B^\infty)$.*
- (ii) *B^∞ has strict comparison of positive elements by bounded traces.*
- (iii) *Let A be a C^* -algebra. A *-homomorphism $\theta: A \rightarrow B^\infty$ is full if and only if $\tau \circ \theta$ is faithful for all traces $\tau \in T(B^\infty)$.*

Proof. For (i), the affine map $T(q_B)$ is a continuous injection (because q_B is surjective). It is surjective as well — and therefore a homeomorphism — as its image is compact, convex, and contains the limit traces on B_∞ , which have dense convex hull in $T(B_\infty)$ by Proposition 4.16.

Part (ii) follows since if $a, b \in M_k(B^\infty)$ are positive elements with $d_\tau(a) < d_\tau(b)$ for all $\tau \in T(B^\infty)$, then taking positive lifts $\tilde{a}$ and $\tilde{b}$ of a and b in $M_k(B_\infty)$, the previous part gives $d_\tau(\tilde{a}) < d_\tau(\tilde{b})$ for all $\tau \in T(B_\infty)$. As B_∞ has strict comparison of positive elements by bounded traces (Proposition 4.17(ii)), $\tilde{a} \precsim \tilde{b}$, and hence $a \precsim b$.

Part (iii) is an immediate consequence of (ii) and Lemma 4.10. $\square$

We will need to work with matrix amplifications of the trace-kernel extension in Sections 6.3 and 7.4. The following observation is implicitly used (in the setting of ultrapowers) in [268, Section 4]. It boils down to the fact that $\tau \mapsto \tau_{M_k} \otimes \tau$ gives an affine homeomorphism $T(B) \rightarrow T(M_k \otimes B)$.

Remark 6.4. For a C^* -algebra D , write $\iota_D^{(k)}: D \rightarrow M_k(D)$ for the top-left corner embedding of D into $M_k(D)$. If B is a unital C^* -algebra and $T(B) \neq \emptyset$, then the following diagram commutes, with natural maps forming the isomorphism of extensions between the bottom two rows:

$$\begin{array}{ccccccccc}
 0 & \longrightarrow & J_B & \xrightarrow{j_B} & B_\infty & \xrightarrow{q_B} & B^\infty & \longrightarrow & 0 \\
 & & \downarrow \iota_{J_B}^{(k)} & & \downarrow \iota_{B_\infty}^{(k)} & & \downarrow \iota_{B^\infty}^{(k)} & & \\
 (6.4) \quad 0 & \longrightarrow & M_k(J_B) & \longrightarrow & M_k(B_\infty) & \longrightarrow & M_k(B^\infty) & \longrightarrow & 0 \\
 & & \downarrow \cong & & \downarrow \cong & & \downarrow \cong & & \\
 0 & \longrightarrow & J_{M_k(B)} & \xrightarrow{j_{M_k(B)}} & M_k(B)_\infty & \xrightarrow{q_{M_k(B)}} & M_k(B)^\infty & \longrightarrow & 0.
 \end{array}$$

Note too that if B satisfies our standing codomain hypotheses (unital, simple, separable, nuclear and $\mathcal{Z}$ -stable), then so too do all matrix amplifications $M_k(B)$.

6.2. Classification into the trace-kernel quotient. As mentioned in the introduction, the intuition behind the classification of maps into the trace-kernel quotient is most readily seen with the ultrapower version of the trace-kernel extension $0 \rightarrow J_{B,\omega} \rightarrow B_\omega \rightarrow B^\omega \rightarrow 0$, for some free ultrafilter ω on $\mathbb{N}$, and when B has unique trace, so that B^ω is the tracial von Neumann ultrapower $(\pi_\tau(B)'', \tau)^\omega$, where π_τ is the GNS-representation associated to τ (see [178], for example). Here, Connes' theorem implies a classification of maps from separable nuclear C^* -algebras into type II₁ von Neumann algebras (see [71, Corollary 10 and Theorem 5], or more generally, [41, Proposition 2.1]).

In general, B^ω will not be a von Neumann algebra, but under the right conditions, classification of maps into B^ω (or B^∞) is possible by a tracial gluing partition of unity technique. This technique was developed in stages. When $T(B)$ is a Bauer simplex (i.e., $\partial_e T(B)$ is compact), this gluing can be achieved through Ozawa's concept of W^* -bundles and his trivialization theorem ([228, Theorem 15]; see [21, Proposition 3.23] for the relevant partition of unity argument). Outside the Bauer simplex setting, the situation is more complicated, and a general technique using nuclearity and $\mathcal{Z}$ -stability of B was developed in [37], for use in the Toms–Winter conjecture. See the second half of the introduction to [37] for more details. The use of [37] for the results in this and the next subsection is the main reason for requiring our codomains to be nuclear (we also need that quasitraces are traces on the codomain).

The exact form of the classification of maps by traces that we need is in [34] (a short sequel to [37]). Note that [34] contains a small error that is corrected in [36].

Theorem 6.5 (Classification of maps into B^∞ by traces; [34, Theorem A]). *Let A be a unital separable nuclear C^* -algebra and let B be a unital separable nuclear $\mathcal{Z}$ -stable C^* -algebra with $T(B) \neq \emptyset$. For every positive unital linear map $\gamma: \text{Aff } T(A) \rightarrow \text{Aff } T(B^\infty)$, there is a unital $*$ -homomorphism $\theta: A \rightarrow B^\infty$ satisfying $\text{Aff } T(\theta) = \gamma$. Moreover, θ is unique up to unitary equivalence.*

Proof. By Kadison duality (see the second paragraph of Section 2.1), there is a natural bijection between positive unital linear functions $\text{Aff } T(A) \rightarrow \text{Aff } T(B^\infty)$ and continuous affine functions $T(B^\infty) \rightarrow T(A)$. The existence and uniqueness of a not-necessarily-unital map θ follow from [34, Theorem A]. Further, if $\theta: A \rightarrow B^\infty$ is a $*$ -homomorphism with $\tau \circ \theta \in T(A)$ for all $\tau \in T(B^\infty)$, (i.e., $\tau \circ \theta$ is a tracial

state not just a tracial functional) then $1_{B^\infty} - \theta(1_A)$ is a positive element of B^∞ which vanishes on all traces on B^∞ . Since a general positive element of B^∞ that vanishes on all traces is automatically zero, this implies θ is unital. $\square$

6.3. K -theory of the trace-kernel ideal and quotient. We now turn to the computation of $K_1(J_B)$ when B is an allowed codomain in Theorem B. Firstly, the gluing procedures used to obtain the classification of maps into B^∞ also show that every unitary in B^∞ is an exponential.

Proposition 6.6 ([34, Proposition 2.1]). *Let B be a unital separable nuclear $\mathcal{Z}$ -stable C^* -algebra with $T(B) \neq \emptyset$. If $u \in B^\infty$ is a unitary, then $u = e^{ih}$ for some self-adjoint $h \in B^\infty$ of norm at most π . In particular, every unitary in B^∞ is the image of a unitary in B_∞ under q_B .*

Combining Proposition 6.6 with the classification of maps into B^∞ by traces gives the following K -theory calculation, analogous to the fact that the K -theory of a II_1 factor is $(\mathbb{R}, 0)$.

Proposition 6.7. *Let B be a unital separable nuclear $\mathcal{Z}$ -stable C^* -algebra with $T(B) \neq \emptyset$. Then $(K_0(B^\infty), K_1(B^\infty)) \cong (\text{Aff } T(B^\infty), 0)$, with the pairing map ρ_{B^∞} inducing the isomorphism in the first entry, and $K_*(B^\infty; \mathbb{Z}/n) = 0$ for all $n \geq 2$.*

Proof. The computation of $K_0(B^\infty)$ is a consequence of Theorem 6.5 with $A := \mathbb{C}^2$, as follows. Given $f \in \text{Aff } T(B^\infty)_+$, fix $n \in \mathbb{N}$ such that $f \leq n$, which exists by the compactness of $T(B^\infty)$. Let $\tilde{f} \in \text{Aff } T(M_n \otimes B^\infty)$ be given by $\tilde{f}(\tau_{M_n} \otimes \tau) := f(\tau)$ for $\tau \in T(B^\infty)$. Define $\gamma: \mathbb{R} \oplus \mathbb{R} \rightarrow \text{Aff } T(M_n \otimes B^\infty)$ by $\gamma(s, t) := \frac{s}{n} \tilde{f} + t(1 - \frac{1}{n} \tilde{f})$. Then existence in Theorem 6.5 (working with the allowed codomain $M_n \otimes B$ in place of B , as noted at the end of Remark 6.4) gives us a unital $*$ -homomorphism $\phi: \mathbb{C} \oplus \mathbb{C} \rightarrow M_n \otimes B^\infty$ realizing γ , and thus a projection $p := \phi(1, 0) \in (M_n \otimes B)^\infty \cong M_n(B^\infty)$ (see Remark 6.4) such that $(\tau_{M_n} \otimes \tau)(p) = \frac{1}{n} f(\tau)$ for all $\tau \in T(B^\infty)$. This means that $\rho_{B^\infty}([p]_0) = f$. Moreover, uniqueness in Theorem 6.5 tells us that any projection $q \in M_n \otimes B^\infty$ with the same property is unitarily equivalent to p , and thus $[p]_0 = [q]_0$ in $K_0(B^\infty)$. Taking linear combinations shows that the pairing map induces an isomorphism $K_0(B^\infty) \cong \text{Aff } T(B^\infty)$.

Proposition 6.6, applied to $M_n \otimes B^\infty \cong (M_n \otimes B)^\infty$ (via Remark 6.4) implies $K_1(B^\infty) = 0$. For the last statement, the map $K_0(B^\infty) \rightarrow K_0(B^\infty)$ of multiplication by n is an isomorphism for all $n \geq 2$ (as $K_0(B^\infty)$ is a real vector space). Using this, and $K_1(B^\infty) = 0$, in the exact sequence (2.28) implies $K_*(B^\infty; \mathbb{Z}/n) = 0$ for all $n \geq 2$. $\square$

Now we turn both to the proof of Theorem 1.5 and the setup for Theorem 1.6. Note that when B is unital, simple, separable, exact, and $\mathcal{Z}$ -stable, we have that $\text{Aff } T(q_B): \text{Aff } T(B_\infty) \rightarrow \text{Aff } T(B^\infty)$ is an isomorphism by Proposition 6.3(i). In the following diagram, the first row is an extract of the six-term exact sequence for the trace-kernel extension, while the second row is exact as it is Thomsen's extension for B_∞ (Proposition 4.18). Commutativity of the left-hand square is naturality of the pairing map.

$$(6.5) \quad \begin{array}{ccccccc} K_0(B_\infty) & \xrightarrow{K_0(q_B)} & K_0(B^\infty) & \xrightarrow{\partial} & K_1(J_B) & \xrightarrow{K_1(j_B)} & K_1(B_\infty) \xrightarrow{K_1(q_B)} K_1(B^\infty) \\ \parallel & & \downarrow (\text{Aff } T(q_B))^{-1} \circ \rho_{B^\infty} & & \downarrow \omega_B & & \parallel \\ K_0(B_\infty) & \xrightarrow{\rho_{B^\infty}} & \text{Aff } T(B_\infty) & \xrightarrow{\text{Th}_{B^\infty}} & \overline{K}_1^{\text{alg}}(B_\infty) & \xrightarrow{\not\#_{B^\infty}} & K_1(B_\infty) \longrightarrow 0 \end{array}$$

A diagram chase gives a natural map

$$(6.6) \quad \omega'_B: \text{im } \partial \rightarrow \text{im } \text{Th}_{B^\infty} = \ker \not\#_{B^\infty}: \partial([p]_0) \mapsto [e^{2\pi i a}]_{\text{alg}},$$

where $a \in M_k(B_\infty)$ is a self-adjoint lifting the projection $p \in M_k(B^\infty)$. For our calculation of $KL(A, J_B)$ in Theorem 8.9 (the precise version of Theorem 1.6), we need to know that ω'_B is an isomorphism and describe it explicitly in terms of elements of $\ker K_1(j_B) = \text{im } \partial$. The explicit description we give extends to a natural map $\omega_B: K_1(J_B) \rightarrow \overline{K}_1^{\text{alg}}(B_\infty)$. Moreover, both this and ω'_B are isomorphisms when B^∞ has the K -theory provided by Proposition 6.7.

Theorem 6.8 (Calculating $K_1(J_B)$). *Let B be a unital separable exact $\mathcal{Z}$ -stable C^* -algebra with $T(B) \neq \emptyset$.*

- (i) *There is a natural map $\omega_B: K_1(J_B) \rightarrow \overline{K}_1^{\text{alg}}(B_\infty)$ such that (6.5) commutes, given explicitly by*

$$(6.7) \quad \omega_B([u]_1) := [u]_{\text{alg}} - [s(u)]_{\text{alg}}, \quad u \in U_\infty(J_B^\dagger),$$

where $s: J_B^\dagger \rightarrow \mathbb{C}1_{B_\infty} \subseteq B_\infty$ is the canonical character. This map extends the natural map ω'_B from (6.6).

- (ii) *When B is also simple and nuclear, ω_B and ω'_B are isomorphisms.*

Proof. (i): The map $\tilde{\omega}_B: U_\infty(J_B^\dagger) \rightarrow \overline{K}_1^{\text{alg}}(B_\infty)$ given by $\tilde{\omega}_B(u) := [u]_{\text{alg}} - [s(u)]_{\text{alg}}$ for $u \in U_\infty(J_B^\dagger)$ is a group homomorphism. To show that this induces a well-defined map ω_B on $K_1(J_B)$, it suffices to check that $\tilde{\omega}_B(e^{2\pi i h}) = 0$ in $\overline{K}_1^{\text{alg}}(B_\infty)$ for any self-adjoint $h \in M_n(J_B^\dagger)$. To this end, write $h = h_0 + s(h)$, where $h_0 = h_0^* \in M_n(J_B)$, so that $\tilde{\omega}_B(e^{2\pi i h}) = [e^{2\pi i h_0}]_{\text{alg}} \in \ker \not\#_{B^\infty}$. Using exactness and $\mathcal{Z}$ -stability of B , all traces on B_∞ vanish on J_B (Proposition 4.16), and so $\hat{h}_0 = 0$ in $\text{Aff } T(B_\infty)$. Applying the Thomsen map from (2.8), $\tilde{\omega}_B(e^{2\pi i h}) = \text{Th}_{B^\infty}(\hat{h}_0) = 0$, as required. The explicit formula (6.7) confirms that ω_B is natural in B .

By construction, the induced map ω_B makes the third square of (6.5) commute. For the second square, consider a projection $p \in M_n(B^\infty)$, which we lift to a positive contraction $a \in M_n(B_\infty)$. This satisfies $\text{Aff } T(q_B)(\hat{a}) = \hat{p}$, $e^{2\pi i a} \in U_n(J_B^\dagger)$ and $\partial([p]_0) = [e^{2\pi i a}]_1$. Noting that $s(e^{2\pi i a}) = 1_{M_n(B_\infty)}$, we have $(\omega_B \circ \partial)([p]_0) = [e^{2\pi i a}]_{\text{alg}} = \text{Th}_{B^\infty}(\hat{a})$ (from the definition of ω_B and that of the Thomsen map in (2.8)). As $(\text{Aff } T(q_B))^{-1} \circ \rho_{B^\infty}([p]_0) = \text{Aff } T(q_B)^{-1}(\hat{p}) = \hat{a}$, the commutativity follows. Note that commutativity of the second square shows that ω_B extends ω'_B from (6.6).

(ii): Proposition 6.3(i) ensures that $\text{Aff } T(q_B)$ is an isomorphism and, using the nuclearity hypothesis, the K -theory computation of Proposition 6.7 shows that ρ_{B^∞}

is an isomorphism and $K_1(B^\infty) = 0$. The second row of (6.5) is exact by Proposition 4.18, and as the four outermost vertical arrows of (6.5) are isomorphisms, the result follows from the five lemma. $\square$

6.4. Separable stability of the trace-kernel ideal J_B . In order to work with $KK(A, J_B)$ in Sections 7 and 9, it would be most convenient if the trace-kernel ideal J_B was stable. However, provided $T(B) \neq \emptyset$ and $J_B \neq 0$, this is never the case as a consequence of Ghasemi's result ([122]) since J_B is an SAW^* -algebra (which can be proven by an ϵ -test argument similar to [178, Proposition 4.6]). This is similar to B_∞ not being $\mathcal{Z}$ -stable (see Section 4.4), and the solution is the same: to separabilize (pass to separable subalgebras with the desired property).

Definition 6.9 (cf. [268, Definition 1.4]). A C^* -algebra D is *separably stable* if for every separable C^* -subalgebra $D_0 \subseteq D$, there exists a stable separable C^* -subalgebra $E \subseteq D$ with $D_0 \subseteq E$.

Our objective is to show in Lemma 6.11 that J_B is separably stable whenever B satisfies the codomain assumptions in the classification of embeddings. A version of this lemma was obtained in [268, Propositions 3.2 and 3.3] under much more restrictive hypotheses, which ensure that B_ω is real rank zero. In our setting, we must use Cuntz semigroup techniques rather than projections. To do this, we note that Hjelmborg and Rørdam's characterization of stability for σ -unital C^* -algebras via positive elements extends to characterize separable stability. We say that a C^* -algebra D satisfies the *Hjelmborg–Rørdam criterion* if, for every $a \in D_+$ and $\epsilon > 0$, there exists $x \in D$ satisfying $\|a - xx^*\| < \epsilon$ and $\|(x^*x)(xx^*)\| < \epsilon$ (this is equivalent to condition [148, Proposition 2.2(b)] since the set $F(D)$ of [148] is dense in D_+ as noted on [148, Page 154]). A σ -unital C^* -algebra D is stable if and only if it satisfies the Hjelmborg–Rørdam criterion by [148, Theorem 2.1 and Proposition 2.2].

Proposition 6.10. *Let D be a C^* -algebra. Then D is separably stable if and only if it satisfies the Hjelmborg–Rørdam criterion.*

Proof. It is immediate that separable stability implies the Hjelmborg–Rørdam criterion. To see the converse, it suffices to show that if D is a C^* -algebra satisfying the Hjelmborg–Rørdam criterion and D_0 is a separable subalgebra of D , then there exists a separable subalgebra $E \subseteq D$ satisfying the Hjelmborg–Rørdam criterion with $D_0 \subseteq E$. Fix a dense sequence $(d_l)_{l=1}^\infty$ in $(D_0)_+$. By the Hjelmborg–Rørdam criterion, for each $k, l \in \mathbb{N}$, there exists $x_{k,l} \in D$ such that $\|d_l - x_{k,l}^* x_{k,l}\| < 1/k$ and $\|(x_{k,l}^* x_{k,l})(x_{k,l} x_{k,l}^*)\| < 1/k$. Let $D_1 := C^*(D_0, \{x_{k,l} : k, l \in \mathbb{N}\})$. This ensures that for any $a \in (D_0)_+$, and $\epsilon > 0$, there exists $x \in D_1$ with $\|a - xx^*\| < \epsilon$ and $\|(x^*x)(xx^*)\| < \epsilon$. Continuing in this way, we obtain a nested sequence $D_0 \subseteq D_1 \subseteq \dots$ of separable C^* -subalgebras of D so that the Hjelmborg–Rørdam criterion for positive elements a in D_n can be witnessed by elements $x \in D_{n+1}$. Then $E := \overline{\bigcup_{n=1}^\infty D_n}$ will satisfy the Hjelmborg–Rørdam criterion. $\square$

With the Hjelmborg–Rørdam criterion in place, we now deduce separable stability of the relevant trace-kernel ideals. While we only need the following result for nuclear B , we state it for B exact — the exactness only being used to apply Haagerup's result that quasitraces are traces ([137]).

Lemma 6.11. *Let B be a unital simple separable exact $\mathcal{Z}$ -stable C^* -algebra with $T(B) \neq \emptyset$. Then J_B is separably stable.*

Proof. We check that J_B satisfies the Hjelmberg–Rørdam criterion of Proposition 6.10. Let $a \in (J_B)_+$ and $\epsilon > 0$. By perturbing a slightly and using functional calculus, we may assume that there is a contraction $e \in (J_B)_+$ such that $ea = a$. Set $b := 1_{B_\infty} - e \in (B_\infty)_+$, and note that $ab = 0$. Since $e \in J_B$, we have $\tau(b) = 1$ for all $\tau \in T_\infty(B)$, and therefore, for all $\tau \in T(B_\infty)$ by density of the convex hull of the limit traces in $T(B_\infty)$ (Proposition 4.16). Since $b \leq 1$, $d_\tau(b) \geq \tau(b) = 1$ for all $\tau \in T(B_\infty)$. On the other hand (again using Proposition 4.16), $d_\tau(a) = \lim_{r \rightarrow \infty} \tau(a^{1/r}) = 0$ for all $\tau \in T(B_\infty)$, since $a^{1/r} \in J_B$ for all r . As B_∞ has strict comparison of positive elements by bounded traces (Proposition 4.17(ii)), it follows that $a \precsim b$. By [176, Proposition 2.7(iii)] there exists $x \in B_\infty$ such that $x^*x = (a - \epsilon)_+$ and $xx^* \in \overline{bB_\infty b}$. Because $x^*x = (a - \epsilon)_+ \in J_B$, we get $x \in J_B$ and $\|x^*x - a\| < \epsilon$. Also, since $xx^* \in \overline{bB_\infty b}$ and $ab = 0$, we have $xx^*a = 0$, so $(xx^*)(x^*x) = 0$. $\square$

Remark 6.12. As mentioned in the introduction, it is natural to consider the extension associated with the finite part of the bidual B_{fin}^{**} . However, when B has infinitely many extreme traces, the kernel of the surjection $B_\omega \rightarrow (B_{\text{fin}}^{**})^\omega$ is not separably stable. To see this, we can take an element $b = (b_n)_{n=1}^\infty \in (B_\omega)_+$ such that $\|b_n\|_{2,\tau} \rightarrow 0$ for all τ yet $\sup_{\tau \in T(B)} \|b_n\|_{2,\tau} \rightarrow 1$ (i.e., the convergence is pointwise but not uniform). This means that the image of b in $(B_{\text{fin}}^{**})^\omega$ is zero, but the trace condition makes it impossible to find $x^*x \in B_\omega$ close to b with $\|(xx^*)(x^*x)\|$ small.

7. CLASSIFYING LIFTS ALONG THE TRACE-KERNEL EXTENSION

The main objective of this section is to classify lifts from the trace-kernel quotient in terms of $KL(A, J_B)$. There are two versions of this result. We first establish Theorem 7.1, which classifies lifts of unitizably full maps $\theta: A \rightarrow B^\infty$, and then we follow a “de-unitization” procedure to convert this result into the classification of unital lifts (Theorem 7.10, which is the precise formulation of Theorem 1.2).

The classification of lifts has a long history, dating back to the theory of extensions of C^* -algebras developed by Brown, Douglas, and Fillmore ([28]). Our results build on more recent developments by CS ([267, 268]); see Remarks 7.3 and 7.4.

7.1. Classifying lifts of unitizably full maps. In the following theorem, we follow the definitions of KK - and KL -groups for non- σ -unital codomain C^* -algebras (such as J_B) from Section 5.1. We defer the proof to Section 7.3, allowing us to first address preliminaries.

Theorem 7.1 (Classification of lifts). *Let A be a separable nuclear C^* -algebra, let B be a unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebra with $T(B) \neq \emptyset$, and suppose $\theta: A \rightarrow B^\infty$ is a unitizably full $*$ -homomorphism.*

- (i) *For any $\kappa \in KK(A, B_\infty)$ with $KK(A, q_B)(\kappa) = [\theta]_{KK(A, B_\infty)}$, there exists a $*$ -homomorphism $\phi: A \rightarrow B_\infty$ that lifts θ and such that $[\phi]_{KK(A, B_\infty)} = \kappa$.*
- (ii) *Given a $*$ -homomorphism $\psi: A \rightarrow B_\infty$ that lifts θ and any $\lambda \in KL(A, J_B)$, there exists a $*$ -homomorphism $\phi: A \rightarrow B_\infty$ that also lifts θ and such that $[\phi, \psi]_{KL(A, J_B)} = \lambda$.*
- (iii) *If ϕ_1 and ϕ_2 are $*$ -homomorphisms that lift θ and satisfy $[\phi_1, \phi_2]_{KL(A, J_B)} = 0$, then they are unitarily equivalent. In particular, the map ϕ in (ii) is unique up to unitary equivalence.*

The data from Theorem 7.1 is summarized in the following diagram: given θ and a KK -class κ we can lift θ to a map $A \rightarrow B_\infty$. Further, for a fixed lift ψ of θ , the lifts ϕ are classified up to unitary equivalence by the class $[\phi, \psi]_{KL(A, J_B)}$.

$$(7.1) \quad \begin{array}{ccccccc} & & & \lambda = [\phi, \psi]_{KL} & & & \\ & & & \swarrow & & \searrow & \\ & & & A & & & \\ & & & \downarrow \theta \text{ unitizably full} & & & \\ 0 & \longrightarrow & J_B & \longrightarrow & B_\infty & \xleftarrow{\phi, \psi} & B^\infty \longrightarrow 0 \end{array}$$

Remark 7.2. It is important in Theorem 7.1(i) that we are given a KK -lift κ of θ . This is a genuine assumption — for example, it certainly fails when A has a non-trivial projection and B does not. When A satisfies the UCT, the existence of a KK -lift of θ is equivalent to the existence of K_0 -lift of θ — that is, a map $\alpha_0: K_0(A) \rightarrow K_0(B_\infty)$ such that $K_0(q_B) \circ \alpha_0 = K_0(\theta)$ (by Proposition 8.6 and surjectivity of $\Gamma^{(A, B_\infty)}: KK(A, B_\infty) \rightarrow \text{Hom}(K_*(A), K_*(B))$). When we apply Theorem 7.1(i) in the classification of full approximate embeddings, we will use the UCT to produce a KK -lift of θ compatible with a specified morphism in total K -theory.

Remark 7.3. CS's proof in [267] of the quasidiagonality theorem of [287], that every faithful trace on a separable nuclear C^* -algebra satisfying the UCT is quasidiagonal, can be viewed as a special case of Theorem 7.1(i) (in the setting of ultrapowers, which are more natural in the unique trace setting of [267]). Indeed, a standard computation (see Section 3 of [267], for example) shows that $K_0(q_Q): K_0(Q_\infty) \rightarrow K_0(Q^\infty)$ is a surjective linear map of rational vector spaces and hence splits. Now, if A is a nuclear C^* -algebra with a faithful trace, one obtains a unitizably full $*$ -homomorphism $\theta: A \rightarrow Q^\infty$ from the amenability of this trace (a naive construction yields a full but not unitizably full map; embedding into the top-left entry of $M_2(Q)^\infty \cong Q^\infty$ provides enough space so that the composition is unitizably full). This necessarily admits a K_0 -lift since $K_0(q_Q)$ is a split surjection, so when A satisfies the UCT, θ admits a KK -lift as well by Remark 7.2. Now, Theorem 7.1(i) implies there is an embedding $A \hookrightarrow Q_\infty$, which allows access to Voiculescu's local characterization of quasidiagonality ([302, Theorem 1]) (since A is nuclear, and so this map has a c.p.c. lift to $\ell^\infty(Q)$ by the Choi–Effros lifting theorem).

Remark 7.4. CS implicitly establishes versions of Theorem 7.1(ii) and (iii) in the proofs of [268, Propositions 4.2 and 4.3], in the case that B has a unique (quasi)-trace and is Q -stable (but is not necessarily nuclear). These results, and CS's approach to the quasidiagonality theorem in [267], allow for exact domains with appropriate amenability conditions on the trace induced by θ and on ϕ and ψ (in [267], this recaptures the more general version of the quasidiagonality theorem from [117]). In comparison, Theorem 7.1 relaxes Q -stability to $\mathcal{Z}$ -stability and allows for general trace simplices but only covers nuclear domains and codomains. The additional machinery to allow for exact domains and general $\mathcal{Z}$ -stable codomains (whose quasitraces are traces) will be found in the second paper of this series.

Remark 7.5. The uniqueness in Theorem 7.1(iii) means that, given ψ as in (ii), any two lifts $\phi_1, \phi_2: A \rightarrow B_\infty$ of θ with $[\phi_1, \psi]_{KL(A, J_B)} = [\phi_2, \psi]_{KL(A, J_B)}$ are unitarily equivalent. In order to view this as a classification of lifts up to unitary equivalence by $KL(A, J_B)$, we should check that $[\cdot, \psi]_{KK(A, J_B)}$ (and hence $[\cdot, \psi]_{KL(A, J_B)}$) is invariant under unitary equivalence: a priori, it is only invariant under conjugation

by unitaries in $J_B^\dagger$ (see Proposition 5.5). Given $\phi_1: A \rightarrow B_\infty$ lifting θ and a unitary $u \in B_\infty$, set $\phi_2 := \text{Ad}(u) \circ \phi_1$. By [34, Proposition 2.1], the unitary group of $B^\infty \cap q_B(\phi_1(A))'$ is path-connected, and hence Lemma 5.18(i) implies that there is a continuous path of unitaries $(v_t)_{t \geq 0} \subseteq J_B^\dagger$ such that $\text{Ad}(v_t) \circ \phi_1 \rightarrow \phi_2$ point-norm (taking the continuous path $(u_t)_{t \geq 0}$ of the lemma to be the constant path u). Then a standard application of Kirchberg's ϵ -test (Lemma 3.14) provides a unitary $v \in J_B^\dagger$ with $\text{Ad}(v) \circ \phi_1 = \phi_2$. This implies $[\phi_1, \psi]_{KK(A, J_B)} = [\phi_2, \psi]_{KK(A, J_B)}$, as claimed.

Pullback arguments relating extensions to the standard multiplier-corona extension are repeatedly used in the proof of Theorem 7.1. We set out our notation for this below, which will be used freely throughout the rest of the section.

Notation 7.6. Given an extension $\mathbf{e}: 0 \rightarrow I \xrightarrow{j_e} E \xrightarrow{q_e} D \rightarrow 0$, we let $\mu_e: E \rightarrow \mathcal{M}(I)$ be the canonical map of E into the multiplier algebra of I , and write $\bar{\mu}_e: D \rightarrow \mathcal{Q}(I)$ for the Busby map of $\mathbf{e}$. Recall that these maps fit into the following commuting diagram, in which the right-hand square is a pullback:

$$(7.2) \quad \begin{array}{ccccccc} \mathbf{e}: 0 & \longrightarrow & I & \xrightarrow{j_e} & E & \xrightarrow{q_e} & D \longrightarrow 0 \\ & & \parallel & & \downarrow \mu_e & & \downarrow \bar{\mu}_e \\ \bar{\mathbf{e}}: 0 & \longrightarrow & I & \xrightarrow{\bar{j}_e} & \mathcal{M}(I) & \xrightarrow{\bar{q}_e} & \mathcal{Q}(I) \longrightarrow 0. \end{array}$$

That is, $*$ -homomorphisms $\theta: A \rightarrow D$ and $\bar{\phi}: A \rightarrow \mathcal{M}(I)$ with $\bar{q}_e \circ \bar{\phi} = \bar{\mu}_e \circ \theta$ induce a unique $*$ -homomorphism $\phi: A \rightarrow E$ with $\mu_e \circ \phi = \bar{\phi}$ and $q_e \circ \phi = \theta$. More concretely, $E \cong \{(x, d) \in \mathcal{M}(I) \oplus D : \bar{q}_e(x) = \bar{\mu}_e(d)\}$ via the isomorphism $e \mapsto (\mu_e(e), q_e(e))$.

Lemma 7.7. *Let A be a separable nuclear C^* -algebra and let*

$$(7.3) \quad \mathbf{e}: 0 \longrightarrow I \longrightarrow E \longrightarrow D \longrightarrow 0$$

be a unital separable extension such that I is stable and $\mathcal{Z}$ -stable. If $\theta: A \rightarrow D$ is a unitizably full $$ -homomorphism, then $\bar{\mu}_e \circ \theta: A \rightarrow \mathcal{Q}(I)$ is an absorbing $*$ -homomorphism, and if $\phi: A \rightarrow E$ is a $*$ -homomorphism that lifts θ , then $\mu_e \circ \phi: A \rightarrow \mathcal{M}(I)$ is an absorbing $*$ -homomorphism.*

Proof. Since $\bar{\mu}_e \circ \theta$ is the composition of a unitizably full $*$ -homomorphism followed by a unital $*$ -homomorphism, it is unitizably full, so absorbing by Theorem 5.15. The second part follows from the first and Corollary 5.11. $\square$

7.2. Separabilization. Theorem 7.1 will be proved using the KK -existence and uniqueness results of Section 5. As such, it is vital to separabilize — pass to a separable subextension — so that one can obtain the absorption needed to apply these results. We do this in Lemma 7.8. The idea is to use the separable stability of J_B and separable $\mathcal{Z}$ -stability of B_∞ to factorize the data in the diagram (7.1) through a unital separable subextension. This is illustrated schematically (the actual factorizations are slightly different) below, where E is $\mathcal{Z}$ -stable and I is stable:

$$\begin{array}{ccccccc}
& & & \lambda' = [\phi|_E, \psi|_E]_{KL} & & & \\
& & & \swarrow & \searrow & & \\
(7.4) \quad 0 & \longrightarrow & I & \longrightarrow & E & \xleftarrow{\phi|_E, \psi|_E} & D \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \downarrow \\
& & J_B & \longrightarrow & B_\infty & \longrightarrow & B^\infty \longrightarrow 0.
\end{array}$$

$\begin{array}{c} A \\ \downarrow \theta|_D \end{array}$

The proof of [268, Proposition 1.9], used below, employs separably inheritable properties (in the sense of Blackadar, [11, II.8.5]), and the fact that they are closed under countable intersections.

Lemma 7.8. *Let A , B , and θ be as in Theorem 7.1. There is a unital separable subextension*

$$(7.5) \quad \mathbf{e}: 0 \longrightarrow I \xrightarrow{j_e} E \xrightarrow{q_e} D \longrightarrow 0$$

of the trace-kernel extension $\mathbf{e}_B$ such that θ corestricts to a unitizably full map $\theta|_D: A \rightarrow D$, E is unital and $\mathcal{Z}$ -stable, and I is stable. Furthermore:

- (i) Given $\kappa \in KK(A, B_\infty)$ such that $KK(A, q_B)(\kappa) = [\theta]_{KK(A, B^\infty)}$, $\mathbf{e}$ can be chosen so that κ factorizes as $KK(A, \iota_{E \subseteq B_\infty})(\kappa')$ with $\kappa' \in KK(A, E)$ and $KK(A, q_e)(\kappa') = [\theta|_D]_{KK(A, D)}$.
- (ii) Given $\psi: A \rightarrow B_\infty$ and $\kappa \in KK(A, J_B)$, $\mathbf{e}$ can be chosen so that both $\psi(A) \subseteq E$ and κ factorizes as $KK(A, \iota_{I \subseteq J_B})(\kappa')$ for some $\kappa' \in KK(A, I)$.
- (iii) Given two maps $\phi_1, \phi_2: A \rightarrow B_\infty$, $\mathbf{e}$ can be chosen so that $\phi_1(A) \cup \phi_2(A) \subseteq E$. Moreover, if each of ϕ_1 and ϕ_2 lift θ and $[\phi_1, \phi_2]_{KL(A, J_B)} = 0$ and $\phi_1|_E$ and $\phi_2|_E$ denote the corestrictions of ϕ_1 and ϕ_2 to E , then $\mathbf{e}$ can be chosen so that $[\phi_1|_E, \phi_2|_E]_{KL(A, I)} = 0$.

Proof. Note that if $\mathbf{e}$ is a unital separable subextension of $\mathbf{e}_B$ satisfying (i), (ii), or (iii), then the same holds for any larger subextension of $\mathbf{e}_B$. Similarly, if $\mathbf{e}$ is such that $\theta|_D$ is unitizably full, then the same property is satisfied for any larger subextension of $\mathbf{e}_B$ since $1_{B^\infty} \in D$. We start with any separable subextension $\mathbf{e}_0: 0 \rightarrow I_0 \rightarrow E_0 \rightarrow D_0 \rightarrow 0$ such that E_0 contains 1_{B_∞} and then enlarge the algebras while keeping them separable as follows.

Applying [268, Proposition 1.9] to the forced unitization $\theta^\dagger: A^\dagger \rightarrow B^\infty$, D_0 may be enlarged to arrange that θ corestricts to a unitizably full map $A \rightarrow D_0$. We make the corresponding enlargement of E_0 so that it contains lifts of a countable dense subset of D_0 , and enlarge I_0 to $J_B \cap E_0$.

For conditions (i), (ii), and (iii), we use the characterizations of $KK(A, C)$ and $KL(A, C)$ as inductive limits of $KK(A, C_0)$ and $KL(A, C_0)$ over separable subalgebras C_0 of C (this is Definition 5.4 for KK and Proposition 5.7 for KL). In particular, given κ as in (i), considering the identity $KK(A, q_B)(\kappa) = [\theta]_{KK(A, B^\infty)}$, we can find suitable separable enlargements of E_0 and D_0 and $\kappa' \in KK(A, E_0)$ so that $q_B(E_0) = D_0$, $\kappa = KK(A, \iota_{E_0 \subseteq B_\infty})(\kappa')$, and $KK(A, q_B|_{E_0}^{D_0})(\kappa') = [\theta|_{D_0}]_{KK(A, D_0)}$.

Given ψ and κ as in (ii), I_0 may be enlarged so that $\psi(A) \subseteq E_0$ and κ factorizes through I_0 ; this forces corresponding enlargements to E_0 and D_0 . Likewise, given ϕ_1 and ϕ_2 as in (iii), we can enlarge E_0 (and hence also I_0 and D_0) so that $\phi_1(A) \cup \phi_2(A) \subseteq E_0$. As $[\phi_1, \phi_2]_{KL(A, J_B)} = 0$, by the characterization of KL as an inductive

limit, a suitable separable enlargement of I_0 and E_0 satisfies $[\phi_1|^{E_0}, \phi_2|^{E_0}]_{KL(A, I_0)} = 0$. We enlarge D_0 to the image $q_B(E_0)$.

Finally, the $\mathcal{Z}$ -stability hypothesis on B ensures that B_∞ is separably $\mathcal{Z}$ -stable by Proposition 4.17(i), and, together with nuclearity, this hypothesis also gives that J_B is separably stable by Lemma 6.11. As both stability and $\mathcal{Z}$ -stability are preserved under sequential inductive limits of separable C^* -algebras with injective connecting maps ([148, Corollary 4.1] and [292, Corollary 3.4]), we can enlarge $0 \rightarrow I_0 \rightarrow E_0 \rightarrow D_0 \rightarrow 0$ to $\mathfrak{e}$ as in (7.5) in such a way that I is stable and E is $\mathcal{Z}$ -stable by [268, Proposition 1.6]. $\square$

7.3. Proof of Theorem 7.1. We start the proof of Theorem 7.1 with the following strengthening of part (ii) of that theorem, which modifies lifts to realize classes in $KK(A, J_B)$ (rather than just classes in $KL(A, J_B)$); we will also use this to correct the KK -class of a lift in the proof of part (i).

Proposition 7.9. *In the notation of Theorem 7.1, if $\psi: A \rightarrow B_\infty$ is a lift of θ and $\kappa \in KK(A, J_B)$, then there exists a $*$ -homomorphism $\phi: A \rightarrow B_\infty$ that also lifts θ and such that $[\phi, \psi]_{KK(A, J_B)} = \kappa$.*

Proof. Use Lemma 7.8(ii) to find a unital separable subextension $\mathfrak{e}$ of the trace-kernel extension such that E is $\mathcal{Z}$ -stable, I is stable, $\psi(A) \subseteq E$, $\theta|^D$ is unitizably full, and $\kappa = KK(A, \iota_{I \subseteq J_B})(\kappa')$ for some $\kappa' \in KK(A, I)$. By Lemma 7.7, $\mu_{\mathfrak{e}} \circ \psi$ is absorbing. Therefore, by KK -existence (Theorem 5.16(i)), κ' can be realized as $[\bar{\phi}, \mu_{\mathfrak{e}} \circ \psi]_{KK(A, I)}$ for some absorbing $*$ -homomorphism $\bar{\phi}: A \rightarrow \mathcal{M}(I)$ so that $(\bar{\phi}, \mu_{\mathfrak{e}} \circ \psi)$ forms a Cuntz pair. In particular, $\bar{\phi}$ lifts $\bar{\mu}_{\mathfrak{e}} \circ \theta|^D$, and so the pullback square in (7.2) induces a $*$ -homomorphism $\phi: A \rightarrow E$ lifting $\theta|^D$ and satisfying $\bar{\phi} = \mu_{\mathfrak{e}} \circ \phi$. Then, by definition (see also (5.6)),

$$\begin{aligned} [\phi, \psi]_{KK(A, I)} &= [\mu_{\mathfrak{e}} \circ \phi, \mu_{\mathfrak{e}} \circ \psi]_{KK(A, I)} \\ (7.6) \quad &= [\bar{\phi}, \mu_{\mathfrak{e}} \circ \psi]_{KK(A, I)} \\ &= \kappa'. \end{aligned}$$

Therefore, viewing ϕ as a map into B_∞ , $[\phi, \psi]_{KK(A, J_B)} = \kappa$. $\square$

Proof of Theorem 7.1. (i): Use Lemma 7.8(i) to produce a unital separable subextension $\mathfrak{e}$ of the trace-kernel extension $\mathfrak{e}_B$ such that I is stable, E is $\mathcal{Z}$ -stable, $\theta(A) \subseteq D$, $\theta|^D$ is unitizably full, and so that κ factorizes as $\kappa = KK(A, \iota_{E \subseteq B_\infty})(\kappa')$ for some $\kappa' \in KK(A, E)$ with $KK(A, q_{\mathfrak{e}})(\kappa') = [\theta|^D]_{KK(A, D)}$. Since κ' is a KK -lift of $\theta|^D$ and the pullback square in (7.2) commutes, it follows that $[\bar{\mu}_{\mathfrak{e}} \circ \theta|^D]_{KK(A, \mathcal{Q}(I))}$ factorizes through $KK(A, \mathcal{M}(I))$. As I is stable, $KK(A, \mathcal{M}(I)) = 0$ by [62, Proposition 4.1], so $[\bar{\mu}_{\mathfrak{e}} \circ \theta|^D]_{KK(A, \mathcal{Q}(I))} = 0$. By the KK -existence theorem (Theorem 5.16(ii)), $\bar{\mu}_{\mathfrak{e}} \circ \theta|^D: A \rightarrow \mathcal{Q}(I)$ lifts to a $*$ -homomorphism $\bar{\psi}: A \rightarrow \mathcal{M}(I)$. By the pullback (7.2), $(\bar{\psi}, \theta|^D)$ induces a $*$ -homomorphism $\psi: A \rightarrow E$. Regarding ψ as taking values in B_∞ , it follows that ψ lifts θ .

There is no reason to expect that ψ provides the KK -class we need. This must be corrected. By construction, in $KK(A, B^\infty)$, we have

$$\begin{aligned} KK(A, q_B)(\kappa) &= [\theta]_{KK(A, B^\infty)} \\ (7.7) \quad &= [q_B \circ \psi]_{KK(A, B^\infty)} \\ &= KK(A, q_B)([\psi]_{KK(A, B_\infty)}), \end{aligned}$$

(the last equality is recorded in Proposition 5.5(iv)). Accordingly, $\kappa - [\psi]_{KK(A, B_\infty)}$ lies in the kernel of $KK(A, q_B)$. By half-exactness of $KK(A, \cdot)$ when A is nuclear (see Proposition 5.5(v)), there exists $\kappa' \in KK(A, J_B)$ with

$$(7.8) \quad KK(A, j_B)(\kappa') = \kappa - [\psi]_{KK(A, B_\infty)}.$$

By Proposition 7.9, there exists $\phi: A \rightarrow B_\infty$ also lifting θ such that $[\phi, \psi]_{KK(A, J_B)} = \kappa'$. Using Proposition 5.5(iv), we have

$$(7.9) \quad \begin{aligned} [\phi]_{KK(A, B_\infty)} - [\psi]_{KK(A, B_\infty)} &= KK(A, j_B)(\kappa') \\ &\stackrel{(7.8)}{=} \kappa - [\psi]_{KK(A, B_\infty)}, \end{aligned}$$

and so $[\phi]_{KK(A, B_\infty)} = \kappa$.

(ii): This is an immediate consequence of Proposition 7.9, since $KL(A, J_B)$ is a quotient of $KK(A, J_B)$.

(iii): Fix a subextension $\mathbf{e}$ as in Lemma 7.8(iii). Then both the maps

$$(7.10) \quad \mu_{\mathbf{e}} \circ \phi_1|_E, \mu_{\mathbf{e}} \circ \phi_2|_E: A \rightarrow \mathcal{M}(I)$$

are absorbing by Lemma 7.7. The extension $\mathbf{e}$ is constructed so that the Cuntz pair $(\mu_{\mathbf{e}} \circ \phi_1|_E, \mu_{\mathbf{e}} \circ \phi_2|_E)$ represents the trivial class in $KL(A, I)$. Therefore, the $\mathcal{Z}$ -stable KL -uniqueness theorem (Theorem 5.17(ii)) gives a sequence $(u_n)_{n=1}^\infty$ of unitaries in $(I \otimes \mathcal{Z})^\dagger \subseteq E \otimes \mathcal{Z}$ such that

$$(7.11) \quad \|\text{Ad}(u_n)((\mu_{\mathbf{e}} \otimes \text{id}_{\mathcal{Z}})(\phi_1(a) \otimes 1_{\mathcal{Z}})) - (\mu_{\mathbf{e}} \otimes \text{id}_{\mathcal{Z}})(\phi_2(a) \otimes 1_{\mathcal{Z}})\| \rightarrow 0$$

for all $a \in A$. Since $\text{Ad}(u_n)$ commutes with $\mu_{\mathbf{e}} \otimes \text{id}_{\mathcal{Z}}$, and $(\mu_{\mathbf{e}} \otimes \text{id}_{\mathcal{Z}})|_{I \otimes \mathcal{Z}}$ is injective, it follows that $\text{Ad}(u_n)(\phi_1(a) \otimes 1_{\mathcal{Z}}) \rightarrow \phi_2(a) \otimes 1_{\mathcal{Z}}$.

We now have that $\phi_1|_E \otimes 1_{\mathcal{Z}}$ and $\phi_2|_E \otimes 1_{\mathcal{Z}}$ are approximately unitarily equivalent. Since E is $\mathcal{Z}$ -stable, Proposition 4.3 implies ϕ_1 and ϕ_2 are approximately unitarily equivalent. Therefore, ϕ_1 and ϕ_2 are unitarily equivalent by Lemma 3.15. $\square$

7.4. Classification of unital lifts. We end this section by proving Theorem 1.2, a unital version of Theorem 7.1 which is needed in the proof of the classification of full approximate embeddings (Theorem 1.1). The map $\Gamma_0^{(A, B_\infty)}: KK(A, B_\infty) \rightarrow \text{Hom}(K_0(A), K_0(B_\infty))$ below was defined in (5.17).

Theorem 7.10 (Classification of unital lifts). *Let A be a unital separable nuclear C^* -algebra, let B be a unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebra with $T(B) \neq \emptyset$, and suppose $\theta: A \rightarrow B^\infty$ is a unital *-homomorphism such that $\tau \circ \theta$ is a faithful trace on A for all $\tau \in T(B^\infty)$.*

- (i) *For any $\kappa \in KK(A, B_\infty)$ satisfying $KK(A, q_B)(\kappa) = [\theta]_{KK(A, B_\infty)}$ and $\Gamma_0^{(A, B_\infty)}(\kappa)([1_A]_0) = [1_{B_\infty}]_0$, there exists a unital *-homomorphism $\phi: A \rightarrow B_\infty$ that lifts θ and satisfies $[\phi]_{KK(A, B_\infty)} = \kappa$.*
- (ii) *Given any unital *-homomorphism $\psi: A \rightarrow B_\infty$ that lifts θ and any $\lambda \in \ker KL(A, j_B)$, there exists a unital *-homomorphism $\phi: A \rightarrow B_\infty$ that also lifts θ and satisfies $[\phi, \psi]_{KL(A, J_B)} = \lambda$.*
- (iii) *Any two unital *-homomorphisms ϕ_1 and ϕ_2 that both lift θ and satisfy $[\phi_1, \phi_2]_{KL(A, J_B)} = 0$ are unitarily equivalent.*

A unital *-homomorphism can never be unitizably full, so Theorem 7.1 cannot be applied directly to such maps. Instead, we use a standard “de-unitization” trick (seen in [267] and [268], for example), working in a 2×2 matrix amplification to

provide additional room for maps to become unitizably full. Recall that $\iota_D^{(2)}: D \rightarrow M_2(D)$ denotes the top left-hand corner embedding (see Remark 6.4).

Proposition 7.11. *Let A and D be unital C^* -algebras. If $\theta: A \rightarrow D$ is a unital and full $*$ -homomorphism, then $\iota_D^{(2)} \circ \theta$ is unitizably full.*

Proof. An element $x \in A^\dagger$ can be written as $x = a + \lambda(1_{A^\dagger} - 1_A)$ for some $a \in A$ and $\lambda \in \mathbb{C}$. Thus

$$(7.12) \quad (\iota_D^{(2)} \circ \theta)^\dagger(x) = \begin{pmatrix} \theta(a) & 0 \\ 0 & \lambda 1_D \end{pmatrix}.$$

If $a \neq 0$ or $\lambda \neq 0$, then this generates $M_2(D)$ as an ideal since θ is full. $\square$

Given a unital map $\theta: A \rightarrow B^\infty$, we will apply Theorem 7.1 to $\iota_{B^\infty}^{(2)} \circ \theta$. Recall that $M_2(B)$ is also unital, simple, separable, nuclear, and $\mathcal{Z}$ -stable, so we can apply our earlier results with $M_2(B)$ in place of B (see Remark 6.4).

Proof of Theorem 7.10. (i): The hypothesis that $\tau \circ \theta$ is faithful for every $\tau \in T(B^\infty)$ ensures that θ is full by Lemma 4.10. Accordingly, $\iota_{B^\infty}^{(2)} \circ \theta: A \rightarrow M_2(B^\infty)$ is unitizably full by Proposition 7.11 (which uses strict comparison of B^∞ coming from Proposition 6.3(ii)). Applying commutativity of the top-right square of (6.4) to $KK(A, q_B)(\kappa) = [\theta]_{KK(A, B^\infty)}$ gives

$$(7.13) \quad KK(A, q_{M_2(B)}) (KK(A, \iota_{B^\infty}^{(2)})(\kappa)) = [\iota_{B^\infty}^{(2)} \circ \theta]_{KK(A, M_2(B)^\infty)};$$

i.e., $KK(A, \iota_{B^\infty}^{(2)})(\kappa)$ is a KK -lift of $\iota_{B^\infty}^{(2)} \circ \theta$. Therefore, Theorem 7.1(i), applied to $M_2(B)$ in place of B , gives a map $\phi^{(2)}: A \rightarrow M_2(B_\infty) \cong M_2(B)_\infty$ (see Remark 6.4) lifting $\iota_{B^\infty}^{(2)} \circ \theta$ with

$$(7.14) \quad [\phi^{(2)}]_{KK(A, M_2(B_\infty))} = KK(A, \iota_{B^\infty}^{(2)})(\kappa).$$

As $\Gamma_0^{(A, B^\infty)}(\kappa)([1_A]_0) = [1_{B^\infty}]_0$, it follows from naturality of $\Gamma_0^{(A, \cdot)}$ and (5.18) that $[\phi^{(2)}(1_A)]_0 = [\iota_{B^\infty}^{(2)}(1_{B^\infty})]_0$. Applying Proposition 4.17(i) to $M_2(B)$ in place of B , we see that $M_2(B_\infty) \cong M_2(B)_\infty$ has stable rank one, which implies it has cancellation and so projections which agree in K_0 are unitarily equivalent ([10, Proposition 6.5.1], for example). Hence $\phi^{(2)}(1_A)$ is unitarily equivalent to $\iota_{B^\infty}^{(2)}(1_{B^\infty})$ and so $\phi^{(2)}$ is unitarily equivalent to a map of the form $\iota_{B^\infty}^{(2)} \circ \tilde{\phi}$ for a unital $*$ -homomorphism $\tilde{\phi}: A \rightarrow B_\infty$.

Fix a trace $\tau \in T(B^\infty)$. Recall that τ_2 is the canonical non-normalized tracial functional on $M_2(B^\infty)$ extending τ (so, $\tau_2(1_{M_2(B^\infty)}) = 2$). Then

$$(7.15) \quad \begin{aligned} \tau \circ q_B \circ \tilde{\phi} &= \tau_2 \circ \iota_{B^\infty}^{(2)} \circ q_B \circ \tilde{\phi} \\ &\stackrel{\text{Rem. 6.4}}{=} \tau_2 \circ q_{M_2(B)} \circ \iota_{B^\infty}^{(2)} \circ \tilde{\phi} \\ &= \tau_2 \circ q_{M_2(B)} \circ \phi^{(2)} \\ &= \tau_2 \circ \iota_{B^\infty}^{(2)} \circ \theta \\ &= \tau \circ \theta, \end{aligned}$$

where the third equality follows from unitary invariance of the trace. As this holds for all $\tau \in T(B^\infty)$, we have $\text{Aff } T(q_B \circ \tilde{\phi}) = \text{Aff } T(\theta)$. Therefore, by the classification of maps into B^∞ (Theorem 6.5), $q_B \circ \tilde{\phi}$ is unitarily equivalent to θ . The von Neumann-like behavior of B^∞ ensures that unitaries in B^∞ lift to unitaries in B_∞

(Proposition 6.6), so we can find a unitary $u \in B_\infty$ such that $\text{Ad}(q_B(u)) \circ q_B \circ \tilde{\phi} = \theta$. Define $\phi := \text{Ad}(u) \circ \tilde{\phi}$. Since $\phi^{(2)}$ is unitarily equivalent to $\iota_{B_\infty}^{(2)} \circ \phi$, (7.14) gives

$$(7.16) \quad KK(A, \iota_{B_\infty}^{(2)})([\phi]_{KK(A, B_\infty)}) = KK(A, \iota_{B_\infty}^{(2)})(\kappa).$$

But $KK(A, \iota_{B_\infty}^{(2)})$ is an isomorphism (Proposition 5.5(i)), and so $[\phi]_{KK(A, B_\infty)} = \kappa$.

(ii): Using the commutativity of the top-right square of (6.4) and making the identification $M_2(B_\infty) \cong M_2(B)^\infty$ (see Remark 6.4) it follows that $\iota_{B_\infty}^{(2)} \circ \psi$ lifts $\iota_{B_\infty}^{(2)} \circ \theta$, and the latter map is unitizably full, as noted in the proof of part (i).

Applying Theorem 7.1(ii) to $M_2(B)$, $\iota_{B_\infty}^{(2)} \circ \psi$, $\iota_{B_\infty}^{(2)} \circ \theta$, and $KL(A, \iota_{J_B}^{(2)})(\lambda)$, there is a *-homomorphism $\phi^{(2)}: A \rightarrow M_2(B)_\infty \cong M_2(B)_\infty$ lifting $\iota_{B_\infty}^{(2)} \circ \theta$ and with $[\phi^{(2)}, \iota_{B_\infty}^{(2)} \circ \psi]_{KL(A, M_2(J_B))} = KL(A, \iota_{J_B}^{(2)})(\lambda)$. Then using Proposition 5.5(iv) in the first step, and commutativity of the top-left square of (6.4) at the third step, we have

$$(7.17) \quad \begin{aligned} & [\phi^{(2)}]_{KL(A, M_2(B_\infty))} - [\iota_{B_\infty}^{(2)} \circ \psi]_{KL(A, M_2(B_\infty))} \\ &= KL(A, j_{M_2(B)})([\phi^{(2)}, \iota_{B_\infty}^{(2)} \circ \psi]_{KL(A, M_2(J_B))}) \\ &= KL(A, j_{M_2(B)} \circ \iota_{J_B}^{(2)})(\lambda) \\ &= KL(A, \iota_{B_\infty}^{(2)}) \circ KL(A, j_B)(\lambda) \\ &= 0. \end{aligned}$$

Therefore, the induced map on K_0 vanishes, so $[\phi^{(2)}(1_A)]_0 = [(\iota_{B_\infty}^{(2)} \circ \psi)(1_A)]_0 = [\iota_{B_\infty}^{(2)}(1_{B_\infty})]_0$.

Using that $M_2(B_\infty) \cong M_2(B)_\infty$ has stable rank one (by Proposition 4.17(i)), there exists a unitary $u \in M_2(B_\infty)$ such that $\text{Ad}(u) \circ \phi^{(2)}(1_A) = \iota_{B_\infty}^{(2)}(1_{B_\infty})$. As $\phi^{(2)}$ lifts $\iota_{B_\infty}^{(2)} \circ \theta$, we have

$$(7.18) \quad (\text{Ad}(q_{M_2(B)}(u)) \circ \iota_{B_\infty}^{(2)})(1_{B_\infty}) = \iota_{B_\infty}^{(2)}(1_{B_\infty}),$$

so we can write

$$(7.19) \quad q_{M_2(B)}(u) = \bar{v} \oplus \bar{w} := \begin{pmatrix} \bar{v} & 0 \\ 0 & \bar{w} \end{pmatrix}$$

for some unitaries $\bar{v}, \bar{w} \in B^\infty$. These lift to unitaries $v, w \in B_\infty$ respectively by Proposition 6.6. Since $(\text{Ad}((v \oplus w)^*u) \circ \phi^{(2)})(1_A) = \iota_{B_\infty}^{(2)}(1_{B_\infty})$, we can write $\text{Ad}((v \oplus w)^*u) \circ \phi^{(2)} = \iota_{B_\infty}^{(2)} \circ \phi$ for some unital *-homomorphism $\phi: A \rightarrow B_\infty$. By construction, $q_{M_2(B)}((v \oplus w)^*u) = 1_{M_2(B_\infty)}$, so ϕ lifts θ . As $(v \oplus w)^*u \in J_{M_2(B)}^\dagger$, we have

$$(7.20) \quad [\iota_{B_\infty}^{(2)} \circ \phi, \iota_{B_\infty}^{(2)} \circ \psi]_{KL(A, M_2(J_B))} = [\phi^{(2)}, \iota_{B_\infty}^{(2)} \circ \psi]_{KL(A, M_2(J_B))}$$

(by Proposition 5.5(ii)). Therefore,

$$(7.21) \quad \begin{aligned} KL(A, \iota_{J_B}^{(2)})([\phi, \psi]_{KL(A, J_B)}) &= [\iota_{B_\infty}^{(2)} \circ \phi, \iota_{B_\infty}^{(2)} \circ \psi]_{KL(A, M_2(J_B))} \\ &= KL(A, \iota_{J_B}^{(2)})(\lambda). \end{aligned}$$

Since $KL(A, \iota_{J_B}^{(2)})$ is an isomorphism (Proposition 5.8(i)), $[\phi, \psi]_{KL(A, J_B)} = \lambda$.

(iii): Given two unital lifts $\phi_1, \phi_2: A \rightarrow B_\infty$ of θ such that $[\phi_1, \phi_2]_{KL(A, J_B)} = 0$, Proposition 5.8(iv) yields

$$(7.22) \quad [\iota_{B_\infty}^{(2)} \circ \phi_1, \iota_{B_\infty}^{(2)} \circ \phi_2]_{KL(A, M_2(J_B))} = KL(A, \iota_{J_B}^{(2)})([\phi_1, \phi_2]_{KL(A, J_B)}) = 0.$$

Just as at the beginning of (i), $\iota_{B_\infty}^{(2)} \circ \theta$ is unitizably full, so we can apply Theorem 7.1(iii) with $M_2(B)$ in place of B , $\iota_{B_\infty}^{(2)} \circ \theta$ in place of θ , $\iota_{B_\infty}^{(2)} \circ \phi_1$ in place of ϕ_1 , $\iota_{B_\infty}^{(2)} \circ \phi_2$ in place of ϕ_2 , and $\lambda = 0$ to learn that $\iota_{B_\infty}^{(2)} \circ \phi_1$ and $\iota_{B_\infty}^{(2)} \circ \phi_2$ are unitarily equivalent. Since ϕ_1 and ϕ_2 are unital, a unitary $u \in M_2(B_\infty)$ with $\text{Ad}(u) \circ \iota_{B_\infty}^{(2)} \circ \phi_1 = \iota_{B_\infty}^{(2)} \circ \phi_2$ must be of the form $u = \begin{pmatrix} v & 0 \\ 0 & w \end{pmatrix}$ for unitaries $v, w \in B_\infty$. Hence $\text{Ad}(v) \circ \phi_1 = \phi_2$. $\square$

We end the section by establishing Theorem 1.2(ii). Recall that $\tilde{\Gamma}_0^{(A, J_B)}$ is the K_0 -component of the map from (5.21).

Theorem 7.12. *Let A be a unital separable nuclear C^* -algebra and B be a unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebra with $T(B) \neq \emptyset$. Suppose that $\psi: A \rightarrow B_\infty$ is a unital full embedding and set $\theta := q_B \circ \psi: A \rightarrow B^\infty$. Then $\phi \mapsto [\phi, \psi]_{KL(A, J_B)}$ induces a bijection between unitary equivalence classes of unital full lifts ϕ of θ and elements λ of $KL(A, J_B)$ satisfying $\tilde{\Gamma}_0^{(A, J_B)}(\lambda)([1_A]_0) = 0$.*

Proof. Certainly the map $\phi \mapsto [\phi, \psi]_{KL(A, J_B)}$ is well defined on unital lifts ϕ of θ , and invariant under unitary conjugation. Moreover, for any such lift ϕ , we have $\tilde{\Gamma}_0^{(A, J_B)}([\phi, \psi]_{KL(A, J_B)})([1_A]_0) = [\phi(1_A)]_0 - [\psi(1_A)]_0 = 0$ by Proposition 5.8 and (5.18).

Theorem 7.10(iii) shows that the induced map is injective. To show it is surjective, consider $\lambda \in KL(A, J_B)$ satisfying $\tilde{\Gamma}_0^{(A, J_B)}(\lambda)([1_A]_0) = 0$. Lift λ to an element $\kappa \in KK(A, J_B)$. Then consider $[\psi]_{KK(A, B_\infty)} + KK(A, j_B)(\kappa) \in KK(A, B_\infty)$. This element satisfies

$$(7.23) \quad KK(A, q_B) \left([\psi]_{KK(A, B_\infty)} + KK(A, j_B)(\kappa) \right) = [\theta]_{KK(A, B^\infty)}$$

by Proposition 5.5(iv), and

$$(7.24) \quad \Gamma_0^{(A, B_\infty)} \left([\psi]_{KK(A, B_\infty)} + KK(A, j_B)(\kappa) \right) ([1_A]_0) = [1_{B_\infty}]_0,$$

by naturality of $\Gamma_0^{(A, -)}$.

By Theorem 7.10(i), there exists a unital full lift ϕ_0 of θ such that

$$(7.25) \quad [\phi_0]_{KK(A, B_\infty)} = [\psi]_{KK(A, B_\infty)} + KK(A, j_B)(\kappa).$$

Descending to KL and using Proposition 5.8(iii), we have

$$(7.26) \quad \lambda - [\phi_0, \psi]_{KL(A, J_B)} \in \ker KL(A, j_B).$$

Therefore by Theorem 7.10(ii) (with ϕ_0 in place of ψ), there exists a unital lift ϕ of θ satisfying

$$(7.27) \quad [\phi, \phi_0]_{KL(A, J_B)} = \lambda - [\phi_0, \psi]_{KL(A, J_B)}.$$

Therefore,

$$(7.28) \quad [\phi, \psi]_{KL(A, J_B)} = [\phi, \phi_0]_{KL(A, J_B)} + [\phi_0, \psi]_{KL(A, J_B)} = \lambda,$$

as required. $\square$

8. THE UNIVERSAL (MULTI)COEFFICIENT THEOREM AND COMPUTING $KL(A, J_B)$

Our main goal in this section is to prove Theorem 1.6, computing $KL(A, J_B)$ in terms of the invariant $\underline{KT}_u$. This is the point in the argument at which the universal (multi)coefficient theorem becomes crucial. We review the UCT and UMCT in Sections 8.1 and 8.2 and turn to the calculation of $KL(A, J_B)$ in Section 8.3.

8.1. The universal coefficient theorem. Inspired by Brown's earlier UCT for Brown–Douglas–Fillmore theory ([27]), Rosenberg and Schochet's UCT ([260]) establishes a class of C^* -algebras A for which $KK(A, I)$ can be computed using K -theory.

Definition 8.1. A separable C^* -algebra A is said to *satisfy the universal coefficient theorem (UCT)*, or *belong to the UCT class* if, for every σ -unital C^* -algebra I , the map $\Gamma^{(A, I)}$ from (5.17) is surjective and a natural map

$$(8.1) \quad \ker \Gamma^{(A, I)} \rightarrow \text{Ext}(K_*(A), K_{1-*}(I)).$$

is bijective (see [10, Section 23.1] for a definition of this map).

Accordingly, when A satisfies the UCT, there is a short exact sequence

$$(8.2) \quad 0 \rightarrow \text{Ext}(K_*(A), K_{1-*}(I)) \rightarrow KK(A, I) \xrightarrow{\Gamma^{(A, I)}} \text{Hom}(K_*(A), K_*(I)) \rightarrow 0$$

where the first map is the inverse of the map in (8.1). It is through this sequence that $KK(A, I)$ can be calculated using K -theory.

As discussed in the introduction, the fundamental question is which C^* -algebras satisfy the UCT. Shortly after [260], Skandalis gave examples of C^* -algebras outside the class ([274, Théorème 4.1 and Corollaire 4.2]), but it remains a major open problem whether all separable nuclear C^* -algebras satisfy the UCT. Building on Higson and Kasparov's deep work on the Baum–Connes conjecture ([144]), Tu established the UCT for C^* -algebras associated to second countable locally compact Hausdorff amenable groupoids ([295, Proposition 10.7 and Lemme 3.5]). This was used to show that nuclear C^* -algebras containing a Cartan masa satisfy the UCT in [5]. Many permanence properties for the UCT date back to [260], but the state of the art for crossed products is more recent.

Remark 8.2. The problem of whether the crossed product $A \rtimes G$ satisfies the UCT has a very different status depending on whether G has torsion.

- (i) Let A be a separable C^* -algebra which satisfies the UCT and let G be a countable discrete amenable group acting on A . When G is torsion-free, it follows from [216, Corollary 9.4] that $A \rtimes G$ satisfies the UCT. In fact, the same result implies $A \rtimes G$ satisfies the UCT if $A \rtimes H$ does for every finite subgroup H of G . To see this, note that the notation $\langle \star \rangle$ in [216] denotes the class of separable C^* -algebras which satisfy the UCT (since it is closed under KK -equivalence and contains the bootstrap class described above, and hence all separable abelian C^* -algebras). The dual Dirac morphism hypothesis of [216, Corollary 9.4] follows from [144], and when G is discrete, all compact subgroups of G are finite.
- (ii) For finite groups, the UCT problem (whether every separable nuclear C^* -algebra satisfies the UCT) is equivalent to whether the UCT is preserved by crossed products of nuclear C^* -algebras by finite groups. More precisely, [6, Theorem 4.17] shows that the UCT problem is equivalent to whether

every crossed product of $\mathcal{O}_2$ by $\mathbb{Z}/2$ or $\mathbb{Z}/3$ satisfies the UCT. This is related to results announced by Kirchberg (cf. [10, Exercise 23.15.12]).

We record the following explicit computation of $\Gamma_1^{(A,I)}$ for use in the calculation of $\ker KL(A, j_B)$ in Section 8.3. This is well-known to experts, but isn't easily pinned down in the literature; the approach we take borrows ideas from [140].

Proposition 8.3. *Suppose A is a unital separable C^* -algebra, I is a C^* -algebra, and $(\phi, \psi): A \rightrightarrows E \triangleright I$ is a unital Cuntz pair (i.e., E , ϕ , and ψ are all unital). Then $\Gamma_1^{(A,I)}([\phi, \psi]_{KK(A,I)}): K_1(A) \rightarrow K_1(I)$ is the map*

$$(8.3) \quad [u]_{K_1(A)} \mapsto [\phi(u)\psi(u)^*]_{K_1(I)},$$

for all unitaries $u \in U_n(A)$ and $n \in \mathbb{N}$.

Note that the right side of (8.3) is indeed in $K_1(I)$, since $\phi(u)\psi(u)^* \in U_n(I^\dagger)$ by the hypothesis that ϕ and ψ are unital.

Proof. Consider the pullback diagram

$$(8.4) \quad \begin{array}{ccccccc} 0 & \longrightarrow & I & \xrightarrow{\tilde{j}} & C & \xrightarrow{\tilde{q}} & A \longrightarrow 0 \\ & & \parallel & & \downarrow \pi & & \downarrow q \circ \phi = q \circ \psi \\ 0 & \longrightarrow & I & \xrightarrow{j} & E & \xrightarrow{q} & E/I \longrightarrow 0. \end{array}$$

By the pullback property, there are unital $*$ -homomorphisms $\tilde{\phi}, \tilde{\psi}: A \rightarrow C$ so that

$$(8.5) \quad \pi \circ \tilde{\phi} = \phi, \quad \pi \circ \tilde{\psi} = \psi, \quad \text{and} \quad \tilde{q} \circ \tilde{\phi} = \tilde{q} \circ \tilde{\psi} = \text{id}_A.$$

By Proposition 5.5(iv) (with $J := I$, $F := C$, $\theta := \text{id}_I$, and $\bar{\theta} := \pi$), $[\phi, \psi]_{KK(A,I)} = [\tilde{\phi}, \tilde{\psi}]_{KK(A,I)}$. Using Proposition 5.5(iv) again, we have

$$(8.6) \quad KK(A, \tilde{j})([\phi, \psi]_{KK(A,I)}) = [\tilde{\phi}]_{KK(A,C)} - [\tilde{\psi}]_{KK(A,C)}.$$

Using (8.6) and naturality of $\Gamma_1^{(A, \cdot)}$, we obtain

$$(8.7) \quad \begin{aligned} K_1(\tilde{j}) \circ \Gamma_1^{(A,I)}([\phi, \psi]_{KK(A,I)}) &= \Gamma_1^{(A,C)}([\tilde{\phi}]_{KK(A,C)} - [\tilde{\psi}]_{KK(A,C)}) \\ &\stackrel{(5.18)}{=} K_1(\tilde{\phi}) - K_1(\tilde{\psi}). \end{aligned}$$

For $n \in \mathbb{N}$, and a unitary $u \in U_n(A)$, $\pi(\tilde{\phi}(u)\tilde{\psi}(u)^*) = \phi(u)\psi(u)^*$, and so in $U_n(I^\dagger)$,

$$(8.8) \quad \tilde{\phi}(u)\tilde{\psi}(u)^* = \phi(u)\psi(u)^*.$$

Consequently, we get

$$(8.9) \quad (K_1(\tilde{\phi}) - K_1(\tilde{\psi}))([u]_{K_1(A)}) = [\tilde{\phi}(u)\tilde{\psi}(u)^*]_{K_1(C)} = K_1(\tilde{j})([\phi(u)\psi(u)^*]_{K_1(I)}).$$

As the top row of (8.4) splits by (8.5), it follows that $K_1(\tilde{j})$ is injective. Combining this with (8.7) and (8.9) we get

$$(8.10) \quad \Gamma_1^{(A,I)}([\phi, \psi]_{KK(A,I)})([u]_{K_1(A)}) = [\phi(u)\psi(u)^*]_{K_1(I)}.$$

□

8.2. The universal multicoefficient theorem. We now turn to the universal multicoefficient theorem obtained by Dadarlat and Loring in [69]. From a modern viewpoint, this computes KL as homomorphisms of total K -theory.

At this point we need to change pictures of total K -theory from the definition $K_i(D; \mathbb{Z}/n) := K_{1-i}(D \otimes \mathbb{I}_n)$ we gave in Section 2.3 to the definition $K_0(D; \mathbb{Z}/n) := KK(\mathbb{I}_n, D)$ and $K_i(D; \mathbb{Z}/n) := KK^i(\mathbb{I}_n, D)$ used by Dadarlat and Loring (KK^0 is just the usual KK -group; KK^1 is defined by suspending in one of the variables). As noted in [69], it is not so difficult to provide a natural isomorphism between $K_{1-i}(D \otimes \mathbb{I}_n)$ and $KK^i(\mathbb{I}_n, D)$; we do so for completeness in Proposition A.1 (the only proof we know of in the literature is somewhat indirect, given in [161] as a consequence of Spanier–Whitehead duality). However, as Dadarlat and Loring also observe, it would be awkward to explicitly transfer the Bockstein operations through such an isomorphism, so they construct a family of Bockstein operations directly in their picture using the Kasparov product. As experts are undoubtedly aware, any two such families generate the same ring and so give rise to the same morphisms in total K -theory (see Lemma A.4(i)). We formalize this in the following proposition (proved in Appendix A.2), the upshot of which is that it is legitimate for us to apply the universal multicoefficient theorem with either definition of total K -theory.

Proposition 8.4. *The definition of total K -theory from [69] is naturally isomorphic to the definition in Section 2.3. Precisely, given any natural isomorphism of the functors $KK^i(\mathbb{I}_n, \cdot)$ with $K_i(\cdot; \mathbb{Z}/n)$, the resulting morphisms of total K -theory coincide.*

The maps $\Gamma^{(A,I)}: KK(A, I) \rightarrow \text{Hom}(K_*(A), K_*(I))$ from (5.17) extend to actions of $KK(A, I)$ on total K -theory. For separable A , working in the $KK^i(\mathbb{I}_n, D)$ picture of total K -theory, the Kasparov product provides a map

$$(8.11) \quad \Gamma_\Lambda^{(A,I)}: KK(A, I) \rightarrow \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(I))$$

that is natural in both variables (associativity of the Kasparov product shows that the induced maps do intertwine the Bockstein operations). As with $\Gamma^{(A,I)}$, the map $\Gamma_\Lambda^{(A,I)}$ is defined in the non-separable setting as a limit over separable subalgebras; see (B.34). Moreover, as the Kasparov product descends by continuity to KL , $\Gamma_\Lambda^{(A,I)}$ factors through $KL(A, I)$ (for the non-separable case, this is shown in the proof of Theorem 8.5 in Appendix B).

Assuming that A satisfies the UCT (and assuming I is separable), Dadarlat and Loring’s universal multicoefficient theorem identifies the kernel of the map in (8.11) with the subgroup $\text{PExt}(K_*(A), K_{1-*}(I))$ of $\text{Ext}(K_*(A), K_{1-*}(I))$ consisting of *pure extensions* (i.e., group extensions where every element of finite order in the quotient lifts to an element of the same order). Using his work on topologies on KK -theory, Dadarlat subsequently identified $\text{PExt}(K_*(A), K_{1-*}(I))$ naturally with $Z_{KK(A,I)} \subseteq KK(A, I)$ from (5.19) when A satisfies the UCT and I is σ -unital ([64], following work by Schochet in the nuclear setting, [272]). Combining these results gives the modern viewpoint on the Dadarlat–Loring universal multicoefficient theorem (which is equivalent to the UCT for separable nuclear C^* -algebras; this is [64, Theorem 5.5], noting that Lin’s approximate UCT from [194] is equivalent to the UMCT, as the topology used in [194] is equivalent to that in Definition 5.6; see [64, Section 5]). In Appendix B.4 we note how to extend the universal multicoefficient theorem to general I .

Theorem 8.5 (The universal multicoefficient theorem). *Let A and I be C^* -algebras with A separable. Then the map $\Gamma_\Lambda^{(A,I)}$ in (8.11) induces a map*

$$(8.12) \quad \widetilde{\Gamma}_\Lambda^{(A,I)} : KL(A, I) \rightarrow \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(I)).$$

This map is natural in A and I and is an isomorphism when A satisfies the UCT. For a $$ -homomorphism $\phi : A \rightarrow I$, the map satisfies*

$$(8.13) \quad \widetilde{\Gamma}_\Lambda^{(A,I)}([\phi]_{KL(A,I)}) = \underline{K}(\phi).$$

In the special case when the codomain is the trace-kernel quotient of one of our allowed codomains, the K -theory computations of Section 6.3 combine with the universal (multi)coefficient theorem to show that both $\Gamma_0^{(A,B^\infty)}$ and $\Gamma_\Lambda^{(A,B^\infty)}$ compute $KK(A, B^\infty)$ (and not just $KL(A, B^\infty)$).

Proposition 8.6. *Let A be a separable C^* -algebra satisfying the UCT and let B be a unital simple separable nuclear finite $\mathcal{Z}$ -stable C^* -algebra. Then the canonical maps*

$$(8.14) \quad \begin{aligned} \Gamma_0^{(A,B^\infty)} : KK(A, B^\infty) &\rightarrow \text{Hom}(K_0(A), K_0(B^\infty)), \text{ and} \\ \Gamma_\Lambda^{(A,B^\infty)} : KK(A, B^\infty) &\rightarrow \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(B^\infty)) \end{aligned}$$

are isomorphisms.

Proof. The K -theory of B^∞ was computed in Proposition 6.7: we have $K_0(B^\infty) \cong \text{Aff } T(B^\infty)$ and $K_1(B^\infty) = 0$. In particular, $K_0(B^\infty)$ is divisible. It follows that $\text{Ext}(K_*(A), K_{1-*}(B^\infty)) = 0$ (by [303, Corollary 2.3.2 and Exercise 2.5.1], for example). Therefore, as A satisfies the UCT, $\Gamma_0^{(A,B^\infty)}$ is an isomorphism by the exactness of (8.2).

As $\Gamma_0^{(A,B^\infty)}$ factors through $\Gamma_\Lambda^{(A,B^\infty)}$, the latter is necessarily injective. Combining this with the surjectivity of $\Gamma_\Lambda^{(A,B^\infty)}$ from the universal multicoefficient theorem gives the result. $\square$

8.3. Computing $KL(A, J_B)$. Theorem 8.7 below relates the KL -class of a unital (A, J_B) -Cuntz pair $(\phi, \psi) : A \rightrightarrows B_\infty \triangleright J_B$ satisfying $[\phi]_{KK(A, B_\infty)} = [\psi]_{KK(A, B_\infty)}$ and the homomorphisms they induce on $\overline{K}_1^{\text{alg}}(A)$. In this case, the map $\overline{K}_1^{\text{alg}}(\phi) - \overline{K}_1^{\text{alg}}(\psi)$ yields a homomorphism

$$(8.15) \quad K_1(A)/\text{Tor}(K_1(A)) \rightarrow \ker \not\!A_{B_\infty} \subseteq \overline{K}_1^{\text{alg}}(B_\infty).$$

The following says that, under the UCT, the map in (8.15) encodes the class $[\phi, \psi]_{KL(A, J_B)}$. The connection between the map $R_{A,B}$ in the following theorem and the “rotation maps” $R_{\phi, \psi}$ appearing in the Gong–Lin–Niu classification ([131, 132]) is discussed in Remark 8.10.

Theorem 8.7. *Let A be a unital separable C^* -algebra, and let B be a unital simple separable nuclear finite $\mathcal{Z}$ -stable C^* -algebra. Then there is a natural group homomorphism*

$$(8.16) \quad R_{A,B} : \ker KL(A, j_B) \rightarrow \text{Hom}(K_1(A)/\text{Tor}(K_1(A)), \ker \not\!A_{B_\infty})$$

with the following property. For a unital (A, J_B) -Cuntz pair $(\phi, \psi): A \rightrightarrows B_\infty \triangleright J_B$ for which $[\phi]_{KL(A, B_\infty)} = [\psi]_{KL(A, B_\infty)}$, the diagram

$$(8.17) \quad \begin{array}{ccc} \overline{K}_1^{\text{alg}}(A) & \xrightarrow{\overline{K}_1^{\text{alg}}(\phi) - \overline{K}_1^{\text{alg}}(\psi)} & \overline{K}_1^{\text{alg}}(B_\infty) \\ \downarrow \text{\textit{\textbf{A}}}_A & & \uparrow \\ K_1(A) & \xrightarrow{t_A} K_1(A)/\text{Tor}(K_1(A)) \xrightarrow{R_{A,B}([\phi, \psi]_{KL(A, J_B)})} & \ker \text{\textit{\textbf{A}}}_{B_\infty} \end{array}$$

commutes, where $t_A: K_1(A) \rightarrow K_1(A)/\text{Tor}(K_1(A))$ is the quotient map. If, in addition, A satisfies the UCT, then $R_{A,B}$ is an isomorphism.

Proof. Applying naturality of the maps $\widetilde{\Gamma}_\Lambda^{(A, \cdot)}$ in the universal multicoefficient theorem (Theorem 8.5) to j_B induces $\xi: \ker KL(A, j_B) \rightarrow \ker \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(j_B))$ as in the following commutative diagram:

$$(8.18) \quad \begin{array}{ccc} \ker KL(A, j_B) & \xrightarrow{\xi} & \ker \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(j_B)) \\ \downarrow & & \downarrow \\ KL(A, J_B) & \xrightarrow{\widetilde{\Gamma}_\Lambda^{(A, J_B)}} & \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(J_B)) \\ \downarrow KL(A, j_B) & & \downarrow \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(j_B)) \\ KL(A, B_\infty) & \xrightarrow{\widetilde{\Gamma}_\Lambda^{(A, B_\infty)}} & \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(B_\infty)) \end{array}$$

The universal multicoefficient theorem tells us that the horizontal maps $\widetilde{\Gamma}_\Lambda^{(A, J_B)}$ and $\widetilde{\Gamma}_\Lambda^{(A, B_\infty)}$ are isomorphisms when A satisfies the UCT. Therefore, ξ is also an isomorphism in this case.

As B is nuclear and $\mathcal{Z}$ -stable, the K -theory computation of Proposition 6.7 ensures that $K_1(B^\infty) = 0$ and $K_0(B^\infty) \cong \text{Aff } T(B^\infty)$, so that $K_0(B^\infty)$ is uniquely divisible. Thus Lemma 2.18 applies to the trace-kernel extension and gives a natural isomorphism

$$(8.19) \quad \ker \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(j_B)) \xrightarrow{\cong} \text{Hom}(K_1(A)/\text{Tor}(K_1(A)), \ker K_1(j_B))$$

sending $\underline{\alpha}$ to the map $\alpha'_1: K_1(A)/\text{Tor}(K_1(A)) \rightarrow \ker K_1(j_B)$ induced by α_1 . Theorem 6.8 provides a natural isomorphism $\omega'_B: \ker K_1(j_B) \rightarrow \ker \text{\textit{\textbf{A}}}_{B_\infty}$ which is given explicitly by (6.7). We define $R_{A,B}$ to be the composition of these three natural maps, as follows:

$$(8.20) \quad \begin{array}{ccc} \ker KL(A, j_B) & \xrightarrow{\xi} & \ker \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(j_B)) \\ & \searrow R_{A,B} & \downarrow \underline{\alpha} \mapsto \alpha'_1 \\ & & \text{Hom}(K_1(A)/\text{Tor}(K_1(A)), \ker K_1(j_B)) \\ & & \downarrow \text{Hom}(K_1(A)/\text{Tor}(K_1(A)), \omega'_B) \\ & & \text{Hom}(K_1(A)/\text{Tor}(K_1(A)), \ker \text{\textit{\textbf{A}}}_{B_\infty}). \end{array}$$

As the vertical maps above are isomorphisms, $R_{A,B}$ is an isomorphism when ξ is, and so $R_{A,B}$ is an isomorphism when A satisfies the UCT.

Now let $(\phi, \psi): A \rightrightarrows B_\infty \triangleright J_B$ be a unital Cuntz pair such that $[\phi, \psi]_{KK(A, J_B)}$ belongs to $\ker KK(A, j_B)$. The computation of $R_{A, B}([\phi, \psi]_{KL(A, J_B)}) \circ t_A \circ \not\!A$ is summarized in the following commutative diagram (commutativity of the middle square is the definition of $R_{A, B}$):

(8.21)

$$\begin{array}{ccccc} \overline{K}_1^{\text{alg}}(A) & \xrightarrow{t_A \circ \not\!A} & K_1(A)/\text{Tor}(K_1(A)) & \xrightarrow{R_{A, B}([\phi, \psi]_{KL(A, J_B)})} & \ker \not\!B_\infty \subseteq \overline{K}_1^{\text{alg}}(B_\infty) \\ & \searrow \not\!A & \uparrow t_A & & \uparrow \omega'_B \\ & & K_1(A) & \xrightarrow{\tilde{\Gamma}_1^{(A, J_B)}([\phi, \psi]_{KL(A, J_B)})} & \ker K_1(j_B) \subseteq K_1(J_B). \\ & & & & \uparrow \omega_B \end{array}$$

It remains to check that

$$(8.22) \quad \omega_B \circ \tilde{\Gamma}_1^{(A, J_B)}([\phi, \psi]_{KL(A, J_B)}) \circ \not\!A = \overline{K}_1^{\text{alg}}(\phi) - \overline{K}_1^{\text{alg}}(\psi).$$

Let $u \in U_n(A)$. By Proposition 8.3,

$$(8.23) \quad \tilde{\Gamma}_1^{(A, J_B)}([\phi, \psi]_{KL(A, J_B)})([u]_1) = [\phi(u)\psi(u)^*]_1.$$

Letting $s: J_B^\dagger \rightarrow \mathbb{C}1_{B_\infty} \subseteq B_\infty$ be the canonical scalar map, so that $s(\phi(u)\psi(u)^*) = 1_{M_n(B_\infty)}$ since (ϕ, ψ) is a unital Cuntz pair. Therefore, the explicit formula for ω_B from Theorem 6.8(i) gives

$$(8.24) \quad \begin{aligned} \omega_B(\tilde{\Gamma}_1^{(A, J_B)}([\phi, \psi]_{KL(A, J_B)})([u]_1)) &= [\phi(u)\psi(u)^*]_{\text{alg}} - [1_{B_\infty}]_{\text{alg}} \\ &= [\phi(u)]_{\text{alg}} - [\psi(u)]_{\text{alg}} \\ &= (\overline{K}_1^{\text{alg}}(\phi) - \overline{K}_1^{\text{alg}}(\psi))([u]_{\text{alg}}), \end{aligned}$$

as required. $\square$

The map in Theorem 8.7 can be directly related to algebraic K_1 .

Remark 8.8. Let A be a unital separable C^* -algebra and B a unital separable exact $\mathcal{Z}$ -stable C^* -algebra with $T(B) \neq \emptyset$. Then there is a natural homomorphism

$$(8.25) \quad KL(A, J_B) \rightarrow \text{Hom}(\overline{K}_1^{\text{alg}}(A), \overline{K}_1^{\text{alg}}(B_\infty)), \quad \lambda \mapsto \omega_B \circ \tilde{\Gamma}_1^{(A, J_B)}(\lambda) \circ \not\!A,$$

where $\omega_B: K_1(J_B) \rightarrow \overline{K}_1^{\text{alg}}(B_\infty)$ is the map from Theorem 6.8. The same computation as in the end of the proof of Theorem 8.7 shows that if $(\phi, \psi): A \rightrightarrows B_\infty \triangleright J_B$ is a unital (A, J_B) -Cuntz pair, then

$$(8.26) \quad [\phi, \psi]_{KL(A, J_B)} \mapsto \overline{K}_1^{\text{alg}}(\phi) - \overline{K}_1^{\text{alg}}(\psi)$$

is the map in (8.25).

We conclude this section by proving Theorem 1.6, explicitly describing $KL(A, J_B)$ in terms of $\underline{K}T_u(\cdot)$.

Theorem 8.9. *Let A be a unital separable C^* -algebra satisfying the UCT and let B be a unital simple separable nuclear finite $\mathcal{Z}$ -stable C^* -algebra. The map*

$$(8.27) \quad \Theta := \tilde{\Gamma}_\Lambda^{(A, B_\infty)} \circ KL(A, j_B): KL(A, J_B) \rightarrow \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(B))$$

$$(8.28) \quad \begin{array}{c} 0 \longrightarrow \ker \operatorname{Hom}_{\Lambda}(\underline{K}(A), \underline{K}(j_B)) \longrightarrow KL(A, j_B) \\ \qquad \qquad \qquad \searrow \Theta \\ \longrightarrow \operatorname{Hom}_{\Lambda}(\underline{K}(A), \underline{K}(B_{\infty})) \longrightarrow \operatorname{Hom}_{\Lambda}(\underline{K}(A), \underline{K}(B^{\infty})), \end{array}$$

(i) The range of Θ consists of all $\underline{\alpha}$ for which $\rho_{B_\infty} \circ \alpha_0 = 0$ and the kernel of Θ is $\ker KL(A, j_B)$.

(ii) Let $\psi: A \rightarrow B_\infty$ be a unital full $*$ -homomorphism. There is a bijection Ω taking $\ker \Theta \subseteq KL(A, j_B)$ to the set of $\beta: \overline{K_1}^{\text{alg}}(A) \rightarrow \overline{K_1}^{\text{alg}}(B_\infty)$ such that $(\underline{K}(\psi), \beta, \text{Aff } T(\psi))$ is a $\underline{KT}_u(\cdot)$ -morphism, given by

Proof. The exactness at the first two entries of (8.28) follows from the naturality of $\tilde{\Gamma}_\Lambda^{(A, \cdot)}$. Indeed, after using $\tilde{\Gamma}_\Lambda^{(A, J_B)}$ to identify $KL(A, J_B)$ with $\text{Hom}_\Lambda(\underline{K}(A), \underline{K}(J_B))$ via the universal multicoefficient theorem, the second arrow becomes the inclusion, Θ becomes $\text{Hom}_\Lambda(\underline{K}(A), \underline{K}(j_B))$ and (8.28) is certainly exact at this entry. For exactness at $\text{Hom}_\Lambda(\underline{K}(A), \underline{K}(B_\infty))$, it suffices to show that $\text{im } \Theta$ is contained in $\ker \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(q_B))$ since the reverse inclusion follows from $q_B \circ j_B = 0$. The issue is that, in general, neither $KL(A, \cdot)$ nor $\text{Hom}_\Lambda(\underline{K}(A), \underline{K}(\cdot))$ is half-exact, so we instead use half-exactness of $KK(A, \cdot)$ together with the control on $KK(A, B^\infty)$ given by the von Neumann-like behavior of B^∞ .

$$(8.30) \quad \begin{array}{ccccc} KK(A, J_B) & \longrightarrow & KK(A, B_\infty) & \longrightarrow & KK(A, B^\infty) \\ \downarrow \Gamma_\Lambda^{(A, J_B)} & & \downarrow \Gamma_\Lambda^{(A, B_\infty)} & & \downarrow \Gamma_\Lambda^{(A, B^\infty)} \\ \mathrm{Hom}_\Lambda(\underline{K}(A), \underline{K}(J_B)) & \rightarrow & \mathrm{Hom}_\Lambda(\underline{K}(A), \underline{K}(B_\infty)) & \rightarrow & \mathrm{Hom}_\Lambda(\underline{K}(A), \underline{K}(B^\infty)) \end{array}$$

By naturality of $\tilde{\Gamma}_\Lambda^{(A, \cdot)}$,

$$(8.31) \quad \Theta = \mathrm{Hom}_{\Lambda}(\underline{K}(A), \underline{K}(j_B)) \circ \tilde{\Gamma}_{\Lambda}^{(A, J_B)},$$

so that (using that $\tilde{\Gamma}_{\Lambda}^{(A, j_B)}$ is an isomorphism by the universal multicoefficient theorem) $\text{im } \Theta = \text{im Hom}_{\Lambda}(\underline{K}(A), \underline{K}(j_B))$. Therefore, (8.28) is exact.

(i): The exactness of (8.28) implies that the image of Θ consists of the set of all $\underline{\alpha}: \underline{K}(A) \rightarrow \underline{K}(B_\infty)$ for which $\underline{K}(q_B) \circ \underline{\alpha} = 0$. By Proposition 6.7, this condition is equivalent to $\rho_{B_\infty} \circ \alpha_0 = 0$. The kernel of Θ is $\ker KL(A, j_B)$ as $\tilde{\Gamma}_\Lambda^{(A, B_\infty)}$ is an isomorphism.

(ii): Fix $\lambda \in \ker \Theta$ and define $\beta := \Omega(\lambda)$. We first check that $(\underline{K}(\psi), \beta, \text{Aff } T(\psi))$ is a morphism $\underline{K}T_u(A) \rightarrow \underline{K}T_u(B_\infty)$. This boils down to the fact that $\underline{K}T_u(\psi)$ is a morphism (Proposition 3.6) and the correcting term $R_{A,B}(\lambda) \circ t_A \circ \not\partial_A$ in (8.29) is annihilated in the $\underline{K}T_u(\cdot)$ -compatibility relations.

Precisely, the first square of (3.14) commutes for free since it does not involve the map β . For the second square, since $\not\alpha_A \circ \text{Th}_A = 0$, we have

$$(8.32) \quad \beta \circ \mathrm{Th}_A \stackrel{(8.29)}{=} \overline{K}_1^{\mathrm{alg}}(\psi) \circ \mathrm{Th}_A \stackrel{\mathrm{Prop.} \ 3.6}{=} \mathrm{Th}_{B_\infty} \circ \mathrm{Aff} \, T(\psi).$$

For the third square of (3.14), since $\text{im } R_{A,B}(\lambda) \subseteq \ker \not{A}_{B_\infty}$, we similarly have

$$(8.33) \quad \not{A}_{B_\infty} \circ \beta \stackrel{(8.29)}{=} \not{A}_{B_\infty} \circ \overline{K}_1^{\text{alg}}(\psi) \stackrel{\text{Prop. 3.6}}{=} K_1(\psi) \circ \not{A}_A.$$

Finally, for commutativity of (3.15), note that by (3.8), we have $\sharp_A \circ \zeta_A^{(n)} = \nu_{0,A}^{(n)}$, so the range of this composition is contained in $\text{Tor}(K_1(A), \mathbb{Z}/n)$. Consequently, $t_A \circ \sharp_A \circ \zeta_A^{(n)} = 0$, so similar to the previous calculations, we have

$$(8.34) \quad \beta \circ \zeta_A^{(n)} \stackrel{(8.29)}{=} \overline{K}_1^{\text{alg}}(\psi) \circ \zeta_A^{(n)} \stackrel{\text{Prop. 3.6}}{=} \zeta_B^{(n)} \circ K_0(\psi; \mathbb{Z}/n).$$

This shows that $(\underline{K}(\psi), \beta, \text{Aff } T(\psi))$ is a $\underline{KT}_u(\cdot)$ -morphism.

Suppose now that $\lambda, \lambda' \in \ker \Theta$ satisfy $\Omega(\lambda) = \Omega(\lambda')$. By (8.29), we get $R_{A,B}(\lambda - \lambda') \circ t_A \circ \not\phi_A = 0$. Since $t_A \circ \not\phi_A$ is surjective, and $R_{A,B}$ is injective (from Theorem 8.7), it follows that $\lambda = \lambda'$. This shows that Ω is injective.

Finally, let us show that Ω is surjective; that is, if $\beta : \overline{K}_1^{\text{alg}}(A) \rightarrow \overline{K}_1^{\text{alg}}(B_\infty)$ is such that $(\underline{K}(\psi), \beta, \text{Aff } T(\psi))$ is a morphism $\underline{K}T_u(A) \rightarrow \underline{K}T_u(B_\infty)$, then there exists $\lambda \in \ker \Theta$ such that $\Omega(\lambda) = \beta$. Using Lemma 3.11 with $\beta' := \overline{K}_1^{\text{alg}}(\psi)$, there is a homomorphism $r : K_1(A)/\text{Tor}(K_1(A)) \rightarrow \ker \not\!A_{B_\infty}$ such that

$$(8.35) \quad \begin{array}{ccc} \overline{K}_1^{\text{alg}}(A) & \xrightarrow{\beta - \overline{K}_1^{\text{alg}}(\psi)} & \overline{K}_1^{\text{alg}}(B_\infty) \\ \downarrow \not\cong_A & & \uparrow \\ K_1(A) & \xrightarrow{t_A} K_1(A)/\text{Tor}(K_1(A)) \xrightarrow{r} & \ker \not\cong_{B_\infty} \end{array}$$

commutes. By surjectivity of $R_{A,B}$ (from Theorem 8.7), it follows that there exists $\lambda \in \ker KL(A, j_B) = \ker \Theta$ such that $r = R_{A,B}(\lambda)$. Hence,

$$\begin{aligned}
 \Omega(\lambda) &\stackrel{(8.29)}{=} \overline{K}_1^{\text{alg}}(\psi) + R_{A,B}(\lambda) \circ t_A \circ \not\! x_A \\
 &= \overline{K}_1^{\text{alg}}(\psi) + r \circ t_A \circ \not\! x_A \\
 &\stackrel{(8.35)}{=} \overline{K}_1^{\text{alg}}(\psi) + (\beta - \overline{K}_1^{\text{alg}}(\psi)) \\
 &= \beta,
 \end{aligned}
 \tag{8.36}$$

☐

Remark 8.10. The map $R_{A,B}$ in Theorem 8.7 is related to the rotation maps appearing in early classification results of *-homomorphism up to asymptotic unitary equivalence, dating back to [182]. For the sake of comparison, we recall the version of the rotation map from [131, Definition 2.21].

Suppose A and B are unital C^* -algebras with A separable and $\phi, \psi: A \rightarrow B$ are *-homomorphisms with $[\phi]_{KK(A,B)} = [\psi]_{KK(A,B)}$ and $\text{Aff } T(\phi) = \text{Aff } T(\psi)$. Define the *mapping torus* of ϕ and ψ by

$$(8.37) \quad M_{\phi,\psi} := \{(f, a) \in C([0, 1], B) \oplus A : \phi(a) = f(0) \text{ and } \psi(a) = f(1)\}.$$

There is a corresponding extension

$$(8.38) \quad 0 \longrightarrow SB \longrightarrow M_{\phi,\psi} \longrightarrow A \longrightarrow 0$$

with the first map being the inclusion into the first factor and the second map being the projection onto the second factor. Since the boundary map in the associated six-term exact sequence of KK -groups is naturally identified with $KK(A, \phi) - KK(A, \psi) = 0$ (as $[\phi]_{KK(A,B)} = [\psi]_{KK(A,B)}$), any preimage of $[\text{id}_A]_{KK(A,A)}$ in $KK(A, M_{\phi,\psi})$ is a KK -splitting $\kappa \in KK(A, M_{\phi,\psi})$ of the extension (8.38). Hence there is an induced split exact sequence

$$(8.39) \quad 0 \longrightarrow K_0(B) \longrightarrow K_1(M_{\phi,\psi}) \xleftarrow{\kappa_1} K_1(A) \longrightarrow 0,$$

where $\kappa_1 := \Gamma_1^{(A, M_{\phi,\psi})}(\kappa)$.

Using that $\text{Aff } T(\phi) = \text{Aff } T(\psi)$, the de la Harpe–Skandalis determinant produces a well-defined map $R_{\phi,\psi}: K_1(M_{\phi,\psi}) \rightarrow \text{Aff } T(B)$ by

$$(8.40) \quad R_{\phi,\psi}([(u, v)]_1) := \tilde{\Delta}_B(u), \quad (u, v) \in U_\infty(M_{\phi,\psi}),$$

called the *rotation map* corresponding to ϕ and ψ . The composition

$$(8.41) \quad K_1(A) \xrightarrow{\kappa_1} K_1(M_{\phi,\psi}) \xrightarrow{R_{\phi,\psi}} \text{Aff } T(B) \longrightarrow \text{Aff } T(B)/\overline{\text{im } \rho_B},$$

which is denoted $\tilde{R}_{\phi,\psi}$, is independent of the choice of splitting κ (see [196, Lemma 3.3]).

If $(\phi, \psi): A \rightrightarrows B_\infty \supset J_B$ is a Cuntz pair as in Theorem 8.7, and, in addition, $[\phi]_{KK(A, B_\infty)} = [\psi]_{KK(A, B_\infty)}$, so that both $\tilde{R}_{\phi,\psi}$ and $R_{A,B}([\phi, \psi]_{KL(A, J_B)})$ are defined, then they are related by the equation

$$(8.42) \quad \tilde{R}_{\phi,\psi} = -\overline{\det}_{B_\infty} \circ R_{A,B}([\phi, \psi]_{KL(A, J_B)}) \circ t_A.$$

To see this, by (2.17) and (8.41), it suffices to show that

$$(8.43) \quad \text{Th}_{B_\infty} \circ R_{\phi,\psi} \circ \kappa_1 = -R_{A,B}([\phi, \psi]_{KL(A, J_B)}) \circ t_A.$$

For $v \in U_\infty(A)$, write $\kappa_1([v]_1) = [(u, v')]_1$, so that $[v']_1 = [v]_1$ and u is a path in $U_\infty(B_\infty)$ from $\phi(v')$ to $\psi(v')$. Then

$$(8.44) \quad \begin{aligned} \text{Th}_{B_\infty}(R_{\phi,\psi}(\kappa_1([v]_1))) &= \text{Th}_{B_\infty}(R_{\phi,\psi}([(u, v')]_1)) \\ &\stackrel{(8.40)}{=} \text{Th}_{B_\infty}(\tilde{\Delta}_{B_\infty}(u)) \\ &\stackrel{(2.13), (2.8)}{=} [u(1)u(0)^*]_{\text{alg}} \\ &= [\psi(v')]_{\text{alg}} - [\phi(v')]_{\text{alg}} \\ &= (\overline{K}_1^{\text{alg}}(\psi) - \overline{K}_1^{\text{alg}}(\phi))([v']_{\text{alg}}) \\ &\stackrel{(8.17)}{=} -R_{A,B}([\phi, \psi]_{KL(A, J_B)})(t_A([v']_1)) \\ &= -R_{A,B}([\phi, \psi]_{KL(A, J_B)})(t_A([v]_1)), \end{aligned}$$

establishing (8.42).

9. CLASSIFICATION OF UNITAL *-HOMOMORPHISMS

We now have all the pieces in place to establish the main classification theorems: Theorems A, B, and 1.1. First we prove the classification of unital full approximate embeddings, and then we use intertwining arguments to deduce Theorem B in Section 9.2 and Theorem A in Section 9.3.

9.1. Classification of unital full approximate embeddings. We start with the precise statement of Theorem 1.1.

Theorem 9.1 (Classification of unital full approximate embeddings). *Let A and B be unital separable nuclear C^* -algebras such that A satisfies the UCT and B is simple and $\mathcal{Z}$ -stable. If $(\underline{\alpha}, \beta, \gamma): \underline{KT}_u(A) \rightarrow \underline{KT}_u(B_\infty)$ is a faithful morphism, then there exists a unital full *-homomorphism $\phi: A \rightarrow B_\infty$ such that $\underline{KT}_u(\phi) = (\underline{\alpha}, \beta, \gamma)$, and this ϕ is unique up to unitary equivalence.*

As mentioned in the introduction, by means of Kirchberg's dichotomy (Theorem 4.4), B is either purely infinite or stably finite. While the stably finite case is our main result, let us first discuss the purely infinite case, which is a refined version of the Kirchberg–Phillips Theorem. This could be proved using Kirchberg's (unpublished) techniques from the 1990s ([170]); we will deduce the result from the purely infinite classification theorem in [115].

Proof of Theorem 9.1 when $T(B) = \emptyset$. B is purely infinite and hence is $\mathcal{O}_\infty$ -stable by [175, Theorem 3.15]. As noted in Remark 3.7, $\underline{KT}_u$ -morphisms from $\underline{KT}_u(A)$ to $\underline{KT}_u(B_\infty)$ (all of which are vacuously faithful) correspond to $\underline{K}$ -morphisms from $\underline{K}(A)$ to $\underline{K}(B_\infty)$ sending $[1_A]_0$ to $[1_{B_\infty}]_0$. Hence by [115, Theorem 8.12], these correspond bijectively to full $\mathcal{O}_\infty$ -stable (in the sense of [115, Definition 4.1]) unital *-homomorphisms $A \rightarrow B_\infty$, up to approximate unitary equivalence.

By [268, Proposition 1.12], B_∞ is separably $\mathcal{O}_\infty$ -stable, and consequently any *-homomorphism $A \rightarrow B_\infty$ factorizes through a separable $\mathcal{O}_\infty$ -stable subalgebra of B_∞ , and is therefore $\mathcal{O}_\infty$ -stable. Note also that approximate unitary equivalence is the same as unitary equivalence (as recorded in Lemma 3.15). This proves the theorem in this case. $\square$

Now we return to the stably finite case, beginning with the existence portion of Theorem 9.1. The strategy is to first realize γ by a map into B^∞ , obtained from the classification of maps into B^∞ , and lift this to a map into B_∞ realizing $\underline{\alpha}$ using the classification of unital lifts. Finally, we adjust this lift by combining the second part of the classification of unital lifts with the KL computations in Section 8.

Proof of existence in Theorem 9.1 when $T(B) \neq \emptyset$. Suppose $(\underline{\alpha}, \beta, \gamma): \underline{KT}_u(A) \rightarrow \underline{KT}_u(B)$ is a faithful morphism. By the classification of maps into B^∞ by traces (Theorem 6.5), there is a unital *-homomorphism $\theta: A \rightarrow B^\infty$ such that

$$(9.1) \quad \text{Aff } T(\theta) = \text{Aff } T(q_B) \circ \gamma: \text{Aff } T(A) \rightarrow \text{Aff } T(B^\infty).$$

Since $(\underline{\alpha}, \beta, \gamma)$ is faithful, every trace $\tau \in T(B^\infty)$ gives rise to a faithful trace $\tau \circ \theta = \gamma^*(\tau \circ q_B)$ on A .

Proof of uniqueness in Theorem 9.1 when $T(B) \neq \emptyset$. Let $\phi, \psi: A \rightarrow B_\infty$ be two unital full $*$ -homomorphisms such that $\underline{KT}_u(\phi) = \underline{KT}_u(\psi)$. Then, in particular, $\text{Aff } T(\phi) = \text{Aff } T(\psi)$, and we have

$$(9.8) \quad \text{Aff } T(q_B \circ \phi) = \text{Aff } T(q_B \circ \psi): \text{Aff } T(A) \rightarrow \text{Aff } T(B^\infty).$$

Hence $q_B \circ \phi$ and $q_B \circ \psi$ are unitarily equivalent by the classification of maps into B^∞ by traces (Theorem 6.5). Since every unitary in B^∞ can be lifted to a unitary in B_∞ by Proposition 6.6, we have that $q_B \circ \phi = q_B \circ \text{Ad}(u) \circ \psi$ for some unitary $u \in B_\infty$. By replacing ψ with $\text{Ad}(u) \circ \psi$ (which leaves $\underline{KT}_u(\cdot)$ unchanged — see Proposition 3.6), we may assume that $q_B \circ \phi = q_B \circ \psi$. Therefore, (ϕ, ψ) forms an (A, J_B) -Cuntz pair, and defines an element $[\phi, \psi]_{KL(A, J_B)}$ in $KL(A, J_B)$.

Letting Θ be as in Theorem 8.9(i), we have

$$(9.9) \quad \begin{aligned} \Theta([\phi, \psi]_{KL(A, J_B)}) &\stackrel{\text{Thm. 8.9(i)}}{=} (\tilde{\Gamma}_\Lambda^{(A, B_\infty)} \circ KL(A, j_B))([\phi, \psi]_{KL(A, J_B)}) \\ &\stackrel{\text{Prop. 5.8(iv)}}{=} \tilde{\Gamma}_\Lambda^{(A, B_\infty)}([\phi]_{KL(A, B_\infty)} - [\psi]_{KL(A, B_\infty)}) \\ &\stackrel{(8.13)}{=} \underline{K}(\phi) - \underline{K}(\psi) \\ &= 0. \end{aligned}$$

This shows that $[\phi, \psi]_{KL(A, J_B)} \in \ker \Theta = \ker KL(A, j_B)$.

Since $\overline{K}_1^{\text{alg}}(\phi) = \overline{K}_1^{\text{alg}}(\psi)$ and both t_A and $\mathfrak{A}_A$ are surjective, it follows by Theorem 8.9 that $R_{A, B}([\phi, \psi]_{KL(A, J_B)}) = 0$. Since $R_{A, B}$ is injective (by Theorem 8.9), we conclude that $[\phi, \psi]_{KL(A, J_B)} = 0$.

Hence by the classification of unital lifts (Theorem 7.10(iii)), it follows that ϕ and ψ are unitarily equivalent. $\square$

9.2. Classification of embeddings. Next we use the classification of unital full approximate embeddings to prove the classification of embeddings theorem (Theorem B). Whereas uniqueness up to approximate unitary equivalence in Theorem B is an immediate consequence of the uniqueness up to unitary equivalence in Theorem 9.1, an intertwining argument is needed for existence. In the literature (such as [189, 195, 131, 132]), this intertwining argument is often applied in an approximate form, by reformulating the approximate classification theorem in terms of approximately multiplicative maps, requiring the use of approximately defined maps on the invariant. In contrast, we opt for an approach that uses intertwining by reparameterization to identify those maps $A \rightarrow B_\infty$ that are unitarily equivalent to maps factoring through B . To the best of our knowledge, the idea first appears (albeit in a continuous form) in Phillips' approach to the Kirchberg–Phillips theorem in [236, Proposition 1.3.7], where it is attributed to Kirchberg.

To set this up, given a function $r: \mathbb{N} \rightarrow \mathbb{N}$ define a reparameterization map $r^*: \ell^\infty(B) \rightarrow \ell^\infty(B)$ by $r^*((b_n)_{n=1}^\infty) := (b_{r(n)})_{n=1}^\infty$. When $\lim_{n \rightarrow \infty} r(n) = \infty$, this also induces a map $B_\infty \rightarrow B_\infty$, which we continue to denote by r^* . Recall from (3.45) that $\iota_B: B \rightarrow B_\infty$ denotes the canonical embedding.

Proposition 9.2 (Intertwining by reparameterization, [118, Theorem 4.3]). *Let A and B be C^* -algebras, with A separable and B unital. Let $\phi_\infty: A \rightarrow B_\infty$ be a $*$ -homomorphism. There exists a $*$ -homomorphism $\phi: A \rightarrow B$ such that ϕ_∞ and $\iota_B \circ \phi$ are unitarily equivalent if and only if for every $r: \mathbb{N} \rightarrow \mathbb{N}$ with $\lim_{n \rightarrow \infty} r(n) = \infty$, the maps ϕ_∞ and $r^* \circ \phi_\infty$ are unitarily equivalent.*

With this additional ingredient in place we can prove the classification of embeddings (Theorem B).

Theorem 9.3 (Classification of embeddings). *Let A and B be unital separable nuclear C^* -algebras such that A satisfies the UCT and B is simple and $\mathcal{Z}$ -stable. If $(\underline{\alpha}, \beta, \gamma): \underline{KT}_u(A) \rightarrow \underline{KT}_u(B)$ is a faithful morphism, then there exists a unital injective *-homomorphism $\phi: A \rightarrow B$ such that $\underline{KT}_u(\phi) = (\underline{\alpha}, \beta, \gamma)$, and this ϕ is unique up to approximate unitary equivalence.*

Proof. Uniqueness is essentially an immediate consequence of Theorem 9.1. Given unital injective *-homomorphisms $\phi, \psi: A \rightarrow B$, the compositions $\iota_B \circ \phi, \iota_B \circ \psi: A \rightarrow B_\infty$ are full as B is simple. Accordingly, if $\underline{KT}_u(\phi) = \underline{KT}_u(\psi)$, then $\underline{KT}_u(\iota_B \circ \phi) = \underline{KT}_u(\iota_B \circ \psi)$ and so $\iota_B \circ \phi$ and $\iota_B \circ \psi$ are unitarily equivalent by Theorem 9.1. As unitaries in B_∞ can be lifted to unitaries in $\ell^\infty(B)$, ϕ and ψ are approximately unitarily equivalent.

For existence, consider a faithful $\underline{KT}_u$ -morphism

$$(9.10) \quad (\underline{\alpha}, \beta, \gamma): \underline{KT}_u(A) \rightarrow \underline{KT}_u(B).$$

Then $\underline{KT}_u(\iota_B) \circ (\underline{\alpha}, \beta, \gamma)$ is also faithful, so the existence part of Theorem 9.1 gives a unital full *-homomorphism $\phi_\infty: A \rightarrow B_\infty$ such that

$$(9.11) \quad \underline{KT}_u(\phi_\infty) = \underline{KT}_u(\iota_B) \circ (\underline{\alpha}, \beta, \gamma).$$

Fix $r: \mathbb{N} \rightarrow \mathbb{N}$ with $\lim_{n \rightarrow \infty} r(n) = \infty$. Since $r^* \circ \iota_B = \iota_B$, it follows that

$$(9.12) \quad \underline{KT}_u(r^* \circ \phi_\infty) = (\underline{KT}_u(r^*) \circ \underline{KT}_u(\iota_B))(\underline{\alpha}, \beta, \gamma) = \underline{KT}_u(\phi_\infty).$$

Moreover, $r^* \circ \phi_\infty$ is full as r^* is unital and ϕ_∞ is full. Therefore, $r^* \circ \phi_\infty$ and ϕ_∞ are unitarily equivalent by the uniqueness portion of Theorem 9.1. Then intertwining by reparameterizations (Proposition 9.2) gives a *-homomorphism $\phi: A \rightarrow B$ such that $\iota_B \circ \phi$ and ϕ_∞ are unitarily equivalent. As ϕ_∞ is unital and faithful, so is ϕ . Note that

$$(9.13) \quad \underline{KT}_u(\iota_B) \circ \underline{KT}_u(\phi) = \underline{KT}_u(\iota_B \circ \phi) \stackrel{\text{Prop. 3.6}}{=} \underline{KT}_u(\phi_\infty) = \underline{KT}_u(\iota_B) \circ (\underline{\alpha}, \beta, \gamma).$$

By Lemma 3.16, $\underline{KT}_u(\iota_B)$ is a monomorphism, so $\underline{KT}_u(\phi) = (\underline{\alpha}, \beta, \gamma)$. $\square$

Remark 9.4. We use sequence algebras B_∞ rather than ultrapowers B_ω in our classification of full approximate embeddings (Theorem 1.1) in order to access the intertwining by reparameterization technique; reparameterizations do not make sense for ultrapowers. Prompted by the discrepancy between sequence algebras and ultrapowers, Farah proved that an existence result into the ultrapower is equivalent to an existence result into the sequence algebra ([110, Theorem A]).

Corollary C is an immediate consequence of the classification of embeddings.

Corollary 9.5. *Let A and B be unital separable nuclear C^* -algebras with non-empty tracial state spaces, and suppose that A satisfies the UCT and that B is simple and $\mathcal{Z}$ -stable. Given a faithful unital compatible pair*

$$(9.14) \quad (\alpha_*, \gamma): (K_*(A), \text{Aff } T(A)) \rightarrow (K_*(B), \text{Aff } T(B)),$$

there is a unital embedding $\phi: A \rightarrow B$ such that $K_(\phi) = \alpha_*$ and $\text{Aff } T(\phi) = \gamma$.*

Proof. By Proposition 2.13, we can extend the pair (α_*, γ) to a compatible triple $(\underline{\alpha}, \beta, \gamma): \underline{KT}_u(A) \rightarrow \underline{KT}_u(B)$. The classification of embeddings theorem (Theorem 9.3) then gives a unital embedding $\phi: A \rightarrow B$ with $\underline{KT}_u(\phi) = (\underline{\alpha}, \beta, \gamma)$. Hence $K_*(\phi) = \alpha_*$ and $\text{Aff } T(\phi) = \gamma$. $\square$

Note that in the previous corollary the embedding ϕ is not necessarily unique up to approximate unitary equivalence. In general, uniqueness requires the total invariant. See Example 9.11 below.

Remark 9.6. Let us note how Theorem 9.3 relates to other classification results for embeddings. In [210, Corollary 6.8 and Theorem 7.1], Matui proves uniqueness results for embeddings $A \rightarrow B$, where A is either a unital AH algebra (including $C(X)$) or a unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebra that is rationally TAF and satisfies the UCT. Matui allows unital simple separable stably finite $\mathcal{Z}$ -stable codomains B that are either rationally TAF or exact with finitely many extreme traces that are separated by projections. For comparing maps $\phi, \psi: A \rightarrow B$, Matui uses an invariant consisting of $\underline{K}(\cdot)$, $\text{Aff } T(\cdot)$ and (vanishing of) an induced homomorphism $\Theta_{\phi, \psi}: K_1(A) \rightarrow \text{Aff } T(B)/\overline{\text{im } \rho_B}$ (defined when $K_1(\phi) = K_1(\psi)$ and $T(\phi) = T(\psi)$ — see [210, Lemma 3.1]) given by

$$(9.15) \quad \Theta_{\phi, \psi}([u]_1) = \det_B(\phi(u)^* \psi(u)), \quad u \in U_\infty(A).$$

Using this, one can compute that $\overline{\text{Th}}_B \circ \Theta_{\phi, \psi} \circ \mathcal{A}_A = \overline{K}_1^{\text{alg}}(\psi) - \overline{K}_1^{\text{alg}}(\phi)$ so (since $\mathcal{A}_A$ is surjective and $\overline{\text{Th}}_B$ is injective) $\Theta_{\phi, \psi}$ vanishes if and only if $\overline{K}_1^{\text{alg}}(\phi) = \overline{K}_1^{\text{alg}}(\psi)$. Note that this map Θ coincides with the map $\tilde{R}_{\phi, \psi}$ of Remark 8.10 in the case $[\phi]_{KK(A, B)} = [\psi]_{KK(A, B)}$ (so that $\tilde{R}_{\phi, \psi}$ is defined). Accordingly, in the case when the codomains are additionally assumed to be nuclear, Matui's uniqueness theorems are encompassed within Theorem 9.3 above.

While this paper was being prepared, Gong, Lin, and Niu independently obtained a classification of embeddings ([130, Theorems 4.3 and 6.5]), building on prior work of Lin and Niu in the rationally tracial rank one setting ([203]). Compared with Theorem 9.3, the main difference in their result is a simplicity hypothesis on the domain algebra.

Whereas our approach establishes Theorem 9.3 as an ingredient towards classification of C^* -algebras, Gong, Lin, and Niu use their earlier classification theorems ([131, 132]) and the additional simplicity hypothesis on A to obtain internal tracial approximation structure on its UHF-stabilizations. They also use a variation of Winter's localization technique.

9.3. The unital classification theorem. The stably finite half of the unital classification theorem is now obtained by the standard strategy of symmetrizing the assumptions on the domain and codomain in the classification of embeddings and using the following form of Elliott's two-sided intertwining argument (see [255, Corollary 2.3.4], for example).

Proposition 9.7 (Elliott's two-sided intertwining). *Let A and B be unital separable C^* -algebras, and let $\phi_0: A \rightarrow B$ and $\psi_0: B \rightarrow A$ be $*$ -homomorphisms. Suppose that $\psi_0 \circ \phi_0$ is approximately unitarily equivalent to id_A and $\phi_0 \circ \psi_0$ is approximately unitarily equivalent to id_B . Then ϕ_0 is approximately unitarily equivalent to an isomorphism between A and B .*

When A and B are as in the unital classification theorem (Theorem A), an isomorphism of the total invariant can be lifted to an isomorphism of the algebra.

Theorem 9.8. *Let A and B be unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebras satisfying the UCT. Suppose*

$$(9.16) \quad (\underline{\alpha}, \beta, \gamma): \underline{KT}_u(A) \rightarrow \underline{KT}_u(B)$$

is a $\underline{KT}_u$ -isomorphism. Then there exists a *-isomorphism $\phi: A \rightarrow B$, unique up to approximate unitary equivalence, such that $\underline{KT}_u(\phi) = (\underline{\alpha}, \beta, \gamma)$.

Proof. Uniqueness follows from the uniqueness portion of the classification of embeddings theorem (Theorem 9.3), noting that faithfulness of the invariant is automatic since all traces on A are faithful.

The existence portion of the classification of embeddings theorem (Theorem 9.3) gives unital *-homomorphisms $\phi_0: A \rightarrow B$ and $\psi_0: A \rightarrow B$ such that $\underline{KT}_u(\phi_0) = (\underline{\alpha}, \beta, \gamma)$ and $\underline{KT}_u(\psi_0) = \underline{KT}_u(\phi_0)^{-1}$. By the uniqueness aspect of the classification of embeddings theorem (Theorem 9.3), $\phi_0 \circ \psi_0$ is approximately unitarily equivalent to id_B and $\psi_0 \circ \phi_0$ is approximately unitarily equivalent to id_A . Therefore, Elliott's two-sided intertwining argument (Proposition 9.7) provides a *-isomorphism $\phi: A \rightarrow B$ which is approximately unitarily equivalent to ϕ_0 . Hence $\underline{KT}_u(\phi) = \underline{KT}_u(\phi_0) = (\underline{\alpha}, \beta, \gamma)$ since $\underline{KT}_u$ is invariant under approximate unitary equivalence (Proposition 3.6). $\square$

We now establish Theorem A as a consequence.

Theorem 9.9. *Let A and B be unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebras satisfying the UCT. Suppose*

$$(9.17) \quad (\alpha_*, \gamma): (K_*(A), \text{Aff } T(A)) \rightarrow (K_*(B), \text{Aff } T(B))$$

*is a $\underline{KT}_u$ -isomorphism. Then there exists a *-isomorphism $\phi: A \rightarrow B$ such that $K_*(\phi) = \alpha_*$ and $\text{Aff } T(\phi) = \gamma$.*

Proof. By Proposition 2.13, there is a unital faithful invertible compatible triple $(\underline{\alpha}, \beta, \gamma): \underline{KT}_u(A) \rightarrow \underline{KT}_u(B)$ extending (α_*, γ) . By Proposition 9.8, there is an isomorphism $\phi: A \rightarrow B$ with $\underline{KT}_u(\phi) = (\underline{\alpha}, \beta, \gamma)$. Hence $K_*(\phi) = \alpha_*$ and $\text{Aff } T(\phi) = \gamma$. $\square$

The classification of automorphisms modulo approximate unitary equivalence has been well-studied in operator algebras. For example, Connes' theorem builds on his prior work on this problem for II_1 factors (e.g., [47, 45]). Given a unital C^* -algebra A , write $\text{Aut}(A)$ for the automorphism group of A , and $\overline{\text{Inn}}(A)$ for the subgroup of approximately inner automorphisms. When A is a unital Kirchberg C^* -algebra satisfying the UCT, the quotient group $\text{Aut}(A)/\overline{\text{Inn}}(A)$ is isomorphic to the group of unital automorphisms of $\underline{K}(A)$ (cf. [106, Theorem 5.6]). For unital finite classifiable C^* -algebras, unital automorphisms of the total invariant encode this information.

Corollary 9.10. *Let A be a unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebra satisfying the universal coefficient theorem. Then the quotient $\text{Aut}(A)/\overline{\text{Inn}}(A)$ is isomorphic to the group of automorphisms of $\underline{KT}_u(A)$.*

Proof. This follows immediately from Theorem 9.8. $\square$

We found the following Bernoulli shift example, where the $\overline{K}_1^{\text{alg}}$ -class of an automorphism is particularly discernible, informative during our work on this paper.

Example 9.11 (Automorphisms of $\mathcal{Z} \wr \mathbb{Z}$). Consider the infinite tensor product $\mathcal{Z}^{\otimes \mathbb{Z}}$, let $\mathbb{Z}$ act on this algebra by shifting, and let $A := \mathcal{Z}^{\otimes \mathbb{Z}} \rtimes \mathbb{Z}$ denote the crossed product. As we review below, it is well-known that such a construction gives rise to a C^* -algebra covered by Theorem A.

It is immediate that A is separable and unital. Since $\mathcal{Z}^{\otimes \mathbb{Z}}$ is isomorphic to $\mathcal{Z}$ ([154, Corollary 8.8]), it is nuclear and satisfies the UCT. Both properties are preserved by crossed products by $\mathbb{Z}$, and therefore A is also nuclear and satisfies the UCT. The shift action on $\mathcal{R}^{\otimes \mathbb{Z}}$ is outer (see [232, Lemma 2.5]), which means that the shift action on $\mathcal{Z}^{\otimes \mathbb{Z}}$ is strongly outer. Therefore, A is simple by [181, Theorem 3.1], $\mathcal{Z}$ -stable by [212, Corollary 4.10], and has unique trace by [284, Theorem 4.3(4) $\Rightarrow$ (1)] (cf. [297, Theorem 1.11]).

Now we turn to the invariant of A . Using $K_0(\mathcal{Z}) \cong \mathbb{Z}$ with unit corresponding to $1 \in \mathbb{Z}$ and $K_1(\mathcal{Z}) = 0$, the Pimsner–Voiculescu sequence ([239]) implies $K_0(A) \cong \mathbb{Z}$ with the unit corresponding to $1 \in \mathbb{Z}$ and $K_1(A) \cong \mathbb{Z}$. Combining this with the fact that A has unique trace, we may therefore (by (2.19)) identify

$$(9.18) \quad \overline{K}_1^{\text{alg}}(A) \cong \mathbb{T} \oplus \mathbb{Z},$$

and so (as explained below) Corollary 9.10 gives

$$(9.19) \quad \text{Aut}(A)/\overline{\text{Inn}}(A) \cong \mathbb{T} \rtimes \mathbb{Z}/2.$$

To do this concretely, let $u \in A$ denote the canonical unitary implementing the group action. Then $[u]_{\text{alg}}$ is a generator of the copy of $\mathbb{Z}$. We get a group monomorphism from $\mathbb{T}$ to $\overline{K}_1^{\text{alg}}(A)$ given by

$$(9.20) \quad z \mapsto [z1_A]_{\text{alg}},$$

giving a complementary copy of $\mathbb{T}$.

Since $K_*(A)$ is torsion-free, we have, by (2.33), that $K_i(A; \mathbb{Z}/n) \cong K_i(A) \otimes \mathbb{Z}/n = \mathbb{Z}/n$ for all n . We also see that for an automorphism ϕ of A , its action on K -theory determines its action on total K -theory; since ϕ also fixes $[1_A]_0$, $\underline{KT}_u(\phi)$ is entirely determined by $\overline{K}_1^{\text{alg}}(\phi)$.

An automorphism of $\overline{K}_1^{\text{alg}}(\phi)$ is compatible with the Thomsen map Th_A if and only if it acts as the identity on $\mathbb{T}$. Hence, by Corollary 9.10, $\text{Aut}(A)/\overline{\text{Inn}}(A)$ identifies with the group of automorphisms of $\overline{K}_1^{\text{alg}}(A) \cong \mathbb{T} \oplus \mathbb{Z}$ that fix $\mathbb{T}$. Each such automorphism is determined by its action on the generator $[u]_{\text{alg}}$ of the copy of $\mathbb{Z}$. This group of automorphisms is $\mathbb{T} \rtimes \mathbb{Z}/2$, where an element $z \in \mathbb{T}$ corresponds to the automorphism α_z taking $[u]_{\text{alg}}$ to $[zu]_{\text{alg}}$, and the generator of $\mathbb{Z}/2$ corresponds to the automorphism ω taking $[u]_{\text{alg}}$ to $-[u]_{\text{alg}}$.

We can find explicit representatives for α_z and ω as follows:

- (i) For $z \in \mathbb{T}$, let $\phi_z: A \rightarrow A$ denote the gauge action. In other words, ϕ_z fixes $\mathcal{Z}^{\otimes \mathbb{Z}}$ and

$$(9.21) \quad \phi_z(u) := zu.$$

Then ϕ_z induces the automorphism α_z .

- (ii) Let $\psi: A \rightarrow A$ be the $*$ -homomorphism determined by

$$(9.22) \quad \begin{aligned} \psi(u) &:= u^*, \\ \psi(\cdots \otimes a_{-1} \otimes a_0 \otimes a_1 \otimes \cdots) &:= \cdots \otimes a_1 \otimes a_0 \otimes a_{-1} \otimes \cdots, \end{aligned}$$

where $\cdots \otimes a_{-1} \otimes a_0 \otimes a_1 \otimes \cdots$ denotes an elementary tensor in $\mathcal{Z}^{\otimes \mathbb{Z}}$. Then we can see that $[\psi(u)]_{\text{alg}} = [u^*]_{\text{alg}} = -[u]_{\text{alg}}$, so ψ induces the automorphism ω .

Since the gauge actions act as the identity on $KT_u(A)$, their non-innerness is not detected by that invariant.

Interestingly, the automorphisms ϕ_z and ψ just described generate a copy of $\mathbb{T} \rtimes \mathbb{Z}/2$ inside $\text{Aut}(A)$ (that is, without taking the quotient by $\overline{\text{Inn}(A)}$). Consequently,

$$(9.23) \quad \text{Aut}(A) \cong \overline{\text{Inn}(A)} \rtimes (\mathbb{T} \rtimes \mathbb{Z}/2) \cong \overline{\text{Inn}(A)} \rtimes \text{Aut}(\underline{KT}_u(A)).$$

In particular, if G is a discrete group acting on $\underline{KT}_u(A)$, then the action lifts to an action on A .

Connes' deep analysis of automorphisms of factors from [45, 47, 49] was instrumental in his equivalence of injectivity and hyperfiniteness and subsequent classification results ([48, 46]). Among other results, Connes showed that there is a unique outer action of a finite cyclic group on $\mathcal{R}$ up to conjugacy [49, Theorem 5.1]. This was subsequently generalized by Jones to all finite groups ([155]) and by Ocneanu to all discrete amenable groups ([225]). These results instigated a vast body of work examining actions of amenable groups on classifiable C^* -algebras (see Izumi's survey [151]). A recent capstone result is JG and Szabó's dynamical Kirchberg-Phillips theorem ([119]), determining when outer actions of discrete amenable groups on Kirchberg algebras are cocycle conjugate in terms of equivariant KK -theory.

Note that any countable discrete group G acts by outer automorphisms on any classifiable C^* -algebra A : identify A with $A \otimes \mathcal{Z}^{\otimes G}$ and consider $\text{id}_A \otimes \sigma$, where $\sigma: G \curvearrowright \mathcal{Z}^{\otimes G}$ is the Bernoulli shift action obtained by permuting the tensor factors. However, it is far from clear what possible actions are induced at the level of invariants. In view of the classification of automorphisms modulo approximately inner automorphisms (Corollary 9.10), this problem takes the following form (see the introduction to [6]).

Question 9.12. Let A be a classifiable C^* -algebra (that is, one satisfying the hypotheses of Theorem A) and let G be a discrete amenable group. When does a group action $G \curvearrowright \underline{KT}_u(A)$ lift to an action $G \curvearrowright A$?

State-of-the-art results for Kirchberg algebras can be found in [166] (extending [7, 275]). In particular, Katsura shows that actions of $\mathbb{Z}/n$ on $KT_u(A)$ lift to actions on A when A is a unital Kirchberg algebra satisfying the UCT. However Question 9.12 is open for a general discrete amenable groups acting on Kirchberg algebras, and even for arbitrary finite groups. Likewise, general results are lacking in the stably finite setting. Indeed, even for a simple unital AF-algebra A , it appears to be open whether any action $\mathbb{Z}/n \curvearrowright \text{Aut}(K_0(A), K_0(A)_+, [1_A]_0)$ lifts to an action on A (see [10, Exercise 10.11.3], and [315]). Some constructions for simple AT algebras (akin to [7] in the Kirchberg setting) have recently been given in [316].

For a finite group G , a strategy was given by Barlak and Szabó in [6]. In the presence of enough UHF-absorption, they are able to lift a homomorphism $G \rightarrow \text{Aut}(A)/\overline{\text{Inn}(A)}$ to a group action by means of a model action of G on the UHF-algebra $M_{|G|^\infty}$. Their method produces actions with the *Rokhlin property*: a strong C^* -algebraic condition inspired by the non-commutative Rokhlin lemmas used by Connes and Ocneanu which goes back to [180, 139] and was extensively developed by Izumi ([149, 150]).

Definition 9.13 ([149, Definition 3.1]). For a finite group G , an action $G \curvearrowright A$ has the Rokhlin property if there is an equivariant *-homomorphism $C(G) \rightarrow A_\infty \cap A'$

Barak and Szabó used their technique to answer Question 9.12 for those $M_{|G|^\infty}$ -stable C^* -algebras where there is a classification of automorphisms up to approximate unitary equivalence ([6, Corollaries 2.8 and 2.13]). Combining their work with Theorem 9.8 yields the following result.

Corollary 9.14. *If A is a unital simple separable nuclear C^* -algebra satisfying the UCT, G is a finite group, and $A \otimes M_{|G|^\infty} \cong A$, then every group action $G \curvearrowright \underline{KT}_u(A)$ lifts to an action $G \curvearrowright A$ with the Rokhlin property.*

In particular, Question 9.12 has a positive answer when A is Q -stable and G is finite.

Proof. Fix an action $\Phi: G \curvearrowright \underline{KT}_u(A)$. By Theorem 9.8, there exist automorphisms $(\theta_g)_{g \in G}$ of A such that $\underline{KT}_u(\theta_g) = \Phi_g$ for each $g \in G$. By the uniqueness in Theorem 9.8, $\theta_g \circ \theta_h$ is approximately unitarily equivalent to θ_{gh} , and so [6, Theorem 2.3] provides the required Rokhlin action. $\square$

The Rokhlin property is a strong condition giving rise to a lot of rigidity. Indeed, in [150, Section 3], Izumi analyzes K -theoretic restrictions imposed when an action has the Rokhlin property. Moreover he shows how to use the Rokhlin property to run an Evans–Kishimoto type intertwining argument ([109]) to obtain uniqueness theorems whenever one has a classification of morphisms ([149, Theorem 3.5]). Combining Izumi’s result with Theorem 9.8 gives the uniqueness counterpart to Corollary 9.14. We thank Gabor Szabó for bringing this to our attention, through seminar talks in 2020.

Corollary 9.15. *Let A be a unital simple separable nuclear $\mathcal{Z}$ -stable C^* -algebra satisfying the UCT and let G be a finite group. Two actions $G \curvearrowright A$ with the Rokhlin property are conjugate via an automorphism of A if and only if the induced actions $G \curvearrowright \underline{KT}_u(A)$ are conjugate via an automorphism of $\underline{KT}_u(A)$.*

Proof. Suppose that $\alpha, \beta: G \curvearrowright A$ are Rokhlin actions such that there is an automorphism Θ of $\underline{KT}_u(A)$ with $\Theta \circ \underline{KT}_u(\alpha_g) \circ \Theta^{-1} = \underline{KT}_u(\beta_g)$ for all $g \in G$. By Theorem 9.8, there exists an automorphism θ of A inducing Θ . Another application of Theorem 9.8 gives that $\theta \circ \alpha_g \circ \theta^{-1} \approx_u \beta_g$ for all $g \in G$. The result then follows from [149, Theorem 3.5]. $\square$

APPENDIX A. TOTAL K -THEORY

In this appendix we collect some technical facts regarding total K -theory that are not easily available in the literature.

A.1. Two definitions of $\mathbb{Z}/n$ -coefficients. Our first objective is to give a direct proof of the following proposition which relates the two pictures of total K -theory we use in this paper. To the best of our knowledge, the only proof in the literature is found in the recent paper [161, Theorem 8.1], where it is derived from a Spanier–Whitehead duality theory developed therein.

Proposition A.1. *There exists a natural isomorphism between the two functors $K_0(\cdot; \mathbb{Z}/n) := K_1(\mathbb{I}_n \otimes \cdot)$ and $KK(\mathbb{I}_n, \cdot)$ (defined on the category of separable C^* -algebras).*

Before proving the proposition, we isolate some notation. Here, and throughout this section, all tensor products are spatial. For separable C^* -algebras C , D , and E there is a natural map

$$(A.1) \quad \mathcal{T}_E^{(C,D)}: KK(C, D) \rightarrow KK(E \otimes C, E \otimes D),$$

defined by

$$(A.2) \quad \mathcal{T}_E^{(C,D)}([\phi, \psi]_{KK(C,D)}) = [\text{id}_E \otimes \phi, \text{id}_E \otimes \psi]_{KK(E \otimes C, E \otimes D)}$$

for a Cuntz pair $(\phi, \psi): C \rightrightarrows \mathcal{M}(D \otimes \mathcal{K}) \triangleright D \otimes \mathcal{K}$. The following standard lemma is a KK -theoretic analogue of the fact that $\theta \otimes \text{id}_C$ and $\text{id}_E \otimes \phi$ commute for $*$ -homomorphisms $\theta: E \rightarrow F$ and $\phi: C \rightarrow D$. To see the analogy, consider the effect of the diagram (A.3) on $[\phi]_{KK(C,D)}$. This can be found in the Fredholm module picture as [10, Proposition 17.8.6], though we note that there is a typo in that statement (the equality there should be of maps $KK(A, B) \rightarrow KK(A \hat{\otimes} D_1, B \hat{\otimes} D_2)$).

Lemma A.2. *Let C , D , E and F be separable C^* -algebras and let $\theta: E \rightarrow F$ be a $*$ -homomorphism. Then the diagram*

$$(A.3) \quad \begin{array}{ccc} KK(C, D) & \xrightarrow{\mathcal{T}_E^{(C,D)}} & KK(E \otimes C, E \otimes D) \\ \downarrow \mathcal{T}_F^{(C,D)} & & \downarrow KK(E \otimes C, \theta \otimes \text{id}_D) \\ KK(F \otimes C, F \otimes D) & \xrightarrow{KK(\theta \otimes \text{id}_C, F \otimes D)} & KK(E \otimes C, F \otimes D) \end{array}$$

commutes.

Proof of Proposition A.1. It suffices to show $K_0(\mathbb{I}_n \otimes \cdot) \cong KK^1(\mathbb{I}_n, \cdot)$ as the result follows by taking suspensions. By Bott periodicity and the stability of KK -theory, it is enough to show

$$(A.4) \quad KK(\mathbb{C}, \mathbb{I}_n \otimes \cdot) \cong KK(\mathbb{I}_n, SM_n \otimes \cdot).$$

This is what we will show.

Recall that for $n \geq 2$,

$$(A.5) \quad \mathbb{I}_n := \{f \in C([0, 1], M_n) : f(0) \in \mathbb{C}1_{M_n}, f(1) = 0\}.$$

Let $\mu_n: SM_n \rightarrow \mathbb{I}_n$ and $\nu_n: \mathbb{I}_n \rightarrow \mathbb{C}$ denote the canonical $*$ -homomorphisms given by the inclusion and evaluation at 0, respectively. Define

$$(A.6) \quad \theta: \mathbb{I}_n \otimes \mathbb{I}_n \longrightarrow SM_n$$

to be the $*$ -homomorphism given on elementary tensors by

$$(A.7) \quad \theta(f \otimes g)(t) := \begin{cases} f(1 - 2t)\nu_n(g), & t \in [0, \frac{1}{2}]; \\ g(2t - 1)\nu_n(f), & t \in [\frac{1}{2}, 1] \end{cases}$$

for $f, g \in \mathbb{I}_n$ and $t \in [0, 1]$. Define a natural transformation

$$(A.8) \quad \Omega: KK(\mathbb{C}, \mathbb{I}_n \otimes \cdot) \rightarrow KK(\mathbb{I}_n, SM_n \otimes \cdot)$$

by

$$(A.9) \quad \Omega_D := KK(\mathbb{I}_n, \theta \otimes \text{id}_D) \circ \mathcal{T}_{\mathbb{I}_n}^{(\mathbb{C}, \mathbb{I}_n \otimes D)}$$

for a separable C^* -algebra D . We will show Ω_D is an isomorphism.

The composition $\theta \circ (\text{id}_{\mathbb{I}_n} \otimes \mu_n) : \mathbb{I}_n \otimes SM_n \rightarrow SM_n$ is given on elementary tensors by

$$(A.10) \quad \theta((\text{id}_{\mathbb{I}_n} \otimes \mu_n)(f \otimes g))(t) = \begin{cases} 0, & t \in [0, \frac{1}{2}); \\ g(2t-1)\nu_n(f), & t \in [\frac{1}{2}, 1] \end{cases}$$

for $f \in \mathbb{I}_n$, $g \in SM_n$, and $t \in [0, 1]$. It follows that $\theta \circ (\text{id}_{\mathbb{I}_n} \otimes \mu_n)$ is homotopic to

$$(A.11) \quad \nu_n \otimes \text{id}_{SM_n} : \mathbb{I}_n \otimes SM_n \longrightarrow \mathbb{C} \otimes SM_n = SM_n.$$

In particular, if D is a separable C^* -algebra, then homotopy invariance of KK -theory implies

$$(A.12) \quad KK(\mathbb{I}_n, (\theta \circ (\text{id}_{\mathbb{I}_n} \otimes \mu_n)) \otimes \text{id}_D) = KK(\mathbb{I}_n, \nu_n \otimes \text{id}_{SM_n} \otimes \text{id}_D).$$

For any separable C^* -algebra D , we get

$$(A.13) \quad \begin{aligned} & \Omega_D \circ KK(\mathbb{C}, \mu_n \otimes \text{id}_D) \\ & \stackrel{(A.9)}{=} KK(\mathbb{I}_n, \theta \otimes \text{id}_D) \circ \mathcal{T}_{\mathbb{I}_n}^{(\mathbb{C}, \mathbb{I}_n \otimes D)} \circ KK(\mathbb{C}, \mu_n \otimes \text{id}_D) \\ & = KK(\mathbb{I}_n, \theta \otimes \text{id}_D) \circ KK(\mathbb{I}_n, \text{id}_{\mathbb{I}_n} \otimes \mu_n \otimes \text{id}_D) \circ \mathcal{T}_{\mathbb{I}_n}^{(\mathbb{C}, SM_n \otimes D)} \\ & = KK(\mathbb{I}_n, (\theta \circ (\text{id}_{\mathbb{I}_n} \otimes \mu_n)) \otimes \text{id}_D) \circ \mathcal{T}_{\mathbb{I}_n}^{(\mathbb{C}, SM_n \otimes D)} \\ & \stackrel{(A.12)}{=} KK(\mathbb{I}_n, \nu_n \otimes \text{id}_{SM_n} \otimes \text{id}_D) \circ \mathcal{T}_{\mathbb{I}_n}^{(\mathbb{C}, SM_n \otimes D)} \\ & \stackrel{\text{Lem. A.2}}{=} KK(\text{id}_{\mathbb{C}} \otimes \nu_n, \mathbb{C} \otimes SM_n \otimes D) \circ \mathcal{T}_{\mathbb{C}}^{(\mathbb{C}, SM_n \otimes D)} \\ & = KK(\nu_n, SM_n \otimes D), \end{aligned}$$

using the naturality of $\mathcal{T}_{\mathbb{I}_n}^{(\mathbb{C}, \cdot)}$ for the second equality and the functoriality of $KK(\mathbb{I}_n, \cdot)$ for the third. It follows that the diagram

$$(A.14) \quad \begin{array}{ccc} KK(\mathbb{C}, SM_n \otimes D) & \xlongequal{\quad} & KK(\mathbb{C}, SM_n \otimes D) \\ \downarrow \times n & & \downarrow \times n \\ KK(\mathbb{C}, SM_n \otimes D) & \xlongequal{\quad} & KK(\mathbb{C}, SM_n \otimes D) \\ \downarrow KK(\mathbb{C}, \mu_n \otimes \text{id}_D) & & \downarrow KK(\nu_n \otimes \text{id}_{SM_n}) \\ KK(\mathbb{C}, \mathbb{I}_n \otimes D) & \xrightarrow{\Omega_D} & KK(\mathbb{I}_n, SM_n \otimes D) \end{array}$$

commutes.

Similarly, $\theta \circ (\mu_n \otimes \text{id}_{\mathbb{I}_n})$ is homotopic to

$$(A.15) \quad \sigma \otimes \nu_n : SM_n \otimes \mathbb{I}_n \rightarrow SM_n \otimes \mathbb{C} = SM_n,$$

where $\sigma : SM_n \rightarrow SM_n$ is the $*$ -homomorphism $\sigma(f)(t) := f(1-t)$ for $t \in [0, 1]$ and $f \in SM_n$. This yields

$$(A.16) \quad KK(SM_n, (\theta \circ (\mu_n \otimes \text{id}_{\mathbb{I}_n})) \otimes \text{id}_D) = KK(SM_n, \sigma \otimes \nu_n \otimes \text{id}_D)$$

for every separable C^* -algebra D . Further, note that $KK(SM_n, \sigma) = -\text{id}_{KK(SM_n, SM_n)}$: this follows, for example, by using the UCT to identify the ring $KK(SM_n, SM_n)$ with $\text{End}(K_1(SM_n))$ and noting that there is an isomorphism $K_1(SM_n) \rightarrow \mathbb{Z}$ given by sending a loop of unitaries to the winding number of the determinant. Bott periodicity then implies that tensoring with σ induces a natural isomorphism in KK -theory; i.e.,

$$(A.17) \quad \eta^{(C, D)} := KK(SM_n \otimes C, \sigma \otimes \text{id}_D) \circ \mathcal{T}_{SM_n}^{(C, D)}$$

is an isomorphism $KK(C, D) \xrightarrow{\cong} KK(SM_n \otimes C, SM_n \otimes D)$ for all C^* -algebras C and D .

Fix a separable C^* -algebra D . We compute

$$\begin{aligned}
 (A.18) \quad & KK(\mu_n, SM_n \otimes D) \circ \Omega_D \\
 & \stackrel{(A.9)}{=} KK(\mu_n, SM_n \otimes D) \circ KK(\mathbb{I}_n, \theta \otimes \text{id}_D) \circ \mathcal{T}_{\mathbb{I}_n}^{(\mathbb{C}, \mathbb{I}_n \otimes D)} \\
 & = KK(SM_n, \theta \otimes \text{id}_D) \circ KK(\mu_n, \mathbb{I}_n \otimes \mathbb{I}_n \otimes D) \circ \mathcal{T}_{\mathbb{I}_n}^{(\mathbb{C}, \mathbb{I}_n \otimes D)} \\
 & \stackrel{\text{Lem. A.2}}{=} KK(SM_n, \theta \otimes \text{id}_D) \circ KK(SM_n, \mu_n \otimes \text{id}_{\mathbb{I}_n} \otimes \text{id}_D) \circ \mathcal{T}_{SM_n}^{(\mathbb{C}, \mathbb{I}_n \otimes D)} \\
 & = KK(SM_n, (\theta \circ (\mu_n \otimes \text{id}_{\mathbb{I}_n})) \otimes \text{id}_D) \circ \mathcal{T}_{SM_n}^{(\mathbb{C}, \mathbb{I}_n \otimes D)} \\
 & \stackrel{(A.16)}{=} KK(SM_n, \sigma \otimes \nu_n \otimes \text{id}_D) \circ \mathcal{T}_{SM_n}^{(\mathbb{C}, \mathbb{I}_n \otimes D)} \\
 & = KK(SM_n, \sigma \otimes \text{id}_D) \circ KK(SM_n, \text{id}_{SM_n} \otimes \nu_n \otimes \text{id}_D) \circ \mathcal{T}_{SM_n}^{(\mathbb{C}, \mathbb{I}_n \otimes D)} \\
 & = KK(SM_n, \sigma \otimes \text{id}_D) \circ \mathcal{T}_{SM_n}^{(\mathbb{C}, D)} \circ KK(\mathbb{C}, \nu_n \otimes \text{id}_D) \\
 & \stackrel{(A.17)}{=} \eta^{(\mathbb{C}, D)} \circ KK(\mathbb{C}, \nu_n \otimes \text{id}_D),
 \end{aligned}$$

where the second equality follows from bifunctionality of $KK(\cdot, \cdot)$, the fourth and sixth equality follow from the functoriality of $KK(SM_n, \cdot)$, and the seventh equality follows from the naturality of $\mathcal{T}_{SM_n}^{(\mathbb{C}, \cdot)}$. This implies

$$\begin{array}{ccc}
 KK(\mathbb{C}, \mathbb{I}_n \otimes D) & \xrightarrow{\Omega_D} & KK(\mathbb{I}_n, SM_n \otimes D) \\
 \downarrow KK(\mathbb{C}, \nu_n \otimes \text{id}_D) & & \downarrow KK(\mu_n, SM_n \otimes D) \\
 (A.19) \quad KK(\mathbb{C}, D) & \xrightarrow[\eta^{(\mathbb{C}, D)}]{\cong} & KK(SM_n, SM_n \otimes D) \\
 \downarrow \times n & & \downarrow \times n \\
 KK(\mathbb{C}, D) & \xrightarrow[\eta^{(\mathbb{C}, D)}]{\cong} & KK(SM_n, SM_n \otimes D)
 \end{array}$$

commutes.

When concatenating the left columns of (A.14) and (A.19), this becomes part of the six-term exact sequence (2.28) for the Bockstein maps, and thus this concatenation is exact. Similarly, concatenating the right columns gives an exact sequence, using the Puppe exact sequence in $KK(\cdot, SM_n \otimes D)$ applied to the diagonal inclusion $\mathbb{C} \rightarrow M_n$ (see [11, Theorem 19.4.3]). This follows by considering the six-term exact sequences in $KK(\cdot, SM_n \otimes D)$ corresponding to the rows of

$$\begin{array}{ccccccc}
 0 & \longrightarrow & SM_n & \longrightarrow & \mathbb{I}_n & \longrightarrow & \mathbb{C} \longrightarrow 0 \\
 & & \parallel & & \downarrow & & \downarrow \\
 (A.20) \quad 0 & \longrightarrow & SM_n & \longrightarrow & CM_n & \longrightarrow & M_n \longrightarrow 0.
 \end{array}$$

Using that the boundary map in the second row is an isomorphism, since the cone is contractible, we may identify the boundary map in the top row with multiplication by n .

By the five lemma, it follows that Ω_D is an isomorphism. $\square$

A.2. Two definitions of total K -theory. In this section, we prove Proposition 8.4. This amounts to showing that for any two sets of Bockstein operations,

the corresponding maps are integer multiples of each other. To that end, we consider an arbitrary collection of natural transformations

$$(A.21) \quad K_i(\cdot) \xrightarrow{\mu_i'^{(n)}} K_i(\cdot; \mathbb{Z}/n) \xrightarrow{\nu_i'^{(n)}} K_{1-i}(\cdot)$$

and

$$(A.22) \quad K_i(\cdot; \mathbb{Z}/n) \xrightarrow{\kappa_i'^{(nm,n)}} K_i(\cdot; \mathbb{Z}/nm) \xrightarrow{\kappa_i'^{(n,nm)}} K_i(\cdot; \mathbb{Z}/n)$$

indexed by $i = 0, 1$ and $n, m \geq 2$.

Remark A.3. If natural transformations as above are only defined on the functors restricted to the category of separable C^* -algebras or σ -unital C^* -algebras, then they can canonically be extended to the functors defined on the category of all C^* -algebras since the functors K_i and $K_i(\cdot; \mathbb{Z}/n)$ preserve inductive limits. Hence we may assume, for example, that the Bockstein operations in [69] (which are only defined there for σ -unital C^* -algebras) are defined on all C^* -algebras.

Lemma A.4. *Given a collection of natural transformations as in (A.21) and (A.22), suppose that $k\mu_i'^{(n)}$, $k\nu_i'^{(n)}$, $k\kappa_i'^{(nm,n)}$, and $k\kappa_i'^{(n,nm)}$ are non-zero for all $i = 0, 1$, $n, m \geq 2$, and $k = 1, \dots, n-1$.*

- (i) *Each map $\mu_i'^{(n)}$, $\nu_i'^{(n)}$, $\kappa_i'^{(nm,n)}$, and $\kappa_i'^{(n,nm)}$ and its corresponding Bockstein map $\mu_i^{(n)}$, $\nu_i^{(n)}$, $\kappa_i^{(nm,n)}$, and $\kappa_i^{(n,nm)}$ are integer multiples of each other.*
- (ii) *For any C^* -algebras D and E , a collection of homomorphisms $\alpha_i: K_i(D) \rightarrow K_i(E)$ and $\alpha_i^{(n)}: K_i(D; \mathbb{Z}/n) \rightarrow K_i(E; \mathbb{Z}/n)$ forms a Λ -homomorphism if and only if these homomorphisms intertwine all the natural transformations $\mu_i'^{(n)}$, $\nu_i'^{(n)}$, $\kappa_i'^{(nm,n)}$, and $\kappa_i'^{(n,nm)}$.*
- (iii) *These natural transformations induce natural exact sequences as in (2.28) and (2.30).*

Proof. Part (ii) follows from (i). For example, to show that if α_0 and $\alpha_1^{(n)}$ intertwine $\nu_1'^{(n)}$ then they intertwine $\nu_1^{(n)}$, we start with $\alpha_0 \circ \nu_{1,D}'^{(n)} = \nu_{1,E}'^{(n)} \circ \alpha_1^{(n)}$, and, from (i), obtain some $j \in \mathbb{Z}$ such that $\nu_1^{(n)} = j\nu_1'^{(n)}$. Then

$$(A.23) \quad \alpha_0 \circ \nu_{1,D}^{(n)} = j\alpha_0 \circ \nu_{1,D}'^{(n)} = j\nu_{1,E}'^{(n)} \circ \alpha_1^{(n)} = \nu_{1,E}^{(n)} \circ \alpha_1^{(n)}.$$

The other cases are all similar.

Similarly, part (iii) follows from (i). For example, as above, if $\nu_1^{(n)} = j\nu_1'^{(n)}$ for some $j \in \mathbb{Z}$, then for any C^* -algebra D we get $\ker \nu_{i,D}'^{(n)} \subseteq \ker \nu_{i,D}^{(n)}$ and $\text{im } \nu_{i,D}'^{(n)} \subseteq \text{im } \nu_{i,D}^{(n)}$. By symmetry, we obtain equalities. The same type of argument works for the other Bockstein operations.

Now, to prove (i), we focus on showing that $\nu_1^{(n)}$ is an integer multiple of $\nu_1'^{(n)}$; a similar argument proves the other direction. Indeed, the hypothesis holds for $\nu_1^{(n)}$ in place of $\nu_1'^{(n)}$ by noting that

$$\nu_{1, S\mathbb{I}_n}^{(n)}: K_1(S\mathbb{I}_n; \mathbb{Z}/n) \rightarrow K_0(S\mathbb{I}_n) \cong \mathbb{Z}/n$$

is an isomorphism by exactness of (2.28), using $K_1(\mathbb{I}_n) \cong \mathbb{Z}/n$ and $K_0(\mathbb{I}_n) = 0$.

By continuity of the functors K_0 and $K_1(\cdot; \mathbb{Z}/n)$, it suffices to find $j \in \mathbb{N}$ such that $\nu_{1,D}^{(n)} = j\nu_{1,D}'^{(n)}$ when D is a separable C^* -algebra. When restricted to this setting,

by associativity of the Kasparov product, the functors K_0 and $K_1(\cdot; \mathbb{Z}/n)$ factorize through the KK -category, in which we have natural isomorphisms $K_1(\cdot; \mathbb{Z}/n) \cong KK(S\mathbb{I}_n, \cdot)$ given in Proposition A.1. (Recall that the KK -category has all separable C^* -algebras as objects and $KK(D, E)$ as the set of morphisms from D to E .) For any separable C^* -algebra D , the functor $KK(D, \cdot)$ is a Hom-functor in the KK -category, so by the Yoneda lemma, the group of natural transformations from $KK(D, \cdot)$ to F is naturally isomorphic to $F(D)$ for any additive functor F from the KK -category to the category of abelian groups. Hence the group of natural transformations from $K_1(\cdot; \mathbb{Z}/n)$ to K_0 is naturally isomorphic to $K_0(S\mathbb{I}_n) \cong \mathbb{Z}/n$. In particular, a natural transformation $\eta: K_1(\cdot; \mathbb{Z}/n) \rightarrow K_0$ generates the entire group of natural transformation between these functors if and only if $k\eta \neq 0$ for every $k = 1, \dots, n-1$. This is true for $\nu_1'^{(n)}$ by assumption. Hence $\nu_1'^{(n)}$ is an integer multiple of $\nu_1'^{(n)}$. $\square$

Proof of Proposition 8.4. After identifying $KK^i(\mathbb{I}_n, \cdot)$ with $K_i(\cdot; \mathbb{Z}/n)$ via natural isomorphisms (which exist by Proposition A.1), the Bockstein operations in [69] induce natural transformations satisfying the criterion of Lemma A.4. $\square$

A.3. A split exact sequence relating ordinary and total K -theory. We end this appendix by showing how to use Bökigheimer's work ([19, 20]) to prove Proposition 2.17 by means of an unnatural splitting of an exact sequence relating total K -theory with ordinary K -theory. To ease notation, in this section, we use $K_i(D)/nK_i(D)$ rather than the isomorphic group $K_i(D) \otimes \mathbb{Z}/n$ as it allows us to more easily describe maps between such groups.

Recall that for an abelian group G and $n \geq 2$, we identify $\text{Tor}(G, \mathbb{Z}/n)$ with the n -torsion group, $\{g \in G : ng = 0\}$. In particular, for any m, n , we have a natural inclusion $\text{Tor}(G, \mathbb{Z}/n) \rightarrow \text{Tor}(G, \mathbb{Z}/mn)$, as well as a natural map $\text{Tor}(G, \mathbb{Z}/mn) \rightarrow \text{Tor}(G, \mathbb{Z}/n)$ given by multiplication by m .

Let $n \geq 2$, and let $n = p_1^{r_1} \cdots p_k^{r_k}$ be its prime factorization. For any C^* -algebra D , $i = 0, 1$, and $j = 1, \dots, k$, using commutativity of (2.30), we get a commuting diagram

$$(A.24) \quad \begin{array}{ccccc} K_i(D)/p_j^{r_j} K_i(D) & \xrightarrow{\bar{\mu}_{i,D}^{(p_j^{r_j})}} & K_i(D; \mathbb{Z}/p_j^{r_j}) & \xrightarrow{\bar{\nu}_{i,D}^{(p_j^{r_j})}} & \text{Tor}(K_{1-i}(D), \mathbb{Z}/p_j^{r_j}) \\ \downarrow \times n/p_j^{r_j} & & \downarrow \kappa_{i,D}^{(n, p_j^{r_j})} & & \downarrow \\ K_i(D)/nK_i(D) & \xrightarrow{\bar{\mu}_{i,D}^{(n)}} & K_i(D; \mathbb{Z}/n) & \xrightarrow{\bar{\nu}_{i,D}^{(n)}} & \text{Tor}(K_{1-i}(D), \mathbb{Z}/n). \end{array}$$

where the rows are as in (2.33). Taking the direct sum of the top row for $j = 1, \dots, k$, we get a commutative diagram

$$(A.25) \quad \begin{array}{ccccc} \bigoplus_j K_i(D)/p_j^{r_j} K_i(D) & \xrightarrow{\bigoplus_j \bar{\mu}_{i,D}^{(p_j^{r_j})}} & \bigoplus_j K_i(D; \mathbb{Z}/p_j^{r_j}) & \xrightarrow{\bigoplus_j \bar{\nu}_{i,D}^{(p_j^{r_j})}} & \bigoplus_j \text{Tor}(K_{1-i}(D), \mathbb{Z}/p_j^{r_j}) \\ \downarrow \cong & & \downarrow \cong \bigoplus_j \kappa_{i,D}^{(n, p_j^{r_j})} & & \downarrow \cong \\ K_i(D)/nK_i(D) & \xrightarrow{\bar{\mu}_{i,D}^{(n)}} & K_i(D; \mathbb{Z}/n) & \xrightarrow{\bar{\nu}_{i,D}^{(n)}} & \text{Tor}(K_{1-i}(D), \mathbb{Z}/n), \end{array}$$

where the left and right vertical maps are isomorphisms by the Chinese remainder theorem ($\mathbb{Z}/n \cong \bigoplus_j \mathbb{Z}/p_j^{r_j}$) together with additivity of $\otimes$ and Tor , and the middle vertical map is therefore an isomorphism by the five lemma.

Lemma A.5 (Bödigheimer). *For any C^* -algebra D there are (unnatural) homomorphisms*

$$(A.26) \quad \text{Tor}(K_{1-i}(D), \mathbb{Z}/n) \xrightarrow{s_{i,D}^{(n)}} K_i(D; \mathbb{Z}/n) \xrightarrow{\iota_{i,D}^{(n)}} K_i(D)/nK_i(D)$$

for $n \geq 2$ such that

$$(A.27) \quad \bar{\nu}_{i,D}^{(n)} \circ s_{i,D}^{(n)} = \text{id}_{\text{Tor}(K_{1-i}(D), \mathbb{Z}/n)}, \quad \iota_{i,D}^{(n)} \circ \bar{\mu}_{i,D}^{(n)} = \text{id}_{K_i(D)/nK_i(D)},$$

and the following diagram commutes:

$$(A.28) \quad \begin{array}{ccccc} \text{Tor}(K_{1-i}(D), \mathbb{Z}/n) & \xrightarrow{s_{i,D}^{(n)}} & K_i(D; \mathbb{Z}/n) & \xrightarrow{\iota_{i,D}^{(n)}} & K_i(D)/nK_i(D) \\ \downarrow & & \downarrow \kappa_{i,D}^{(nm,n)} & & \downarrow \times m \\ \text{Tor}(K_{1-i}(D), \mathbb{Z}/nm) & \xrightarrow{s_{i,D}^{(nm)}} & K_i(D; \mathbb{Z}/nm) & \xrightarrow{\iota_{i,D}^{(nm)}} & K_i(D)/nmK_i(D) \\ \downarrow \times m & & \downarrow \kappa_{i,D}^{(n,nm)} & & \downarrow \\ \text{Tor}(K_{1-i}(D), \mathbb{Z}/n) & \xrightarrow{s_{i,D}^{(n)}} & K_i(D; \mathbb{Z}/n) & \xrightarrow{\iota_{i,D}^{(n)}} & K_i(D)/nK_i(D). \end{array}$$

Proof. We start with the maps $s_{i,D}^{(n)}$. Recall that for $n, m \geq 2$, the maps

$$(A.29) \quad \begin{aligned} \kappa_{*,D}^{(nm,n)} : K_*(D; \mathbb{Z}/n) &\longrightarrow K_*(D; \mathbb{Z}/nm) \text{ and} \\ \kappa_{*,D}^{(n,nm)} : K_*(D; \mathbb{Z}/nm) &\longrightarrow K_*(D; \mathbb{Z}/n) \end{aligned}$$

are induced by the canonical inclusions $\mathbb{I}_n \rightarrow \mathbb{I}_{nm}$ and $\mathbb{I}_{nm} \rightarrow \mathbb{I}_n \otimes M_m$, respectively. The composition $\mathbb{I}_n \rightarrow \mathbb{I}_{nm} \rightarrow \mathbb{I}_n \otimes M_m$ is the diagonal embedding, and so $\kappa_{*,D}^{(n,nm)} \circ \kappa_{*,D}^{(nm,n)}$ is multiplication by m . The composition $\mathbb{I}_{nm} \rightarrow \mathbb{I}_n \otimes M_m \rightarrow \mathbb{I}_{nm} \otimes M_m$ is homotopic to the diagonal embedding and thus $\kappa_{*,D}^{(nm,n)} \circ \kappa_{*,D}^{(n,nm)}$ also induces multiplication by m . To see this, note that this composition takes values in $\mathbb{I}_{nm^2}$. Every unitary in M_{nm^2} induces by conjugation an automorphism on $\mathbb{I}_{nm^2}$ homotopic to the identity, so by composing the previous composition with a unitary flipping the two copies of M_m , we obtain the desired homotopy.

Further, note that $nx = 0$ for all $x \in K_*(D; \mathbb{Z}/n)$. It follows that for a prime p , the diagram (A.30)

$$\begin{array}{ccccc}
 \vdots & & \vdots & & \vdots \\
 \updownarrow & & \updownarrow & & \updownarrow \\
 K_i(D)/p^k K_i(D) & \xrightarrow{\bar{\mu}_{i,D}^{(p^k)}} & K_i(D; \mathbb{Z}/p^k) & \xrightarrow{\bar{\nu}_{i,D}^{(p^k)}} & \text{Tor}(K_{1-i}(D), \mathbb{Z}/p^k) \\
 \uparrow \times p & & \uparrow \kappa_{i,D}^{(p^k, p^{k+1})} & & \uparrow \times p \\
 K_i(D)/p^{k+1} K_i(D) & \xrightarrow{\bar{\mu}_{i,D}^{(p^{k+1})}} & K_i(D; \mathbb{Z}/p^{k+1}) & \xrightarrow{\bar{\nu}_{i,D}^{(p^{k+1})}} & \text{Tor}(K_{1-i}(D), \mathbb{Z}/p^{k+1}) \\
 \updownarrow & & \updownarrow & & \updownarrow \\
 \vdots & & \vdots & & \vdots
 \end{array}$$

satisfies the hypotheses of the (purely algebraic) lemma in [20, Section 2] for $i = 0, 1$ (note that $G[n] := \text{Tor}(G, \mathbb{Z}/n)$ in [20]), where the vertical maps on the left and right are induced by the canonical maps between $\mathbb{Z}/p^k$ and $\mathbb{Z}/p^{k+1}$. This lemma implies that the maps $s_{i,D}^{(n)}$ exist satisfying (A.27) and such that (A.28) commutes when n and m above are powers of the same prime p ; we therefore fix such $s_{i,D}^{(p^r)}$ for all primes p and $r \geq 1$.

We now use the identifications in the right-hand square of (A.25) to define maps $s_{i,D}^{(n)}$ for general n . For $n \geq 2$ arbitrary, let $n = p_1^{r_1} \cdots p_k^{r_k}$ be its prime factorization. Then we define $s_{i,D}^{(n)}: \text{Tor}(K_{1-i}(D); \mathbb{Z}/n) \rightarrow K_i(D; \mathbb{Z}/n)$ such that the following commutes

$$\begin{array}{ccc}
 \bigoplus_j \text{Tor}(K_{1-i}(D), \mathbb{Z}/p_j^{r_j}) & \xrightarrow{\bigoplus_j s_{i,D}^{(p_j^{r_j})}} & \bigoplus_j K_i(D; \mathbb{Z}/p_j^{r_j}) \\
 \downarrow \cong & & \downarrow \cong \\
 \text{Tor}(K_{1-i}(D), \mathbb{Z}/n) & \xrightarrow{s_{i,D}^{(n)}} & K_i(D; \mathbb{Z}/n)
 \end{array}
 \quad (A.31)$$

Since the top map, $\bigoplus_j s_{i,D}^{(p_j^{r_j})}$, is a right splitting for the top row of (A.25), it follows that the $s_{i,D}^{(n)}$ is a right splitting for the bottom row, i.e., $\bar{\nu}_{i,D}^{(n)} \circ s_{i,D}^{(n)} = \text{id}_{\text{Tor}(K_{1-i}(D), \mathbb{Z}/n)}$.

To check commutativity of the left side of (A.28), we consider $m, n \geq 2$. Let $n = p_1^{r_1} \cdots p_k^{r_k}$ and $m = p_1^{t_1} \cdots p_k^{t_k}$ be their prime factorizations, allowing some exponents to possibly be zero. For convenience, we take $K_i(D; \mathbb{Z}/1)$ to be 0 by convention; then (A.25) and the definition of $s_{i,D}^{(n)}$ from (A.31) are still correct with some r_j equal to 0.

Since $\kappa_{i,D}^{(nm,p_j^{r_j+t_j})} \circ \kappa_{i,D}^{(p_j^{r_j+t_j},p_j^{r_j})} = \kappa_{i,D}^{(nm,p_j^{r_j})} = \kappa_{i,D}^{(nm,n)} \circ \kappa_{i,D}^{(n,p_j^{r_j})}$ (by the definition of these maps), the following diagram commutes:

$$(A.32) \quad \begin{array}{ccc} \bigoplus_j K_i(D; \mathbb{Z}/p_j^{r_j}) & \xrightarrow{\bigoplus_j \kappa_{i,D}^{(p_j^{r_j+t_j}, p_j^{r_j})}} & \bigoplus_j K_i(D; \mathbb{Z}/p_j^{r_j+t_j}) \\ \cong \downarrow \bigoplus_j \kappa_{i,D}^{(n,p_j^{r_j})} & & \cong \downarrow \bigoplus_j \kappa_{i,D}^{(nm,p_j^{r_j+t_j})} \\ K_i(D; \mathbb{Z}/n) & \xrightarrow{\kappa_{i,D}^{(nm,n)}} & K_i(D; \mathbb{Z}/nm). \end{array}$$

From this and (A.31), one can deduce commutativity of the top-left square of (A.28) for general n and m from Bödighheimer's commutativity of this square for the case that n and m are powers of the same prime.

Fix $j \in \{1, \dots, k\}$. Then we have

$$(A.33) \quad \kappa_{i,D}^{(n,nm)} \circ \kappa_{i,D}^{(nm,p_j^{r_j+t_j})} = \frac{m}{p_j^{t_j}} \kappa_{i,D}^{(n,p_j^{r_j})} \circ \kappa_{i,D}^{(p_j^{r_j}, p_j^{r_j+t_j})}$$

since the left side comes from the inclusion $\mathbb{I}_{p_j^{r_j+t_j}} \rightarrow \mathbb{I}_n \otimes M_m$ given by $f \mapsto f \otimes 1_{M_{mn/p_j^{r_j+t_j}}}$, while the right side (ignoring the factor $m/p_j^{t_j}$) comes from the inclusion $\mathbb{I}_{p_j^{r_j+t_j}} \rightarrow \mathbb{I}_n \otimes M_{p_j^{t_j}}$ given by $f \mapsto f \otimes 1_{M_{n/p_j^{r_j}}}$.

Therefore, for an element x of the direct summand $\text{Tor}(K_i(D), \mathbb{Z}/p_j^{r_j+t_j})$ of $\text{Tor}(K_i(D), \mathbb{Z}/nm)$, we have

$$(A.34) \quad \begin{aligned} \kappa_{i,D}^{(n,nm)}(s_{i,D}^{(nm)}(x)) &\stackrel{(A.31)}{=} \kappa_{i,D}^{(n,nm)}\left(\kappa_{i,D}^{(nm,p_j^{r_j+t_j})}(s_{i,D}^{(p_j^{r_j+t_j})}(x))\right) \\ &\stackrel{(A.33)}{=} \frac{m}{p_j^{t_j}} \left(\kappa_{i,D}^{(n,p_j^{r_j})} \left(\kappa_{i,D}^{(p_j^{r_j}, p_j^{r_j+t_j})} (s_{i,D}^{(p_j^{r_j+t_j})}(x)) \right) \right) \\ &= \frac{m}{p_j^{t_j}} \kappa_{i,D}^{(n,p_j^{r_j})} \left(s_{i,D}^{(p_j^{r_j})}(p_j^{t_j} x) \right) \\ &= \kappa_{i,D}^{(n,p_j^{r_j})} \left(s_{i,D}^{(p_j^{r_j})}(mx) \right) \\ &\stackrel{(A.31)}{=} s_{i,D}^{(n)}(mx), \end{aligned}$$

where the third equation comes from Bödighheimer's commutativity of the bottom-left square of (A.28) in the case that n and m are powers of p_j . Since

$$(A.35) \quad \text{Tor}(K_i(D), \mathbb{Z}/nm) = \bigoplus_j \text{Tor}(K_i(D), \mathbb{Z}/p_j^{r_j+t_j})$$

and the above shows commutativity of the bottom-left square of (A.28) on each of these summands, it follows that the bottom-left square of (A.28) commutes.

The splitting lemma provides the maps $\iota_{i,D}^{(n)}$. A version that suffices for our purposes says that given a commutative diagram of abelian groups

$$(A.36) \quad \begin{array}{ccccccc} 0 & \longrightarrow & G_1 & \longrightarrow & H_1 & \longrightarrow & L_1 \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & G_2 & \longrightarrow & H_2 & \longrightarrow & L_2 \longrightarrow 0 \end{array}$$

with exact rows, if there is a splitting for the quotient maps in each row that commutes with the downward maps, then the corresponding splittings for the inclusion maps also commute with the downward maps.

This last claim follows from the construction of the splitting for the inclusion map in the proof of the splitting lemma. In more detail, if $0 \rightarrow G \xrightarrow{i} H \xrightarrow{p} L \rightarrow 0$ is an exact sequence of abelian groups and $q: L \rightarrow H$ is a splitting of p , then a splitting $j: H \rightarrow G$ of i is given explicitly as follows: the map $\text{id}_H - q \circ p: H \rightarrow H$ composed with p vanishes. Since $\ker p = \text{im } i$ and i is injective, there is a unique homomorphism $j: H \rightarrow G$ such that $i \circ j = \text{id}_H - q \circ p$. Since i is injective and $i \circ j \circ i = i$, it follows that $j \circ i = \text{id}_G$. $\square$

Proof of Proposition 2.17. Fix homomorphisms $\iota_{i,A}^{(n)}$ and $s_{i,B}^{(n)}$ as in Lemma A.5. Given $\alpha_*: K_*(A) \rightarrow K_*(B)$ we define group homomorphisms $\alpha_i^{(n)}: K_i(A; \mathbb{Z}/n) \rightarrow K_i(B; \mathbb{Z}/n)$ by

$$(A.37) \quad \alpha_i^{(n)} := \bar{\mu}_{i,B}^{(n)} \circ (\alpha_i \otimes \text{id}_{\mathbb{Z}/n}) \circ \iota_{i,A}^{(n)} + s_{i,B}^{(n)} \circ \text{Tor}(\alpha_{1-i}, \mathbb{Z}/n) \circ \bar{\nu}_{i,A}^{(n)}.$$

It follows from the construction that the collection of maps α_i and $\alpha_i^{(n)}$ for $i = 0, 1$ and $n \geq 2$ intertwines the natural transformations μ_i and ν_i , and by Lemma A.5, they also intertwine the κ_i . $\square$

APPENDIX B. KK -THEORY OUTSIDE THE σ -UNITAL SETTING

In this appendix we give some details regarding the inductive limit description of KK -theory in the case when the second variable is not σ -unital, extending some standard results from the σ -unital case to the general framework.

B.1. Cuntz's picture of KK -theory. We will make use of Cuntz's picture of KK -theory from [59]. This (like its precursor in [57]) works smoothly for non- σ -unital algebras in the second variable and allows for a very clean description of $KK(A, \theta)$ for a *-homomorphism θ .

Definition B.1 (Cuntz, [59]). Let A and I be C^* -algebras with A separable. Let qA be the kernel of the map $\text{id}_A * \text{id}_A: A * A \rightarrow A$ (where $*$ denotes the full free product). Define

$$(B.1) \quad KK_c(A, I) := [qA, I \otimes \mathcal{K}],$$

(i.e., homotopy classes of *-homomorphisms from qA to $I \otimes \mathcal{K}$). Write $[\rho]_{KK_c(A, I)} \in KK_c(A, I)$ for the homotopy class of $\rho: qA \rightarrow I \otimes \mathcal{K}$.

Given another C^* -algebra J and a *-homomorphism $\theta: I \rightarrow J$, the induced map $KK_c(A, \theta): KK_c(A, I) \rightarrow KK_c(A, J)$ is given by

$$(B.2) \quad KK_c(A, \theta)([\rho]_{KK_c(A, I)}) := [(\theta \otimes \text{id}_{\mathcal{K}}) \circ \rho]_{KK_c(A, J)}.$$

The above definition of $KK_c(A, \theta)$ makes $KK_c(A, \cdot)$ a covariant functor, and indeed $KK_c(\cdot, \cdot)$ is a bifunctor (though we do not need an explicit description of functoriality in the first variable). Cuntz used Higson's characterization of KK -theory for separable C^* -algebras ([140]) to show that $KK_c(\cdot, \cdot)$ gives another realization of KK -theory in the separable setting. This was extended to the case when the second variable is σ -unital in [152], where the isomorphism between the Cuntz picture and Cuntz-Thomsen picture is described explicitly (the proof reveals that the isomorphism is natural) as in the following proposition. To set this up, note

that given a Cuntz pair $(\phi, \psi): A \rightrightarrows E \triangleright I \otimes \mathcal{K}$, we have $(\phi * \psi)(qA) \subseteq I \otimes \mathcal{K}$ (see [152, Lemma 5.14]); we write

$$(B.3) \quad [\phi, \psi]_{KK_c(A, I)} := [(\phi * \psi)|_{qA}^{I \otimes \mathcal{K}}]_{KK_c(A, I)}.$$

Proposition B.2 (cf. [152, Theorem 5.2.4]). *Let A and I be C^* -algebras with A separable and I σ -unital. Then $[\phi, \psi]_{KK(A, I)} \mapsto [\phi, \psi]_{KK_c(A, I)}$ defines an isomorphism $KK(A, I) \cong KK_c(A, I)$ that is natural in both variables.*

The clean description of $KK_c(A, \theta)$ above facilitates the following computation. This calculation was performed in Kasparov's original Fredholm module picture, in the case of $I_2 = E_1 = E_2$ separable, in [268, Proposition 2.1]. Comparing the computations demonstrates the power of Cuntz's picture.

Proposition B.3. *Let A be a separable C^* -algebra and let $\theta: I_1 \rightarrow I_2$ be a $*$ -homomorphism between C^* -algebras. Suppose that E_i is a C^* -algebra containing I_i as an ideal for $i = 1, 2$ and that θ extends to a $*$ -homomorphism $\bar{\theta}: E_1 \rightarrow E_2$. Given a Cuntz pair $(\phi, \psi): A \rightrightarrows E_1 \triangleright I_1$, we have*

$$(B.4) \quad KK_c(A, \theta)([\phi, \psi]_{KK_c(A, I_1)}) = [\bar{\theta} \circ \phi, \bar{\theta} \circ \psi]_{KK_c(A, I_2)}.$$

In particular, if I_1 and I_2 are σ -unital, then

$$(B.5) \quad KK(A, \theta)([\phi, \psi]_{KK(A, I_1)}) = [\bar{\theta} \circ \phi, \bar{\theta} \circ \psi]_{KK(A, I_2)}.$$

Proof. To shorten notation, we may assume (by stabilizing everything) that I_1, I_2 are stable. One may verify (using [152, Lemma 5.1.2]) that $((\bar{\theta} \circ \phi) * (\bar{\theta} \circ \psi))|_{qA} = \bar{\theta} \circ (\phi * \psi)|_{qA}$. We use this for the third equality below:

$$(B.6) \quad \begin{aligned} & KK_c(A, \theta)([\phi, \psi]_{KK_c(A, I_1)}) \\ & \stackrel{(B.3)}{=} KK_c(A, \theta)([(\phi * \psi)|_{qA}^{I_1}]_{KK_c(A, I_1)}) \\ & \stackrel{(B.2)}{=} [\theta \circ ((\phi * \psi)|_{qA}^{I_1})]_{KK_c(A, I_2)} \\ & = [((\bar{\theta} \circ \phi) * (\bar{\theta} \circ \psi))|_{qA}^{I_2}]_{KK_c(A, I_2)} \\ & = [\bar{\theta} \circ \phi, \bar{\theta} \circ \psi]_{KK_c(A, I_2)}. \end{aligned}$$

The last line of the proposition follows from Proposition B.2 □

B.2. The inductive limit picture of KK -theory. Recall that in Definition 5.4 for C^* -algebras A and I with A separable, we defined

$$(B.7) \quad KK(A, I) := \varinjlim_{I_0 \text{ sep.}} KK(A, I_0),$$

with the limit taken over all separable C^* -subalgebras of I , ordered by inclusion. We now justify the interpretation of the class of an (A, I) -Cuntz pair in this picture.

Proposition B.4. *Let A and I be C^* -algebras with A separable. Given a Cuntz pair $(\phi, \psi): A \rightrightarrows E \triangleright I$, the class $[\phi, \psi]_{KK(A, I)}$ in Definition 5.4 is well-defined. That is,*

- (i) *there exist separable C^* -subalgebras $E_0 \subseteq E$ and $I_0 \subseteq I$ such that $I_0 \triangleleft E_0$, $\phi(A) \cup \psi(A) \subseteq E_0$, and $(\phi - \psi)(A) \subseteq I_0$, and*
- (ii) *the class of $[\phi|^{E_0}, \psi|^{E_0}]_{KK(A, I_0)}$ in $\varinjlim_{I_0 \text{ sep.}} KK(A, I_0)$ does not depend on the choice of I_0 and E_0 in (i).*

Proof. (i): Let E_0 denote the C^* -algebra generated by $\phi(A) \cup \psi(A)$ and let I_0 be the ideal generated by $(\phi - \psi)(A)$ in E_0 .

(ii): Suppose that $I_1 \subseteq I$ and $E_1 \subseteq E$ are separable C^* -subalgebras that also satisfy the conditions in (i). Let E_2 be the C^* -algebra generated by $E_0 \cup E_1$ and let I_2 be the ideal generated by $I_0 \cup I_1$ in E_2 . These are separable C^* -subalgebras of E and I , respectively.

Using Proposition B.3 twice (with θ equal to the inclusions of I_0 and I_1 into I_2 in turn), we have

$$(B.8) \quad \begin{aligned} KK(A, \iota_{I_0 \subseteq I_2})([\phi|^{E_0}, \psi|^{E_0}]_{KK(A, I_0)}) &= [\phi|^{E_2}, \psi|^{E_2}]_{KK(A, I_2)} \\ &= KK(A, \iota_{I_1 \subseteq I_2})([\phi|^{E_1}, \psi|^{E_1}]_{KK(A, I_1)}). \end{aligned}$$

Therefore, $[\phi|^{E_0}, \psi|^{E_0}]_{KK(A, I_0)}$ and $[\phi|^{E_1}, \psi|^{E_1}]_{KK(A, I_1)}$ represent the same class in $\varinjlim_{I_0 \text{ sep.}} KK(A, I_0)$. $\square$

Proposition B.5. $KK(\cdot, \cdot)$ (as in Definition 5.4) is a bifunctor.

Proof. Functoriality in the first variable is entirely straightforward.

For the second variable, given C^* -algebras A , I , and J such that A is separable, and a $*$ -homomorphism $\theta: I \rightarrow J$, we must define a group homomorphism $KK(A, \theta): KK(A, I) \rightarrow KK(A, J)$. First, fix a separable C^* -subalgebra $I_0 \subseteq I$; then there exists a separable C^* -subalgebra $J_0 \subseteq J$ such that $\theta(I_0) \subseteq J_0$. Moreover, by functoriality of $KK(A, \cdot)$ in the separable case, we see that the map

$$(B.9) \quad KK(A, I_0) \rightarrow \varinjlim_{J_0 \text{ sep.}} KK(A, J_0)$$

given by $KK(A, \theta|_{I_0}^{J_0})$ does not depend on the choice of J_0 . We may call this map $KK(A, \theta|_{I_0}): KK(A, I_0) \rightarrow KK(A, J)$.

The system of maps $(KK(A, \theta|_{I_0}))_{I_0 \subseteq I \text{ sep.}}$ is compatible with the connecting maps defining $KK(A, I)$ (by functoriality for separable codomains) and therefore they induce a map $KK(A, \theta): KK(A, I) \rightarrow KK(A, J)$. It is a straightforward check that this definition makes $KK(A, \cdot)$ a covariant functor, using that it is a covariant functor in the separable case. It is likewise straightforward to check that the functoriality in the first and second variable commute, so that $KK(\cdot, \cdot)$ is a bifunctor. $\square$

We end this subsection by noting that Cuntz's picture of KK -theory agrees with the inductive limit definition in the non-separable case. (The left-hand side of (B.10) below is, by definition, the inductive limit over separable subalgebras of I .) This justifies that the inductive limit picture agrees with the usual picture in the case of σ -unital but non-separable algebras I .

Proposition B.6. Let A and I be C^* -algebras with A separable. Then

$$(B.10) \quad KK(A, I) \cong KK_c(A, I),$$

and the isomorphism is natural in both variables.

In particular, if I is σ -unital, then the definitions of $KK(A, I)$ given in Definitions 5.2 and 5.4 agree.

Proof. Using Proposition B.2, there is a natural identification

$$(B.11) \quad KK(A, I) \cong \varinjlim_{I_0 \text{ sep.}} KK_c(A, I_0).$$

We will therefore prove the existence of a natural isomorphism between the right-hand side and $KK_c(A, I)$.

Given separable C^* -subalgebras $I_0 \subseteq I_1 \subseteq I$, we have

$$(B.12) \quad KK_c(A, \iota_{I_1 \subseteq I}) \circ KK_c(A, \iota_{I_0 \subseteq I_1}) = KK_c(A, \iota_{I_0 \subseteq I}),$$

since $KK_c(A, \cdot)$ is designed to be functorial without restriction on the codomain. Therefore, we may define

$$(B.13) \quad \Theta := \varinjlim_{I_0 \text{ sep.}} KK_c(A, \iota_{I_0 \subseteq I}) : \varinjlim_{I_0 \text{ sep.}} KK_c(A, I_0) \rightarrow KK_c(A, I),$$

a homomorphism that is natural in both variables.

For surjectivity, consider $\rho : qA \rightarrow I \otimes \mathcal{K}$. Let I_1 be a separable C^* -subalgebra of I such that $\rho(qA) \subseteq I_1 \otimes \mathcal{K}$. Then ρ corestricts to $I_1 \otimes \mathcal{K}$, thus giving an element $\kappa := [\rho]_{KK_c(A, I_1)}^{I_1 \otimes \mathcal{K}} \in \varinjlim_{I_0 \text{ sep.}} KK_c(A, I_0)$. We have

$$(B.14) \quad \Theta(\kappa) = KK_c(A, \iota_{I_1 \subseteq I})([\rho]_{KK_c(A, I_1)}^{I_1 \otimes \mathcal{K}}) = [\rho]_{KK_c(A, I)}$$

by (B.2).

For injectivity, suppose that I_1 is a separable C^* -subalgebra of I and $\rho : qA \rightarrow I_1 \otimes \mathcal{K}$ is a $*$ -homomorphism such that

$$(B.15) \quad \Theta([\rho]_{KK_c(A, I_1)}) = 0.$$

That is to say, $\iota_{I_1 \subseteq I} \circ \rho$ is homotopic to the zero $*$ -homomorphism within the space of $*$ -homomorphisms $qA \rightarrow I \otimes \mathcal{K}$. Let $(\rho_t : qA \rightarrow I \otimes \mathcal{K})_{t \in [0, 1]}$ be such a homotopy, so that $\rho_0 = \iota_{I_1 \subseteq I} \circ \rho$ and $\rho_1 = 0$. Then let $I_2 \subseteq I$ be a separable C^* -subalgebra containing I_1 and $\rho_t(A) \subseteq I_2 \otimes \mathcal{K}$ for $t \in [0, 1] \cap \mathbb{Q}$. By density, it follows that $(\rho_t)_{t \in [0, 1]}$ corestricts to a homotopy of $*$ -homomorphisms $qA \rightarrow I_2 \otimes \mathcal{K}$, from $\iota_{I_1 \subseteq I_2} \circ \rho$ to 0. Thus

$$(B.16) \quad KK_c(A, \iota_{I_1 \subseteq I_2})([\rho]_{KK_c(A, I_1)}) = [\iota_{I_1 \subseteq I_2} \circ \rho]_{KK_c(A, I_2)} = 0.$$

This shows that $[\rho]_{KK_c(A, I_1)}$ represents 0 in $\varinjlim_{I_0 \text{ sep.}} KK(A, I_0)$, as required.

The last statement follows using Proposition B.2. $\square$

Remark B.7. If A and I are separable C^* -algebras and J is a σ -unital and non-separable C^* -algebra, then the Kasparov product $KK(A, I) \times KK(I, J) \rightarrow KK(A, J)$ agrees with taking the limit of Kasparov products over separable subalgebras J_0 of J . To see this, let $\kappa \in KK(A, I)$ and $\kappa' \in KK(I, J)$. According to Definition 5.4, we may find a separable subalgebra J_0 of J such that κ' comes from $\kappa'_0 \in KK(A, J_0)$ — that is, $\kappa' = KK(A, \iota_{J_0 \subseteq J})(\kappa'_0) = [\iota_{J_0 \subseteq J}]_{KK(J_0, J)} \circ \kappa'_0$. Then

$$(B.17) \quad \begin{aligned} \kappa' \circ \kappa &= ([\iota_{J_0 \subseteq J}]_{KK(J_0, J)} \circ \kappa'_0) \circ \kappa \\ &= [\iota_{J_0 \subseteq J}]_{KK(J_0, J)} \circ (\kappa'_0 \circ \kappa) \\ &= KK(A, \iota_{J_0 \subseteq J})(\kappa'_0 \circ \kappa). \end{aligned}$$

The above argument only uses that the Kasparov product is associative and that $KK(A, \iota_{J_0 \subseteq J})(\cdot) = [\iota_{J_0 \subseteq J}]_{KK(J_0, J)} \circ (\cdot)$, so it holds for the Kasparov product as defined in any picture.

B.3. KL -theory outside the σ -unital setting. Next we turn to KL , aiming to prove Proposition 5.7 that $KL(A, I)$ can also be described as a limit over separable C^* -subalgebras.

Lemma B.8. *Let A and I be C^* -algebras with A separable. Then*

$$(B.18) \quad Z_{KK(A, I)} = \varinjlim_{I_0 \text{ sep.}} Z_{KK(A, I_0)},$$

$$\text{inside } \varinjlim_{I_0 \text{ sep.}} KK(A, I_0).$$

Proof. Suppose $I_0 \subseteq I$ is separable and $\kappa_0 \in Z_{KK(A, I_0)}$. Then there exists an element $\bar{\kappa}_0 \in KK(A, C(\bar{\mathbb{N}}, I_0))$ such that $KK(A, \text{ev}_n)(\bar{\kappa}_0) = 0$ for $n \in \mathbb{N}$ and $KK(A, \text{ev}_\infty)(\bar{\kappa}_0) = \kappa_0$. We see that $KK(A, \iota_{I_0 \subseteq I})(\kappa_0) \in Z_{KK(A, I)}$ by pushing this forward to $C(\bar{\mathbb{N}}, I)$.

On the other hand, suppose that $\kappa \in Z_{KK(A, I)}$. Then there exists an element $\bar{\kappa} \in KK(A, C(\bar{\mathbb{N}}, I))$ such that $KK(A, \text{ev}_n)(\bar{\kappa}) = 0$ for all $n \in \mathbb{N}$ and $KK(A, \text{ev}_\infty)(\bar{\kappa}) = \kappa$.

Using the definition of $KK(A, C(\bar{\mathbb{N}}, I))$ as a limit, there exists a separable subalgebra $J_0 \subseteq C(\bar{\mathbb{N}}, I)$ and $\bar{\kappa}_0 \in KK(A, J_0)$ such that $KK(A, \iota_{J_0 \subseteq C(\bar{\mathbb{N}}, I)})(\bar{\kappa}_0) = \bar{\kappa}$. By increasing J_0 , we can arrange that J_0 has the form $C(\bar{\mathbb{N}}, I_0)$ for some separable $I_0 \subseteq I$, and that $KK(A, \text{ev}_n)(\bar{\kappa}_0) = 0$ in $KK(A, I_0)$ for all $n \in \mathbb{N}$. Setting $\kappa_0 := KK(A, \text{ev}_\infty)(\bar{\kappa}_0) \in Z_{KK(A, I_0)}$, we then have $\kappa = KK(A, \iota_{I_0 \subseteq I})(\kappa_0)$, as required. $\square$

Proof of Proposition 5.7. Functoriality of KL descends from functoriality of KK (Proposition B.5). Given separable C^* -subalgebras $I_0 \subseteq I_1 \subseteq I_2 \subseteq I$, functoriality applied to the inclusions $I_0 \subseteq I_1 \subseteq I_2$ tells us that the limit $\varinjlim_{I_0 \text{ sep.}} KL(A, I_0)$ exist,

and functoriality applied to $I_0 \subseteq I_1 \subseteq I$ provides a map

$$(B.19) \quad \Theta := \varinjlim_{I_0 \text{ sep.}} KL(A, \iota_{I_0 \subseteq I}): \varinjlim_{I_0 \text{ sep.}} KL(A, I_0) \rightarrow KL(A, I).$$

By naturality of the map from KK to KL , we get a commuting diagram

$$(B.20) \quad \begin{array}{ccc} \varinjlim_{I_0 \text{ sep.}} KK(A, I_0) & \xlongequal{\quad} & KK(A, I) \\ \downarrow & & \downarrow \\ \varinjlim_{I_0 \text{ sep.}} KL(A, I_0) & \xrightarrow{\quad \Theta \quad} & KL(A, I). \end{array}$$

In particular, this shows that Θ is surjective.

For injectivity, let $I_0 \subseteq I$ and $\lambda \in KL(A, I_0)$ satisfy $KL(A, \iota_{I_0 \subseteq I})(\lambda) = 0$. Let $\kappa \in KK(A, I_0)$ be a lift of λ . Then $KK(A, \iota_{I_0 \subseteq I})(\kappa) \in Z_{KK(A, I)}$, so by Lemma B.8, there exists $I_1 \subseteq I$ separable such that $I_0 \subseteq I_1$ and $KK(A, \iota_{I_0 \subseteq I_1})(\kappa) \in Z_{KK(A, I_1)}$. Therefore,

$$(B.21) \quad KL(A, \iota_{I_0 \subseteq I_1})(\lambda) = 0,$$

as required. $\square$

Now we recall and prove Propositions 5.5 and 5.8.

Proposition B.9. *Let A and I be C^* -algebras with A separable, let E be a C^* -algebra containing I as an ideal, and let $(\phi, \psi): A \rightrightarrows E \triangleright I$ be an (A, I) -Cuntz pair.*

(i) *Let $\iota_I^{(2)}: I \rightarrow M_2(I)$ be the top-left corner inclusion. Then the induced maps $KK(A, \iota_I^{(2)}): KK(A, I) \rightarrow KK(A, M_2(I))$ and $KL(A, \iota_I^{(2)}): KL(A, I) \rightarrow KL(A, M_2(I))$ are isomorphisms. Moreover,*

$$(B.22) \quad \begin{aligned} KK(A, \iota_I^{(2)})([\phi, \psi]_{KK(A, I)}) &= [\iota_E^{(2)} \circ \phi, \iota_E^{(2)} \circ \psi]_{KK(A, M_2(I))}, \quad \text{and} \\ KL(A, \iota_I^{(2)})([\phi, \psi]_{KL(A, I)}) &= [\iota_E^{(2)} \circ \phi, \iota_E^{(2)} \circ \psi]_{KL(A, M_2(I))}. \end{aligned}$$

(ii) *For any unitary $u \in I^\dagger$,*

$$(B.23) \quad \begin{aligned} [\phi, \psi]_{KK(A, I)} &= [\text{Ad } u \circ \phi, \psi]_{KK(A, I)}, \quad \text{and} \\ [\phi, \psi]_{KL(A, I)} &= [\text{Ad } u \circ \phi, \psi]_{KL(A, I)}. \end{aligned}$$

(iii) *$[\phi, \psi]_{KK(A, I)} + [\psi, \phi]_{KK(A, I)} = 0$ and $[\phi, \psi]_{KL(A, I)} + [\psi, \phi]_{KL(A, I)} = 0$.*

(iv) *If $J \triangleleft F$ and $\theta: I \rightarrow J$ is a $*$ -homomorphism that extends to $\bar{\theta}: E \rightarrow F$, then $KK(A, \theta)([\phi, \psi]_{KK(A, I)}) = [\bar{\theta} \circ \phi, \bar{\theta} \circ \psi]_{KK(A, J)}$ and similarly $KL(A, \theta)([\phi, \psi]_{KL(A, I)}) = [\bar{\theta} \circ \phi, \bar{\theta} \circ \psi]_{KL(A, J)}$.*

(v) *Let A be nuclear, and suppose*

$$(B.24) \quad e: 0 \longrightarrow I \xrightarrow{j_e} E \xrightarrow{q_e} D \longrightarrow 0$$

is an extension of C^ -algebras. Then the sequence*

$$(B.25) \quad KK(A, I) \xrightarrow{KK(A, j_e)} KK(A, E) \xrightarrow{KK(A, q_e)} KK(A, D)$$

is exact. The same holds when A is KK -equivalent to a nuclear C^ -algebra.*

Proof. (i): In the case when I is separable, the KK isomorphism result follows from [10, Example 17.8.2(c)], for example. The formula for $KK(A, \iota_I^{(2)})$ follows from Proposition B.3. For general I , the result follows from the definition of $KK(A, \iota_I^{(2)})$ as a limit of maps $KK(A, \iota_{I_0}^{(2)})$ (where $I_0 \subseteq I$ is separable), in the same spirit as the proof of Proposition B.5.

(ii): First suppose that I is separable. We may assume that the scalar part of u is $1_{I^\dagger}$. Then $u \oplus u^*$ is homotopic to $1_{I^\dagger} \oplus 1_{I^\dagger}$ via the unitary path

$$(B.26) \quad [0, \pi/2] \ni t \mapsto \begin{pmatrix} \cos^2(t)(u - 1_{I^\dagger}) + 1_{I^\dagger} & -\cos(t)\sin(t)(u - 1_{I^\dagger}) \\ \cos(t)\sin(t)(u^* - 1_{I^\dagger}) & \cos^2(t)(u^* - 1_{I^\dagger}) + 1_{I^\dagger} \end{pmatrix}$$

which is constantly $1_{I^\dagger} \oplus 1_{I^\dagger}$ modulo $M_2(I)$, since $u - 1_{I^\dagger} \in I$. Conjugating with this unitary path in the first entry induces a homotopy of $(A, M_2(I))$ -Cuntz pairs from $((\text{Ad } u \circ \phi) \oplus 0, \psi \oplus 0) = (\text{Ad}(u \oplus u^*) \circ (\phi \oplus 0), \psi \oplus 0)$ to $(\phi \oplus 0, \psi \oplus 0)$. Thus (using the identification in (i)), $[\phi, \psi]_{KK(A, I)} = [\text{Ad } u \circ \phi, \psi]_{KK(A, I)}$.

For non-separable I , the result follows from the separable case and the definition $KK(A, I)$ as a direct limit (see Definition 5.4). The KL version follows by taking quotients.

(iii): When I is separable, the KK formula can be found in [152, Proposition 4.1.5], for example. For non-separable I , the result follows from the separable case, and the KL version follows by taking quotients.

(iv): When I and J are separable, this is the last part of Proposition B.3. For the non-separable case, set $E_0 := C^*(\phi(A) \cup \psi(A)) \subseteq E$ and $F_0 := C^*(\theta(I_0)) \subseteq F$.

Then let $I_0 \triangleleft E_0$ and $J_0 \triangleleft F_0$ be the ideals generated by $(\phi - \psi)(A)$ and $\theta(I_0)$, respectively. Then Proposition B.3 gives

$$(B.27) \quad KK(A, \theta|_{I_0}^{J_0})([\phi|_{I_0}^{E_0}, \psi|_{I_0}^{E_0}]_{KK(A, I_0)}) = [(\bar{\theta} \circ \phi)|^{F_0}, (\bar{\theta} \circ \psi)|^{F_0}]_{KK(A, J_0)}.$$

By the definition of $KK(A, \theta)$ in the proof of Proposition B.5, it follows that $KK(A, \theta)([\phi, \psi]_{KK(A, I)}) = [\bar{\theta} \circ \phi, \bar{\theta} \circ \psi]_{KK(A, J)}$. The computation for KL follows by taking quotients.

(v): This is standard when E is separable, being a fragment of the six-term exact sequence (see [10, Theorem 19.5.7 and Example 19.5.2(a)], or [10, Exercise 20.10.2(f)] for the case that A is KK -equivalent to a nuclear C^* -algebra, for example). For the non-separable case, as $\text{im } KK(A, j_e) \subseteq \ker KK(A, q_e)$ by functoriality, the only thing to check is that $\ker KK(A, q_e) \subseteq \text{im } KK(A, j_e)$. Fix $\kappa \in \ker KK(A, q_e)$. As in the proof of Proposition B.5, there exist separable subalgebras E_0 and D_0 of E and D , respectively, such that $q_e(E_0) \subseteq D_0$, κ comes from some element $\kappa_0 \in KK(A, E_0)$, and $KK(A, q_e|_{E_0}^{D_0})(\kappa_0) = 0$ in $KK(A, D_0)$. Setting $I_0 := I \cap E_0$, we have an extension of separable C^* -algebras

$$(B.28) \quad e_0: 0 \rightarrow I_0 \xrightarrow{j_{e_0}} E_0 \xrightarrow{q_{e_0}} D_0 \rightarrow 0.$$

Thus by the separable case, there exists $\kappa'_0 \in KK(A, I_0)$ such that $KK(A, j_{e_0})(\kappa'_0) = \kappa_0$. Letting $\kappa' \in KK(A, I)$ be the image of κ'_0 , it follows that $KK(A, j_e)(\kappa') = \kappa$. $\square$

B.4. The universal multicoefficient theorem. Now we turn to the natural map

$$(B.29) \quad \Gamma^{(A, I)}: KK(A, I) \rightarrow \text{Hom}(K_*(A), K_*(I)).$$

In the case that I is σ -unital, the natural identification of $KK(\mathbb{C}, S^i I)$ with $K_i(I)$ for $i = 0, 1$ ([10, Proposition 17.5.5]) combined with the Kasparov product provides this map. For $\phi: A \rightarrow I$, since $KK(\mathbb{C}, S^i \phi)$ identifies with $K_i(\phi)$ and since taking the Kasparov product with $[\phi]_{KK(A, I)}$ corresponds to applying $KK(\cdot, \phi)$ ([10, Example 18.4.2(a)]), we have

$$(B.30) \quad \Gamma_i^{(A, I)}([\phi]_{KK(A, I)}) = K_i(\phi).$$

For I non-separable, since $K_*(A)$ is countable, we have a natural isomorphism

$$(B.31) \quad \text{Hom}(K_*(A), K_*(I)) \cong \varinjlim_{I_0 \text{ sep.}} \text{Hom}(K_*(A), K_*(I_0)),$$

(see [268, Proposition 1.10], for example), and upon making this identification we may define

$$(B.32) \quad \Gamma^{(A, I)} := \varinjlim_{I_0 \text{ sep.}} \Gamma^{(A, I_0)}.$$

When I is both non-separable and σ -unital, a priori this gives two competing definitions for $\Gamma^{(A, I)}$. By Remark B.7, they agree.

More generally, identifying $KK(\mathbb{I}_n, S^i I)$ naturally with $K_i(I; \mathbb{Z}/n)$ as per the picture in [69] (using continuity of $K_i(\cdot; \mathbb{Z}/n)$ for non-separable I), we likewise have

$$(B.33) \quad \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(I)) \cong \varinjlim_{I_0 \text{ sep.}} \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(I_0)).$$

Injectivity of the map $\varinjlim_{I_0 \text{ sep.}} \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(I_0)) \rightarrow \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(I))$ follows

directly from injectivity of the corresponding map (B.31). For surjectivity, given an element $\underline{\alpha}$ of $\text{Hom}_\Lambda(\underline{K}(A), \underline{K}(I))$, first by surjectivity of the corresponding map (B.31), we can lift $\underline{\alpha}$ to a graded group homomorphism $\underline{K}(A) \rightarrow \underline{K}(I_0)$ for some separable $I_0 \subseteq I$. Then using injectivity of the map (B.31), we can enlarge I_0 to ensure compatibility with each of the countably many Bockstein operations.

This gives a homomorphism

$$(B.34) \quad \Gamma_\Lambda^{(A,I)} : KK(A, I) \rightarrow \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(I)).$$

Just as with $\Gamma^{(A,I)}$, the two possible definitions of $\Gamma_\Lambda^{(A,I)}$ for I σ -unital and non-separable agree. When A satisfies the UCT, Dadarlat and Loring's universal multicoefficient theorem says that $\Gamma_\Lambda^{(A,I)}$ descends to an isomorphism when taking the KL quotient on the left. While their statement only allows separable codomains, the result extends to the non-separable case.

Proof of Theorem 8.5. First let us justify that $\tilde{\Gamma}_\Lambda^{(A,I)}$ is well-defined, without assuming that A satisfies the UCT. When I is separable, this is implicit in the paragraph following [64, Eq. (13)]. The idea is that for $\kappa \in KK(A, C(\mathbb{N}, I))$, the sequence

$$\left(\Gamma_\Lambda^{(A,I)}(KK(A, \text{ev}_n)(\kappa))_i^{(m)} : K_i(A; \mathbb{Z}/m) \rightarrow K_i(I; \mathbb{Z}/m) \right)_{n=1}^\infty$$

of homomorphisms is eventually equal to $\Gamma_\Lambda^{(A,I)}(KK(A, \text{ev}_\infty)(\kappa))_i^{(m)}$. Hence any element of $Z_{KK(A,I)}$ is mapped to zero by $\Gamma_\Lambda^{(A,I)}$.

For I non-separable, since $Z_{KK(A,I)} = \varinjlim_{I_0 \text{ sep.}} Z_{KK(A, I_0)}$ (Lemma B.8), it follows

from the separable case that $\Gamma_\Lambda^{(A,I)}$ vanishes on $Z_{KK(A,I)}$.

Now, suppose that A satisfies the UCT. If I is separable, then $\tilde{\Gamma}_\Lambda^{(A,I)}$ is an isomorphism by [64, Theorem 4.1] (see also [69]). For the non-separable case, we have

$$(B.35) \quad \tilde{\Gamma}_\Lambda^{(A,I)} = \varinjlim_{I_0 \text{ sep.}} \tilde{\Gamma}_\Lambda^{(A, I_0)}.$$

Since $\tilde{\Gamma}_\Lambda^{(A, I_0)} : KL(A, I_0) \rightarrow \text{Hom}_\Lambda(\underline{K}(A), \underline{K}(I_0))$ is an isomorphism for each I_0 , it follows that $\tilde{\Gamma}_\Lambda^{(A,I)}$ is as well. $\square$

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